Rosco®

A LeeBoy Company

OPERATIONS, SERVICE AND PARTS MANUAL

RB50 BROOM

Manual No. 1017258-03

Rev. B

This manual applies to Serial Number 255027 and above.

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Disclaimer

All information, illustrations and specifications in this manual are based on the latest information available at the time of publishing. The illustrations used in this manual are intended as representative reference views only. Moreover, because of our continuous product improvement policy, we may modify information, illustrations and/or specifications to explain and/or exemplify a product, service or maintenance improvement. We reserve the right to make any change at any time without notice. ST Engineering LeeBoy, Inc., dba LeeBoy, ("LeeBoy"), is referred to as LeeBoy throughout this manual.

Title 40, Code of Federal Regulations (CFR) 1068

This product meets certified-emission requirements set by the EPA (Environmental Protection Agency), governed by Title 40 CFR 1068, which specifies actions that are prohibited by law and lists civil penalties for noncompliance. As part of those regulations, modification or rendering inoperative any emission-related component can subject you to government penalties (and void your warranty). Tampering with emission controls is in violation of federal law, and can result in civil penalties of up to $3,750 each day an engine or piece of equipment is operated in violation.

Please be aware that you are responsible for maintaining the machine and the certified emission engine installation. Failure to comply to comply could result in penalties as listed above and void the warranty on this engine and this machine.

For more information, visit: https://www.epa.gov/laws-regulations/regulations

California Proposition 65 Warning

Diesel engine exhaust and some of its constituents are known to the State of California to cause cancer, birth defects, and other reproductive harm.

Battery posts, terminals and other related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and other reproductive harm. Wash hands after handling.

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NOTES

Thank you for purchasing the Rosco RB50 Broom. We wish you many years of safe and efficient operation of your machine.

READ THIS MANUAL PRIOR TO OPERATING the unit. This is an important part of the machine and should be kept in the dedicated storage container at all times. Though you may be familiar with similar equipment, you MUST read and understand this manual before operating this machine to help prevent injury or damage to the unit. If this manual becomes lost or damaged, contact your authorized Rosco LeeBoy dealer immediately to order a replacement (see **Contact Information** in **Section 2**).

This manual is intended as a guide for the safe and efficient use for proper operation and maintenance of the machine. This manual contains information that was available at the time of printing and are subject to change without notice.

Use this manual along with all related supplemental books, engine and transmission manuals included with your machine. Related Service Bulletins should be reviewed to provide information regarding some of the recent changes. Contact your authorized Rosco LeeBoy dealer for the latest available information.

Section 1 - Safety: Contains general and specific safety guidelines for product and safety label locations.

Section 2 - Information and Specifications: Contains warranty, contact information, machine specification tables, and machine dimensions.

Section 3 - Component Location: Contains overview of major component locations and functions.

Section 4 - Operation: Contains instructions for safe operation and information for optional equipment.

Section 5 - Maintenance: Contains routine maintenance procedures, mechanical adjustments, component replacement and troubleshooting charts for common problems and corrections. (For specific engine maintenance procedures, refer to the engine manufacturer manual.)

Section 6 - Schematics: Contains electrical and hydraulic schematics for product functionality.

Section 7 - Illustrated Parts List (IPL): Contains parts numbers and illustrations for serviceable components.

LeeBoy is proud to be ISO 9001 certified. The International Standards Organization (ISO) establishes guidelines to ensure that products and services are safe, reliable, and of good quality. ISO certifies companies who demonstrate compliance with all aspects of product safety, customer satisfaction, efficiency, environmental stewardship and social responsibility. Our teams work hard to deliver quality industrial machines that exceed customer expectations and we strive for continuous improvement in everything we do. The LeeBoy family of companies is committed to total quality management with a strong focus on meeting customer needs.

LeeBoy is also proud to be an accredited ANAB manufacturer, which is a certification process comprised of quality standards established by the American National Standards Institute (ANSI) and the American Society for Quality (ASQ). The ANSI-ASQ National Accreditation Board plays an important role in ensuring the safety and quality of goods and services, along with protecting the environment.

NOTES

Section 1

SAFETY

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NOTES

This manual provides important information to familiarize you with safe operating and maintenance procedures. Even though you may be familiar with similar equipment, you MUST read and understand this manual before operating the Rosco RB50 Broom and follow its instructions when operating the broom.

Safety is everyone's business and our top concern. Knowing the guidelines covered in this section will help ensure your safety, the safety of those around you, as well as proper broom operation.

Keep safety decals in good condition. If safety decals become missing or damaged, replace them immediately. Replacement safety decals are available from your authorized Rosco/LeeBoy dealer.

You can find more information about occupational health and safety in the industry on the internet. A few resources are listed below:

www.osha.gov

cdc.gov

www.safety.fhwa.dot.gov/

LOOK FOR THESE SYMBOLS THROUGHOUT THIS MANUAL. THESE ITEMS ARE EXTREMELY IMPORTANT FOR THE SAFETY OF YOU AND YOUR COWORKERS. READ AND UNDERSTAND THOROUGHLY. HEED THE WARNINGS AND FOLLOW THE INSTRUCTIONS.

⚠ DANGER Indicates a hazardous situation which, if not avoided, *will* result in death or serious injury.

⚠ WARNING Indicates a hazardous situation which, if not avoided, *could* result in death or serious injury.

⚠ CAUTION Indicates a hazardous situation which, if not avoided, *could* result in minor or moderate injury.

NOTICE **Indicates a situation which can cause damage to the equipment, personal property and/or the environment, or cause the machine to operate improperly.**

NOTE: Indicates a procedure, practice or condition that should be followed in order for the machine or component to function in the manner intended.

SAFETY PRECAUTIONS

⚠ CAUTION

The safety messages that follow have CAUTION level hazards.

Pre-Operation Hazard

Read and understand this manual before operating or servicing the engine to ensure that safe operating practices and maintenance procedures are followed.

- Never permit anyone to service or operate the machine without proper training.
- Safety signs and labels are additional reminders for safe operating and maintenance techniques.
- Contact your authorized Rosco/LeeBoy dealer for additional training.
- Make sure you are aware of all laws and regulations that are in effect for the location in which the broom is operated.
- Make sure you have all necessary licenses to operate the broom.

The safety messages that follow have CAUTION level hazards.

Poor Lighting Hazard

The work area must be well lit to ensure safe and proper operation.

- Ensure that the work area is adequately illuminated.
- Always install wire cages on portable safety lights.

Tool Hazard

Always use tools appropriate for the task at hand and use the correct size tool for loosening or tightening machine parts.

⚠ DANGER

The safety messages that follow have DANGER level hazards.

Power Lines Hazard

If your machine comes into contact with electric power lines, observe the following:

- Stay in the operators seat.
- Warn other workers to stay away and do not touch any control or any part of the machine.
- If contact can be broken, drive the machine away from the danger zone.
- If contact cannot be broken, stay in the operators seat until told that power is off.
- Failure to observe these directions could result in electrocution or death.

Electrocution Hazard

Disconnect the battery before welding anywhere on the machine.

Suffocation Hazard

Carbon monoxide poisoning is a serious condition that occurs as a result of improper ventilation.

- Never operate the internal combustion engine on this machine in an enclosed area with poor ventilation. Ensure proper ventilation to reduce risk of carbon monoxide poisoning or death.

Exhaust Hazard

All internal combustion engines create carbon monoxide gas during operation and special precautions are required to avoid carbon monoxide poisoning:

- Never block windows, vents or other means of ventilation.
- Always ensure that all connections are tightened to specifications after repair is made to the exhaust system.

⚠ WARNING

The safety messages that follow have WARNING level hazards.

Crush Hazard

Keep bystanders away from work area before and during operation.

Modification Hazard

Never modify the machine without the written consent of Rosco/LeeBoy . Any modification can affect safe operation and may cause personal injury or even death.

Exposure Hazard

Operators must be aware of their work environment and the equipment needed to work safely.

- Always wear personal protective equipment, including appropriate clothing, gloves, work shoes, and protection for eyes and ears, as required by the task at hand.

Explosion Hazard

While the engine is running or the battery is charging, hydrogen gas is being produced and can be easily ignited. Keep the area around the battery well-ventilated and keep sparks, open flame, and any other form of ignition out of the area.

- Always disconnect the negative (-) battery cable before servicing the broom.
- Do not start the engine by shorting the starter circuit or any other starting method not stated in this manual. Only use the starting procedure as described in this manual to start the engine.
- Never charge a frozen battery. Always slowly warm the battery to room temperature before charging.

Fire and Explosion Hazard

- Diesel fuel is flammable and explosive under certain conditions.
- Never use a shop rag to catch fuel.
- Wipe up all spills immediately.
- Never refuel with the engine running.
- Store any containers containing fuel in a well-ventilated area, away from any combustibles or sources of ignition.

Fire Hazard

When operating machinery there is a risk for fire. Always ensure appropriate safety equipment is available.

- Keep a charged fire extinguisher within reach when working in an environment where a fire may occur.
- Have all fire extinguishers checked periodically for proper operation and/or readiness.
- Always read and follow safety-related precautions found on containers of hazardous substances like parts cleaners, primers, sealants and sealant removers.
- Undersized wiring systems can cause electrical fires.

Entanglement/Sever Hazard

Verify there are no people, obstacles or other equipment near the machine before starting the engine. Sound the horn as a warning before starting the engine.

If the engine must be serviced while it is operating, remove all jewelry and tie back long hair before operating or servicing the machine.

- Keep hands, other body parts, and clothing away from moving/rotating parts.
- Always stop the engine before beginning service. Before maintenance, remove negative battery cable from battery post to ensure vehicle is not operated during maintenance.
- Verify that all guards and covers are properly attached before starting the engine. DO NOT start the engine if any guards or covers are not properly installed on the machine.

- If you must run the engine during maintenance procedures, make sure you have a helper to keep bystanders clear of the machine and make observations of moving parts as requested by the operator.
- Always turn the start switch to the OFF position after operation is complete and remove the key from the switch. Keep the key in your possession when the broom is not operating.
- Attach a “Do Not Operate” tag near the key switch while performing maintenance on the equipment.
- Never operate the engine while wearing a headset to listen to a radio or music because it will be difficult to hear the warning signals.
- Always start the engine and operate the controls while seated in the operators seat.

Alcohol and Drug Hazard

Never operate the machine while under the influence of alcohol, drugs, or when ill.

Piercing Hazard

High-pressure hydraulic fluid or fuel can penetrate your skin and result in serious injury. Avoid skin contact with high-pressure hydraulic fluid or diesel fuel spray caused by a hydraulic or fuel system leak such as a broken hydraulic hose or fuel injection line.

- If you are exposed to high-pressure hydraulic fluid or fuel spray, obtain prompt medical treatment.
- Never check for a hydraulic fluid or fuel leak with your hands. Always use a piece of wood or cardboard. Have your authorized LeeBoy dealer or distributor repair the damaged parts.

Flying Object Hazard

Always wear eye protection when cleaning the machine with compressed air or high-pressure water.

Dust, flying debris, compressed air, pressurized water or steam may cause eye injury.

Coolant Hazard

Coolant must be handled properly to ensure operator safety.

- Wear eye protection and rubber gloves when handling engine coolant.
- If contact with the eyes occurs, flush eyes with clean water for 15 minutes.
- If contact with skin occurs, wash immediately with soap and clean water.

Burn Hazard

Some of the machine’s surfaces become very hot during operation and shortly after shutdown.

- Keep hands and other body parts away from hot machine surfaces.
- Handle hot components with heat-resistant gloves.

NOTICE

The safety messages that follow have NOTICE level hazards.

Any part that is found defective as a result of inspection or any part whose measured value does not satisfy the standard or limit must be replaced.

Always tighten components to the specified torque. Loose parts can cause damage to the machine or cause it to operate improperly.

Only use replacement parts approved by Rosco/ LeeBoy. Other replacement parts may affect warranty coverage.

Follow the guidelines of the EPA or other governmental agencies for the proper disposal of hazardous materials such as engine oil, diesel fuel, and engine coolant. Consult the local authorities or reclamation facility.

Dispose of hazardous materials in accordance with all applicable laws and regulations. Never dispose of hazardous materials by dumping them into a sewer, on the ground, or into groundwater or waterways.

Clean all accumulated dirt and debris away from the body of the broom and its components before you inspect the broom or perform preventive maintenance procedures or repairs. Operating a machine with accumulated dirt and debris will cause premature wear of components.

If any alert indicator illuminates during operation, stop the engine immediately. Determine the cause and repair the problem before continuing to operate.

MACHINE PRECAUTIONS

Hot Material Precautions

- Allow machine to cool before repairing or maintaining working components.
- DO NOT remove radiator cap, drain plugs, service grease fittings, or pressure taps while engine is hot. Add coolant to the radiator and perform other services only when the engine is stopped and fully cooled.

Hydraulic Systems Precautions

- Ensure all components are in good working condition. Replace any worn, cut, abraded, flattened or crimped hoses and metal lines.
- DO NOT attempt makeshift repairs using tape, clamps or cements. The hydraulic system operates under extremely high pressure and such repairs could cause serious injury.
- Wear proper hand and eye protection when checking for a high pressure leak. Use a piece of wood or cardboard as a back stop to isolate and identify leaks.

⚠WARNING Hydraulic oil under pressure can cause serious personal injury. Check for oil leaks with a piece of cardboard. DO NOT expose hands to possible high-pressure oil. Turn off engine before attempting to tighten oil lines and fittings.

- Escaping pressurized hydraulic fluid has force sufficient to penetrate the skin, which could cause serious personal injury. Ensure all pressure is relieved before disconnecting line, hoses or valves.
- If injury from concentrated high pressure steam or hydraulic fluid occurs, seek medical attention immediately. Injuries resulting from hydraulic fluid penetrating the skin's surface can result in serious infections or toxic reactions.

Refueling Precautions

- Do NOT overfill the fuel tank as overflow creates a fire hazard when spilled on hot components.
- DO NOT smoke when refueling and never refuel when the engine is running. Fuel is highly flammable and should be handled with care. Death or serious injury can occur due to explosion and/or fire.
- DO NOT fill tank to capacity. Allow room for expansion to reduce the risk of fuel expanding and spilling from the tank.
- Tighten fuel cap securely. Should fuel cap be lost, replace it with an original manufacturer's approved cap. Pressurization of the tank may result from use of non-approved cap.
- Prevent fires by keeping the machine clean of accumulated debris, grease, and spilled fuel.
- Use ultra-low sulfur diesel fuel (ULSD) only.

Battery Precautions

- Keep all sparks and flames away from batteries, as gas given off by electrolytes is explosive.
- Acid propelled by an explosion can cause blindness if it comes in contact with eyes. Always wear safety glasses when working near batteries.
- If you come in contact with battery electrolyte solution, wash off immediately. Chemicals can cause burns.
- Always disconnect the battery ground cable before working on the electrical system to avoid injury from spark or short circuit. Electrical shock and burns can occur.
- To avoid electrolyte loss, DO NOT tip batteries more than 45 degrees.
- Use jumper cables ONLY as directed in Section 5 (Maintenance). Improper use can result in battery explosion or unexpected machine motion.

Starting and Stopping Precautions

- Check all around the machine to ensure there are no people working on the machine or in its path before starting. DO NOT start until area is clear. Death or serious injury can occur to bystanders from being crushed under a moving machine.

⚠WARNING **DO NOT operate the engine in an enclosed area without proper ventilation. Exhaust gasses are odorless and deadly.**

Parking Precautions

- Park machine on level ground whenever possible and apply the parking brake.
- Before leaving operator's station:
  1. Turn off all accessories.
  2. Set parking brake.
  3. Shut off engine.
- Always remove ignition key when leaving the machine parked or unattended.

Operating Precautions

- Always comply with local regulations regarding industrial equipment on public roads and highways.
- Know and use the hand signals required for a particular job. Know who has the responsibility for signaling.
- Make sure that all lights and reflectors comply with state and local regulations. Ensure that they are clean, in good working order, and can be seen clearly by all traffic.
- DO NOT ride on attachments.
- Check all gauges and warning instruments for proper operation. If malfunctions are found, shut down the machine and report the problem for resolution. If the failure causes loss of steering control, loss of brake control, or loss of engine power, stop motion as quickly as possible. Apply parking brake and turn off the machine. Keep the machine securely parked until the failure is corrected or the machine can be safely transported.

- Drive the machine with care. Make sure speed is compatible with conditions. Use caution on rough ground, slopes, in traffic zones and while turning.
- Be alert for hazards and obstructions such as ditches, trees, cliffs, overhead power lines, and areas where there is danger of a slide.
- Be aware of and understand the job site traffic flow patterns.
- Obey flagmen, road signs, and signals.
- Watch for bystanders. Never allow anyone to be under the machine during operation. Never allow anyone to reach into the machine during use.
- Operator must know how to use signaling devices and understand which circumstances require use of each signal. Use tail lights, slow moving vehicle signs, and warning beacon as needed when traveling on public roads. It is recommended that you provide an escort on the road.
- DO NOT tow the machine, except to remove from road or load on a trailer for transport.

Storage Precautions

- Store machine in an area away from human activity.
- DO NOT permit children to play on or around the machine. Serious injury or death can occur from improper/unauthorized use of the machine.
- Make sure the unit is stored on a surface that is firm, level, and free of debris.
- Store the machine inside a building or cover securely with a weatherproof tarpaulin.

Maintenance Precautions

- DO NOT attempt repairs unless trained to do so. Refer to manuals and experienced repair personnel for help.
- Before working on the machine, securely block the machine and any components that may fall. Block any working components to prevent unexpected movement while repairs are being made.
- Always wear safety glasses and other required safety equipment when servicing or making repairs.
- Disconnect battery before working on the electrical system.
- Avoid lubrication or mechanical adjustments while the machine is in motion and engine is operating.
- If lubrication or mechanical adjustment is necessary, use extreme caution.
- Never make repairs on pressurized components such as fluid lines, the gas system, or mechanical items until the pressure has been relieved.
- When servicing or replacing hardened pins, use a brass drift or other suitable material between the hammer and pin.
- Keep brake and steering systems in good operating condition.

Tire Precautions

⚠WARNING DO NOT mount or demount tires without proper training. A violent explosion can occur.

Follow all procedures and safety instructions. Charts containing mounting and demounting instructions are available from OSHA at www.osha.gov/pls/publications.

NOTICE **Contact your authorized dealer or a qualified tire dealer or repair service to perform required tire maintenance.**

- Clean rims and repaint to stop corrosion and facilitate checking air level and tire mounting. Be careful to clean all dirt and rust from the lock ring and gutter. This is important to secure lock ring in its the proper position.
- Check rim components periodically for cracks. Replace all cracked, badly worn, severely rusted, and otherwise damaged components with new parts of the same size and type. If you are not sure about proper mating of tire and wheel parts, consult your authorized dealer or a wheel and rim expert.

⚠WARNING DO NOT attempt to rework, weld, heat or braze any rim components that are cracked, broken or damaged. Replace damaged components with new parts of the same size and type. Mixing parts of one type with those of another is potentially dangerous.

⚠CAUTION DO NOT reinflate a tire that has been run flat without first inspecting the tire, tube, flap, rim, and wheel assembly. Double-check the side ring and O-ring for damage, ensuring they are secured in the gutter before inflation.

⚠WARNING Mismatched rim parts are dangerous and could cause SEVERE injury.

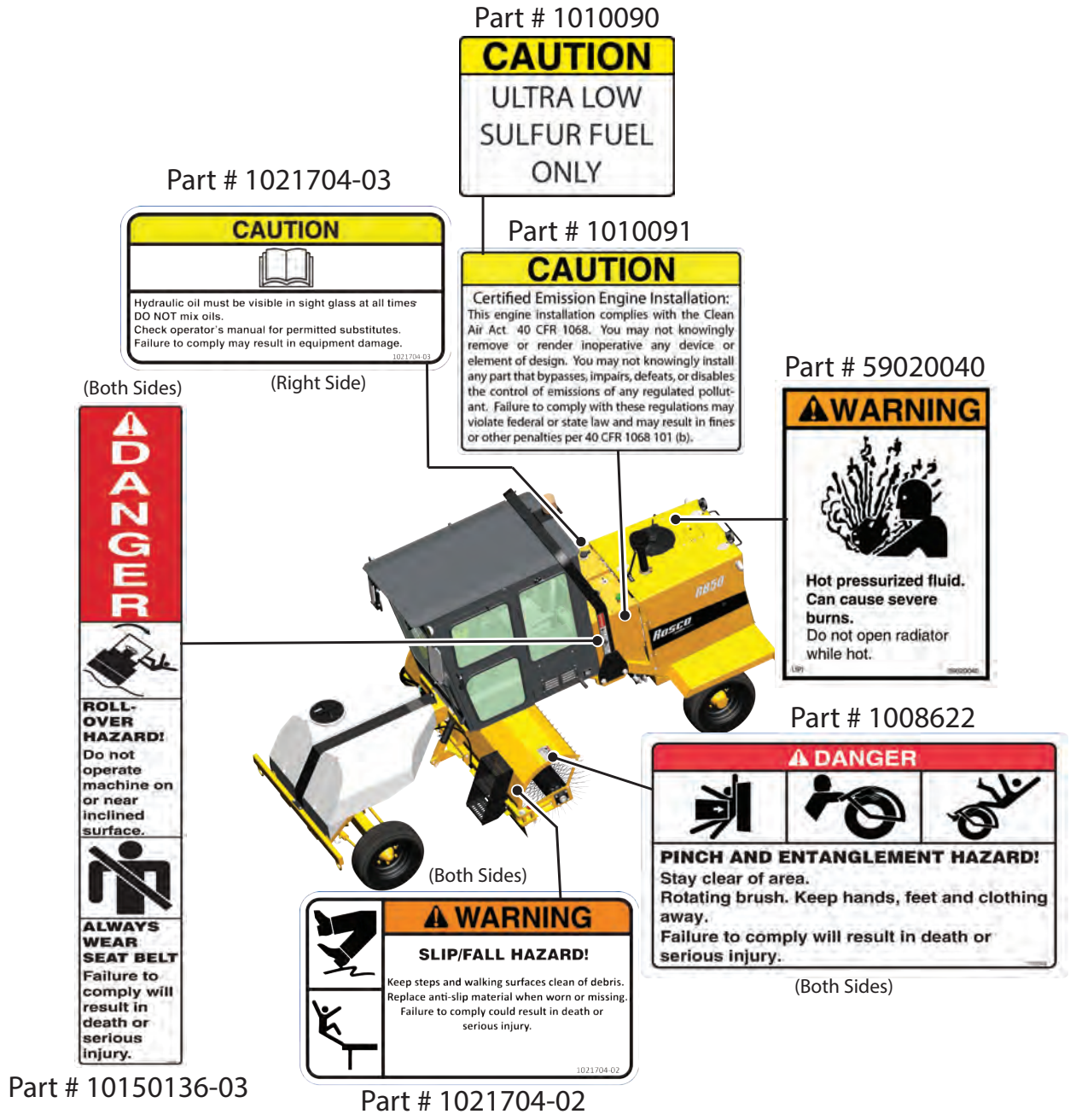
- Remove valve core to exhaust all air from tire. ALWAYS exhaust all air from tire prior to removing rim components or wheel components such as nuts or rim clamps. Check the valve stem by running a piece of wire through the stem to make sure it is not plugged.
- DO NOT attempt to demount a tire unless you have the proper equipment and experience to do the job.
- DO NOT try to seat rings or other components by hammering unless tire is fully deflated. Double-check all components prior to inflation.
- DO NOT inflate a tire before all components are properly in place. Recheck components for proper assembly. If improperly assembled, deflate and correct.
- Never hammer on a tire/rim assembly that is not fully deflated. If assembly is proper at 10 psi (68,947.572 Pa), continue to inflate to fully seat the tire beads. Then completely deflate the tire to prevent overly stretching the tube. Reinflate to recommended operating pressure.
- NEVER sit on or stand in front of a tire/rim assembly during inflation. Use a clip-on chuck and make sure inflation hose is long enough permit the person inflating the tire to stand to the side of the tire.
- DO NOT hammer components with steel hammers. Use rubber, lead, plastic, or brass-faced mallets to tap components together.
- DO NOT try to drive an assembled or partially-assembled tire or rim over a cast spoke wheel by hammering. DEFLATE and examine to determine the reason for the improper fit. Look for distortion or components that are not properly locked or sealed.
- Block the tire and wheel on opposite sides of the vehicle before you place the jack in position. Always put hardwood blocks under the jack and crib up the vehicle in case the jack slips.
- DO NOT inflate tires beyond the maximum recommended inflation pressure.
- NEVER operate a vehicle on a single tire of a dual assembly. The carrying capacity of the single tire and rim is dangerously exceeded and operating a vehicle in this manner can result in damage to the rim and tire.

SAFETY DECALS

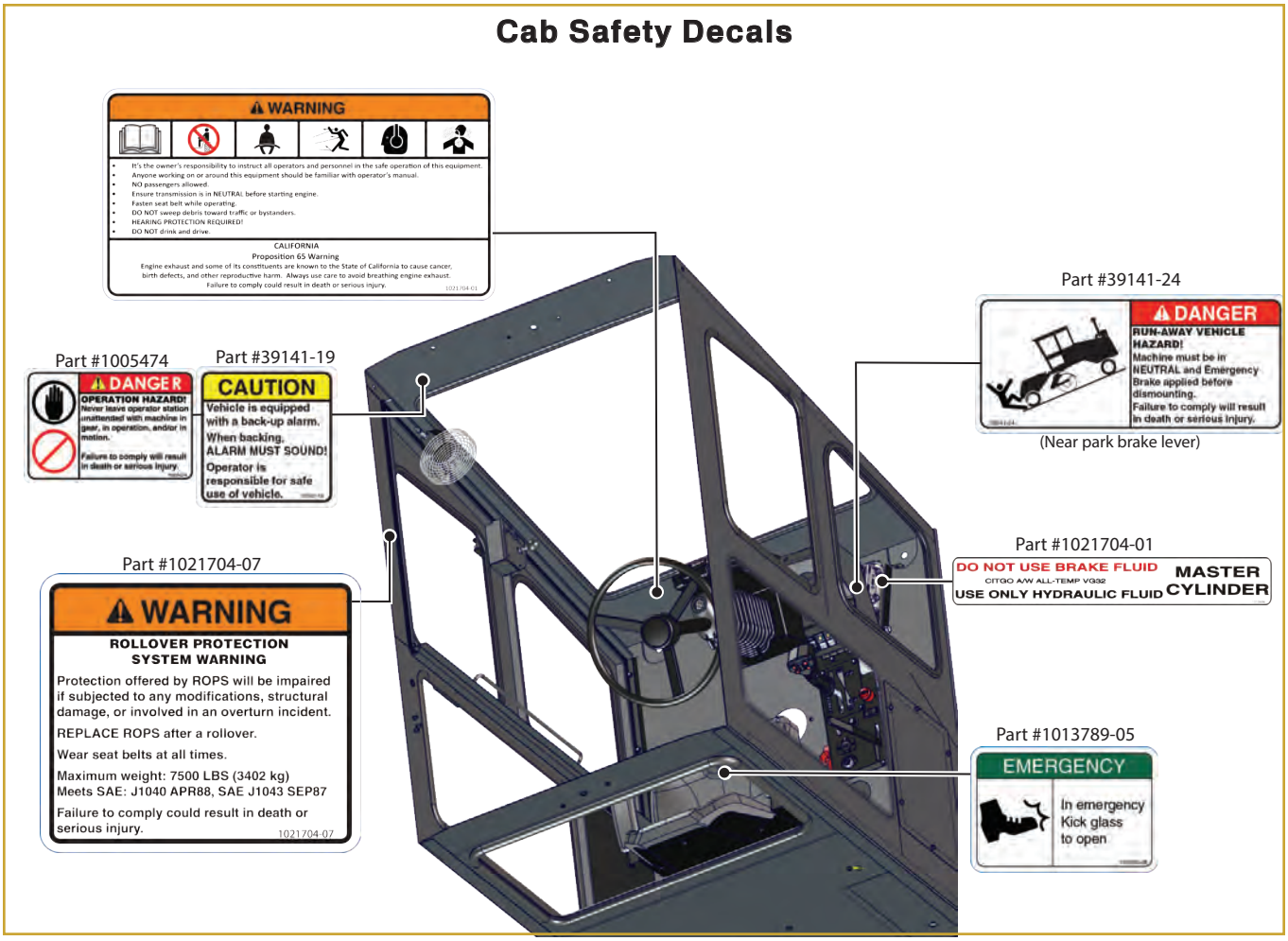
If your machine is repainted, it is extremely important that you replace all the CAUTION, WARNING and DANGER safety decals as shown in the following graphics. For additional help, refer to the parts listing in **Section 7** and contact your authorized LeeBoy dealer to order a replacement kit.

NOTE: It is the responsibility of the owner and operator to make sure that all safety labels are readable and located on the machine as designated by LeeBoy.

Safety Decal Locations



Cab Safety Decals



Safety Decals Care

- Keep safety decals and signs clean and legible at all times.
- Replace decals and signs that are missing or become impossible to read.
- Become familiar with the content and position of each safety decal.
- When replacing parts that display a safety decal, ensure that the new part is fitted with the correct decal.
- Obtain replacement safety decals or signs from your authorized dealer.

Decal Application

1. Be sure the installation area is clean and dry. Use hot, soapy water to clean the surface.
2. Thoroughly dry the surface.
3. Measure and test fit before removing decal paper backing.
4. Remove the smallest split backing paper and align decal over the specified area, carefully pressing exposed portion into place.
5. Slowly remove remaining backing and carefully smooth into place.
6. Small air pockets can be pierced with a pin and smoothed using a piece of decal backing.

Section 2

INFORMATION AND SPECIFICATIONS

2

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TWO YEAR LIMITED WARRANTY

Limited Warranty

If a manufacturing defect in factory supplied materials or factory workmanship is found and the authorized LeeBoy dealer ("Dealer") is notified during the warranty period, LeeBoy will be responsible for repairing or replacing any part or component of the unit or part that fails to conform to the warranty during the warranty period.

Terms and Conditions

The warranty applies for a period of two years (for unlimited hours of usage on the unit during the two years), beginning on the date that the Dealer from which the unit is purchased submits the warranty registration form to LeeBoy (typically on the date of purchase).

- The warranty is not transferable and applies only to the original purchaser of a new unit from a Dealer.
- Warranty work must be performed at an authorized Dealer location.
- LeeBoy has the right to repair any component or part before replacing it with a new one.
- Replacement parts furnished by LeeBoy are covered for the remainder of the warranty period applicable to the unit or component in which such parts are installed.
- All new replacement parts purchased from a Dealer will carry a six (6) month warranty.

Exclusions and Limitations

LeeBoy has no obligation under this warranty for any defects or failures caused by:

- Misuse, misapplication, negligence, accident or failure to maintain or use in accordance with the most current operating instructions;
- Unauthorized alterations or modifications;
- Any replacement parts or attachments not manufactured by or expressly approved by LeeBoy;
- Failure to conduct normal maintenance and operating service, including without limitation, providing lubricants, coolant, fuel, tune-ups, inspections or adjustments; or
- Unreasonable delay, as determined by LeeBoy, in making the applicable units or parts available upon notification of a service notice ordered by LeeBoy.

LeeBoy is not responsible for any of the following:

- Defects or failures with respect to any engine or truck chassis;
- Defects or failures covered by a warranty provided by a third-party, including without limitation warranties with respect to engines or truck chassis;
- Costs related to travel time, mileage or overtime;
- Costs related to transporting the unit to and from the place at which warranty work is performed;
- Airfreight charges related to transporting repair parts to the place at which warranty work is performed;
- Used units or used parts of any kind;
- Repairs due to normal wear and tear, or brought about by abuse or lack of maintenance of the equipment, except for premature failures;
- Attachments not manufactured or installed by LeeBoy; or
- Miscellaneous charges.

Exclusion of Other Warranties/Limitation of Liability

OTHER THAN THE EXPRESS WARRANTY PROVIDED IN SECTION A. ABOVE, LEEBOY GIVES NO EXPRESS, STATUTORY, OR IMPLIED WARRANTY OR GUARANTEE AND HEREBY DISCLAIMS ALL SUCH WARRANTIES, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, SUITABILITY, DURABILITY, CONDITION, QUALITY, OR FREEDOM FROM CLAIMS OF ANY PERSON BY WAY OF INTERFERENCE, INFRINGEMENT, OR THE LIKE.

IN NO EVENT OR UNDER ANY THEORY OF RECOVERY, WHETHER AS A RESULT OF BREACH OF CONTRACT OR WARRANTY, ALLEGED NEGLIGENCE, OR LIABILITY WITHOUT FAULT, SHALL LEEBOY BE LIABLE FOR ANY SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE SALE, USE, OR LOSS OF USE OF ANY PRODUCT, INCLUDING WITHOUT LIMITATION LOST PROFITS OR REVENUES, COST OF CAPITAL, COST OF SUBSTITUTED EQUIPMENT, FACILITIES OR SERVICES, DOWNTIME COSTS, LABOR COSTS, OR DIMINUTION OF VALUE.

CONTACT INFORMATION

For information regarding parts and repairs about your LeeBoy product, contact your authorized Rosco/ LeeBoy dealer. If your dealer is unable to resolve the problem, contact LeeBoy directly.

Record dealer information in the space provided. For additional information about LeeBoy, please visit: www.leeboy.com.

Sales Representative: _____

Dealership Name: _____

Dealership Address: _____

Dealership Phone: _____

Record of Ownership

Please complete the following information for use if you need to contact Rosco/LeeBoy for service, parts or literature.

Machine Model Number: _____

Product Serial Number: _____

Date of Purchase: _____

Nameplate

The nameplate contains the model and serial numbers used to identify the machine and its components for parts or service information. Refer to the Engine Operator's Manual for the location of the engine nameplate.

Figure 2-1. Nameplate Location

SPECIFICATION CHARTS

The specifications provided in this section include the machine weights, dimensions, performance, and torque values for both metric and standard inch fasteners.

⚠CAUTION **Replace original equipment only with components approved by Rosco/LeeBoy.**

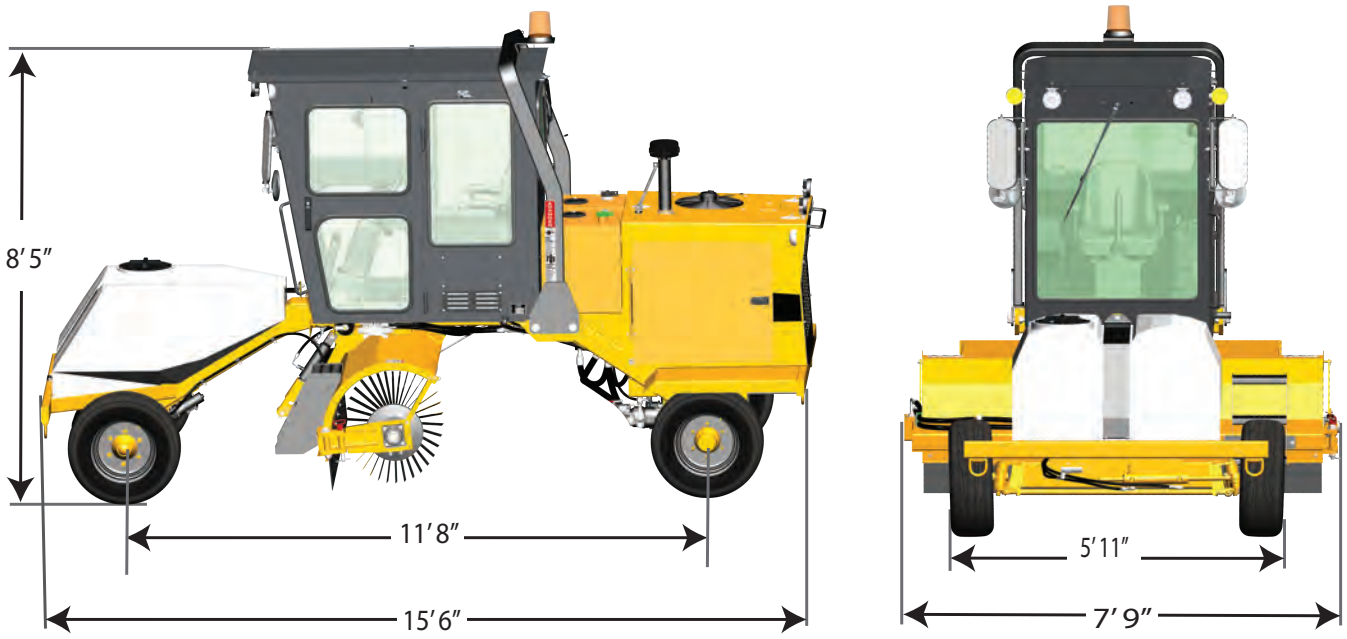

Table 2-1. Machine Dimensions

ITEM	SPECIFICATION
Overall Length	15 ft 6 in
Overall Height (with Roll-Over Protection System [ROPS])	8 ft 5 in
Overall Height (with ROPS and Beacon Light)	9 ft 5 in
Overall Width	7 ft 9 in
Tread Width (Front and Rear)	5 ft 11 in
Overall Weight (without Water)	5250 lb
Wheelbase	11 ft 8 in
Outside Turning Radius	18 ft 9 in
Ground Clearance	8.75 in
Cab	Includes: 12-Volt, Variable Speed Rear Defroster Fan; 12-Volt, 2-Speed Windshield Wiper; Windshield Washer (Front and Rear). Climate-Control Cab Options: 16,000 BTU/Hr (4690 Watt) Air Conditioning; 28,000 BTU/Hr (8206 Watt) Heater

Table 2-2. System Capacity Specifications

ITEM	SPECIFICATION
Fuel	30 gal
Engine Lubrication Oil Refill Capacity	7.4 qt
Engine Coolant	14 qt
Hydraulic Oil Tank	25 gal
Axle	4.7 qt

Table 2-3. Fuel and Lubricant Specifications

ITEM	SPECIFICATION
Engine Oil	15W-40, API CH-4, CI-4
Hydraulic Oil	All Weather All Temperature VG32
Fuel	Ultra-Low Sulfer Diesel
Grease	Shell Avania EP Grease or Equivalent
Drive Axle	80W-90LS GL5
Engine Coolant	Ethylene/Glycol or Propylene Glycol
Master Cylinder	All Weather All Temperature VG32

Table 2-4. Engine Specifications

ITEM	SPECIFICATION
Manufacturer and Model	Hatz 4H50TIC
Engine Type	74 HP, 4-Cycle Liquid Cooled Diesel, Tier 4 Final
Peak Torque	184 ft. lb. @ 1600 RPMs
Displacement	119 cu in
Combustion System; Intake System	Direct Injection; Turbocharged, Air Cooled
Power Rating HP (kW)	74 HP @ 2500 RPMs
Maximum Speed	2500 RPMs
Fuel	Low Sulfer Diesel (do not use any other type of fuel)

Table 2-5. Drive System Specifications

ITEM	SPECIFICATION
Transmission	2-Speed Hydrostatic
Steering	Hydraulic, Orbital Motor, 3.7 GPM. Priority flow at 103 Bar, 1500 PSI.
Rear Axle	Comer 068 Series
Tire Size	ST225x75-15R, Load Range E
Tire Inflation Pressure	50 PSI, 3.45 Bar (345 kPa)
Travel Speed	LOW: 0 - 12 mpg HIGH: 0 to 22 mpg
Hydrostatic Pump Displacement	2.8 CIR
Hydrostatic Motor Displacement	2.8 CIR
Brakes, Limited Slip	2-Wheel Wet Disc, Foot-Operated with Hand-Operated Park Brake

Table 2-6. Electrical Specifications

ITEM	SPECIFICATION
Battery	12 Vdc Maintenance Free, Negative Ground
Cold Cranking Amps (CCA)	770 CCA
Alternator Output Amperage	110 Amps @ 2500 RPMs
Alternator Fan Belt Tension	Manual belt tension mechanism keeps serpentine belt under tension at all times.

Table 2-7. Brush Operating System

ITEM	SPECIFICATION
Brush Angle	45° Left or Right
Brush Speed	200 RPMs maximum (variable with engine RPMs)
Brush Core (Welded Steel)	7 ft 6 in x 32 in, OD Poly Bristles
Brush Down Pressure Control	Hydraulic, Fully Adjustable
Brush Cover (Steel)	Full Length: 160°
Brush Watering System	Poly Tank, 1 @ 150 gal. Strainer, 12-Volt Diaphragm Pump
Motor	Direct Drive
Pump (Gear)	19 GPM, 207 Bar, 3000 PSI

Table 2-8. Miscellaneous Standard Equipment

ITEM	SPECIFICATION
Horn	12-Volt, Automotive
Back-Up Alarm	97 db +/- 4 db at 4 ft
Lights	Turns Signals, Tail Lights, Work Lights, Stop Lights, Beacon Light
12-Volt Outlet	12-Volt Electrical Outlet Connection
Suspension Seat	Suspension Adjustable with Seat Belt

Table 2-9. Optional Equipment

ITEM	SPECIFICATION
Dust Suppression System	150-Gal (568 L) Water Tank; Spray Bar with Corrosion-Resistant Spray Nozzles
Brush Filler (Water or Tube Type)	7 ft 6 in x 32 in
Strike-Off Blade	Front-Mounted
Curb Brush	Side-Mounted (Front Right Bumper)
Radio	CD/Weather Band Radio
Back-Up Camera	LCD Color Monitor

TORQUE SPECIFICATIONS

The following tables list torque values for standard hardware. This is a guide for average application involving typical stresses and machined surfaces. Values are based upon physical limitations of clean, plated and lubricated hardware. Under more extreme conditions, individual torque value should be followed.

Conversion formulas are provided below:

Conversion	Formula
ft-lb to N•m	[ft-lb]*1.3558 = [N•m)
ft-lb to in-lb	[ft-lb]*12 = [in-lb]
N•m to in-lb	[N•m]*8.8508 = [in-lb]

Standard Inch Fasteners

Table 2-1.Torque Specifications For Standard Inch Fasteners

SIZE	THREAD	CAPSCREWS: SAE GRADE 5				CAPSCREWS: SAE GRADE 8			
		TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
		Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
1/4	20 UNC	8	6	11	8	12	9	16	12
	28 UNF	10	7	14	9	14	10	19	14
5/16	18 UNC	17	13	23	18	25	18	34	24
	24 UNF	19	15	26	20	27	20	37	27
3/8	16 UNC	31	23	42	31	44	33	60	45
	24 UNF	35	26	47	35	49	37	66	50
7/16	14 UNC	49	37	66	50	70	52	95	71
	20 UNF	55	41	75	56	78	58	106	79
1/2	13 UNC	75	57	102	77	106	80	144	108
	20 UNF	85	64	115	87	120	90	163	122
9/16	12 UNC	109	82	148	111	154	115	209	156
	18 UNF	121	91	164	123	171	128	232	174
5/8	11 UNC	150	113	203	153	212	159	287	216
	18 UNF	170	127	230	172	240	180	325	244
3/4	10 UNC	267	200	362	271	376	282	510	382
	16 UNF	297	223	403	302	420	315	569	427
7/8	9 UNC	429	322	582	437	606	455	822	617
	14 UNF	474	355	643	481	669	502	907	681
1	8 UNC	644	483	873	655	909	681	1232	923
	14 UNF	722	542	979	735	1020	765	1383	1037
1-1/4	7 UNC	1121	840	1520	1139	1817	1363	2464	1848
	12 UNF	1241	930	1683	1261	2012	1509	2728	2046
1-1/2	6 UNC	1950	1462	2644	1982	3162	2371	4287	3215
	12 UNF	2194	1645	2975	2230	3557	2668	4823	3617

Metric Fasteners

Table 2-12. Torque Specifications for Metric Fasteners

NOMINAL SIZE AND PITCH	CLASS 8.8 [GRADE 5 EQUIVALENT]				CLASS 10.9 [GRADE 8 EQUIVALENT]			
	TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
	Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
M4 x 0.7	2	2	3	2	3	2	4	3
M5 x 0.8	5	3	7	4	7	5	9	7
M6 x 1	8	6	11	8	11	8	15	11
M8 x 1.25	19	14	26	19	27	20	37	27
M10 x 1.5	37	28	50	38	53	40	72	54
M12 x 1.75	65	49	88	66	93	70	126	95
M14 x 2	104	78	141	106	148	111	201	150
M16 x 2	161	121	218	164	230	173	312	235
M18 x 2.5	222	167	301	226	318	239	431	324
M20 x 2.5	314	236	426	320	449	337	609	457
M22 x 2.5	428	321	580	435	613	460	831	624
M24 x 3	543	407	736	552	777	582	1053	789
M27 x 3	796	597	1079	809	1139	854	1544	1158
M30 x 3.5	1079	809	1463	1097	1544	1158	2093	1570

Hydraulic Fittings

Tightening Flare-Type Tube Fittings

1. Check the flare and flare seat for defects that might cause leakage.
2. Align tube with fitting before tightening.
3. Lubricate connection.
4. Hand tighten swivel nut until snug.
5. To prevent twisting the tube(s), use two wrenches. Place one wrench on the connector body and tighten the swivel nut with the second to the torque shown in the following table:

NOTE: The torque values shown are based upon lubricated connections.

Table 2-13. Torque Specifications for Steel Flare Type Tube Fittings

TUBE SIZE OUTER DIAMETER	NUT SIZE ACROSS FLATS	TORQUE VALUE	
(IN)	(IN)	(LB FT)	(N•m)
3/16	7/16	8	11
1/4	9/16	12	16
5/16	5/8	16	22
3/8	11/16	23	31
1/2	7/8	38	52
5/8	1	54	73
3/4	1 1/4	75	102
7/8	1 3/8	83	113

Determining Proper Torque

The only reliable method of creating a consistently leak-free and long-lasting connection is to ensure the coupling is brought to the proper torque. Using a torque wrench with crowfoot is the best method, but the flats method can be used if a torque wrench is not available.

The most straightforward method of determining the correct torque setting is to multiply the desired torque by the length of the wrench from the center of the handle to the center of the drive (L); divided by the length of the wrench from the center of the handle to the crowfoot center (LA) as shown below:

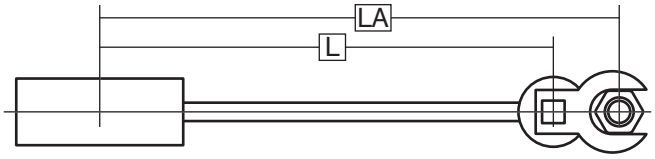

Figure 2-2. Torque Wrench - Crowfoot

NOTICE **The minimum torque values are adequate for sealing most applications. Maximum torque values should never be exceeded.**

There are several methods of determining the correct setting on the torque wrench when using a crowfoot. All of the methods involve making the setting proportional to the effective change in length of the wrench multiplied by the desired final torque. The equations and illustration below describe proper measurements.

Equations

- Torque setting if the crowfoot is placed in line with respect to the wrench:

TS = TD * L / LA

OR

TS = TD * L / (L+E)

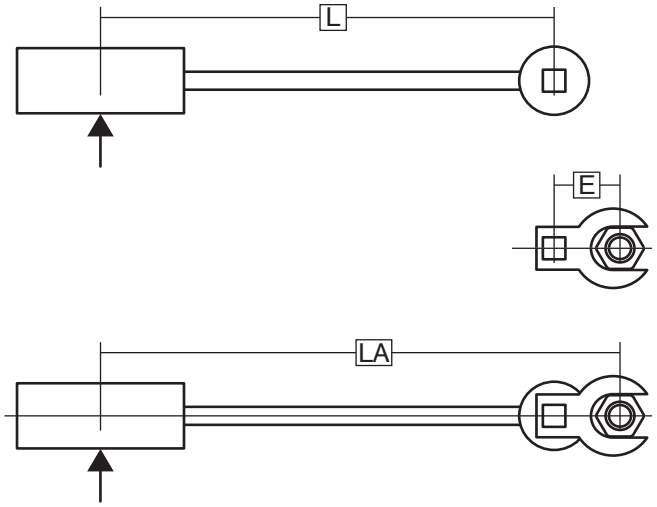

Figure 2-3. Measurements Needed

LEGEND

L = Distance from center of torque wrench handle to the center of socket drive

E = Distance from center of socket drive to the center of crowfoot

LA = Distance from center of torque wrench handle to the center of crowfoot

TD = Desired torque at the fitting

TS = Torque setting indicated on wrench

Coupling Installation

Use the following steps for proper coupling installation:

1. Determine the correct torque value for your coupling.

NOTE: Only use the torque values specified from the manufacturer. DO NOT use SAE torque recommendations.

2. Ensure the seal face and threads are clean and in good condition. O-Rings should be lubricated with light oil, but threads should be completely dry unless making pipe thread connections (interference seal).

NOTE: Attach the male end of the hose onto the equipment first since it may be necessary to rotate the entire hose assembly to tighten the male threads. Then route the hose into position while avoiding twisting the hose.

3. Tighten the connection (by hand), bringing the seal face into contact and rotating the nut until it stops.
4. Mark a line across the coupling nut and backup hex for the flats method verification of coupling torque.
5. Apply a wrench to the backup hex to prevent the coupling and hose from moving while tightening the nut with a torque wrench.

NOTICE **Failure to retain the backup hex during installation will also result in additional clamp load force that could cause damage to the seal face.**

NOTE: The coupling nut must be in motion for an accurate torque reading. If the nut is stopped before final torque value is achieved, it must be loosened and retightened until the torque is attained while the nut is in motion.

If a torque wrench cannot fit into the coupling area or if it is unavailable, the flats method may be used to ensure that the coupling is properly tightened.

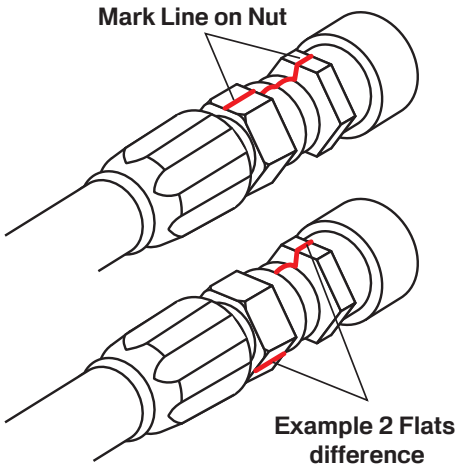

Figure 2-4. Flats Method Tightening

NOTE: The mark placed on the nut and backup hex after tightening by hand should rotate during final tightening according to the table below. The nut and backup hex can then be marked to indicate if the coupling loosens over time.

Table 2-14. Flats Method Values for Selected Terminations

FLATS METHOD VALUES		
Termination Type	**Dash Size**	**Flats**
JIC	-4	1.5 - 1.75
JIC	-6	1.0 - 1.5
JIC	-8	1.5 - 1.75
JIC	-10	1.0 - 1.5
JIC	-12	1.0 - 1.5
JIC	-16	.75 - 1.0
JIC	-20	.75 - 1.0
JIC	-24	.75 - 1.0
JIC	-32	.75 - 1.0
JIS	-4	.5 - 1.5

1. Seal faces must be in contact with the fitting fully tightened by hand before marking flats.
2. The flats method is most accurate for the first assembly cycle. For multiple disassembly and assembly cycles, torque values are more reliable.
3. Tightening two (2) flats or more may damage seal faces.

Table 2-15. Torque Specifications For US Style Coupling Terminations

JIC, SAE 45°, ORFS, O-RING BOSS, GATES ADAPTERLESS AND MEGASEAL										
DASH SIZE	JIC 37°, SAE 45° & Mega-Seal (Steel)		JIC 37°, SAE 45° & Mega-Seal (Brass)		Flat Face O-Ring Seal (Steel)		SAE O-Ring Boss (Steel) & Gates Adapterless ≤ 4000 PSI		SAE O-Ring Boss (Steel) & Gates Adapterless > 4000 PSI	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
-3									8	10
-4	10	11	5	6	10	12	14	16	14	16
-5	13	15	7	9					18	20
-6	17	19	12	15	18	20	24	26	24	26
-8	34	38	20	24	32	40	37	44	50	60
-10	50	56	34	40	46	56	50	60	72	80
-12	70	78	53	60	65	80	75	83	125	135
-14					65	80			160	180
-16	94	104	74	82	92	105	111	125	200	220
-20	124	138	75	83	125	140	133	152	210	280
-24	156	173	79	87	150	180	156	184	270	360
-32	219	243	158	175						

Table 2-16. Torque Specifications for DIN 24, DIN 60, and Inverted Cone Style Coupling Terminations

DIN 24, DIN 60, AND INVERTED CONE			
Size (mm)		Torque (lb ft)	
Light Series Tube OD	Heavy Series Tube OD	Min	Max
6		7	15
8		15	26
10	8	18	30
12	10	22	33
14	12	26	37
15	14	30	52
	16	30	52
18	20	44	74
22	25	59	89
28	30	74	111
	38	74	162
35		133	184
42		148	221

Table 2-17. Torque Specifications for 4-Bolt Flange Connections

Figure 2-1. 4-BOLT FLANGES		
Dash Size	Bolt Size (in)	Torque (lb ft)
-8	0.31	17
-12	0.38	26
-16	0.44	43
-20	0.50	65
-24	0.63	130
-32	0.75	220

1. Align faces and tighten bolts (by hand) before applying final torque in a pattern. The seal faces must be parallel with an even bolt tension to seal properly.
2. Torque values apply to bolts that are plated or coated in light engine oil.
3. Before assembly, lubricate O-Ring with light oil (SAE 10W or 20W).

Table 2-18. Torque Specifications for NPTF Dry Seal Pipe Threads

Figure 2-1. NPTF	
Dash Size	**Max Torque (ft-lb)**
-2	20
-4	25
-6	35
-8	45
-12	55
-16	65
-20	80
-24	95
-32	120

1. The torque values obtained from tightening pipe threads can vary considerably depending upon thread condition. Adequate sealing can occur at values much lower than the maximum values listed above. Only enough torque to achieve adequate sealing should be used.
2. When using a male tapered pipe thread with a female straight or parallel pipe thread, maximum values are 50% of those listed in the table above.
3. If thread sealant is used, maximum values shown should be decreased by 25%.

Table 2-19. Torque Specifications for BSP 30° Inverted Cone and JIS Coupling Terminations

BSP 30° INVERTED CONE AND JIS		
Dash Size	**Torque (ft-lb)**	
	Min	**Max**
-2	7	9
-4	11	18
-6	19	28
-8	30	36
-10	37	44
-12	50	60
-16	79	95
-20	127	152
-24	167	190
-32	262	314

Section 3

COMPONENTS

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NOTES

COMPONENTS OVERVIEW

This section describes the major components for the Rosco RB50 Broom. Become familiar with these components before operating the machine.

Broom Cab

The Rosco RB50 Broom is equipped with a fully-enclosed, pressurized cab for maximum operator comfort on the job site, including air conditioning and heat. Cab options are also available.

The broom is equipped with the PV380 Digital Display on the control panel to the right of the operator that features throttle control, tachometer, fuel gauge, oil pressure gauge, coolant temperature, voltmeter and hour meter.

The main control panel is conveniently located to the right of the operator's seat. Various controls and a joystick equipped with controls for floating brush and swinging the broom left or right enhance efficiency and ease of operation on the job site.

Engine

The Rosco RB50 Broom uses a four-cylinder, turbo-charged diesel engine to drive the hydraulic auxiliary pumps for steering and broom control. The Hatz 74 HP turbo diesel engine performs at 2500 RPMs, and is easily accessible from either side of the machine.

An air cleaner mounted on top of the engine filters fine particles such as dust, sand, chaff, and lint from the air. A pre-cleaner removes larger particles of dirt and debris before the air enters the air filter elements, depositing the trapped materials into the pre-cleaner bowl.

As air is taken in, a cyclone action deposits some of the fine particles in the evacuator mounted on the bottom of the air cleaner housing. The evacuator is held closed during engine operation by suction. This pre-cleaner system relieves the load on the air filter elements and allows longer intervals between servicing.

Primary and secondary fuel filters remove contaminants from diesel fuel before the fuel flows to the injection pump for injection into the engine combustion chamber.

A radiator mounted at the rear of the broom cools the engine. As coolant flows through the radiator, airflow from the engine-driven fan removes heat from the coolant.

Refer to your Engine Operator's Manual for a complete description and specifications of the engine.

Brush Control and Quick-Change Core System

The brush drive speed is regulated by the engine speed and a control dial on the main control panel. Brush lift and brush swing are controlled by push buttons and control dial on the control panel. The buttons operate solenoid-controlled hydraulic valves.

The Quick-Change Core system features a single-motor drive that is suited for applications ranging from road construction and industrial site cleanup to landscaping and snow removal. The quick-change core requires no tools to change the brush.

Electrical System

The electrical system is powered by a 12-volt battery in the engine compartment. The battery maintains 770 cold cranking amperes (CCA). The battery charge rate can be monitored using the voltmeter included in the PV380 Digital Display unit.

An engine-mounted alternator capable of 110 amperes charging capacity keeps the battery charged during normal operation.

Hydrostatic Drive System

The Rosco RB50 Broom is a hydrostatic, self-propelled unit. The hydrostatic drive and hydraulic systems power the steering and broom drives using one common hydraulic oil tank.

This system propels the RB50 Broom forward and reverse, which also acts as dynamic braking. A variable displacement pump provides power to a fixed displacement two-speed motor that drives the rear axle.

The operator controls the drive system via a joystick on the control panel located on the right side of the seat. A foot pedal transmission option is also available. A neutral position, which activates a neutral start switch, is included between forward and reverse directions of the control lever. A toggle switch engages high or low speed.

A hydraulic pump connected to the hydrostatic system pump provides hydraulic flow to the manifold. This manifold provides priority flow to the steering system and the brush lift-and-swing valves at approximately five (5) gallons per minute. Excess flow from the pump provides power for the brush drive motor. The steering flow always receives the priority flow, therefore, it is always powered.

Steering is performed by a hydraulically-powered orbital system operated by a steering wheel.

MACHINE OVERVIEW

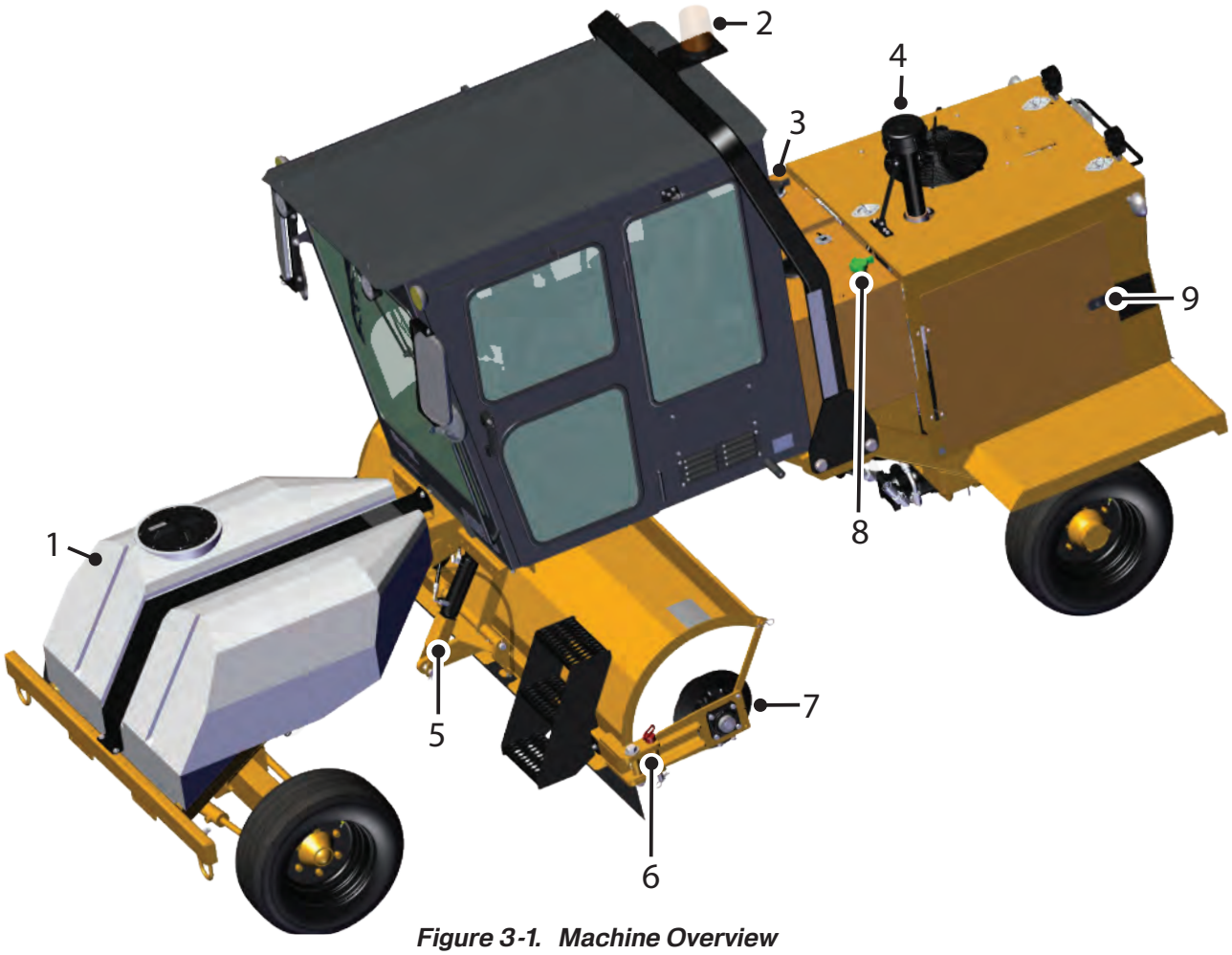

Figure 3-1. Machine Overview

ITEM NO.	COMPONENT NAME	FUNCTION
1	Water Tank (Option)	Water tank mounted on front of machine if equipped with optional Dust Suppression System. Holds 150 gallons.
2	Beacon Light	Flashing beacon light for added safety to alert traffic of broom operation. The beacon light is removable.
3	Hydraulic Oil Filler Cap	Hydraulic oil filler cap contains strainer to prevent debris from entering system when filling hydraulic tank.
4	Air Intake Stack	Rotary pre-cleaner.
5	Brush Lock Arm	Manual lock arm for locking brush into place (position shown) during travel and transport.
6	Brush Core Lock/Release	Lock pin releases and locks brush core into place when changing brush. No tools are required to change brush.
7	Brush	Angles 45 degrees right or left. 32-inch steel brush tube with many different bristle options.
8	Fuel Filler Cap	Fill using only ULSD (Ultra-Low Sulfur Diesel) fuel.
9	Engine Access Panel	Locking panel to access engine (on each side of the machine).

RB50 BROOM CONTROLS

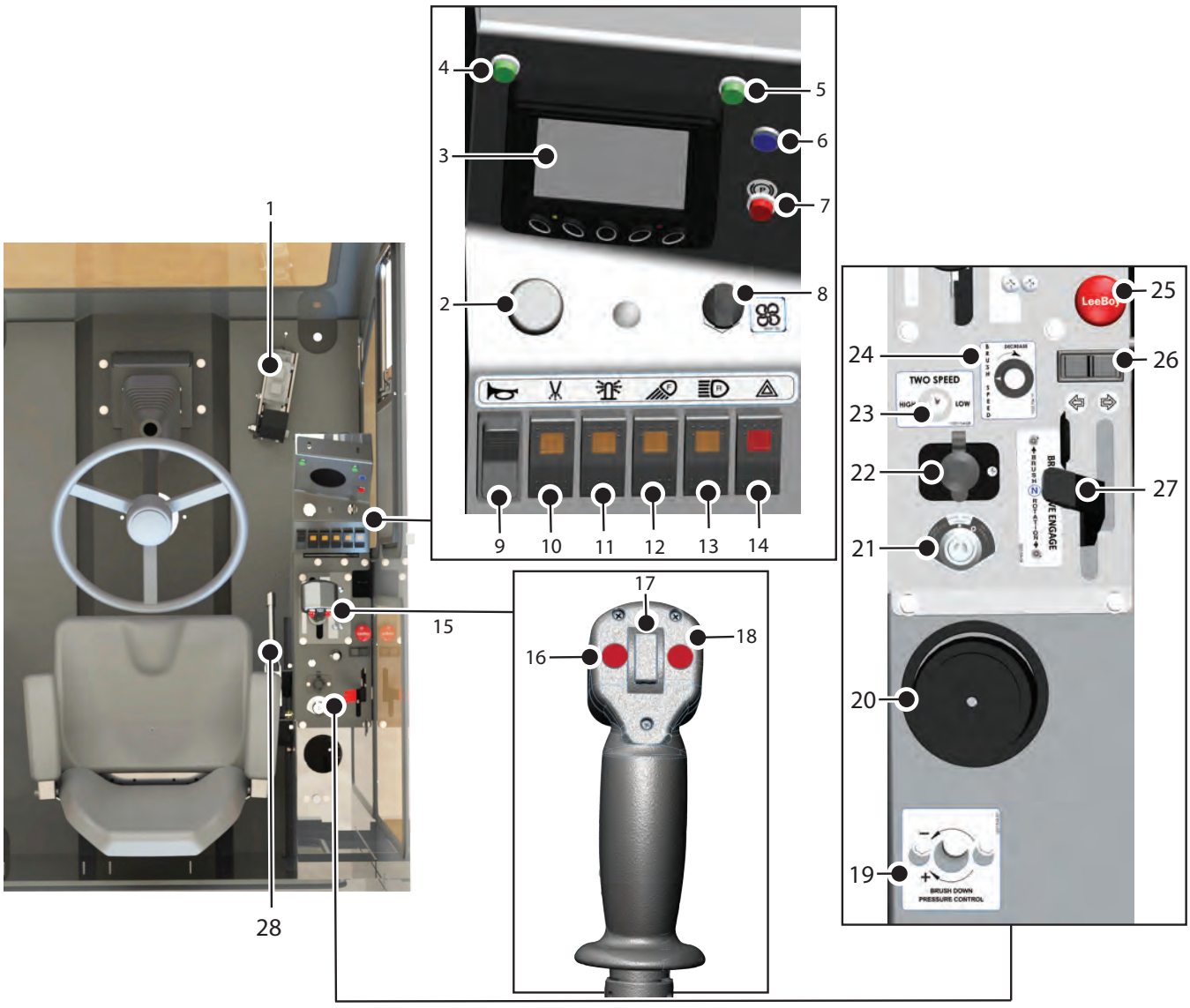

Figure 3-2. RB50 Controls

ITEM NO.	COMPONENT NAME	FUNCTION
1	Foot Brake	GENTLY step on brake to stop the broom.
2	Windshield Wiper/Washer	Turn knob to active wipers. Press knob to spray washing fluid.
3	PV380 Digital Display	Multi-functional display that shows engine temperature, voltage, oil pressure, engine RPMs, machine hours, service reminders, diagnostics, and are alerted to any fault conditions that require attention.
4	Left Signal Light	Illuminates when left turn signal is engaged.
5	Right Signal Light	Illuminates when right turn signal is engaged.
6	High Gear Light	Illuminates when High Gear is engaged.

ITEM NO.	COMPONENT NAME	FUNCTION
7	Park Brake Light	Illuminates when the park brake is engaged.
8	Cab Fan	Turns cab fan on and off. Turn button right to activate.
9	Horn	Press button to sound horn.
10	Water System	Turns the water system on and off.
11	Beacon Light	Turns beacon light on and off.
12	Front Work Lights	Turns front work lights on and off.
13	Rear Work Lights	Turns rear work lights on and off.
14	Hazard Lights	Flashing hazard lights to alert traffic.
15	Transmission Selector Lever / Joystick	Pushing joystick forward increases machine speed while pulling joystick backward slows machine speed. Also features controls for floating and swinging brush left or right.
16	Brush Swing Left	Controls brush swing to the left.
17	Brush Lift/Float	Controls brush lift and float.
18	Brush Swing Right	Controls brush swing to the right.
19	Brush Down Pressure	Controls the brush pressure on the work surface. Turn clockwise to increase pressure and counterclockwise to decrease pressure.
20	Cup Holder	The cup holder is an added convenience for the operator.
21	Ignition Switch	Turn key clockwise to start the engine. Turn key counterclockwise to shut off the engine. Transmission Selector MUST be in the NEUTRAL position to start. Features a Neutral Start system to prevent operation of the engine starter when the transmission is not in neutral.
22	12-Volt Outlet	12-volt electrical outlet.
23	Two Speed Switch	High Gear: Full displacement that allows higher transport speeds. Push switch left for High Gear. Low Gear: Partial displacement providing greater torque and lower speeds. Push switch right for Low Gear. **NOTE: Use ONLY low gear while sweeping.**
24	Brush Speed Dial	Turn dial left to increase brush speed. Turn dial right to decrease brush speed based on engine RPMs.
25	Emergency Stop	This button breaks power to the ignition switch. Push button to lock. Twist button to unlock.
26	Signal Lights	Push button left for left turn signal. Push button right for right turn signal.
27	Brush Drive Engage Lever	Turns brush rotation on and off. Push lever forward to engage. Pull lever backward to reverse brush rotation.
28	Park Brake Lever	Raise to engage park brake. Lower lever to release park brake.

FOOT PEDAL CONTROLS OPTION

The optional forward and reverse foot pedal in lieu of joystick transmission control allows for a broader range of operator driving opportunities. ***(See Page 4-22)*** Switches on the main control panel replace button controls on the joystick while the other control panel items remain the same as shown on ***Pages 3-6 and 3-7.***

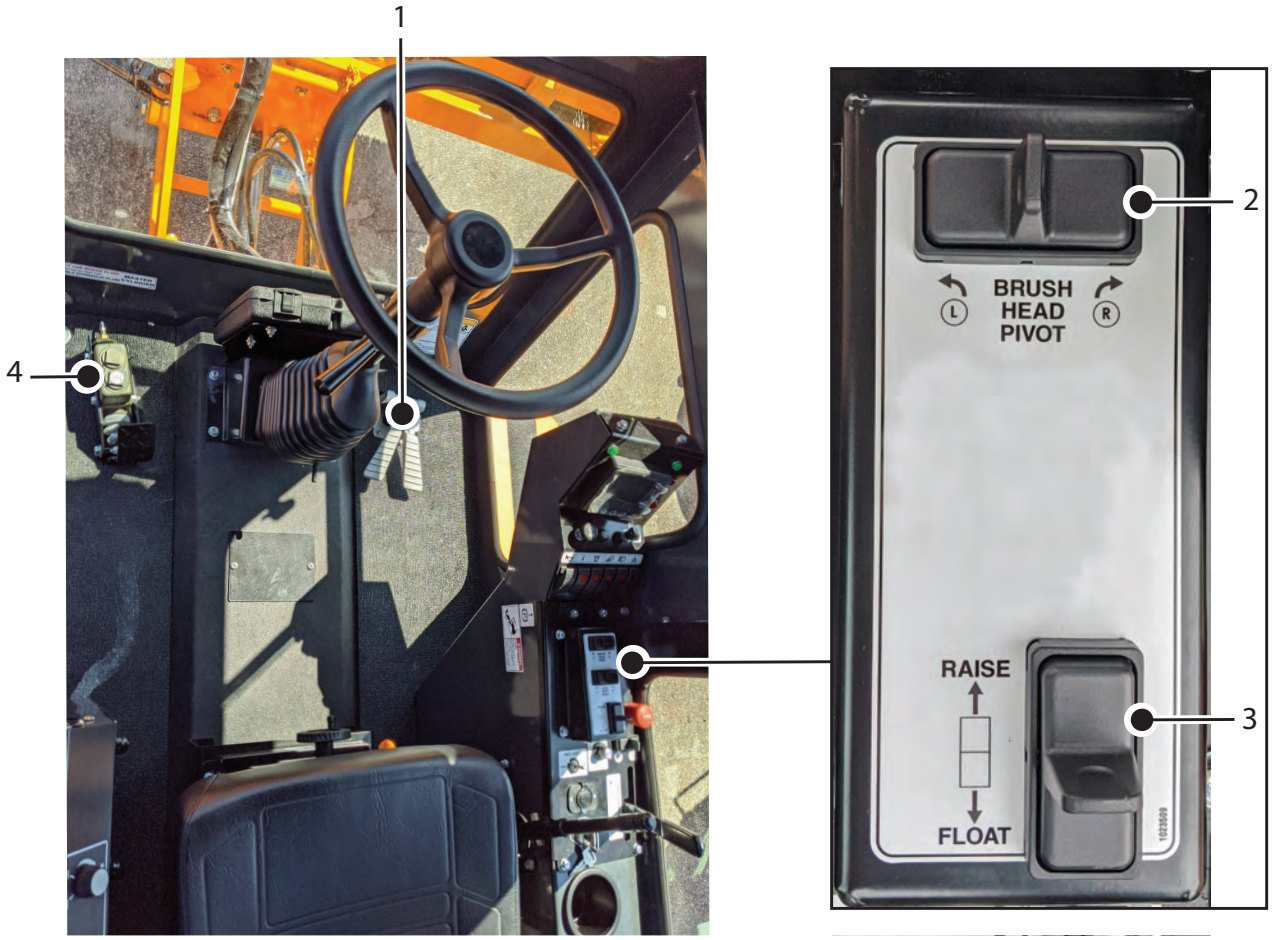

Table 3-3. Foot Pedal Controls Option

ITEM NO.	COMPONENT NAME	FUNCTION
1	Forward/Neutral/Reverse (FNR) Pedal	• Press the right pedal to move forward. • Press the left pedal to move in reverse. • The pedals automatically return to neutral when released.
2	Brush Head Pivot	Push the brush head pivot switch Left to swing the brush left; or right to swing the brush right.
3	Raise/Float	Push the Raise/Float button forward (raise) or back (float) for brush raise and float control.
4	Brake Pedal	Starting: Firmly press pedal to start machine. Braking: GENTLY press pedal to stop machine. The pedal is very sensitive.

Section 4

OPERATION

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GENERAL INFORMATION

The Rosco RB50 Broom is a hydrostatic, self-propelled unit. The broom angles at 45 degrees right or left for efficient general road sweeping or snow removal. The hydraulically-controlled RB50 Broom can brush, scour and remove job site debris with ease.

The RB50 features a Quick-Change Core system with a single-motor drive that is suited for applications ranging from road construction and industrial site cleanup to landscaping and snow removal. The quick-change core requires no tools to change the brush.

The Rosco RB50 is ideal for cleaning parking lots, industrial plants, airport runways, streets and driveways. Because the RB50 can clean large surfaces quickly and efficiently due to its bristle rotation, ability to angle and adjust the brush-down force, it is perfect for a variety of construction, industrial and landscaping applications.

Options include a fully-enclosed cab version, curb brush and snow blade attachments. To loosen particles and reduce dust, an optional Dust Suppression System is available, equipped with its own electric pump, a 150-gallon water tank, corrosion-resistant spray nozzles and piping system.

⚠ DANGER **Only authorized personnel who are properly trained should operate this machine.**

Receiving the Machine

Although the machine has been checked thoroughly by the manufacturer, road hazards or other factors during transport may result in damage. Inspect machine for any damage during transport. Contact your authorized dealer immediately if any damage has occurred.

The RB50 Broom is equipped with ROPS (Roll-Over Protective System). The unit is shipped with ROPS in the stowed position. **The ROPS will need to be placed in its operating position prior to use. See the following page for the procedure to rotate ROPS from shipping to its operational position.**

Be sure to check the machine thoroughly upon receipt:

1. Rotate ROPS from the shipping position to the operational position. ***(Page 4-4)***
2. Check engine oil. ***(Page 5-9)***
3. Check hydraulic fluid level. ***(Page 5-18)***
4. Check machine hoses for any damage or leaks.

Initial and Daily Inspection

The following inspections are essential and should be performed before starting the machine. Visually inspect the unit to check its general condition. Continue with a check of the systems and components shown in **Table 4-1** below:

Table 4-1. Initial and Daily Inspection

INSPECT	PROCEDURE
Engine Oil Level	Maintain oil level between ADD and FULL marks on dipstick.
Fuel Tank	Check for adequate fuel supply. Fill if needed. Always fill tank at end of the day to prevent condensation in the tank.
Hydraulic Tank	Check the tank for oil level and leaks.
Radiator	Check coolant level.
Battery	Ensure all cables are tight and clean. Check for corrosion on the battery terminals.
Air Cleaner	Check the air filter element and connections. Air cleaner has both a primary and secondary filter.
Drain Plugs	Make sure plugs are inserted and tight.
Engine Belt	Check for proper belt tension.
Grease Fittings	Ensure fittings are greased and in good working order.

ROPS

The RB50 Broom is equipped with ROPS (Roll Over Protection System) as an added safety feature to protect the operator and equipment in the event of overturning the vehicle. The ROPS meets strict requirements designated by various agencies.

Use the following procedure for rotating ROPS from the shipping position to the working position:

1. Remove the bolts holding each of the two shipping straps (four bolts on each strap).
2. Loosen the two front ROPS mounting bolts. **DO NOT REMOVE. (Figure 4-1)**
3. Rotate the ROPS forward into position.
4. Retighten the mounting bolts and torque. ***(See Torque Tables in Section 2).***

Figure 4-1. ROPS in Shipping Position

START-UP PROCEDURE

A Neutral Start switch has been installed to prevent operation of the engine starter when the transmission is not in neutral.

You can check that it is functioning properly by:

1. Engage the park brake. **(Figure 4-2)**
2. Place the transmission selector into the NEUTRAL position.
3. Turn the ignition key to the START position. The Wait to Start light will illuminate for a few seconds on the PV380 digital display controller.
4. When the Wait to Start light goes off, turn key to the ON position.

NOTICE **Using starting additives, such as ether, is not recommended as severe engine damage can occur.**

⚠ DANGER **DO NOT leave the operator station unattended while the machine is in gear or in motion.**

⚠ CAUTION **Verify there are no people, obstacles or other equipment in the machine's path before starting the engine.**

To start the Rosco RB50 Broom:

1. Ensure the park brake is engaged before starting the machine.
2. Place the transmission selector lever into the NEUTRAL position.

⚠ WARNING **Be sure the park brake is engaged and transmission selector is in the neutral position before starting the engine. The machine's sudden movement if started in gear can cause serious injury or even death if operator is thrown from the machine.**

3. Insert key into the ignition and turn clockwise to the START position. The "Wait to Start" light will illuminate for a few seconds on the PV380 digital display controller. ***(Page 4-9)***
4. When the Wait to Start banner disappears, turn key to the ON position to crank engine.

NOTICE **DO NOT hold the starter longer than 10 - 15 seconds. If the engine does not start, allow the starter to cool two or three minutes between attempts to prevent damage to the starter.**

5. Set the engine throttle to one-half speed (1800 RPMs) using the PV380 digital controller. ***(Page 4-9)***

6. Check control panel gauges for proper readings.
7. Adjust seat position and mirrors.
8. Allow the engine to warm and fasten your seat belt.
9. Disengage the park brake.

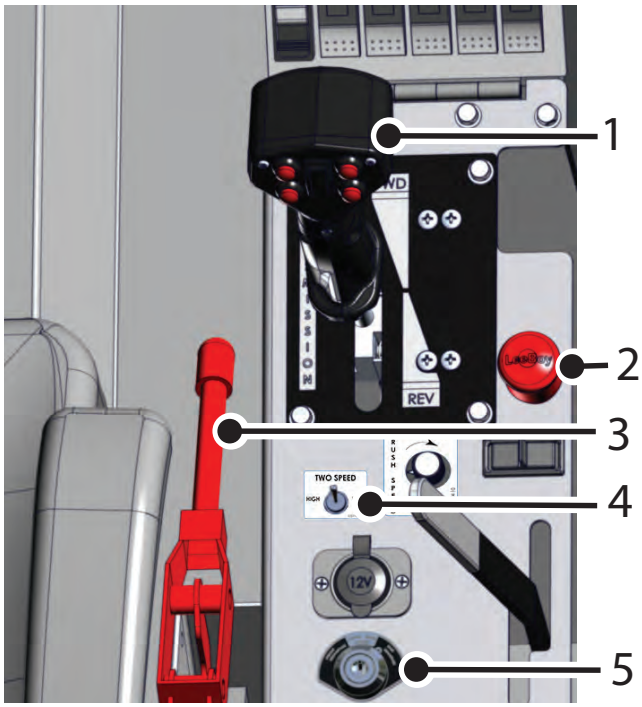


Figure 4-2. Start-Up Controls

1 - Joystick (Transmission Selector)

2 - Emergency Stop Button

3 - Park Brake Lever

4 - Speed Selector Switch

5 - Ignition Switch

Cold Weather Starting

- Engage the parking brake. **(Figure 4-2)**
- Place the transmission selector into the NEUTRAL position.
- Turn the ignition key to the Start position. The "Wait to Start" light will illuminate for a few seconds on the PV380 digital display controller.
- When the Wait to Start banner disappears, turn key to the ON position to crank engine.
- Allow the engine to idle about ten minutes before driving the machine.

Stopping the Engine

1. Adjust throttle to idle using the decrease RPMs feature on the PV380 digital display ***(Page 4-9)*** until idle speed is reached (1200 RPMs).
2. Apply park brake.
3. Turn ignition key counterclockwise to the OFF position and remove key.

NOTE: The electrical system will continue to operate for approximately 20 seconds after the ignition is turned off. This allows the engine ECU (Engine Control Unit) to store critical engine data as it powers down.

Emergency Stop Button

The purpose of the Emergency Stop Button (E-Stop) is to shut down the engine and all machine functions in an emergency situation. This additional safety feature should only be used if the operator needs to shut down all machine operations quickly. **(Figure 4-2)**

Figure 4-3. Driving the RB50

DRIVING THE BROOM

It is important that the operator become familiar with driving the Rosco RB50 Broom before operating it on the job site. Know and understand the control functions before operation. ***(Pages 3-6 and 3-7)***

While the RB50 Broom is easy to operate, operators should become accustomed to using the engine throttle, brakes and hydrostatic transmission before engaging the brush sweeper.

Keep these safety tips in mind when operating the broom:

- Perform an operational safety check before operating the broom, ensuring all equipment functions properly and all safety equipment (i.e., fire extinguisher) is present.
- **ALWAYS wear your seat belt.**
- **ALWAYS use turn signals.**
- While the RB50 Broom is easy to operate, operators should become accustomed to using the engine throttle, brakes and hydrostatic transmission before engaging the brush sweeper. The foot brake pedal is very aggressive!
- Stop at all railroad crossings and look both ways before proceeding.
- Watch your speed, especially if driving in bad road conditions. Use extra caution at intersections.
- Be aware of traffic behind you. Pull over to the side of the road to allow vehicles to safely pass if needed.
- Ensure proper warning flags and/or devices are in place to warn traffic of road construction, slow-moving vehicle or other traffic alerts. If working at night, ensure the work site is well-lit and well-marked.
- Use spotters to look out for danger when loading and unloading equipment from vehicles. Understand all communication signals used between equipment operators and workers on foot.
- Avoid obstacles that can damage the broom or cause an accident, i.e., manholes, signs, and road debris, etc.
- Vehicles parked on inclines should use appropriately-sized blocks placed behind or in front of the tires.
- Always be mindful of what's going on around you. Never allow anyone to walk behind or in front of the broom while operating, especially near the swing radius of the brush.

Cab Door Release

When you open the cab door completely on the RB50 Broom, the door will lock into place for operator convenience. Simply pull the door release knob to release the door from its locked position. **(Figure 4-4)**

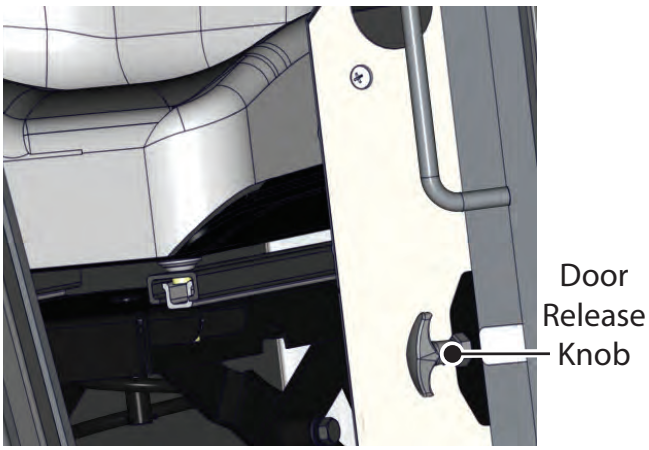


Figure 4-4. Cab Door Release Knob

Adjust Seat

The operator seat suspension provides comfort by absorbing some of the machine movements during operation. It can also be adjusted for individual comfort. **(Figure 4-5)**

Always make seat and seat belt adjustments BEFORE driving the sweeper. The seat belt is designed for safe operator retention if the machine is involved in an accident or roller-over.

- Use the suspension adjustment knob to set the suspension for operator weight. Rotate knob to increase or decrease setting as shown on the knob face.
- Use the seat position adjustment lever to move the seat forward or backward. Pull up on the lever to adjust and release to lock into place.

Figure 4-5. Seat Adjustment

Adjust Steering Console

Figure 4-6. Tilt Lock Toe Pedal

The steering wheel steers like any other steering wheel. Steering takes priority over brush operations (rotation, lift and swing) for operator safety.

You can adjust the steering column using the tilt lock pedal at the bottom of the steering column. **(Figure 4-6)**

To adjust, simply depress the pedal with your foot and tilt the steering column forward or backward to a comfortable position, then release the tilt lock pedal and it locks into place.

Cab AC/Heat

For added operator comfort, an air conditioning and heater option is also available. **(Figure 4-7)**

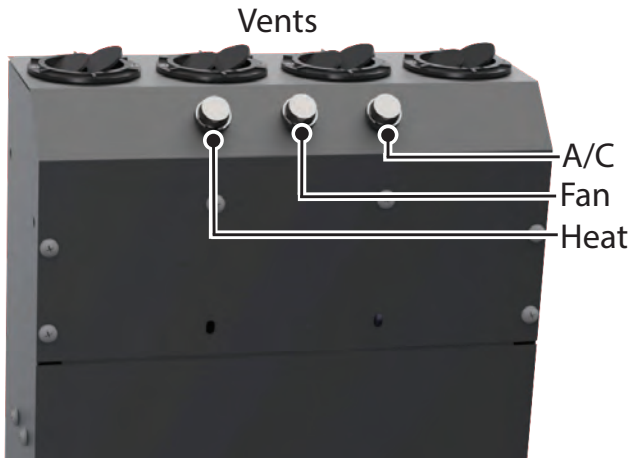


Figure 4-7. Cab AC/Heat

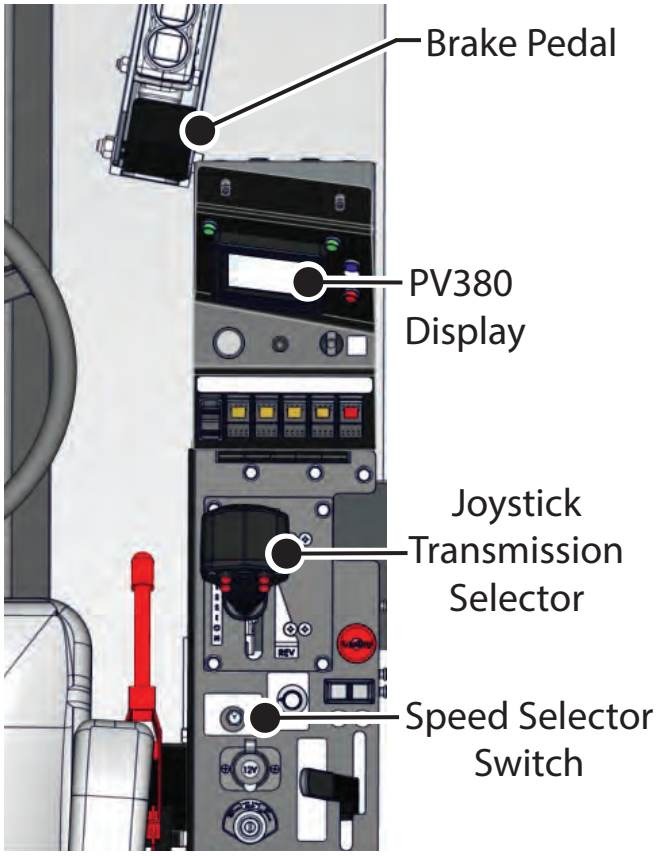


Figure 4-8. Operator Control Panel

Joystick Transmission Selector

The hydrostatic transmission selector joystick is on the control panel to the operator's right side. **(Figure 4-8)** With some experience, an operator can use this joystick lever along with proper throttle control to make smooth movements between forward and reverse. The center position is NEUTRAL.

- Push the joystick forward to move the broom forward.
- Pull the joystick backward to move the broom into reverse.

NOTE: Placing this joystick into reverse activates the back-up alarm to alert others of the machine's reverse motion.

Speed Selector Switch

The hydrostatic transmission is equipped with a HIGH/LOW Speed selector switch located on the right control panel. **(Figure 4-8)**

- Flip the switch left to the HIGH gear position for road travel.
- Flip the switch backward to the LOW position for sweeping operations.

⚠CAUTION NEVER operate in high gear while sweeping. The operator cannot drive and control the machine functions safely in high gear.

Stopping and Braking

The RB50 is equipped with a de-stroke braking system to provide aggressive stopping power in the event of an emergency. When the foot brake is applied, the hydrostatic pump is de-stroked, which redirects the flow and allows the foot brake to engage. **(Figure 4-8*)***

Though a brake pedal is situated on the floorboard to the right of the steering column, most operators seldom use it. Instead, operators use the throttle and hydrostatic transmission to slow down, stop or change direction. Become accustomed to using the hydrostatic transmission to assist in braking.

Regardless of the preferred stopping method, it is important to always put the transmission selector lever into the NEUTRAL position as the foot brake is applied or if throttling down to stop or turn.

⚠CAUTION The foot brake alone is not sufficient to stop the machine if stopping on a grade, especially steep grades. Familiarize yourself with the job site before sweeping to anticipate these variables.

Transmission Foot Pedal Option

Available for those who prefer is the optional forward and reverse foot pedal in lieu of joystick transmission control. **(Figure 4-9)** Adding the forward and reverse transmission foot pedal in lieu of the joystick control allows for a broader range of operator driving opportunities.

If equipped with this option:

- The **brake pedal** is on the left side and must be pressed to start the broom.
- The **transmission pedal** is on the right side and has a left and right pedal:
  - The left pedal moves the transmission into reverse.
  - The right pedal propels the broom forward.
  - Releasing the pedal puts the broom in neutral.

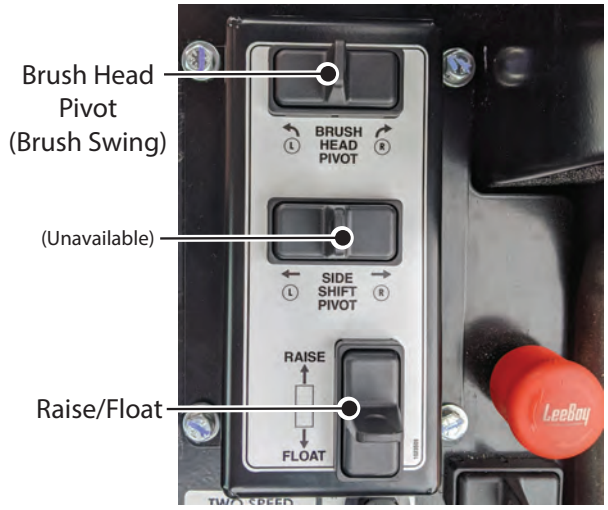


Figure 4-9. Control Panel Switches

Switches on the main control panel replace button controls on the joystick if equipped with this option.

- Push the brush head pivot switch to swing the brush left or right.
- Push the Raise/Float button forward or back for brush raise and float settings. (The side shift feature is not available for this model, therefore the switch is nonfunctional.)

PV380 POWERVIEW™ DISPLAY

The PV380 PowerView™ system is a multi-functional display that enables operators to view standard engine and transmission parameters along with active and stored trouble codes. The digital display screen can show single or multiple text descriptions, i.e., engine temperature, voltage, oil pressure, engine RPMs along with service reminders or fault conditions.

The operator can toggle through different screens for displaying critical data broadcast by the electronic engine and transmission Engine Control Unit (ECU).

Diagnostic codes with text descriptions of fault conditions are enhanced by ultra-bright alarm indicators that illuminate amber or red to alert the operator of a condition requiring attention. The amber warning light illuminates to visually alert the operator that a trouble code or service reminder is being shown on the display. If the red warning light illuminates, the operator should shut down machine and promptly repair the problem to prevent possible damage.

When turning the ignition key to the ON position, you will see a Murphy logo with the “Wait to Start Preheating” banner along the bottom of the screen. **(Figure 4-10)** DO NOT start the engine until this message disappears and the main gauge screen appears. **(Figure 4-11)**

Figure 4-10. Wait to Start Screen

Keypad Functions

The keypad is a touch system with five buttons along the bottom that corresponds to the navigational menu bar above the button (at the bottom of the display screen). **(Figure 4-11)** Button functions can change depending upon the screen displayed. When directed to press a symbol within the procedural steps, it is referring to the button below the displayed symbol on the navigational menu bar.

Figure 4-11. Navigation and Keypad Functions

Engine and transmission parameters displayed include:

- Engine Speed
- Engine Temperature
- Battery Voltage
- Coolant Temperature
- Engine Oil Pressure
- Total Machine Hours
- Engine Fuel Level
- Discharge Pressure
- Suction Pressure
- Active Fault Codes
- Stored Fault Codes

NOTE: Up to four (4) warning symbols can be shown at a time (upper left icon area). The most important symbol will be shown first followed by any others by priority.

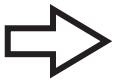
The following tables show functions that each button represents by the symbol appearing on the navigational menu bar and its function, depending upon the screen mode:

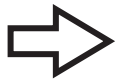
Table 4-4. Keypad Navigation Menu Bar Symbols

KEY 1	
Symbol	**Function**
⇦	Alternates between screen sets or moves the cursor one position to the left.
⇧	Scrolls up in certain screens when in specific menus.
-	Use to decrease numbers in various screens.

KEY 2	
Symbol	**Function**
FAULTS	Displays diagnostic messages (faults) and stored codes.
⇩	Scrolls down in certain screens when in specific menus.
+	Use to increase numbers in various screens.

KEY 3	
Symbol	**Function**
☰	Navigates to the Main Menu entrance point.
⇦	Serves as the Escape or Return to Previous Menu button.

KEY 4	
Symbol	**Function**
	Decreases the throttle set point.
	Decreases various settings.
STORED CODES	Displays the stored codes when in the diagnostic messages screen.
	Moves cursor one slot at a time to the right.

KEY 5	
Symbol	**Function**
	Increases the throttle set point.
ACTIVE FAULTS	Displays active faults.
	Serves as the “Enter” key for menu selections on certain screens.
	Increases various settings.
	Moves cursor one slot at a time to the right and displays alternate menu screens/choices.
	Moves the cursor to the next cell when customizing parameters on the home screen.
ACK	Acknowledges active faults.

Gauge Screens

Once the engine is running, the main gauge screen appears. There are four gauge screens--two electronic and two mechanical. You can scroll through the gauge screens by pushing ⇦ (Button 1). **(Figure 4-12)**

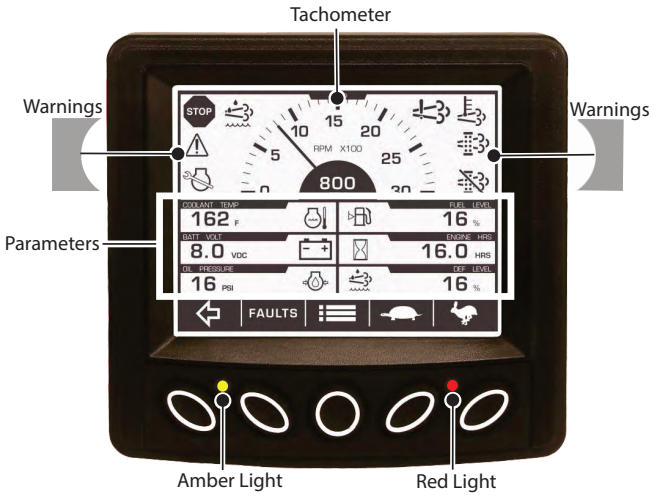


Figure 4-12. Gauge Screen Parameters

The gauge screen also shows the tachometer, engine information parameters and warning alerts.

Tachometer: The tachometer displays the engine speed on a 3000-RPM dial. Engine hours are displayed at the bottom of the tachometer area.

Parameters: The parameters sections of the screen displays engine oil pressure, engine temperature, engine hours, battery voltage, fuel level and other data. (This machine is not equipped with a DEF tank, therefore the DEF Level parameter does not function and always displays 0%.)

NOTE: The target speed gauge appears for five (5) seconds when the user throttles the engine.

Warnings: Alerts and warnings appear in this area of the screen to notify the operator a component needs attention. The most important warning symbols are shown in the upper left corner of the screen. Up to four (4) symbols can be shown at a time.

It is important to heed these warnings and resolve the issue promptly for optimum performance and preventing damage to the engine or other components.

When a warning message is displayed, an amber light will illuminate on the keypad. **(Figure 4-11)** If the red light illuminates, shut off the engine and resolve the problem to prevent possible damage.

User Settings Screen

These features allow you to customize , modify or make adjustments to screen items in your PV380 Digital Display. **(Figure 4-13)**

Navigation

To navigate to the User Settings screens:

1. Press the ☰ button to display the menu.
2. Use the ⇧ and ⇩ arrow buttons to scroll to the User Settings screen. **(Figure 4-13)**

Procedures follow for navigation and adjustment to each of the various User Setting functions.

Figure 4-13. User Settings Screen

Set Brightness and Contrast

You can select the desired screen brightness and contrast intensity using this option. (The display will automatically adjust the contrast when working at night.)

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Brightness and/or Contrast. (The selected function will be highlighted.)
3. Press + or - buttons to make adjustments within that selection.
4. Press ↵ to save setting and exit the menu.

Set Language

The default language is set to English, but can be changed to German, Spanish, French or Italian.

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Language.
3. Press the ⇨ button to scroll to the desired language.
4. Press ↵ to save setting and exit the menu.

Set Units

You may select how unit measurements are displayed using this option: US STD (U.S. Standard Measurements), METRIC KPA or METRIC BAR.

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Units.
3. Press the ⇨ button to scroll to the desired measurement units.
4. Press ↵ to save setting and exit the menu.

Set Time

Follow these steps to set the time:

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Time. The cursor will be beneath the hour.
3. Use the + and - buttons to adjust the hour.
4. Press the ⇩ button to adjust the minutes (cursor will move right).
5. Repeat Steps 3 and 4 to adjust minutes (and seconds).
6. Press ⇦ to save setting and exit the menu.

Set Date

Follow these steps to set the date.

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Date. The cursor will be beneath the day.
3. Use the + and - buttons to adjust the day.
4. Press the ⇩ button and the cursor will move to the month.
5. Repeat Steps 3 and 4 to adjust the month and year.
6. Press ⇦ to save setting and exit the menu.

Show Clock

You can choose to show or hide the clock:

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Units.
3. Press the ⇨ button to choose Show or Hide.
4. Press ⇦ to save setting and exit the menu.

Service Reminders Screen

The Service Reminders screen allows the operator to scroll through the service list to view information about engine oil, hydraulic oil, air filter, fuel filter, service engine and service machine reminders. **(Figure 4-14)**

Figure 4-14. Service Reminders Screen

To navigate to the User Settings screens:

1. Press the ☰ button to display the menu.
2. Press the ⇩ button to scroll to Service Reminders.
3. Press the ↵ button to select Service Reminders.
4. Use the ⇨ and ⇦ buttons to scroll through the "read-only" screens.

Faults Screen

Faults in the system can be displayed by navigating to the Active Faults screen. **(Figure 4-15)**

Figure 4-15. Navigate to Faults Screen

Follow these steps to access the active faults screen.

1. Press the **FAULTS** button on Main Gauge screen. **(Figure 4-12 on *Page 4-11*)**
2. Use the ⇦ and ⇨ buttons to scroll through any messages present.
   - Each saved code shows the SPN (Suspect Parameter Number); FMI (Failure Mode Identifier); and OC (Occurrence Count), which indicates if the same fault occurred more than once. A text explanation of the warning or shutdown may also display.
   - To view the stored codes, press **Stored Codes.** "Requesting" and "Receiving Data" will be displayed while the system obtains the data. (If the data is not received, "TIMEOUT" will display and you will need to press the **Stored Codes** button again.)
3. Use the ⇧ and ⇩ buttons to scroll through any additional messages (if present). Press **Get Faults** to receive any additional stored faults.
4. Press ⇦| to exit and return to the main gauge screen.

PV380 Diagnostic Codes

** **PGN: Parameter Group Number**

Table 4-5. Diagnostic Trouble Codes (DTC) Chart

ICON (or Text)	PGN	DESCRIPTION
STOP	DM1	Stop Diagnostic Lamp: If this icon illuminates, there is an active stop fault. Stop the machine and resolve the problem before operating again to prevent damage.
⚠	DM1	Warning Diagnostic Lamp: If this icon illuminates, an active fault exists that needs immediate attention.
ACCEL PED1	61443	Accelerator Pedal Position 1
	61443	Percent Load at Current RPM
	61444	Actual Engine Torque
	61444	Engine Speed
TRIP DIST	65248	Trip Distance
VEH DIST	65428	Total Vehicle Distance
ENG HRS	65253	Total Engine Hours
TRIP FUEL	65257	Trip Fuel
FUEL USED	65257	Total Fuel Used
	65262	Engine Coolant Temperature
	65262	Fuel Temperature
	65262	Engine Oil Temperature
INTC TEMP	65262	Engine Intercooler Temperature
	65263	Fuel Delivery Pressure
	65263	Engine Oil Level
	65263	Engine Oil Pressure

ICON (or Text)	PGN	DESCRIPTION
	65263	Coolant Pressure
	65263	Coolant Level
	DM1	Service Indicator
VEH SPD	65265	Wheel-Based Vehicle Speed
FUEL RATE	65266	Fuel Rate
FUEL ECON	65266	Instantaneous Fuel Economy
AVG ECON	65266	Average Fuel Economy
BARO PRES	65269	Barometric Pressure
	65269	Air Inlet Temperature
BST PRES	65270	Boost Pressure
	65270	Intake Temperature
	65270	Air Filter Pressure
	65270	Exhaust Gas Temperature
ALT VOLT	65271	Alternator Voltage
	65271	Electrical Voltage
	65271	Battery Voltage
	65272	Transmission Oil Pressure
	65272	Transmission Oil Temperature
	65213	Fan Drive
AUX TEMP	65164	Auxiliary Temperature
AUX PRES	65164	Auxiliary Pressure
SLECT GEAR	61445	Selected Gear

ICON (or Text)	PGN	DESCRIPTION
CURNT GEAR	61445	Current Gear
OUT SFT SP	61442	Output Shaft Speed
IN SFT SPD	61442	Input Shaft Speed
TORQ LOCK	61442	Torque Converter Lockup
AUX IO 1	65241	Auxiliary IO Status 1
PEDAL SWT	61443	Accelerator Pedal Switch
DES ENG SP	65247	Desired Engine Speed
THROTTLE	65266	Throttle Position
	65270	Air Inlet Pressure
ENG TIMING	65159	Actual Engine Timing
ENG REVOLU	65253	Total Engine Revolutions
REQ GEAR	65256	Requested Gear
	65276	Fuel Level
	61448	Hydraulic Pressure
	65128	Hydraulic Temperature
	61444	Machine Hours
	Analog Input 1	Discharge Pressure
	Analog Input 2	Suction Pressure

4

STARTING TO SWEEP

Figure 4-16. Sweeping with the RB50 Broom

The Rosco RB50 Broom features excellent broom control and functionality for a variety of industrial and commercial applications. The broom operator controls are conveniently located on the control panel to the operator's right side. These are the primary controls used while sweeping. Refer to controls on ***Pages 3-6 and 3-7.***

This broom is easy to operate for drivers of all experience levels. Its heavy-gauge, rectangular tube frame and industrial-grade structure make the RB50 a road workhorse from general application sweeping to highway jobs. Add the optional front blade for mild pushing and/or optional curb brush to maximize productivity.

Joystick Controls

The transmission selector joystick is equipped with controls to lift or float the brush, and swing it left or right. **(Figure 4-17)** The operator can control these brush functions along with increasing or decreasing the machine's speed as needed for the job by pushing the joystick lever forward and in reverse. The brush speed is also controlled by changing the engine's RPMs using the joystick transmission control.

Figure 4-17. Joystick

Brush Lift/Float

The Brush Lift/Float Control switch is located in the center of the joystick. **(Figure 4-17)**

- Pushing the button up lifts the brush. Releasing the switch automatically returns the brush to the "hold" position.
- The center position "holds" or locks the brush carrier at the height last used while in the UP position.
- Pushing down on the switch activates the brush "float" mode. When releasing the switch back to center, float is maintained until the UP switch is pressed.

Brush Swing

Push the brush swing switches on the joystick slowly to gradually adjust the brush swing left or right. **(Figure 4-17)** Hold the button down to move the brush to its full swing position.

Brush Rotation Lever

The Brush Rotation lever is located on the control panel. **(Figure 4-18)**

- Pushing the lever forward allows the brush to push debris to the front of the broom. This is the normal sweeping position.
- Pulling the lever back turns the brush rotation off.

Brush Speed and Down Pressure

The Brush Down Pressure knob controls the amount of brush contact pressure to the work surface. **(Figure 4-18)**

- Turn dial left to increase brush speed.
- Turn dial right to decrease brush speed based on engine RPMs.
- Turn the dial clockwise to increase pressure.
- Turn the dial counterclockwise to decrease pressure.
- For best results, operate in the FLOAT position.

NOTICE **Be careful not to use excessive down pressure as the brush can be damaged.**

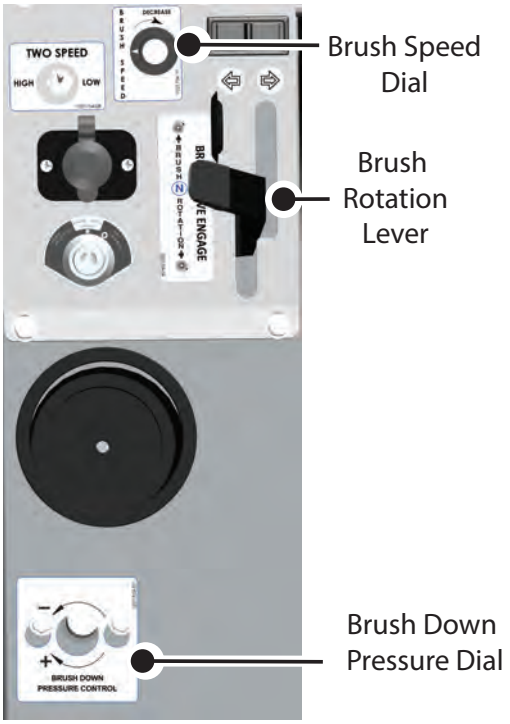


Figure 4-18. Brush Down Pressure Knob

SWEEPING GUIDELINES

An easy way to determine the ideal brush height setting is to sprinkle a little sand or dirt under the brush before beginning. Put transmission in neutral and set the park brake. Engage the brush using the Brush Rotation lever and press the Lift/Float control switch. Turn off the Forward/Reverse and raise the brush. There should be a clean swatch under the entire length of the brush.

Figure 4-19. Sweeping

- Sweep with the tips of the brush bristles for maximum results. Adjust the sweeping action using the Brush Down Pressure control valve to create a 3- to 4-inch wide footprint the width of the sweeper where the broom touches the pavement.

NOTICE **Excessive down pressure can wear and possibly damage the brush.**

- For most sweeping operations, the FLOAT position works best.
- For heavy sweeping, slow the ground speed and increase the brush RPMs. The right combination of ground speed and brush speed will correct ineffective sweeping.
- Engage the dust suppression system when sweeping in extremely dusty areas, especially when sweeping in high-traffic areas.

BRUSH CORE ADJUSTMENT

There are two adjustment screws on the brush frame for adjusting the brush core due to uneven brush wear or "coning." **(Figure 4-20)**

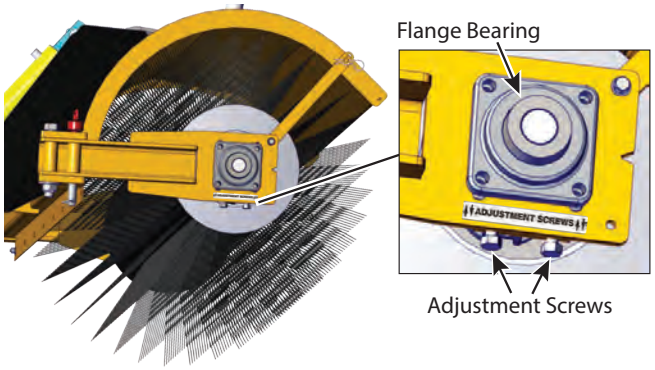


Figure 4-20. Brush Core Adjustment Screws

To make this adjustment:

1. Park vehicle on a flat surface and engage the parking brake.
2. Align the brush frame and brush core parallel to the front bumper.
3. Measure the height from the bottom of the brush bristles to the ground. **(Figure 4-21)**

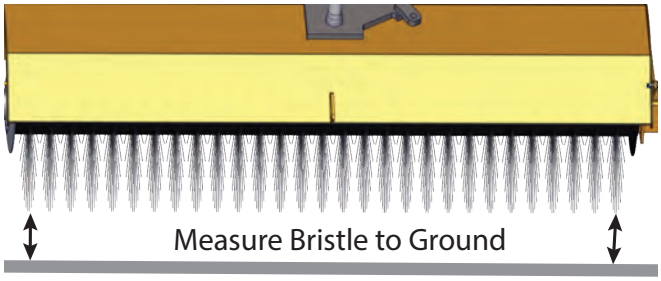


Figure 4-21. Measurement

4. Loosen the four (4) flange-bearing mounting bolts. DO NOT REMOVE.
5. Use the adjustment screws to bring the drive and swing arm sides of the core up or down, ensuring equal distance from the ground on both sides to level the brush core.
   - Turn bolt clockwise to raise height.
   - Turn bolt counterclockwise to lower height.
6. Once adjusted, retighten all four flange-bearing mounting bolts.

QUICK-CHANGE BRUSH CORE

The RB50 features a Quick-Change Core System that requires no tools to change the brush. Be sure to grease the brush core shaft bearings and lift pivot daily using a multi-purpose grease for optimum operation.

1. Park broom on flat surface, place machine in neutral, turn off the engine and set park brake.
2. Remove both cotter pins from brush cover. **(Figure 4-22)**

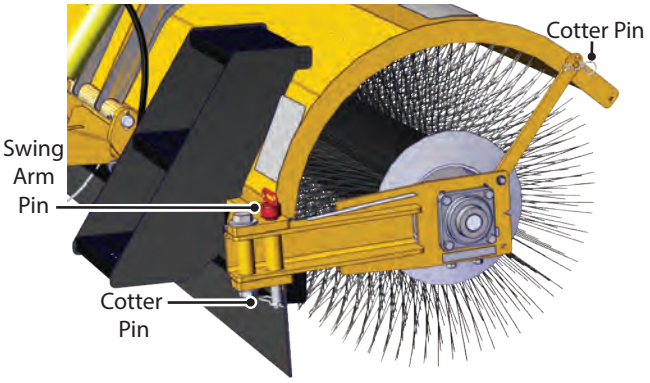


Figure 4-22. Brush Core Pins

3. Remove the swing arm pin.
4. Pull the brush core away from the drive hub and slide out. Set aside.
5. Check hub on both sides for damage. (Consult your authorized dealer if any damage is noted.)
6. Grease the U-Joint on the idle hub.
7. Slide new brush onto the slotted drive hub, ensuring the key stock inside the brush core goes onto the drive hub side.
8. Align the idle hub with the smooth end of the brush core and push the swing arm back into position into the brush core, keeping hands away from pinch points. Carefully guide the hub by hand if necessary to start into the core center.

NOTE: Brush carriage may need to be raised slightly to install a new core if the old brush core was worn.

9. Once the idle hub is fully engaged, reinsert the swing arm pin and cotter pins.

TRANSPORTING THE BROOM

Do not tow the RB50 to prevent damage to the machine. Transporting the broom on a flatbed truck (or trailer) is the preferred method of transporting the broom safely in all driving situations.

1. Clean the broom with a pressure washer, ensuring you remove all loose gravel, mud or debris from the wheels, frame and brush.
2. Ensure the combined height of the truck bed (or trailer) and loaded broom meeting height restrictions for the transport route, including bridges, overpasses and overhead obstructions.
3. Park the transport vehicle on a level surface and set the park brake.
4. Install chocks or blocks against the truck and trailer wheels.
5. Use a ramp or loading dock that has the capacity to support the broom, with a low angle of rise to the trailer bed.
6. Using the joystick controller, center the brush. Raise the brush to its full upright position.
7. Carefully and slowly drive the broom onto the trailer, centering the broom over the center line of the trailer.
8. Set the park brake and block brooms wheels in both directions with blocks or chocks to prevent rolling.
9. Lock brush in the UP position using the brush lock arm as shown below in **Figure 4-23.** To lock, move arm up as shown.

Figure 4-23. Brush Lock

10. Turn off the engine and remove key.
11. Fasten chains (or cables) to the broom frame using the four D-Ring tie-downs located on each side of the frame front and rear. **(Figure 4-24)**
    - Be careful you do not place chains or cables over the broom's hoses or electrical harnesses. Fasten chain or cable only to the D-Rings on the frame.

⚠WARNING To prevent the broom from breaking loose and being damaged in transport, ensure it is properly secured to the transport trailer.

Figure 4-24. D-Ring Tie Down

12. Cover the exhaust opening with heavy gauge plastic to prevent dust and moisture from entering the engine.
13. After transporting the machine, be sure to remove the plastic before operating.

OPTIONS

Brushes

The 7.6-feet brush options are Poly/Steel Tube, Poly Wafer, or Poly/Steel Water brushes. Contact your authorized Rosco LeeBoy dealer for purchase.

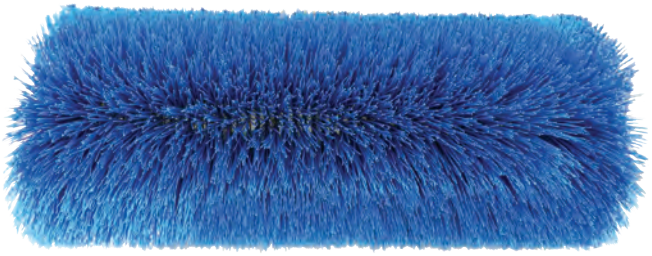


Figure 4-25. Brush Replacement Options

Front Windshield Washer and Rear Wiper/Washer

The front windshield wiper is standard, but the windshield washer is optional. A rear wiper with washer assembly is also an available option. See your authorized dealer for these options.

Rearview Camera System

A rearview camera system is also an available option. **(Figure 4-26)** This additional safety feature assists operators when moving in reverse by using the rear camera to see directly behind the machine.

Figure 4-26. Rearview Camera

Dust Suppression System

In certain conditions, it may be necessary to sweep using the Dust Suppression System to control dust. The Rosco RB50 is equipped with a 150-gallon water tank along with corrosion-resistant nozzles along the spray bar that spray water as the broom sweeps. **(Figure 4-27)** The spray bar runs the length of the broom. Be sure to inspect and clean the water nozzles after each use to ensure the spray pattern is not adversely affected.

Figure 4-27. Dust Suppression System

To operate the Dust Suppression system:

1. Fill up the water tank.
2. Turn on the Water System switch located on the control panel.
3. Follow the sweeping procedures and guidelines contained in this section.
4. Be sure to inspect and clean the water nozzles after each use to ensure the spray pattern is not adversely affected.

Cab Audio and Bluetooth

Equip your broom with AM/FM radio, CD player and Bluetooth audio system for operator enjoyment on the job.

Forward/Reverse Brush Rotation

Adding the ability for reverse brush rotation provides more flexibility for optimal cleaning.

Spare Tire and Mount

An optional spare tire with mount is also available. See your authorized dealer for purchase and installation.

Curb Brush

An optional curb brush attachment hydraulically adjusts for sweeping next to retaining walls, buildings, guardrails and curbs. It conveniently raises to a vertical position when not in use. **(Figure 4-28)** The curb brush features a 27-inch (68.6 cm) diameter heat-treated, steel brush with hydraulic motor drive. Refer to **Section 7,** *Illustrated Parts List (IPL),* for parts diagram.

1. Raise the brush to its full upright position and lock into place.
2. Turn off the engine and set the park brake.

NOTE: Be careful handling the curb brush as the bristles are sharp and could cause minor injury.

Figure 4-28. Curb Brush Attachment Option

Strike-Off Blade

You will need an assistant to attach and remove the optional strike-off blade attachment to the front of the broom. **(Figure 4-29)** Refer to **Section 7,** *Illustrated Parts List (IPL),* for parts diagram.

1. Park the broom on a level surface, turn off the engine and engage the parking brake.
2. Clean the blade-mounting front fender thoroughly before attaching blade.
3. Support the blade with jack stands in front of the broom front fender.

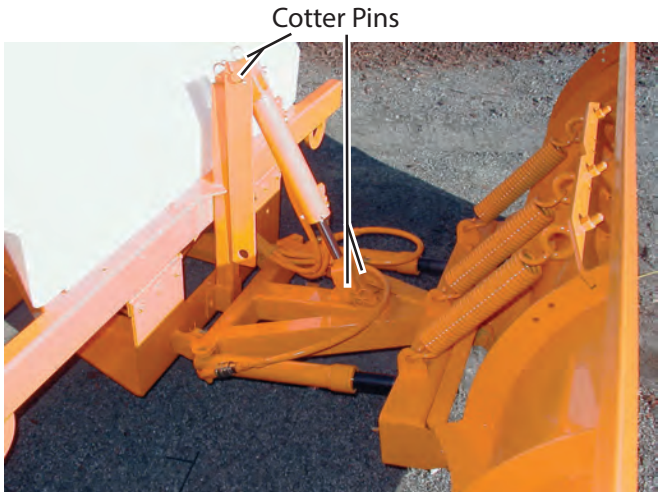


Figure 4-29. Blade Attachment Option

Figure 4-30. Telescoping Steering and Transmission Foot Pedal

Magnetic Cup Holder

Magnetic cup holders can be purchased for use with your road widener (and other machines in your fleet). **(Figure 4-31)**

See your authorized dealer for details. Ask for **Part Number 1020521.**

Figure 4-31. Magnetic Cup Holder

Telescoping Steering

The optional telescoping steering column coupled with the standard tilt column adds maximum cab comfort on the job. **(Figure 4-30)** The telescoping steering column option allows you to tilt the steering column and adjust your steering wheel for a more comfortable ride.

Cab Air Quality Kit

The Cab Air Quality Kit option offers the best line of defense for protecting your operators from exposure to harmful dust and toxins while sweeping. The RESPA®-CF2 delivers a powerful pre-cleaner, filter and pressurizer in a compact, custom unit that sustains positive pressurization and clean air in the cab.

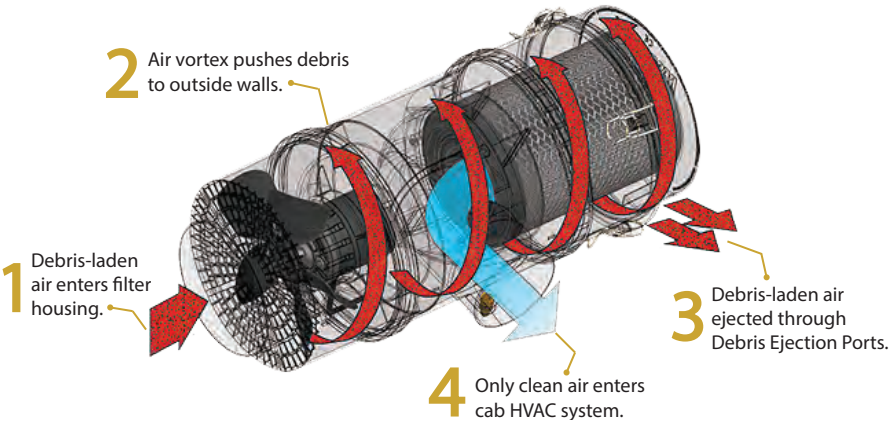


Figure 4-32. Vortex HyperFLOW® Technology

You will save maintenance costs with longer intervals for filter changes and less wear on the HVAC system. The self-cleaning air filter extends filter life up to 1,000 hours. Its air vortex HYPERFLOW® system continuously ejects particulate without collecting in filter housing.

Healthy, efficient and economical, the Cab Air Quality Kit can be easily installed on your broom. The system is recommended by the National Institute of Occupational Safety and Health. Contact your authorized dealer for purchase and retrofit.

Section 5

MAINTENANCE

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MAINTENANCE SCHEDULE

Before performing any maintenance procedures on the Rosco RB50 Broom, review the safety information in **Section 1**.

⚠WARNING Always use the appropriate and correct sized tools for the task at hand to prevent damage or possible injury.

⚠WARNING Always wear heat-resistant gloves when handling hot components.

Following the maintenance schedules and procedures will maintain the machine in top operating condition and provide years of trouble-free operation. Refer to the Engine Operator's Manual accompanying your machine for more detailed engine service information.

When performing any routine maintenance, always include the previous routine maintenance hours in the higher hourly schedule.

NOTICE Changing oil and cleaning the machine should only be done in a designated area where the oil and chemicals can be contained. These by-products should be discarded in accordance with environmental regulations.

Table 5-1. Periodic Maintenance Schedule

SYSTEM	ITEM	Every 10 Hours Daily	Every 50 Hours Weekly	Every 100 Hours Monthly	Every 250 Hours Quarterly	Every 500 Hours Semi-Annually	Every 1000 Hours Annually
Engine	Check oil level.	X					
	Check coolant level.	X					
	Replace coolant.					X	
	Check primary and secondary air filters.		X				
	Change air filters.			X			
	Replace fuel filter.				X		
	Check and drain fuel water separator.				X		
	Change oil and oil filter. **(Initial 50 Hours)**				X		
	Check main engine belt (Change every 3000 hours).			X			
Hydraulic	Check hydraulic oil level and filters.	X					
	Change hydraulic filters.				X		
	Change hydraulic oil.						X
	Clean hydraulic oil strainer.				X		
	Drain hydraulic tank. Replace suction strainer.						X
Mechanical	Lubricate steering axle pivot.			X			
	Lubricate steering axle king pins and tie rods.			X			
	Replace drive axle oil.						X
	Repack wheel bearings.						X
	Check fluid at master cylinder.	X					
	Change filter for crankcase bleeding.					X	

5

ROUTINE MAINTENANCE

Preparing Machine for Maintenance

1. Park the machine on a flat surface.
2. Engage the parking brake.
3. Turn off the broom drive.
4. Run engine at half-speed for three to five minutes, then reduce engine speed to a slow idle.
5. Lock brush in the UP position using the brush cylinder lock as shown below in **Figure 5-1.** To lock, move lock arm up as shown.

Figure 5-1. Cylinder Lock

6. Place ignition switch in the OFF position.

⚠WARNING If any maintenance must be performed while the engine is running, DO NOT leave the machine unattended.

Every 10 Hours or Daily Routine Maintenance

1. Lubricate the required parts. ***(Pages 5-6 to 5-8)***
2. Check battery cables and connections. ***(Page 5-13)***
3. Check coolant level. Fill if needed.
4. Check for indications of oil leakage. Check for loose, worn or damaged hoses, tubes and clamps. Tighten, repair or replace promptly.
5. Keep the fuel tank full to prevent condensation. Always fill at the end of the day.
6. Inspect the hand spray wand nozzle (and spray bar nozzles if equipped) to ensure they are free from clogs and operating properly.
7. Check the tires for proper inflation before operating the broom. ***(Pages 5-22 to 5-23)***
8. Grease the brush core shaft bearings and brush lift bushings. ***(Pages 5-6 to 5-8)***
9. Check engine oil level. ***(Page 5-9)***
10. Check hydraulic oil level and filters. ***(Page 5-18)***
11. Check hydraulic fluid at master cylinder in cab on the floor. ***(Page 5-25)***
12. Perform any other engine preventive maintenance described in the Engine Operator's Manual.

Every 50 Hours or Weekly Routine Maintenance

1. The FIRST 50 HOURS, change engine oil and oil filter, then every 250 hours thereafter. ***(Page 5-11)***
2. Lubricate the steering axle king pins and tie rods. ***(Pages 5-6 to 5-8)***
3. Check all battery connections and remove any corrosion present. ***(Page 5-13)***
4. Check air filter. Clean or replace if needed. ***(Page 5-12)***
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 100 Hour or Monthly Maintenance

1. Check main engine belt. Replace if needed.
2. Change air filter. ***(Page 5-11)***
3. Lubricate steering axle pivot. ***(Pages 5-6 to 5-8)***
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 250 Hours or Quarterly Routine Maintenance

1. Clean hydraulic oil strainer. ***(Page 5-19)***
2. Change the hydraulic oil and charge filter. ***(Page 5-18)***
3. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 500 Hours or Semi-Annual Routine Maintenance

1. Drain and replace coolant.
2. Change engine oil and filter. ***(Page 5-10)***
3. Replace fuel filter. ***(Page 5-12)***
4. Check and drain fuel water separator. ***(Page 5-13)*** Replace filter if needed.
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 1000 Hours or Annual Routine Maintenance

1. Replace strainer in the hydraulic tank. ***(Page 5-19)***
2. Replace drive axle oil. ***(Page 5-25)***
3. Repack front wheel bearings. ***(Page 5-24)***
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

MACHINE LUBRICATION

Proper lubrication is necessary to maintain the machine at top efficiency. Use a high-quality multi-purpose grease for optimum performance.

(See the following pages for lubrication points.)

Be sure to grease the brush core shaft bearing and lift pivot daily.

It is important to grease the king pin bushings, tie rod ends, steering cylinder rod and steering axle pivot bushings every 100 hours of service.

⚠WARNING Use jack stands on a solid, flat surface to prevent the machine from tipping when it is raised. Block the remaining wheels using sturdy wheel chocks. DO NOT lie under the broom while supported only by a jack. Death or serious injury can result from improperly supporting a raised machine.

Use the following procedure for lubricating these components:

1. Locate and clean the single grease fitting at each of the lubrication points.
2. Apply one shot of grease to each fitting with a manual (hand) grease gun.
3. Wipe away any excess grease.

5

Lubrication Points

Lubrication points are shown below:

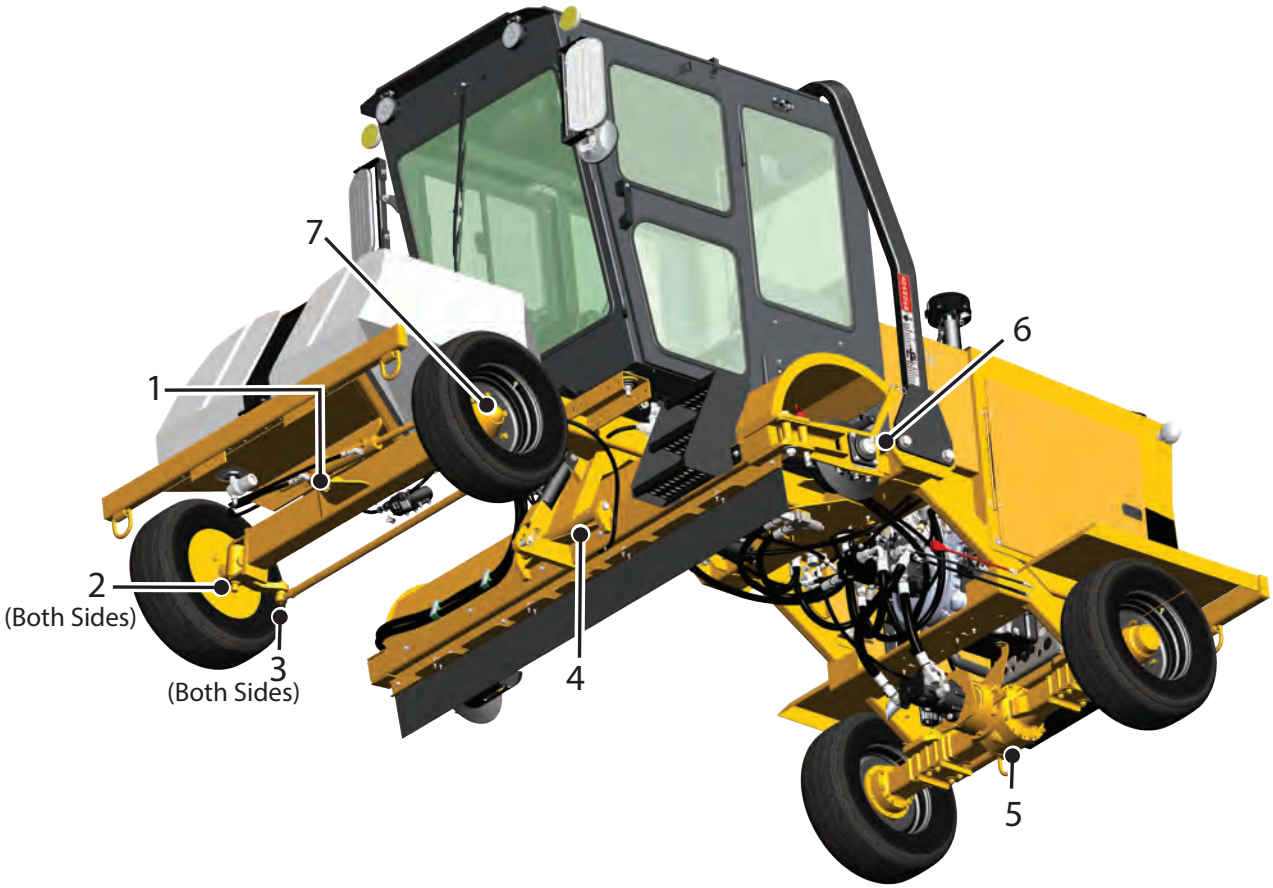
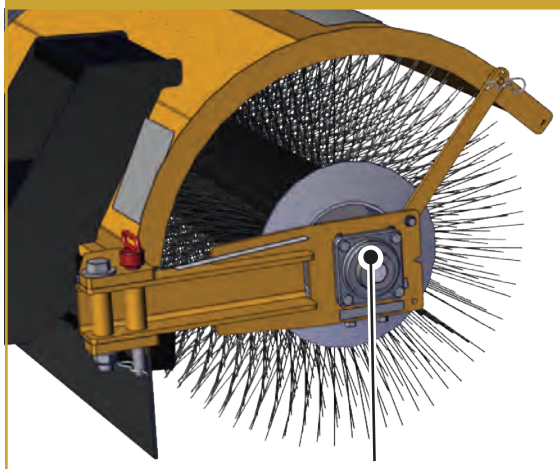
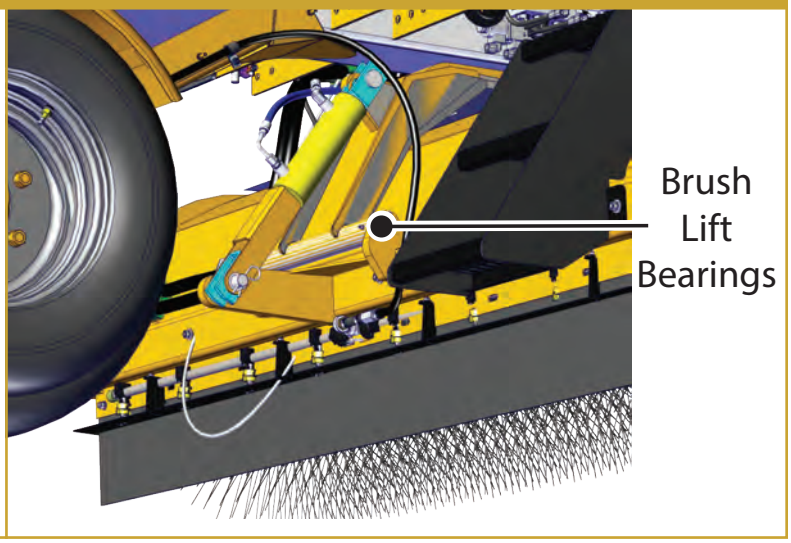


Figure 5-2. Lubrication Points Schedule

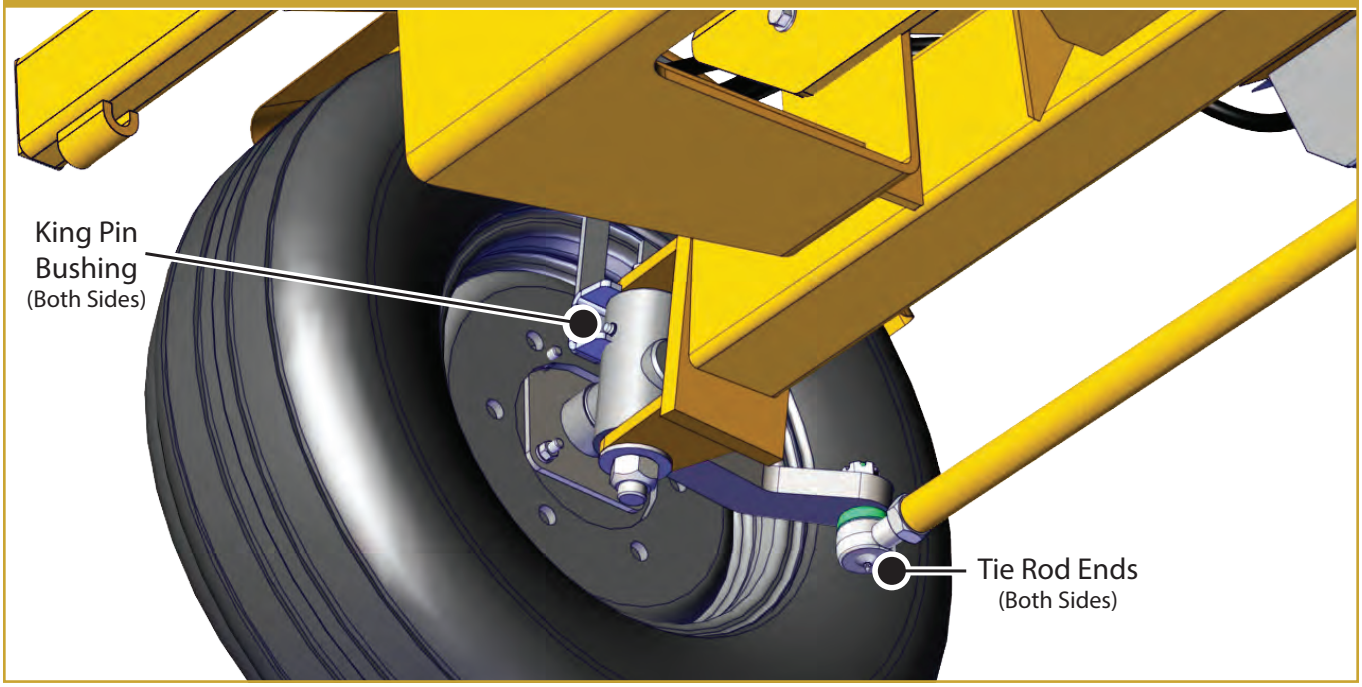
No.	ITEM	TYPE OF LUBRICANT	INTERVAL
1	Steering Axle Pivot	Multi-Purpose Grease	Monthly
2	King Pin Bushings	Multi-Purpose Grease	Weekly
3	Tie Rod Ends	Multi-Purpose Grease	Weekly
4	Brush Lift Bushings	Multi-Purpose Grease	Daily
5	Drive Axle	Gear Oil	Every 6 Months
6	Brush Core Shaft Bearings	Multi-Purpose Grease	Daily
7	Steering Axle Bearings	Multi-Purpose Grease	Every 6 Months

Daily Lubrication





Weekly Lubrication



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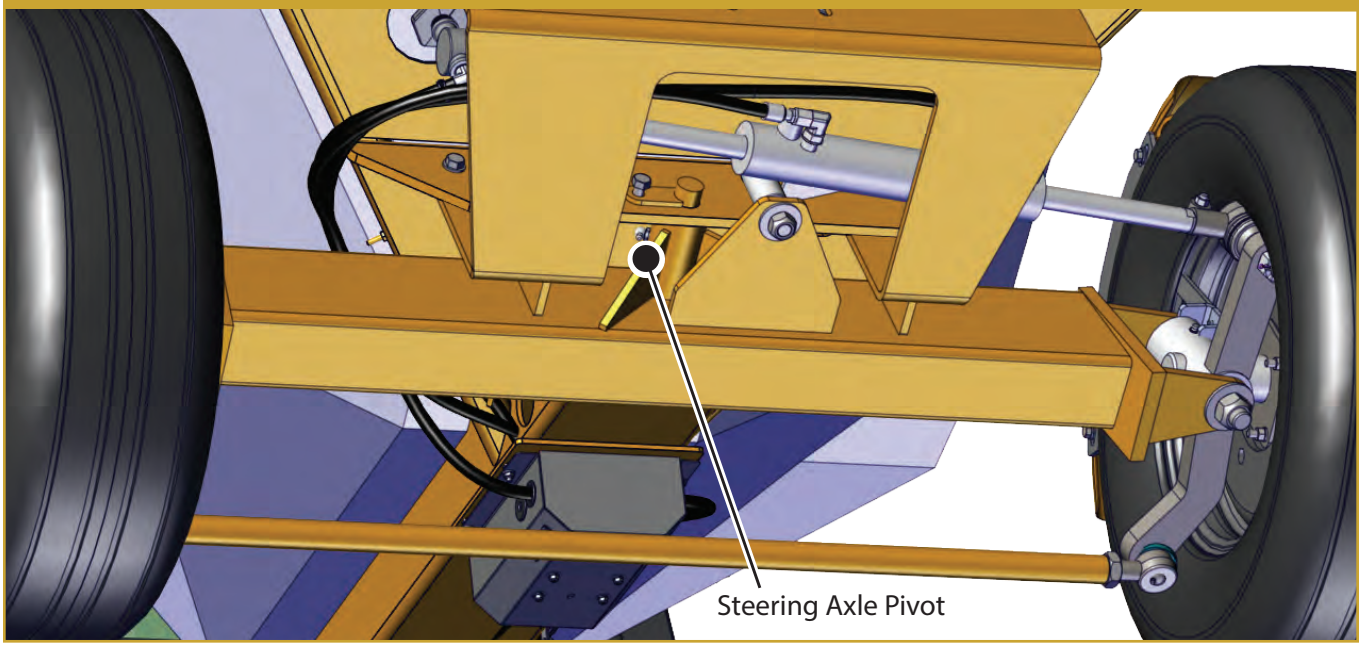
Monthly Lubrication



Biannual and Annual Lubrication

Drive Axle

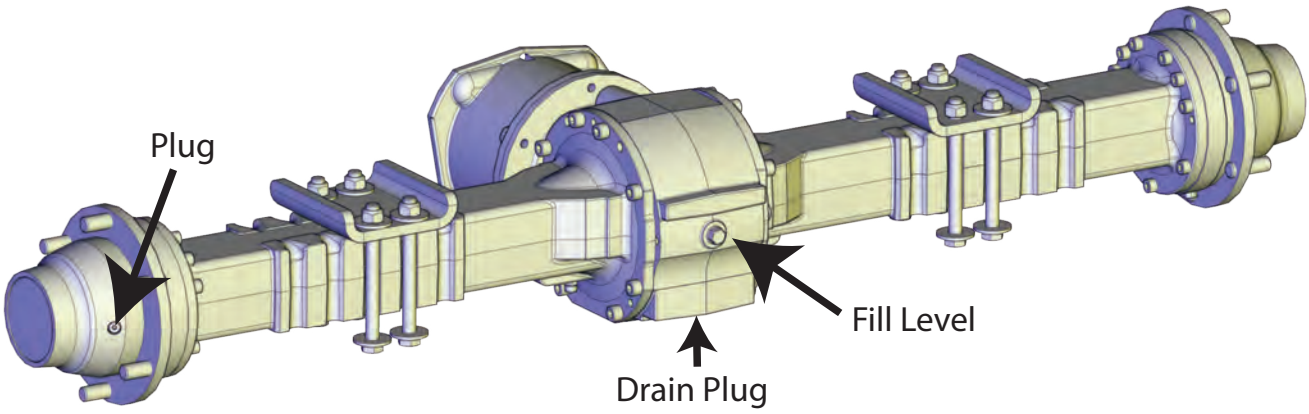
Check the drive axle oil level at least every six months and fill with gear oil if needed.

To fill the oil:

1. Rotate the hub on one side of the front axle until the plug is in the 12 o'clock position.
2. Remove the plug.
3. Add one quart of 80W-90LS GL5 oil..
4. Replace the plug.
5. Follow Steps 1 - 4 on the other side.

Change the drive axle oil every 12 months:

1. Remove the level plug in the center of the housing.
2. Remove the breather plug in the housing.
3. Add oil until it starts coming out of the level plug.
4. Replace the level plug.
5. Allow the axle to sit level for at least five minutes to allow the oil to flow to the hubs.
6. Remove the level plug and check oil level.
7. Add more oil if needed.



ENGINE MAINTENANCE

The Hatz 74 HP is a turbo-charged, four-cylinder diesel engine that also drives the hydraulic auxiliary pumps for steering and broom control. The engine performs at 2500 RPMs with peak torque at 184 ft lb @ 1600 RPMs and displacement at 119 cu in.

The diesel engine manufacturer recommends several engine-specific maintenance inspections every 250 hours, or every three months. You can access the engine and other machine components via the access panel doors on either side of the machine, depending upon the component. **(Figure 5-3)** (Refer to your Engine Operator's Manual for more detailed information regarding engine-related maintenance procedures.)

Check Engine Oil

The engine lubrication oil must be kept at a level above the ADD mark, but not above the FULL mark, on the engine lubrication oil dipstick.

To accurately check the engine lubrication oil level, park the machine on a level surface and stop the engine.

⚠WARNING ALWAYS stop the engine before checking the engine lubrication oil level as hot oil can be thrown, causing serious injury.

NOTICE **Wait five minutes after engine shutdown before removing the dipstick from the engine and checking the oil level to assure accuracy. This also prevents filling the engine with too much lubrication oil. Engine oil should be warm when checked.**

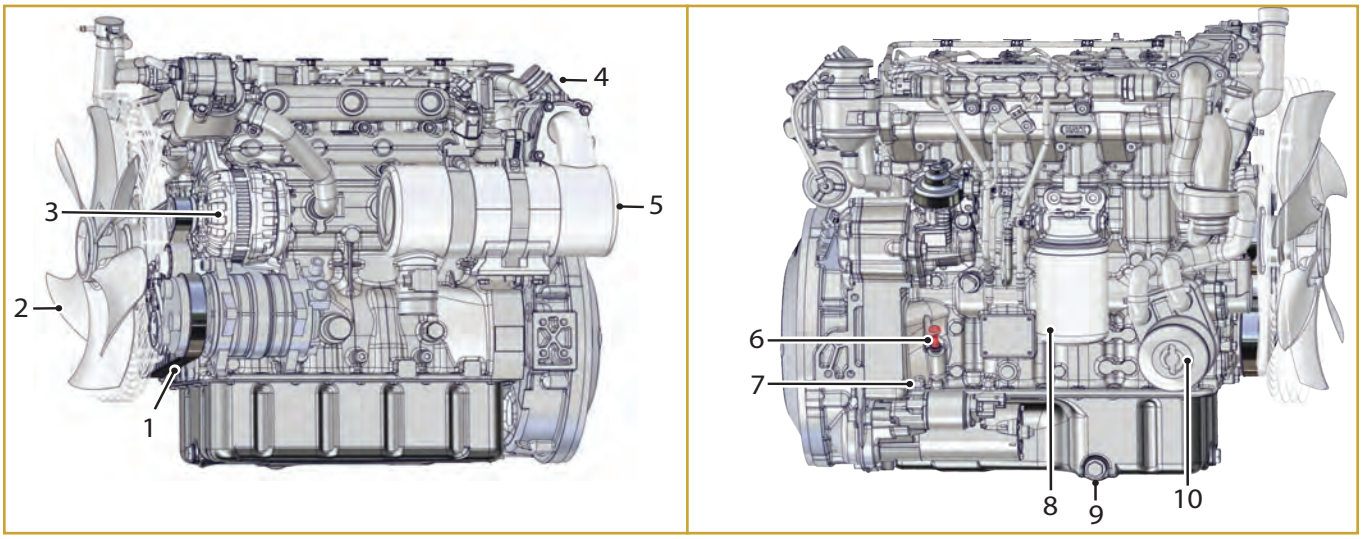


Figure 5-3. Engine Components

1 - Engine Belt

2 - Engine Fan

3 - Alternator

4 - Exhaust Gas Turbocharger

5 - Diesel Oxidation Catalytic Converter (DOC)

6 - Engine Oil Dipstick

7 - Bottom Engine Oil Filler Plug

8 - Fuel Filter

9 - Engine Oil Drain Screw

10 - Oil Filter

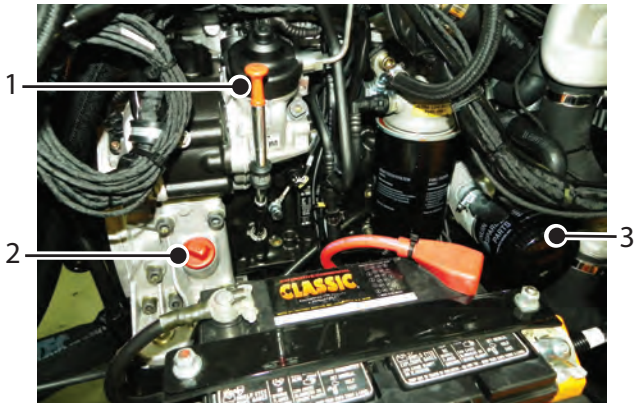


Figure 5-4. Engine Oil Components

1 - Oil Dipstick

2 - Oil Filler Cap

3 - Oil Filter

1. Open left engine access panel to access the oil dipstick. **(Figure 5-4)**
2. Clean around the engine lubrication oil dipstick, then remove it from the engine.
3. Wipe the dipstick with a clean cloth, then reinsert.
4. Pull dipstick out of the engine and check that oil level is near the full mark on the dipstick.
5. To add oil, remove the filler cap located next to the dipstick. Twist to remove cap and twist to install (use 15W40 motor oil).

Change Engine Oil and Filter

⚠WARNING **DO NOT change the engine lubrication oil when the engine and lubrication oil are hot. Hot oil can cause serious injury.**

NOTICE **DO NOT change the engine lubrication oil while the engine is running as serious damage or engine failure can occur.**

NOTICE **The engine lubrication oil filter must be changed when the engine lubrication oil is changed.**

While the engine is warm, proceed as follows:

1. Clean the area around the engine oil dipstick.
2. Place a container that has sufficient capacity to hold the drained oil directly under the engine oil drain plug.
3. Carefully remove the engine lubrication oil drain plug to allow the oil to drain into the container.
4. Wipe the area around the engine lubrication oil filter element and its mounting base with a clean cloth. **(Figure 5-4)**
5. Place the container under the filter element.
6. Use a filter removal wrench to loosen and remove the filter element by turning it in a counterclockwise direction.
7. Drain and discard the removed filter element.

NOTE: Be sure the used rubber gasket is removed and discarded with the filter element.

NOTE: Comply with environmental laws by properly discarding used oil, filter and gasket.

8. Wipe the inside area of the lubrication oil filter head using a clean, lint-free cloth.
9. Rub a little oil on the rubber gasket area of the new filter element and fill the new filter element with the oil.
10. Install the new filter element onto the filter head.
11. Carefully tighten the filter element (by hand only).

NOTE: Tighten the filter element as directed by the filter manufacturer.

12. Clean, reinstall and carefully tighten the lubrication oil drain plug.

NOTICE **DO NOT overtighten the drain plug.**

13. Fill the engine with 7.4 quarts of oil.
14. Reinsert the dipstick.

Inspect Cooling System

1. Open the right engine panel door to access the cooling fan. **(Figure 5-5)**
2. Check engine cooling fan for damaged or bent blades. Replace if damaged.
3. Inspect the fan belt for serviceability, looking for signs of cracking, tearing or separation. Replace if damaged.
4. Check fan belt and pulleys for dust, dirt and debris. Clean before starting the engine.

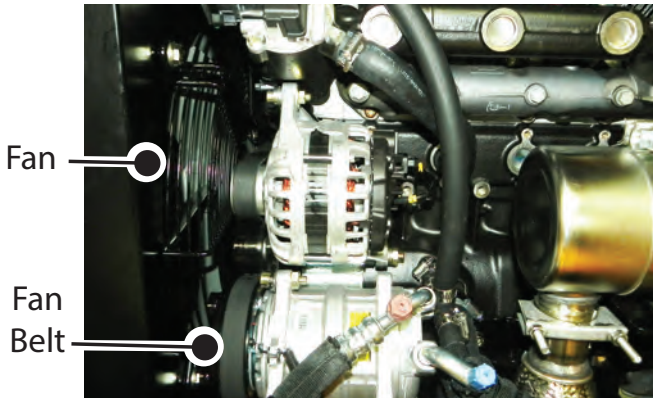


Figure 5-5. Fan and Fan Belt

Air Filter

The heavy-duty engine air filter system is mounted inside the (left) engine compartment. **(Figure 5-6)** To prolong engine life and prevent contaminants from entering the engine, inspect the air intake system components daily for damage, cracked hoses or loose clamps.

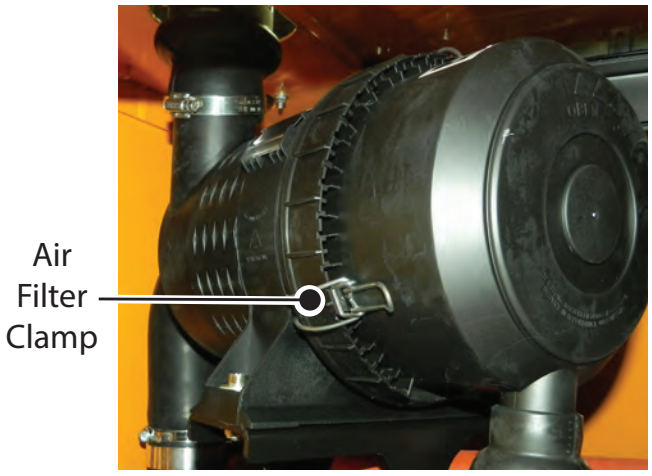


Figure 5-6. Air Filter and Fuel Filter

The air filter must be cleaned or replaced as follows:

1. Open clamps securing the cover on the filter housing, then remove the cover.
2. Remove the filter cartridge.
3. Wipe inside the air filter housing with a damp cloth.
4. Replace with the new filter cartridge if needed.
5. Reinstall cover and secure with the clamps.

NOTICE **Never operate the engine without an air cleaner as this will cause severe engine damage.**

FUEL SYSTEM

Fuel Tank

Always fill the fuel tank at the end of each work day to reduce the possibility of moisture accumulation from condensation in the tank.

⚠WARNING NO SMOKING while filling the fuel tank as all fuels for internal combustion engines are flammable. Fill the fuel tank only in a designated area where there is good ventilation and a fire extinguisher available.

⚠WARNING NEVER overfill the tank, check fuel level or check for fuel leaks near an open flame or near equipment that can create sparks.

⚠WARNING Diesel fuel is very flammable. Use extra caution and DO NOT spill fuel.

Fuel Filter

The fuel filter elements are located inside the left engine access panel. **(Figure 5-7)**

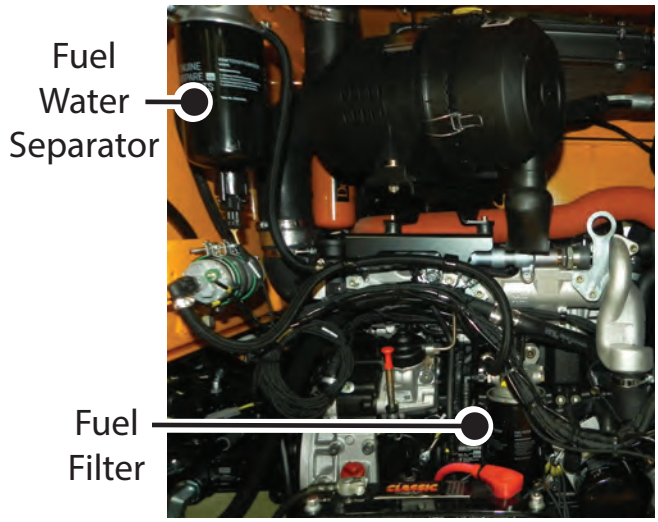


Figure 5-7. Fuel Filter and Water Separator Location

⚠WARNING DO NOT change the fuel filter while the machine is running. DO NOT change the fuel filter in an area near an open flame. DO NOT smoke while changing the fuel filter.

⚠DANGER NEVER loosen a fuel injector line as contents are under high pressure. DO NOT try to bleed air by loosening injector lines. NEVER loosen the bleed screw on the fuel filter. Large quantities of fuel under high pressure can escape and cause serious injury.

Replace the fuel filter using the following general procedures:

1. Stop the engine.
2. Put a container under fuel filters before removing the filter elements.

NOTE: Consider the environment when discarding used filters. Dispose of fuel filters in accordance with safe and lawful practices.

3. Wipe the area around the fuel filter elements and element mounting heads using a clean, lint-free cloth.
4. Use a filter removal wrench to loosen and remove fuel filter elements by turning in a counterclockwise direction. Drain and discard the removed elements.
5. Wipe the inside area of the filter heads with a clean, lint-free cloth. Fill the new fuel filter elements completely with clean fuel.
6. Put clean fuel onto the element rubber gaskets.
7. Install the new fuel filter elements onto the filter heads. Carefully tighten the elements (by hand only). **DO NOT overtighten the fuel filter elements onto the filter heads.**

NOTE: After changing the fuel filter, the ignition key must be set to the ON position for a few seconds sot he feed pump fills the filter with fuel and air is pushed out of the system.

8. Start the engine and check for ANY fuel leaks.

⚠WARNING STOP the engine immediately if any fuel leakage is noted. DO NOT start the engine until the leakage problem is corrected.

Fuel Water Separator

If water gets into the fuel system, it settles into the water separator. **(Figure 5-8)** When the water level reaches a certain level, an error code will appear on the PV380 digital display. It is important to drain it immediately.

NOTICE **If you run out of fuel, it is important to pump the water separator to prime fuel injection lines. Failure to do so can adversely affect performance.**

To drain the water separator:

1. Pull retainer clip from plug to disconnect the harness.
2. Unplug the harness, then unscrew the plug.
3. Drain and reinsert components.

Figure 5-8. Fuel Water Separator

To clean the water separator:

1. Follow the previous steps for draining the water separator.
2. Loosen and remove the filter body and clean it with light oil. Replace if needed.
3. Reinsert and tighten.

ELECTRICAL SYSTEM

Battery Servicing

The electrical system is powered by a 12-volt,negative-ground, maintenance-free battery located in its own compartment on the side of the cab. The master battery On/Off switch is also located in this compartment. **(Figure 5-9)**

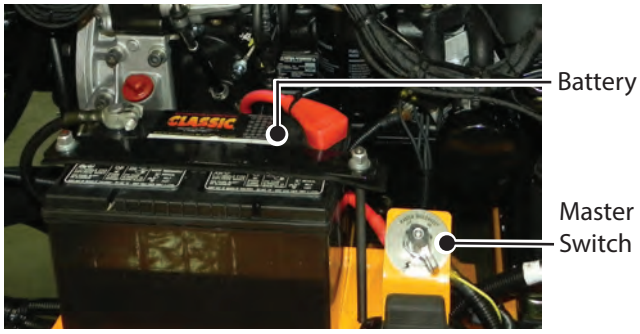


Figure 5-9. Battery

NOTE: The electrical system is a negative-ground system. Connect the positive (+) cable to the positive (+) post of the battery. Connect the ground cable to the negative (-) post of the battery. It is advisable to disconnect the negative (-) cable first and reconnect it last. Reversed polarity can damage the electrical system.

Clean terminals and battery posts thoroughly and check that battery cable terminals are tight. Dirty or loose connections can create high electrical resistance and permit arcing, which will quickly burn and pit terminals and posts.

Keep the battery clean by washing it off whenever dirt buildup is excessive. If corrosion is present around terminal connections, remove them and wash with ammonia solution or a solution consisting of 1/4 lb. (11 kg) baking soda added to one quart of warm water. Make certain the vent caps are tight to prevent solution from entering the cells. After cleaning, pour clean water over the battery and surrounding area to wash the solution away. Check vent cap breather openings to make sure they are open.

⚠WARNING **Batteries contain sulfuric acid and normally produce explosive gases that can cause serious injury. DO NOT allow flames or sparks to come near the battery.**

⚠WARNING **When charging or working near a battery, always shield your face and wear eye protection. In case of acid contact with skin or eyes, flush immediately with water for a minimum of 15 minutes and get prompt medical attention. If acid is swallowed, call a physician immediately.**

⚠CAUTION **When servicing the electrical system or welding on the machine, always turn the Master Disconnect switch to the OFF position to disconnect the ground strap from the battery to prevent damage to the machine electrical system.**

If the battery repeatedly discharges or the engine is difficult to start, check the electrical charging system. Using a battery tester, you can check the voltage following these procedures:

1. Shut down the engine and remove ignition key.
2. Place the positive (red) multimeter lead on the positive (+) battery terminal; and the negative (black) multimeter lead on the grounded negative (-) battery terminal.
3. With the engine off and the multimeter set at 12 VDC, the battery must show a charge of 12 - 12.5 volts.
4. Start the engine.
5. Check the battery charge level. The multimeter should read at least 13.5 volts.

Also check that the circuit breaker has not tripped. The circuit breaker is located inside the engine compartment on the auxiliary panel. **(Figure 5-10)**

If the battery does not charge properly, you may need a new battery or the alternator may need repair or replacement.

If storing the machine for more than 30 days, be sure to remove the battery and store it separately in a cool, dry place. Continue to keep the battery fully charged and check the level of electrolyte regularly to prevent damage or corrosion.

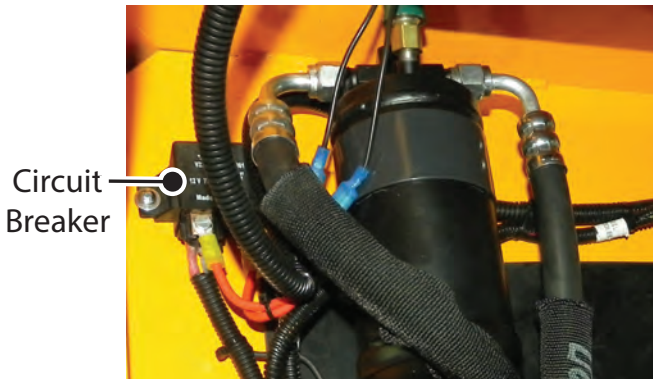


Figure 5-10. Circuit Breaker

Check Battery Electrolyte Level

⚠WARNING **Batteries create explosive gases when charging. DO NOT expose these gases to sparks or open flame during maintenance. Always perform battery service in a well-ventilated area or outside. Serious injury can occur from exploding batteries.**

⚠CAUTION **Do not attempt to charge a frozen battery. Allow a frozen battery to thaw completely before charging.**If using serviceable batteries, fill each cell to specified range with distilled water. DO NOT overfill.

6. Check battery charge and charging rate.
7. Shut down machine by turning ignition switch key counterclockwise to the OFF position.
8. Using a multimeter, check the battery charge by placing the positive (red) multimeter lead on the positive (+) battery terminal, and the negative (black) multimeter lead on the grounded (-) battery terminal. With the multimeter set at the 12 VDC setting, the battery should show a charge of at least 12 volts. If not, charge the battery or perform a load test.
9. Start the engine. From the operator's seat, check the battery charge level. The multimeter should read at least 13.5 volts. If the multimeter does not indicate minimum charge, check the machine charging system.

⚠WARNING **DO NOT dry batteries using compressed air. Allow the batteries and surrounding area to dry. Battery acid can cause severe injury if blown into the face or other areas of the body. If injury occurs, seek medical attention immediately. Flush eyes and skin for 30 minutes with clean water.**

Alternator

The alternator supplies electrical power to electronic controls and charges the battery. Its built-in regulator controls voltage output. **(Figure 5-11)**

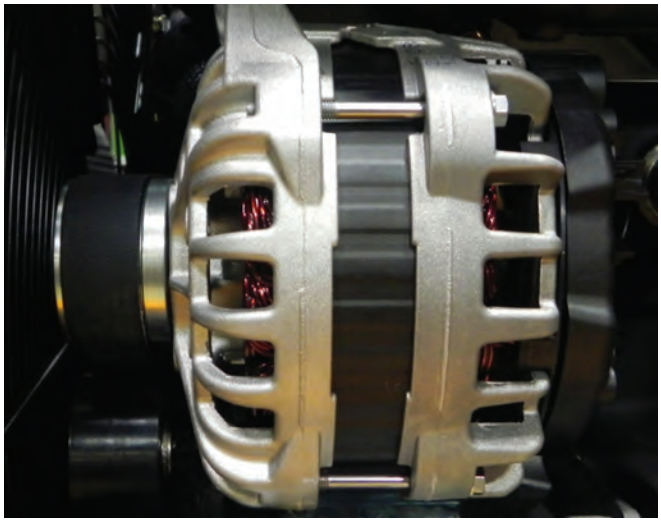


Figure 5-11. Alternator

Use the following precautions to prevent damage to the alternator or regulator:

- Never polarize the alternator.
- Never ground any alternator terminals or circuits.
- Always disconnect the battery before disconnecting or connecting the alternator.
- Never disconnect the alternator while machine is operating.
- Ensure wiring is properly connected before connecting the battery.
- Always connect a booster battery in the proper polarity: negative (-) to negative (-) and positive (+) to positive (+).

Lighting

Inspect lights for proper operation daily. If a light or group of lights does not function:

- Check the fuse box panel for a blown fuse.
- Examine all visible wiring connections, ensuring they are not frayed or damaged, and securely fastened.
- Check lighting mounts for proper ground.
- Remove light lens or casing and check bulb. Replace if needed.
- Inspect wiring harness for damage (see Schematics in **Section 6**).

If broken wires are found, solder together and cover with a shrink-wrap type of plastic covering (preferred) or electrician's tape to prevent contamination of the solder joint by moisture.

If you make any repairs to the wiring harness on the broom, always replace or repair the protective loom covering the wiring to prevent future damage to the wiring harness. Examine the routing of the harness and ensure it is not subjected to excessive movement, which often causes broken wiring.

Cold Weather Starting

NOTICE **Be sure to keep the battery fully charged during cold weather to keep it from freezing. Freezing weather has little effect on a fully charged battery.**

If a booster battery is necessary for cold weather starting:

1. Connect one end of the first jumper cable to the positive (+) terminal of the dead battery and the other end to the positive (+) terminal of the booster battery.
2. Connect one end of the second jumper cable to the negative (-) terminal of the booster battery and the other end to the frame of the machine with the dead battery.

Fuse Box Panel

The electrical system is protected from overload damage by fuses. If an electrical component fails to operate, check the fuse panel first to ensure that a fuse has not "blown." The panel is located on the right side panel in the cab. See the following table for corresponding fuse information. **(Figure 5-12)**

A blown fuse is an indication that there is an overload that needs to be corrected. If you replace a fuse, but it continues to blow, contact your authorized Rosco/LeeBoy dealer.

NOTICE **Always replace a blown fuse with the same rating as specified. Never replace with a higher amperage rating. Damage to the electrical system or fire can occur from improper fuse replacement.**

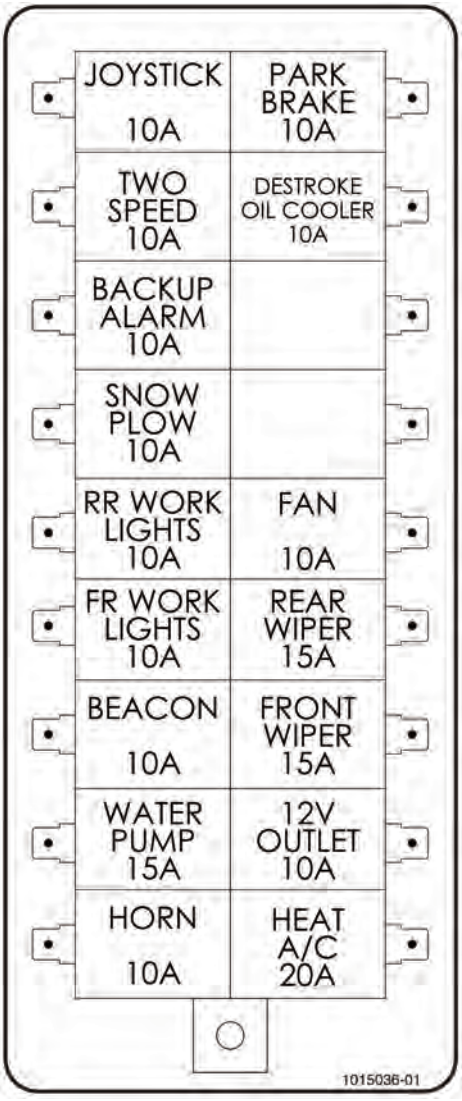


Figure 5-12. Fuse Panel and Fuse Sizes

Dielectric Grease

Dielectric grease is a nonconductive, insulating compound applied to electrical connectors. When applied correctly, a high-quality dielectric grease displaces debris and moisture in the connector. The proper application of dielectric grease fills any gaps between pins or in the connector which prevents debris and moisture from creating corrosion of the electrical components.

⚠CAUTION **Equipment damage can occur if using low-quality, incompatible dielectric grease.**

If not applied or incorrectly applied, corrosion can result that causes rapid increase in resistance, intermittent connections, overheating, and a variety of electrical problems. Numerous electrical issues throughout the machine can be avoided by the proper application and use of dielectric grease.

Figure 5-13. Correct Application of Dielectric Grease

To correctly apply dielectric grease:

- Apply a thin, visible, uniform coat using **ONLY** high-quality dielectric grease (Dow Corning 4 or equivalent) across the **MALE** connector as shown in **Figure 5-13.**
    - Coating too thin may not fill the gaps between pins.
    - Coating too thick may prevent proper connection or force pins or seals out. **(Figure 5-14)**
- Plug in the male connector.

⚠CAUTION DO NOT force connector to lock if too much grease has been applied.

⚠CAUTION Excessive grease application does not allow a proper connection by preventing the plugs from locking into place, and can cause damage to the connector seals.

Figure 5-14. Incorrect Application of Dielectric Grease

HYDRAULIC SYSTEM

The RB50 Broom features a hydraulic system consisting of an engine-mounted gear pump, three hydraulic cylinders, hydraulic reservoir, filter and a four-station valve block with solenoid-operated valves. The hydraulic motors and the hydraulic cylinders use the same hydraulic oil reservoir and oil supply.

The hydraulic pump and motor generally require no regularly scheduled maintenance, but inspect components for leaks or damage, which could be an indicator of the need for service. See your authorized LeeBoy/Rosco dealer if these components need servicing.

You should inspect the machine for hydraulic leaks daily, and check to ensure that hose fittings are secure and tight weekly.

If leaking fluid is found, it is probably on the pressure side of the hydraulic system. Find and repair the leaking component before using the machine to prevent damage to hydraulic components.

Leaks on the suction side of the hydraulic system are more difficult to find. This condition is serious since air or dirt introduced into the hydraulic system causes rapid component wear and eventual failure. Some symptoms of suction leakage are:

- Foaming of hydraulic oil.
- Sluggish hydraulic system operation.
- Unusual noise in the hydraulic pump or motor.

If you suspect a suction leak, verify that all reservoir connectors and fittings are properly tightened. If the problem persists, replace the defective hose assembly or fitting.

Contact your authorized LeeBoy/Rosco dealer for assistance and service repair if needed.

Your machine was factory-filled with Citgo All-Weather All-Temp VG32 hydraulic oil, a multi-grade anti-wear oil for use in equipment where wide temperature ranges are encountered. It is recommended that you use this hydraulic oil, or a comparable brand. Only introduce another brand after completely draining and flushing the hydraulic system, ensuring all hoses, cylinders, valves, motors and pumps have been purged of the factory-filled oil, and changing the oil filters.

⚠CAUTION Using other hydraulic oils could result in substandard performance or failure of the machine's hydraulic components.

⚠CAUTION DO NOT mix manufacturer or grade hydraulic oil weights. Substandard performance or hydraulic component failure can occur.

⚠WARNING NEVER attempt to repair hydraulic hoses with tape, clamps or cements. The hydraulic system operates under extremely high pressure. Temporary repairs will fail, creating a hazardous condition that can result in bodily injury and damage to the machine.

NOTICE Hydraulic oil that has oxidized or contains contamination can shorten the expected service life of the components in the hydraulic system.

Check Hydraulic Oil Level

Wait 10 minutes after engine has been shut down before checking hydraulic oil. If level is low, remove hydraulic oil fill cap strainer to add hydraulic oil to the tank.

Oil level is determined by sight gauge. **(Figure 5-15)** Hydraulic oil should be added if the oil level falls below the fill line. Never fill above the black fill line.

⚠CAUTION Use extreme caution when removing the filler cap to prevent any foreign matter from entering the hydraulic reservoir. This can cause damage to hydraulic components.

- Fill tank to the black fill line on sight gauge.
- Clean the fill cap strainer and reinstall it.

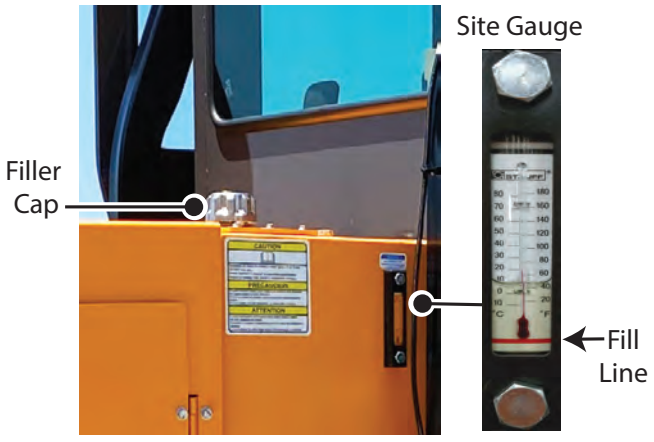


Figure 5-15. Hydraulic Sight Gauge and Filler Cap

Change Hydraulic Oil and Filter

Changing the hydraulic oil removes the accumulation of dirt, water and mechanical wear particles from the hydraulic oil reservoir and system. The chemical structure of hydraulic oil changes after continuous use, therefore, new oil is important to ensure correct operation of the hydraulic system.

NOTICE Never let the hydraulic oil reservoir tank run dry. Pump damage will occur.

Condensation can also build up in the hydraulic system and clog the filter elements. This could lead to insufficient hydraulic oil at the pump and degrade performance, causing damage to components.

Change the hydraulic oil and filter after the first 100 hours of use, then every 250 hours or seasonally, whichever comes first. Also remove and clean the suction strainer at the 500-hour intervals.

Use the following procedures to change the hydraulic oil and filter:

1. Stop the engine. Allow the hydraulic oil to cool until it is at a warm temperature. Slowly loosen and remove the hydraulic oil filler cap **(Figure 5-15)**. Put a clean, lint-free cloth over the reservoir fill tube opening and secure in place with tape.

⚠WARNING DO NOT drain the hydraulic oil from the reservoir when it is HOT. Hot hydraulic oil can cause serious injury. Drain at a warm temperature only.

2. Carefully remove the plug from the hydraulic tank. Use a drain collection container of sufficient capacity to collect the hydraulic oil underneath. Allow all of the hydraulic oil to drain into the container.
3. Check the hydraulic suction strainer in bottom of the hydraulic oil tank. Clean if necessary.
4. Install the hydraulic oil reservoir drain plug and tighten securely.

5. Unscrew hydraulic oil filter and remove. **(Figure 5-16)**

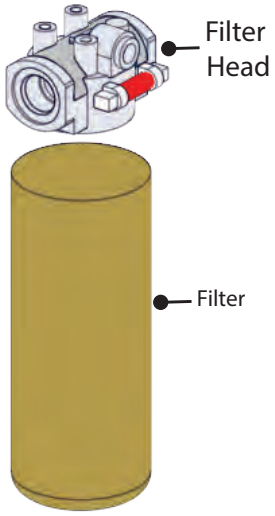


Figure 5-16. Hydraulic Oil Filter Components

6. Discard filter properly in accordance with government environmental regulations.
7. Reinstall new hydraulic oil filter.
8. Carefully remove the cloth from the hydraulic oil reservoir fill tube opening.
9. Refill the hydraulic oil reservoir (recommend Citgo All Weather/All Temperature Multiviscosity hydraulic oil).

NOTICE DO NOT fill the hydraulic oil reservoir with new hydraulic oil until the suction strainer has been cleaned. (See next section.)

NOTICE DO NOT overfill the hydraulic oil reservoir with oil.

10. Check the oil level at the sight gauge and add more if needed.
11. Replace the hydraulic oil fill strainer cap onto the reservoir filler neck and tighten securely.
12. Start the engine and operate hydraulic controls several times.
13. Check the hydraulic system for any leaks.

Hydraulic Pump and Motor

The hydraulic pump and motor generally require no regularly scheduled maintenance. Frequent inspection for leaks will indicate the need for service of these components. See your authorized LB Performance dealer for service if needed.

Change Hydraulic Oil Strainer

The oil strainer is mounted in the hydraulic oil filler opening under the filler cap. **(Figure 5-17)**

⚠WARNING DO NOT remove the hydraulic filler cap from the reservoir when it is hot as this can cause serious injury. Allow hydraulic oil to cool down to a warm temperature.

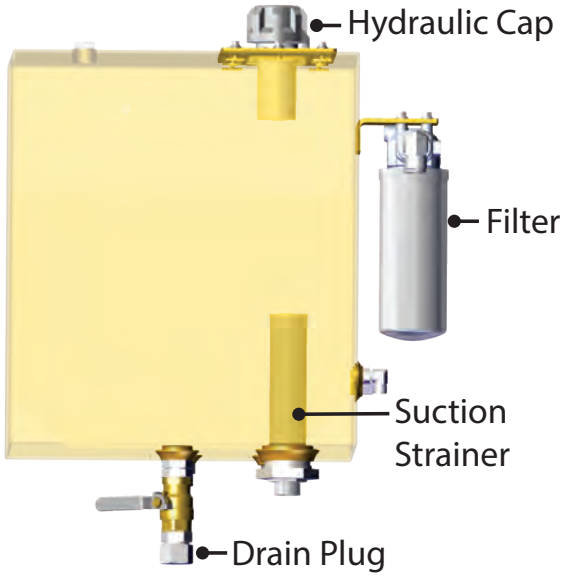


Figure 5-17. Hydraulic Suction Strainer

To change the hydraulic oil strainer:

1. Remove the hydraulic oil filler cap.
2. Remove the six (6) screws securing the strainer, then remove the strainer and gasket.
3. Install a new gasket, aligning the six screw holes in the gasket with the mounting holes on the reservoir.
4. Install the new strainer, aligning the holes in the strainer with the mounting holes of the gasket and secure the strainer with the six screws.
5. Fill the hydraulic oil reservoir with (recommend Citgo All Weather/All Temperature Multiviscosity hydraulic oil).

NOTICE DO NOT overfill the hydraulic oil reservoir with oil.

6. Check the oil level in the hydraulic oil reservoir. Add oil if needed.
7. Reinstall the hydraulic oil filler cap and tighten securely.

Priority Relief Valve

The priority flow divider in the hydraulic manifold is set to supply the power steering with approximately 4 GPM (15 LPM) of hydraulic flow. This flow goes to the steering circuit before any other circuit.

The Priority Relief Valve controls the maximum operating pressure for the power steering, brush lift and swing circuits. **(Figure 5-18)** This valve is located in the hydraulic pump attached to the rear of the hydrostatic pump on the engine flywheel.

These are signs the priority relief valve needs adjusting:

- Total or partial loss of steering functions or hard steering.
- Constant noise from the hydraulic pump when using steering or brush lift and swing.
- Hydraulic oil overheating.

⚠WARNING Only an experienced technician should service the hydraulic manifold (priority relief) as death or serious injury can result from improper valve adjustments. See your authorized LeeBoy/ Rosco dealer for assistance if experiencing steering function issues or hydraulic oil overheating.

To adjust the priority relief valve:

1. Set the park brake and ensure the transmission is in neutral. Use the foot brake as an extra precaution.
2. Plumb a 0 to 5000 PSI pressure gauge into the priority flow circuit.
3. Start the engine and allow the hydraulic oil to warm to at least 100° F.
4. Increase engine speed to 2500 RPMs.
5. Raise the brush until it stops, continuing to hold pressure to the lift cylinder until a gauge reading can be taken.
6. The pressure gauge should read 1500 +/- 50 PSI.
7. Adjust the relief pressure by removing the locknut and turning the adjusting screw:
   - Turn adjusting screw clockwise to increase pressure.
   - Turn adjusting screw counterclockwise to decrease pressure.

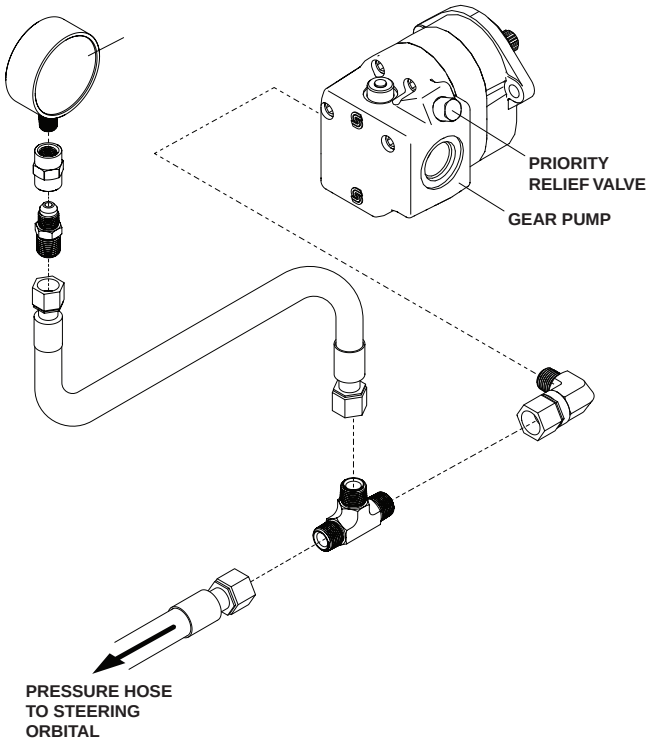


Figure 5-18. Hydraulic Manifold (Priority Relief)

Brush Drive Relief Valve

The brush drive circuit gets the remaining hydraulic flow after the priority divider supplies the power steering. The Brush Drive Relief Valve protects the brush drive circuit and is also located end of the rotation control valve under the cab on the right side of the broom. **(Figure 5-19)**

Adjust the Brush Drive Relief Valve if:

- Brush stalls frequently.
- Brush lacks sweeping power.
- There is a constant noise while sweeping.

⚠WARNING Only an experienced technician should perform this procedure as death or serious injury can result from improper valve adjustments. See your authorized Rosco/LeeBoy dealer for assistance if needed.

To adjust the Brush Drive Relief Valve:

1. Set the park brake and ensure the transmission is in neutral. Use the foot brake as an added precaution.
2. Plumb a 0 - 5000 PSI pressure gauge into the rotation flow circuit.
3. Turn the Brush Float control valve OFF by turning clockwise until stopped.
4. Start the engine and warm the hydraulic oil to at least 100° F.
5. Engage the brush drive control handle to turn brush in the forward direction.
6. Increase engine speed to 2500 RPMs.
7. Using the Brush Forward/Reverse control, turn the brush motor ON to engage the control valve.
8. The pressure gauge should read 2500 PSI +/- 100 PSI.
9. Disengage the brush drive control lever.
10. Adjust relief pressure by removing the Relief Pressure Locknut and turning the adjusting screw clockwise to increase pressure, or counterclockwise to decrease pressure.

NOTE: Turn the adjusting screw in 1/8 turn increments. Repeat turning increments until the correct pressure is obtained.

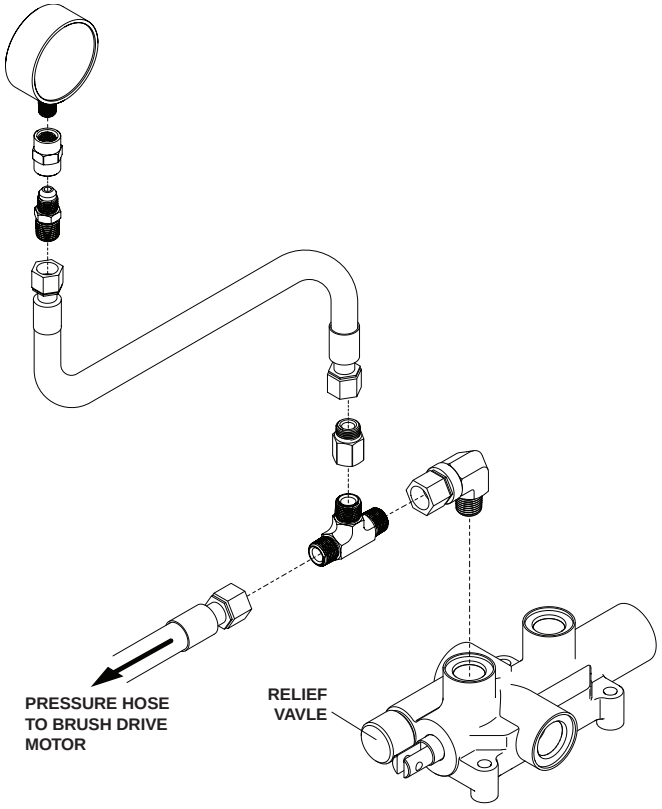


Figure 5-19. Hydraulic Manifold (Brush Drive)

CHASSIS MAINTENANCE

Tires and Wheels

Check the tires using an accurate tire gauge before operating your RB50 every day. Tires should be inflated to 80 PSI when tires are cold.

⚠CAUTION DO NOT exceed maximum inflation pressure for rims or wheels.

- Check tires for excessive wear, correct tire pressure and any damage regularly. ***(Refer to Table 5-3, Tire Wear Chart, on the following page.)***
- Use proper tire mounting equipment and experienced personnel for tire repair service.
- Check tires and wheels for cuts, bubbles, damaged rims or missing wheel bolts or nuts.
- Protect tires from contact with petroleum products and chemicals.
- Use proper tire mounting equipment and only allow experienced personnel to change tires.
- Carefully inspect any tire and rim assembly for damage before inflating tire. Contact a qualified repair service if needed.

NOTE: Use a tire pressure gauge with 1 PSI graduations.

NOTE: Limit air pressure to the proper level.

- It is extremely important to apply and maintain the proper torque to prevent loose wheels, broken studs and possible dangerous separation of wheels from the axle. Be sure to use only fasteners matched to the cone angle of your wheel. **ALWAYS use a torque wrench.**

To check tire air pressure:

1. Install lock on air chuck or stand-off gauge on tire stem. **(Figure 5-20)**
2. Turn on air supply. Stand to one side of the tire during inflation.
3. After proper inflation (the recommended pressure is 0 PSI), release chuck or stand-off gauge.
4. Torque wheel lug nuts to 85 ft lb.

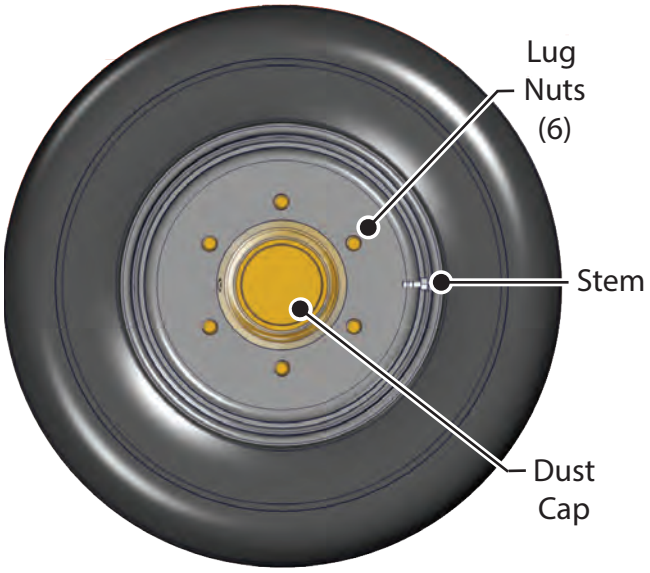


Figure 5-20. Tire

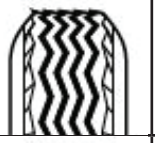
To inflate tires:

1. Install lock on air chuck or stand-off gauge on tire stem. **(Figure 5-20)**
2. Turn on air supply. Stand to one side of the tire during inflation.
3. After proper inflation (the recommended pressure is 0 PSI), release chuck or stand-off gauge.
4. Torque wheel lug nuts to 85 ft lb.
   - Protect tires from contact with petroleum products and chemicals.
   - Repair or replace damaged tires. DO NOT operate the machine with damaged tires.
   - Visually inspect the wheels for damage prior to operating the broom. Replace bent or cracked wheels immediately.

⚠DANGER Explosive separation of a tire and rim parts can cause serious injury or death.

⚠WARNING Never cut or weld on inflated tire/rim assembly. Heat from welding can cause an increase in pressure and may result in tire explosion.

Table 5-3.Tire Wear Chart

WEAR PATTERN		CAUSE	ACTION
	Center Wear	Over Inflation	Deflate tire to correct load level.
	Edge Wear	Under Inflation	Inflate tire to correct load level.
	Side Wear	Loss of camber or overloading.	Get align-ment. Ensure load doesn't exceed axle rating.
	Toe Wear	Incorrect toe-in.	Get alignment service.
	Cupping	Wheels out of balance.	Check balance and bearing adjustment.
	Flat Spots	Wheels locked and tire skidding.	Adjust brakes.

Change Tires

⚠WARNING It is recommended that you use a hydraulic lift if available when raising the vehicle. Otherwise, use jack stands on a solid, flat surface to prevent the machine from tipping when it is raised. Block the remaining wheels using sturdy wheel chocks. DO NOT lie under the machine while supported only by a jack. Death or serious injury can result from improperly supporting a raised machine.

To replace the tires:

1. Loosen lug nuts.
2. Release any air that remains in the tire.
3. Use a jack or hoist to lift the side of the machine.
4. Place supports under the machine.
5. Attach a sling or other support to the tire.
6. Remove the six (6) lug nuts from the wheel.
7. Release any air that remains in the tire.
8. Using standard tire removal tools, remove the tire from the wheel.
9. Install new tire on the wheel.
10. Measure PSI using tire pressure gauge and inflate to the proper PSI.
11. Align the mounting holes with the studs on the hub.
12. Reinstall lug nuts (6), tightening every other lug nut alternately in a star pattern.
13. Torque to 80 ft lbs.
14. Remove supports and lower jack.

Repack Front Wheel Bearings

The wheel bearings secure the wheel and tire to the vehicle on the drive shaft inside the wheel hub. While wheel bearings are designed to last 100,000 miles, they can lose lubricity over time. Dirt and debris can also seep into the wheel bearings. If the wheel bearings aren't cleaned and repacked, they can wear out and possibly break or damage other wheel components.

When any of the following symptoms appear, the wheel bearing is damaged and should be replaced on ***both*** wheels of the same axle.

- Abnormal and excessive tire wear on the inside or outside edge of tire.
- Grinding or roaring noise from a tire. This symptom indicates metal-on-metal contact and definitely requires wheel bearing replacement.
- The steering wheel bounces and vibrates at lower speeds, vibrating more as the machine accelerates.

NOTICE **Cleaning and repacking or replacing wheel bearings is a messy job. Wear gloves and have plenty of shop rags and room to work around the entire wheel hub. Dispose of used grease in accordance with environmental law.**

⚠WARNING If using compressed air for drying wheel hub components after cleaning, wear eye protection! Compressed air can cause serious personal injury if misused. Use compressed air at safe low levels. NEVER point the nozzle at others.

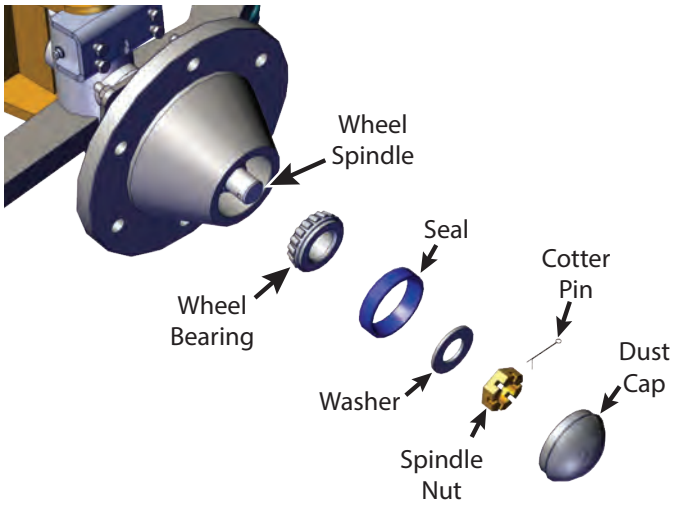


Figure 5-21. Wheel Hub Components

To clean and repack (or replace) wheel bearings:

1. Park the broom on a level surface in a designated area where grease does not touch or run off onto the ground.
2. Using a hydraulic lift (if possible) or floor jack, raise vehicle with a pair of jack stands on the front axle to service wheel bearings. Use wheel chocks on the other wheels for safety.
3. Remove dust cover and lug nuts from one wheel, then remove tire and wheel hub, placing it clearly away from the work area. **(Figure 5-20 *on Page 5-22)***
4. Remove cotter pin to release spindle nut, then remove spindle nut and washer. **(Figure 5-21)**
5. Carefully pull the wheel bearing and seal off the axle spindle. Inspect wheel bearing and seal for wear or damage.
   - Discard bearing if replacement is needed. It is recommended that you discard the seals and replace with new ones.
   - Otherwise, thoroughly clean wheel bearings, wheel hub, axle spindle and mounting hardware with solvent. Once the excess grease is removed, you can spray liberal amounts of brake cleaner inside the wheel bearings and wheel hub for additional cleaning.
6. Work fresh grease into the wheel bearings. You can put the wheel bearing and grease into a plastic zip lock bag to really work the grease into every small wheel and bearing.
7. Work fresh grease inside the wheel hub, ensuring the outer edges are fully covered, and put liberal amounts of grease onto the axle spindle.
8. Reinsert the wheel bearing, ensuring it is oriented as shown in **Figure 5-20**, then new seal, washer, spindle nut. Continue packing grease inside all the parts as you install them.
9. Tighten the spindle nut until the wheel "drags" during hand rotation.
10. Back off the nut 1/8th of a turn and reinsert cotter pin.

- You can spin the hub and rotor to test for noise and smoothness before reinstalling the wheel and tire. When packed correctly, the rotor spins freely without noise.

11. Reinstall tire and wheel. Torque lug nuts. (***See Section 2, Torque Specifications)***
12. Repeat procedure on the other wheel.

Change Drive Axle Oil

Change the drive axle oil every year. Check the oil level and fill if needed quarterly.

To fill the oil:

1. Rotate the hub on one side of the front axle until the plug is in the 12 o'clock position.
2. Remove the plug. **(Figure 5-22)**
3. Add one quart of 80W-90LS GL5 oil.
4. Replace the plug.
5. Follow Steps 1 - 4 on the other side.

To change the drive axle oil:

1. Remove the level plug in the center of the housing.
2. Remove the breather plug in the housing.
3. Add oil until it starts coming out of the level plug.
4. Replace the level plug.
5. Allow the axle to sit level for at least five minutes to allow the oil to flow to the hubs.
6. Remove the level plug and check oil level.
7. Add more oil if needed.

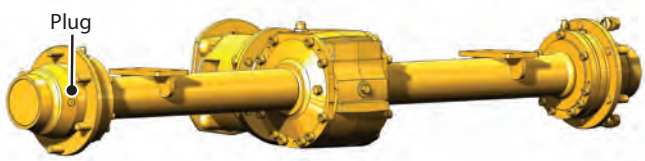


Figure 5-22. Front Drive Axle

Park Brake Adjustment

NOTICE **Only a qualified service technician should adjust the parking brake as serious damage can occur to this important component. See your authorized LeeBoy/Rosco dealer for assistance.**

⚠WARNING **Serious injury or even death can result from an improper park brake adjustment.**

Only a qualified technician should perform the following procedure:

1. Loosen the park brake bolt.
2. Rotate the park brake knob clockwise to tighten the brake cable or counterclockwise to loosen the brake cable.
3. When the desired tension is reached, retighten the park brake bolt.

Brakes

Check the master cylinder fluid daily. The master cylinder fluid filler cap is located on the cab floor behind the brake pedal. **(Figure 5-23)** Only fill one inch below the top of the fill neck with All Weather All Temperature VG32.

NOTICE **If the master cylinder fluid is frequently low, check the hydraulic system for leaks.**

Figure 5-23. Brake Fluid Fill

5

DUST SUPPRESSION SYSTEM (OPTION)

Clean the water system strainer after each use. Simply unscrew the strainer from its mount under the water tank at the rear of the vehicle, clean, and reinsert. Remove the drain plug to drain the water tank. **(Figure 5-24)**

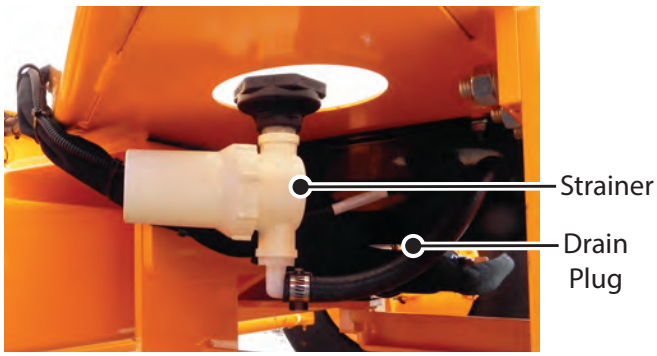


Figure 5-24. Water Strainer and Drain Plug

Inspect the brush water nozzles daily to ensure they are free from clogs and operating properly. If a nozzle becomes clogged:

1. Remove the nozzle by unscrewing it from the spraybar.
2. Clean the nozzle slots, being careful not to damage the nozzle opening as the spray pattern will be adversely affected.
3. Reinsert or replace the nozzle if needed.
4. If the nozzle is still clogged or not spraying properly, contact your authorized Rosco/LeeBoy dealer for a replacement nozzle(s).
   - Flush the spraybar line and nozzles at the end of each operating season (or more often for extended use).
   - Ensure the water tank, strainer, pump and all lines are completely drained to prevent damage from freezing if storing broom in temperatures below freezing.

COMPONENT REPLACEMENT

Change Brush Core

1. Park broom on flat surface, place machine in neutral, turn off the engine and set park brake.
2. Remove both cotter pins from brush cover. **(Figure 5-25)**

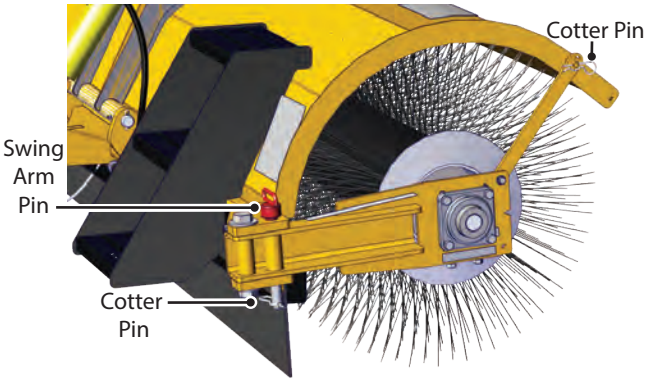


Figure 5-25. Brush Core Pins

3. Remove the swing arm pin.
4. Pull the brush core away from the drive hub and slide out. Set aside.
5. Check hub on both sides for damage. (Consult your authorized dealer if any damage is noted.)
6. Grease the U-Joint on the idle hub.
7. Slide new brush onto the slotted drive hub ensuring the key stock inside the brush core goes onto the drive hub side.
8. Align the idle hub with the smooth end of the brush core and push the swing arm back into position into the brush core, keeping hands away from pinch points. Carefully guide the hub by hand if necessary to start into the core center.

NOTE: Brush carriage may need to be raised slightly to install a new core if the old brush core was worn.

9. Once the idle hub is fully engaged, reinsert the swing arm pin.
10. Reinsert swing arm pin and cotter pins on brush cover.

Replace Batteries

⚠WARNING Batteries create explosive gases when charging. DO NOT expose these gases to sparks or open flame during maintenance. Always perform battery service in a well-ventilated area or outside. Serious injury can occur from exploding batteries.

1. Remove the black ground (-) cable from the negative (-) battery terminal, followed by the red (+) cable from the positive (+) battery terminal. **(Figure 5-26)**

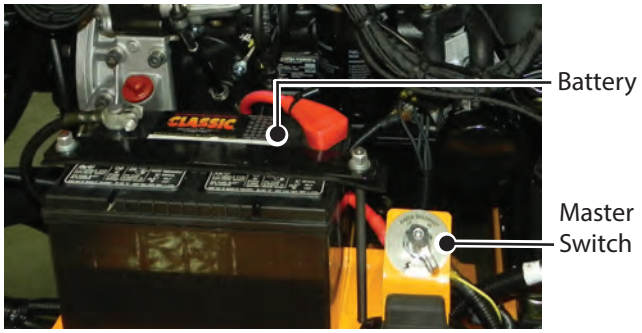


Figure 5-26. Battery System

⚠WARNING If the negative cable is grounded during removal, sparks will not occur. If the positive cable is grounded during removal, sparks will occur and could ignite explosive gases.

2. Remove the hold down bar.
3. Inspect the batteries for cracks, wear and damage that could result in leaks.
4. Carefully remove the batteries from the battery plate, noting battery position during removal.

⚠WARNING DO NOT dry batteries using compressed air. Allow the batteries and surrounding area to dry. Battery acid can cause severe injury if blown into the face or other areas of the body. If injury occurs, seek medical attention immediately. Flush eyes and skin for 30 minutes with clean water.

5. Make sure the battery cells are sealed. Clean the batteries, battery tray, cables, and brackets with baking soda and warm water. Allow to air dry.
6. Place the batteries in proper position in the battery tray.
7. Install hold down bar. Tighten bar to retain battery, but not so tight as to damage battery.

⚠WARNING When installing batteries, always install the positive cable first followed by the negative cable. Reversed polarity can damage the electrical system.

8. Install red (+) battery cable to the positive (+) battery terminal.
9. Install the black (-) battery cable to the negative (-) battery terminal.
10. Reattach the battery box cover and reconnect the battery box locks.

Replace Radiator Hoses

Radiator hoses should be replaced if worn or frayed. Replacement parts are listed in **Section 7, Illustrated Parts List (IPL).**

⚠WARNING **This procedure should only be performed when the equipment is not operating.**

⚠WARNING **DO NOT remove the radiator cap while the engine is still hot. The radiator is under pressure. Removing the cap could cause serious injury.**

1. Turn off ignition, remove key, and allow the engine to cool down before proceeding:
2. Remove the radiator surge tank cap.
3. Drain radiator into a suitable container.
4. To remove upper hose, loosen the two hose clamps and disconnect. If clamps are not damaged, retain for installation.
5. To remove lower hose, loosen the two hose clamps and disconnect. If clamps are not damaged, retain for installation.
6. Install new hoses in reverse order of procedures above.

WINTERIZING FOR STORAGE

Perform the following procedures to prepare the machine for winterizing and storage.

1. Store the machine in a dry, protected area:
   - Cover seat with plastic. Place a “DO NOT OPERATE” tag on the main control panel.
   - Cover all exposed control panels with tarps or plastic.
   - Ensure all cab doors and windows are closed and seal properly.
   - Ensure the tires are properly inflated.
   - Replace worn or damaged decals if needed.
2. Inspect the machine thoroughly and repair worn or damaged parts. Inspect all hoses, couplers, fittings and cylinders. Tighten loose fittings and replace any worn hoses found.
3. Lubricate all grease points:
   - Coat all exposed cylinder shafts with lithium-based grease.
   - Ensure all exposed threads are lubricated.
   - Ensure cables are lubricated and move freely.
   - If the machine has high- and low-speed chains, be sure they are oiled.
4. Check air filter. Clean or replace air filter elements if needed.
5. Check engine oil level and fill if needed.
   - Drain dirty engine oil, fill with new oil and run the engine five minutes to allow the oil to penetrate components.

⚠WARNING **To avoid the danger of exhaust fume poisoning, DO NOT operate the engine in a closed building without proper ventilation. Check hydraulic oil sight gauge and fill if needed.**

NOTICE **Hydraulic oil that is contaminated must be drained, the filter elements replaced and the hydraulic tank refilled. Failure to perform this maintenance procedure if needed could result in premature failure of the pumps or motors.**

6. Ensure the fuel tank is full to prevent condensation in the fuel tank. Add cold-weather additive when appropriate.
7. If machine is equipped with Diesel Exhaust Fluid (DEF), ensure the DEF tank is full.

NOTICE **IF machine is equipped with DEF and is stored for more than one year, replace the DEF fluid before operating to prevent SCR system malfunction and/or damage.**

8. If machine is equipped with a citrus tank, ensure the tank is full to prevent rust.
9. If machine is equipped with water tank(s), the tank should be drained. The pump and filter must also be purged to prevent freezing.
10. If machine is equipped with windshield washer bottle, ensure the bottle is full of window washer fluid with a cold-weather blend.
11. Perform specific gravity test on engine coolant. Drain and replace or fill with antifreeze mixture to prevent freeze damage if needed.

KUBOTA ENGINE NOTICE: If not using anti-freeze, open the cock at the bottom of radiator and remove the pressure cap to drain water completely. Leave the cock open. Place a "No Water" sign on the pressure cap.

12. Disconnect battery, adjust the electrolyte level, and recharge it. Store battery inside in a cool, dry place. Ensure breakers are in the OFF position.
13. Remove steering control boxes and electrical cords. Store inside in a cool, dry place.
    - If equipped with any plugged-in remote sensors (i.e., sonic auger sensor), disconnect, remove and store inside in a cool dry place.
14. Cover the exhaust and intake filter. Be sure to seal the exhaust cover with tape to prevent moisture from entering, especially if storing outside.
15. If the machine is equipped with a propane system, remove the propane gas bottle and store in a safe area away from the machine. Purge the gas system and seal connection(s).
16. Lower machine components and attachments (i.e., screed, broom, blade, etc.) to the lowest position, resting on blocks to prevent ground contact or pavement damage.
17. Clean the machine, including the undercarriage. Paint chipped or rusty areas to prevent further rusting. Lock all lockable panels, cab doors and spare tire. Remove ignition key and store keys in a safe place.
18. If storing machine for an extended time, run the engine five minutes every two to three months (under no load) to keep engine free from rust caused by moisture in the air. If you do not run the engine for six months or longer, apply engine oil to the valve guide and stem seal before starting.

Periodic Maintenance in Storage

If storing the Rosco RB50 more than two months, perform the routine maintenance procedures for every 100 hours.

Also be sure to:

- Keep battery fully charged and check the electrolyte level regularly.
- Check the hydraulic fluid on a regular basis for possible moisture contamination. This is especially important in climates with varying temperatures and humidity as condensation can develop inside the hydraulic tank.

NOTICE **Hydraulic oil that is contaminated must be drained, the filter elements replaced and the hydraulic tank refilled. Failure to perform this maintenance procedure if needed could result in premature failure of the hydraulic pumps or motors.**

- Reinstall the battery to start the engine and allow the engine to warm until all machine functions are at operating temperature at least twice per month.
- Continue to perform routine maintenance procedures as required using the Periodic Maintenance Table on ***Page 5-3.***

Removing from Storage

1. Perform maintenance procedures described above.
2. Check all fluid levels, belt tensions, and bolt torques.
3. Replace battery.
4. Replace alternator belt.
5. Clean any excess grease or rust inhibitor from all exposed cylinder rods, seals and O-rings.
6. Disengage hydraulic brush lift lock valve.
7. Start engine and allow machine to warm to operating before driving out of storage.

Figure 5-27. Welding

WELDMENT

If the machine has been damaged or metal parts worn due to extensive use, minor welding may be necessary for repair. Welding is very dangerous. Only individuals who are properly trained should weld in a dedicated area away from others as even looking at ultraviolet light from the flame can cause serious eye injury.

Welding hazards include toxic fumes, harmful dust, light radiation, burns, combustion from sparks and the potential for extensive damage to the eye. There is also a danger of electric shock. If combustible or flammable materials are nearby, the heat and sparks produced by welding can cause fires or explosions.

DISCONNECT BATTERY TERMINALS BEFORE WELDING ON THE MACHINE.

Electrical current can ignite fire or cause an explosion, therefore it is critical to turn off power to the machine. It is also important to grind off the paint to ensure proper ground on the area being welded.

DO NOT MAKE MODIFICATIONS TO THE MACHINE. ONLY weld if needed for repair purposes. Weldment modifications to the machine can void your warranty.

Before welding on the machine:

PREPARE:

- Clear the area of any fire hazards, flammable liquids and solids, or any items that could catch on fire.
- Ensure a fire extinguisher is available in the work area.
- Hang welding blanket(s) around and underneath the work area.
- Paint and rust should be removed from the welding area to prevent arcing in other areas or components. An indirect ground can cause arcing that will damage machine components. Only connect the ground to the component being welded. DO NOT connect the ground to other bolted assemblies, wheels, bearings, motors, tracks, torque hubs, etc.

DISCONNECT:

- Turn the master battery switch to the OFF position.
- Disconnect the battery positive and negative terminals.
- Disconnect engine ECU(s) controller (if equipped).
- Disconnect engine DEF controller (if equipped).

TROUBLESHOOTING

Hydraulic Motor Installation Start-Up Procedure

Pre-Start

If the hydraulic system is down as a result of a major component failure:

1. Drain and clean the tank and system components (hoses, valves, fittings and cooler) to ensure it is free from metallic debris and other contamination. Failure to do so may result in damage to the pump(s) and/or other components on start-up.
2. Change all the filters.
3. Change the fluid. On large systems where the cost of changing the fluid may be prohibitive, the fluid should be flushed until a cleanliness level of ISO 4406 18/16/13 or better is achieved.

Installation and Start-Up

When installing hydraulic motors, it is important that the mounting flange of the motor makes full contact with the mounting surface of the application. Mounting hardware of the appropriate grade and size must be used:

1. Use Grade 8 socket-head capscrews to attach the motor.
2. Install lock washer over capscrew.
3. Apply Loctite 243 (blue) to the capscrew threads.
4. Install extra thick, hardened H-D thick washers over the capscrew.
5. Install the capscrew-hardened washer and lock washer in the SAE two-bolt flange.
6. Torque the capscrew to the wet torque value defined for the capscrew grade and size.

NOTICE **It is CRITICAL to use the correct mounting hardware.**

Hubs, pulleys, sprockets and couplings must be properly aligned to avoid inducing excessive thrust or radial loads. Although the output device must fit the shaft tightly, a hammer should never be used to install any type of output device onto the shaft. The port plugs should only be removed from the motor when the system connections are ready to be made.

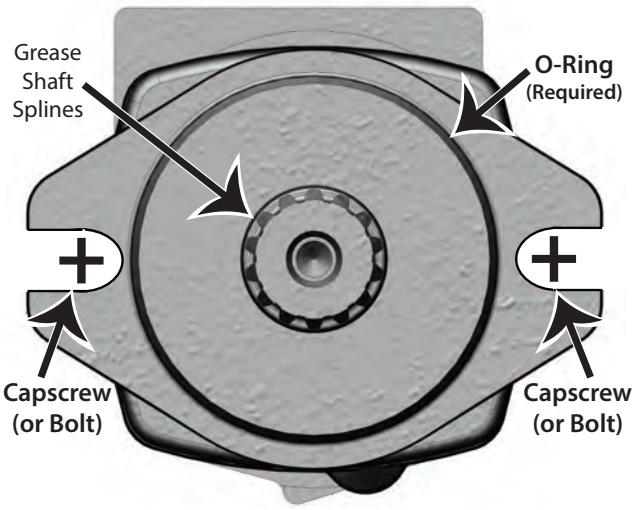


Figure 5-28. Hydraulic Motor Mount

To avoid contamination:

- Remove all matter from around the ports of the motor and the threads of fittings.
- Fill the case of the motor being used (piston-type, gear motor, gerotor motor, geroller motor or vane motor) with clean hydraulic fluid through the highest case drain port and connect the case drain line. Failure to do so will result in damage to the motor through inadequate lubrication on start-up.
- Units that are mounted vertically with the shaft up require special attention to ensure the fluid level in the case is high enough to lubricate the front shaft bearing(s).

Once all system connections are made, the motor should be run for 15 - 30 minutes at no-load and half-speed to bleed air from the hydraulic system.

Troubleshooting Chart

The troubleshooting chart below identifies the most common symptoms of mechanical problems. Fault codes are also shown on the PV380 Digital Display unit on the Operator Control Panel.

If the problem persists, contact your authorized LeeBoy dealer for assistance or repair.

Table 5-4. RB50 Broom Troubleshooting

SYMPTOM	CAUSE	REMEDY
Engine difficult to start or does not start.	Defective battery or low battery charge.	Replace or charge battery as applicable.
	Insufficient fuel supply.	Fill fuel tank.
	Safety switch is faulty.	Replace.
	Solenoid doesn't function properly.	Ensure wires are tight. Clean solenoid plunger. Replace solenoid coil. Replace or rebuild starter or solenoid.
	Switch box unplugged.	Replug switch box.
	Faulty starter.	Repair or replace.
	Engine cranking speed too low.	Clean battery terminal connections. Charge or replace battery.
	Insufficient fuel supply to injector nozzles.	Check fuel system. Clean or replace fuel filter.
	Air in the fuel system.	Check for air leaks in the low-pressure side of the fuel system. Prime fuel system. Refer to your Engine Operator's Manual.
	Fuel supply contaminated.	Verify by operating engine with a known fuel quality.
Engine turns over but does not start.	Low fuel.	Add fuel to fuel tank.
	No voltage to fuel shut-off solenoid.	Voltage should be 9 volts with the ignition switch in the ON or START position.
	No fuel to injector pump.	Check fuel supply system.
	Fuel filter plugged or restricted.	Replace fuel filter.
	Intake or exhaust system restricted.	Check for and remove restrictions.
Engine doesn't shut off.	Electrical wiring fault supplying power to the fuel solenoid when key is in the OFF position.	Repair wiring.
	Faulty diode in engine wire harness at alternator.	Check diode wire connection and/or replace diode.
	Injection pump fuel solenoid inoperative.	Check solenoid for defects or foreign material inhibiting proper operations.
	Engine operating on fumes drawn into air intake.	Locate and isolate source of fumes.
	Low idle set too high.	Set idle to specifications.

SYMPTOM	CAUSE	REMEDY
Engine power output is low.	Operating engine at high altitude.	De-rate engine power output for altitudes above 10,000 feet.
	Air intake piping restricted.	Remove restrictions.
	Air cleaner element(s) dirty.	Clean or replace air cleaner element(s).
	Fuel suction line or filter restricted.	Check fuel line for restrictions. Replace fuel filter element(s).
	Fuel return system restricted.	Correct restricted fuel return system.
	Fuel quality poor.	Verify by operating with a known fuel quality.
	Fuel transfer pump malfunctioning.	Replace fuel transfer pump. Refer to Engine Operator's Manual.
	Throttle improperly adjusted.	Adjust throttle controls.
	Fuel injector malfunctioning.	Replace fuel injector. Refer to Engine Operator's Manual.
Broom doesn't move with engine running or moves in one direction only.	Park brake engaged.	Release park brake.
	Broom rotation control valve is bypassing oil at low pressure.	Repair or replace relief valve cartridge(s).
	Damaged hydrostatic pump.	Repair or replace hydrostatic pump.
	Damaged hydrostatic motor.	Repair or replace hydrostatic motor.
Broom stalls or lacks power.	Brush drive relief valve set too low or defective.	Adjust relief valve to 2500 PSI. Replace if defective.
	Sweeping with too much down pressure.	See Brush Down Pressure Control on **Page 4-14**.
	Ground speed too fast.	Pull back on joystick and increase engine RPMs to increase brush speed.
Steering is difficult.	Low hydraulic fluid level.	Add hydraulic oil.
	Hydraulic filters contain contamination.	Check filters for contamination and clean or replace.
	Hydraulic pump priority relief valve set incorrectly.	Check relief valve setting. It should be 1500 PSI. Adjust setting if necessary.
	Worn hydraulic pump.	Repair or replace.
	Worn steering orbital motor.	Repair or replace.
	Worn steering cylinder.	Repair or replace.
Alternator not charging.	Loose wiring or fault connection.	Repair or replace loose wiring or connector.
	Diode at alternator loose or faulty.	Tighten connection or replace diode.
	Alternator belt loose or broken.	Adjust or replace belt.
	Alternator malfunctioning.	Replace alternator.
	Circuit breaker tripped.	Reset circuit breaker.
Instrument gauges don't work.	Faulty gauge or sender.	Replace gauge or sender.
	Alternator not charging.	Repair wiring or replace alternator.
	Faulty wiring.	Replace wiring or connector.

SYMPTOM	CAUSE	REMEDY
Hydraulic system overheats.	Low hydraulic oil level.	Refill with hydraulic fluid.
	Defective temperature gauge or sender giving wrong temperature reading.	Replace gauge or sender.
	Brush drive relief valve set too low or defective.	Adjust relief valve to 2500 PSI. Replace if defective.
	Excessive ambient air temperature and high-duty cycle.	Operate broom at slower ground speed and maximum engine RPMs during hot weather.
	Plugged fins on fluid cooler.	Clean fins and correct any other problems with cooling air flow.
	Hydrostatic pump bypass valve defective or open.	Repair or replace. If valve is open, turn valve clockwise until seated and torque to 9.5 - 14 N•m (7-10 ft lb). DO NOT over torque as this will damage valve.
	Worn hydrostatic pump.	Repair or replace.
Engine oil pressure is low.	Electrical power not being supplied to gauge.	Check fuse.
	Incorrect oil level. Too high or too low.	Check for leaks. Add or drain engine oil. Check dipstick calibration.
	Oil filter plugged.	Change oil filter.
	Oil diluted with fuel.	Replace fuel injector, fuel transfer pump and injection pump.
	Oil diluted with coolant.	See your authorized Rosco/LeeBoy dealer.
	Incorrect oil specifications.	Change oil.
	Oil pressure sender or gauge malfunctioning.	Replace oil pressure sender or gauge.
Engine coolant temperature above normal.	Radiator find damaged or obstructed.	Inspect radiator fins. Clean, repair or replace.
	Collapsed or restricted radiator hose.	Inspect hoses. Replace if needed.
	Loose fan drive belt.	Check belt tension and tighten or replace as needed.
	Cooling fan shroud damaged or missing.	Inspect shroud. Repair or replace.
	Incorrect or malfunctioning radiator cap.	Check radiator cap and replace if needed.
	Temperature gauge or sender malfunctioning.	Repair or replace.
	Thermostat stuck in closed position.	Test thermostat. Replace if necessary.
	Dirt, scale or rust in the cooling system.	Clean cooling system.

Section 6
SCHEMATICS

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NOTES

MAIN WIRING SCHEMATIC (1 OF 5)

Schematic #1016958

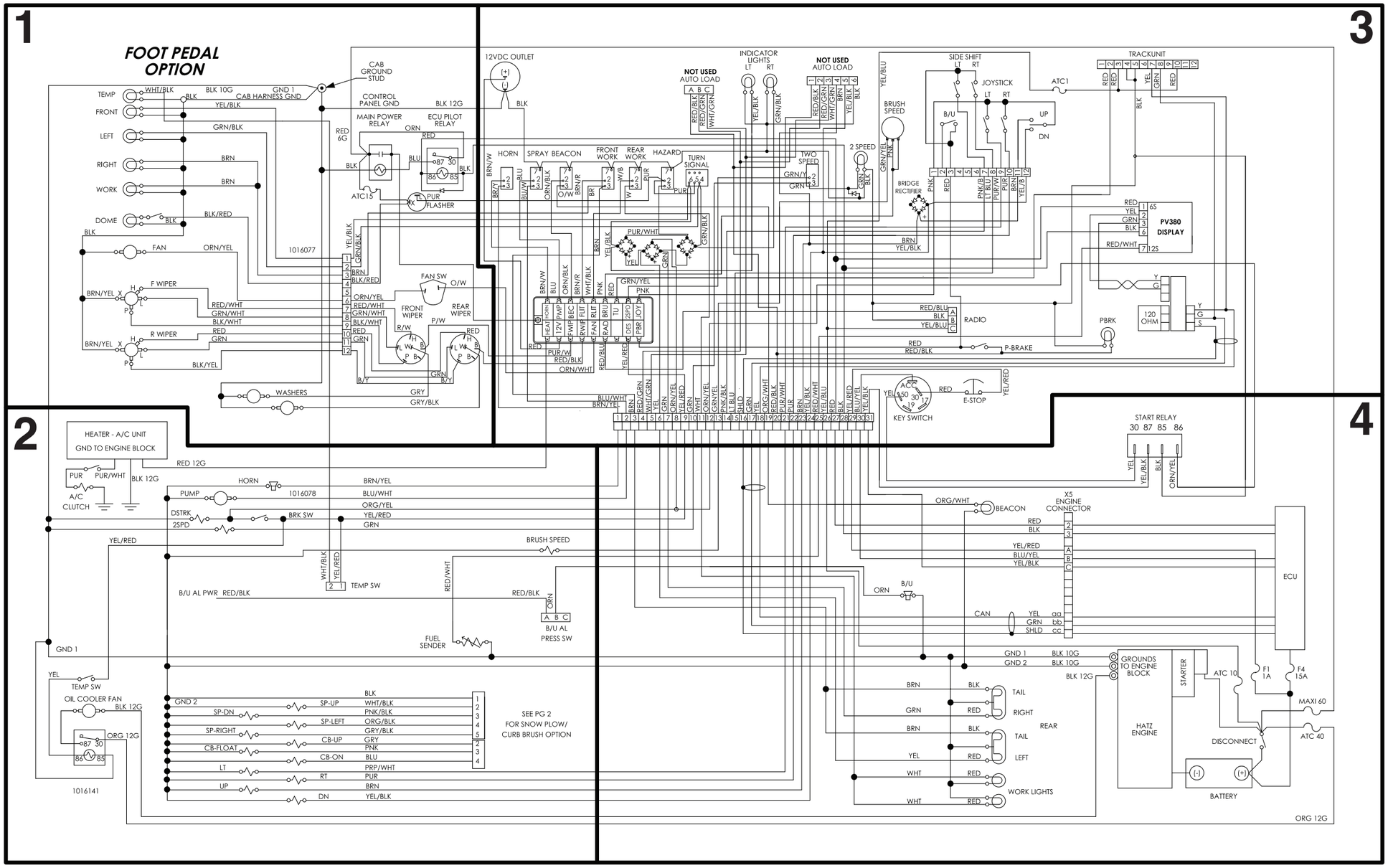


Figure 6-1. Main Wiring Schematic (1 of 5)

NOTES

MAIN WIRING SCHEMATIC (2 OF 5)

Schematic #1016958

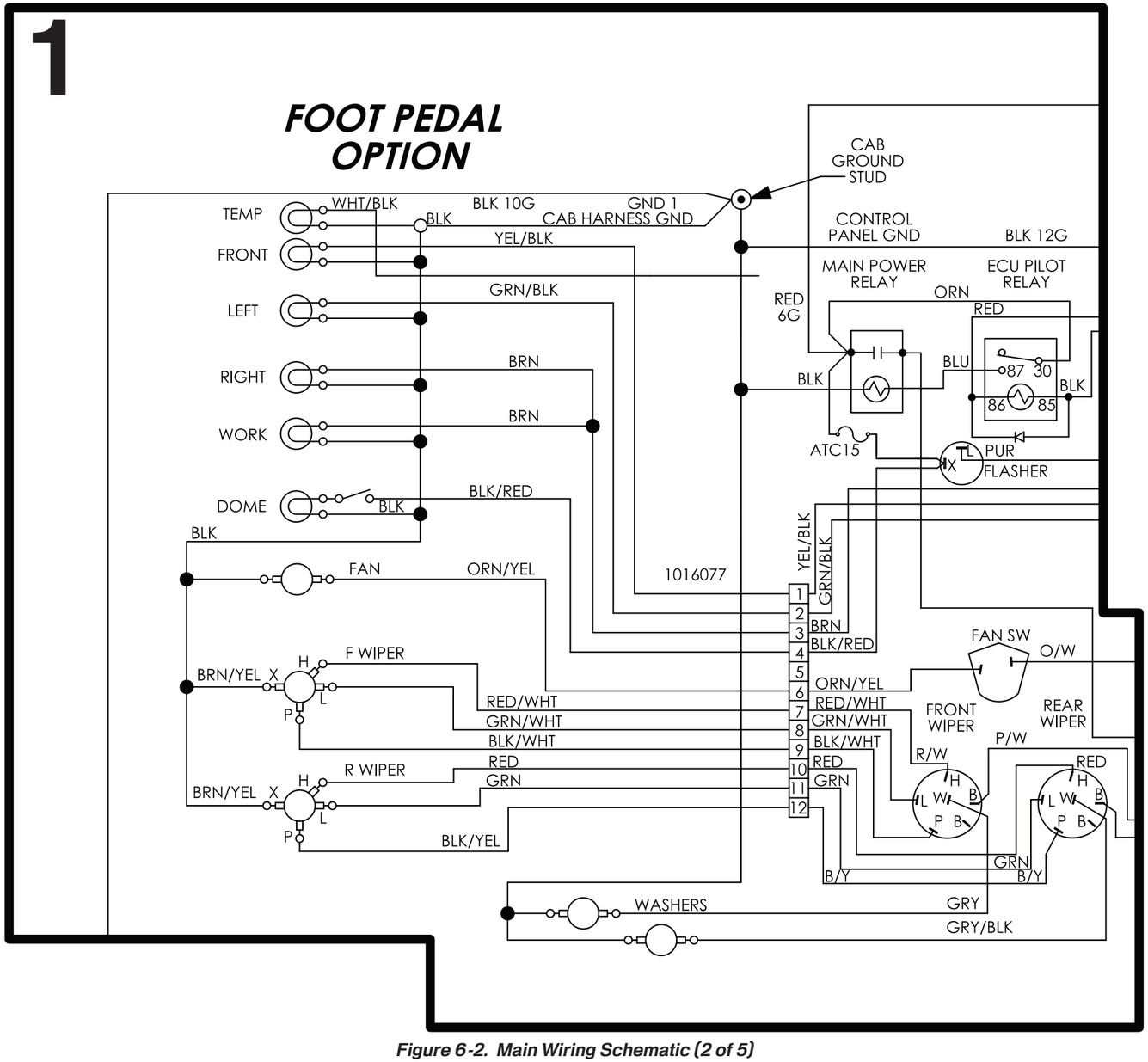


Figure 6-2. Main Wiring Schematic (2 of 5)

NOTES

MAIN WIRING SCHEMATIC (3 OF 5)

Schematic #1016958

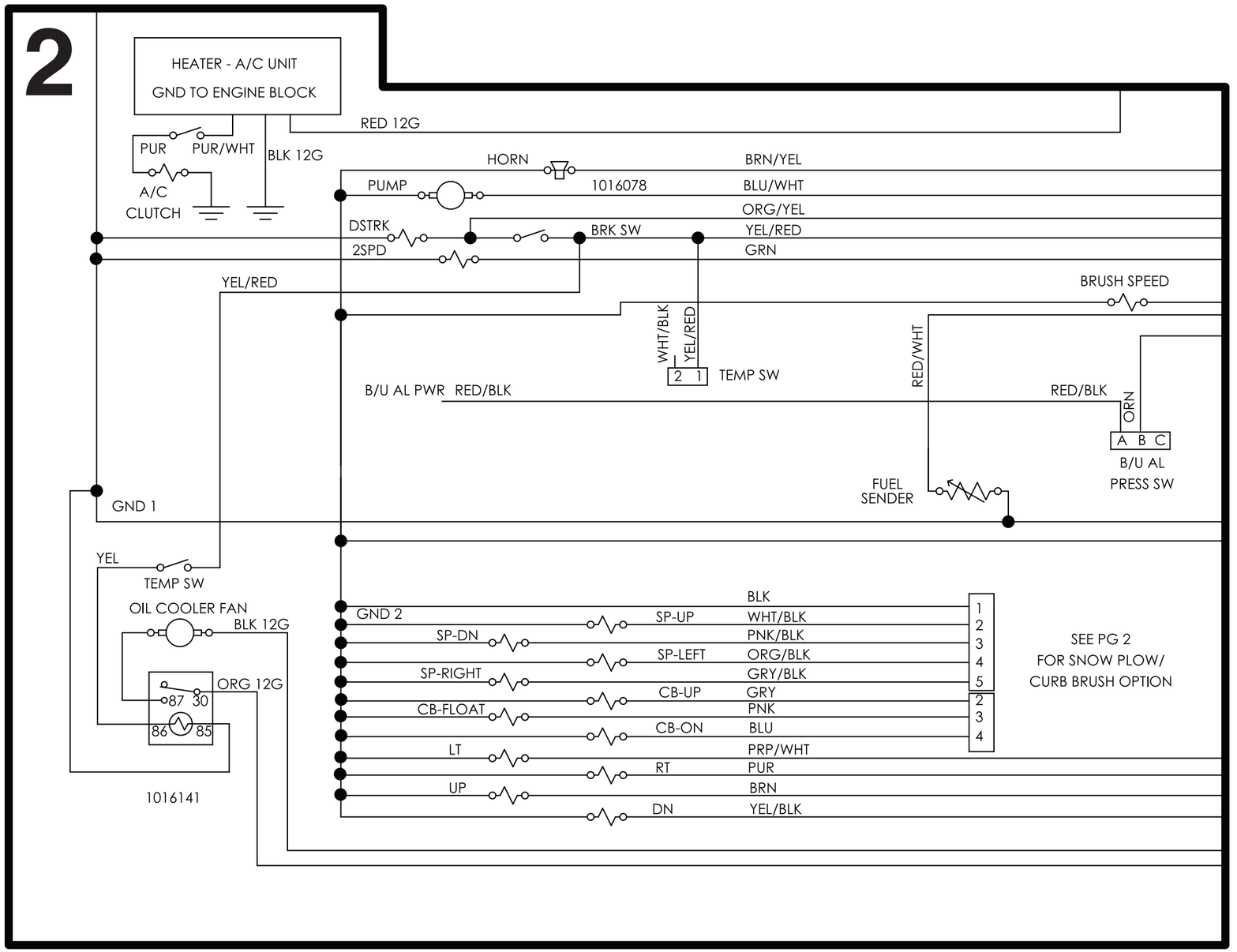


Figure 6-3. Main Wiring Schematic (3 of 5)

NOTES

MAIN WIRING SCHEMATIC (4 OF 5)

Schematic #1016958

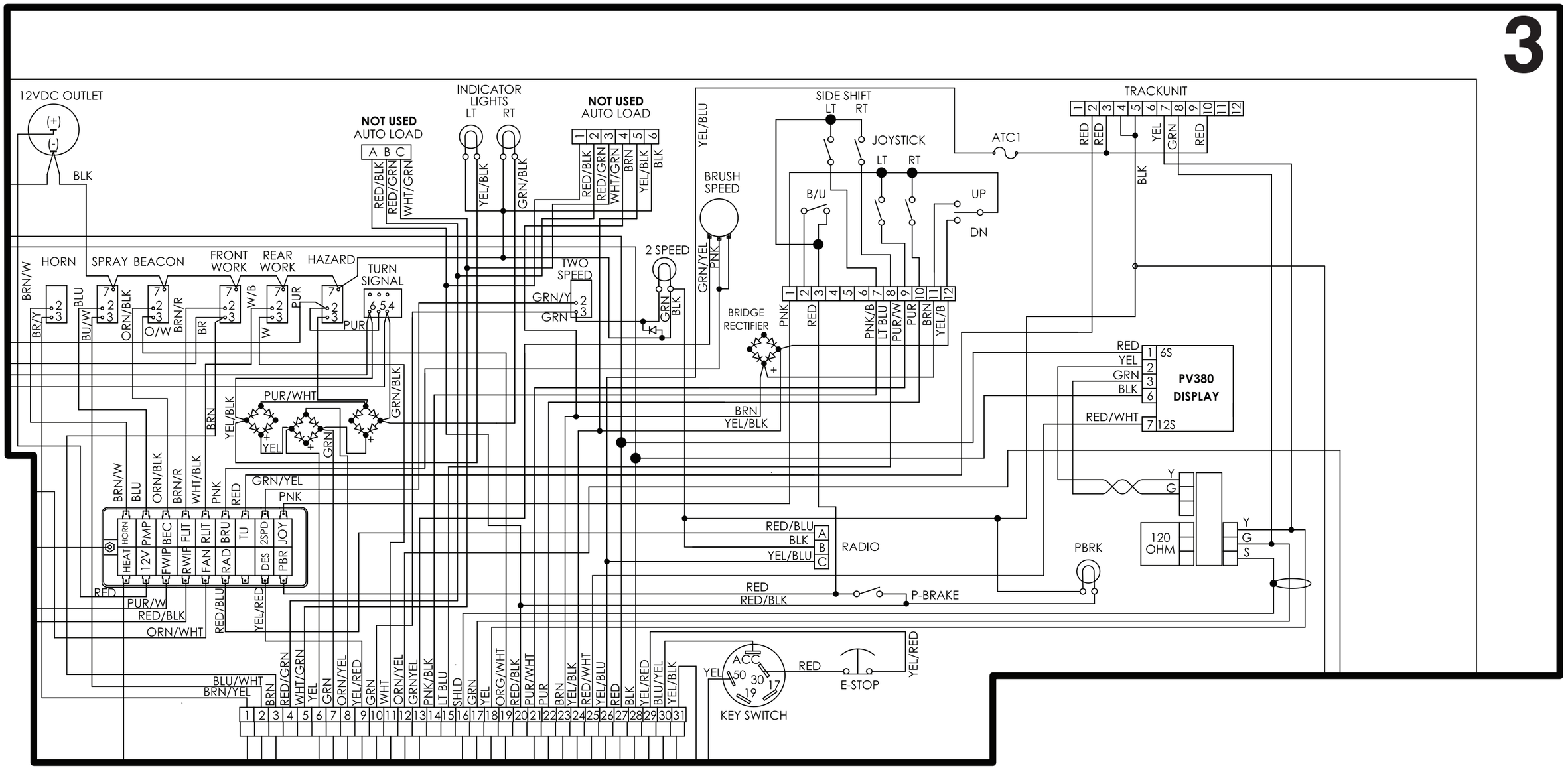


Figure 6-4. Main Wiring Schematic (4 of 5)

NOTES

MAIN WIRING SCHEMATIC (5 OF 5)

Schematic #1016958

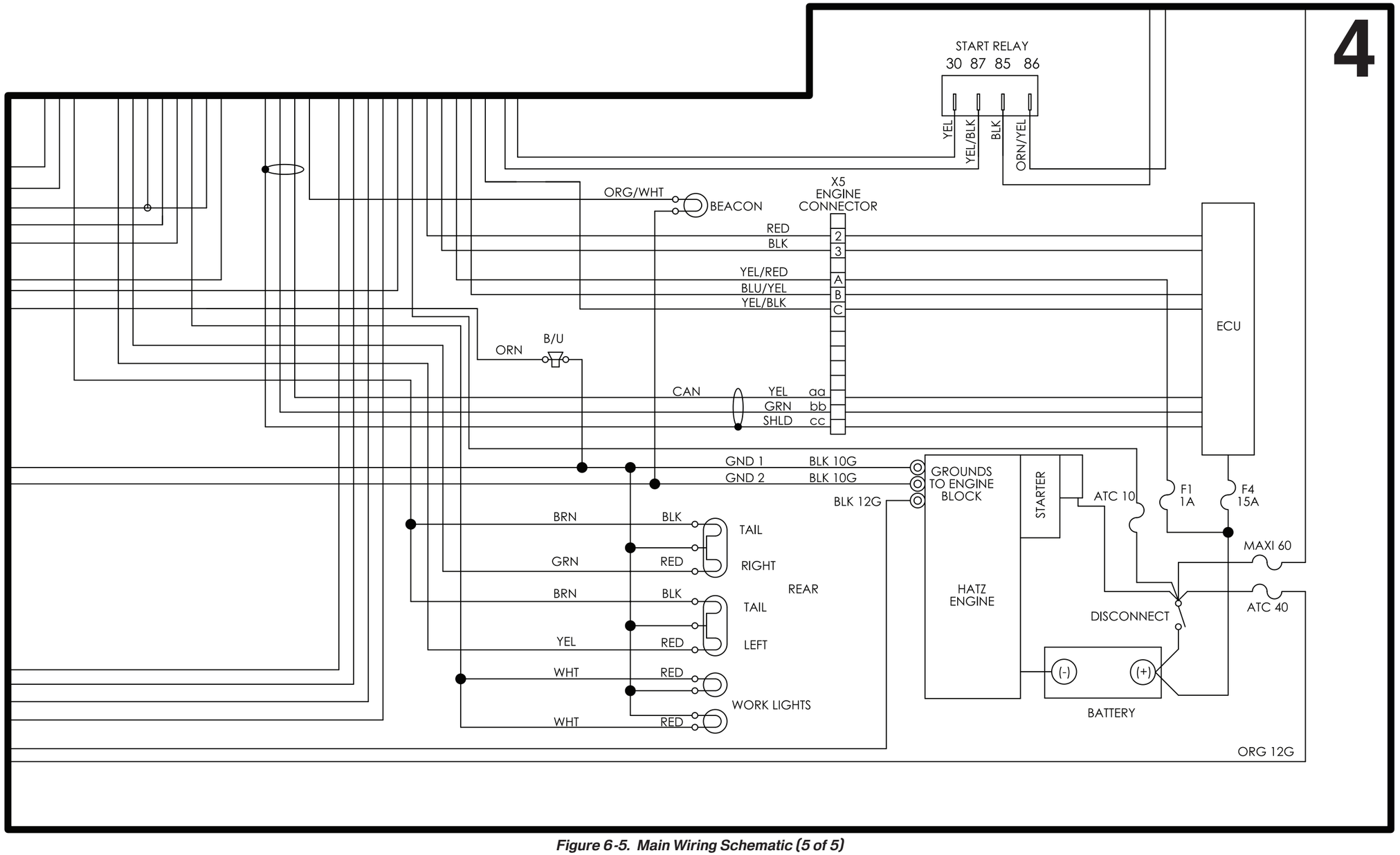


Figure 6-5. Main Wiring Schematic (5 of 5)

NOTES

MAIN WIRING HARNESS (1 OF 2)

Schematic #1022386

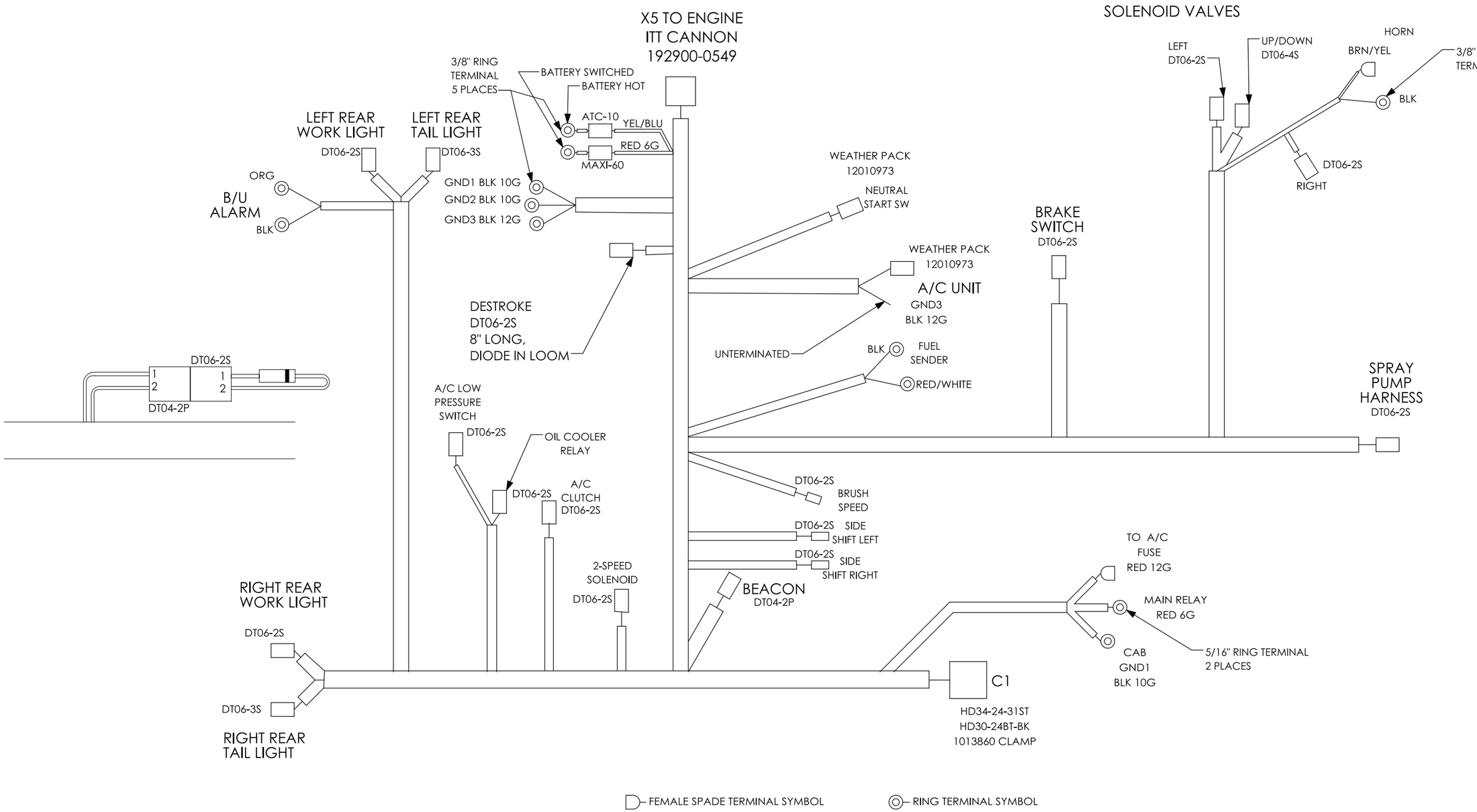


Figure 6-6. Main Wiring Harness (1 of 2)

NOTES

MAIN WIRING HARNESS (2 OF 2)

Schematic #1022386

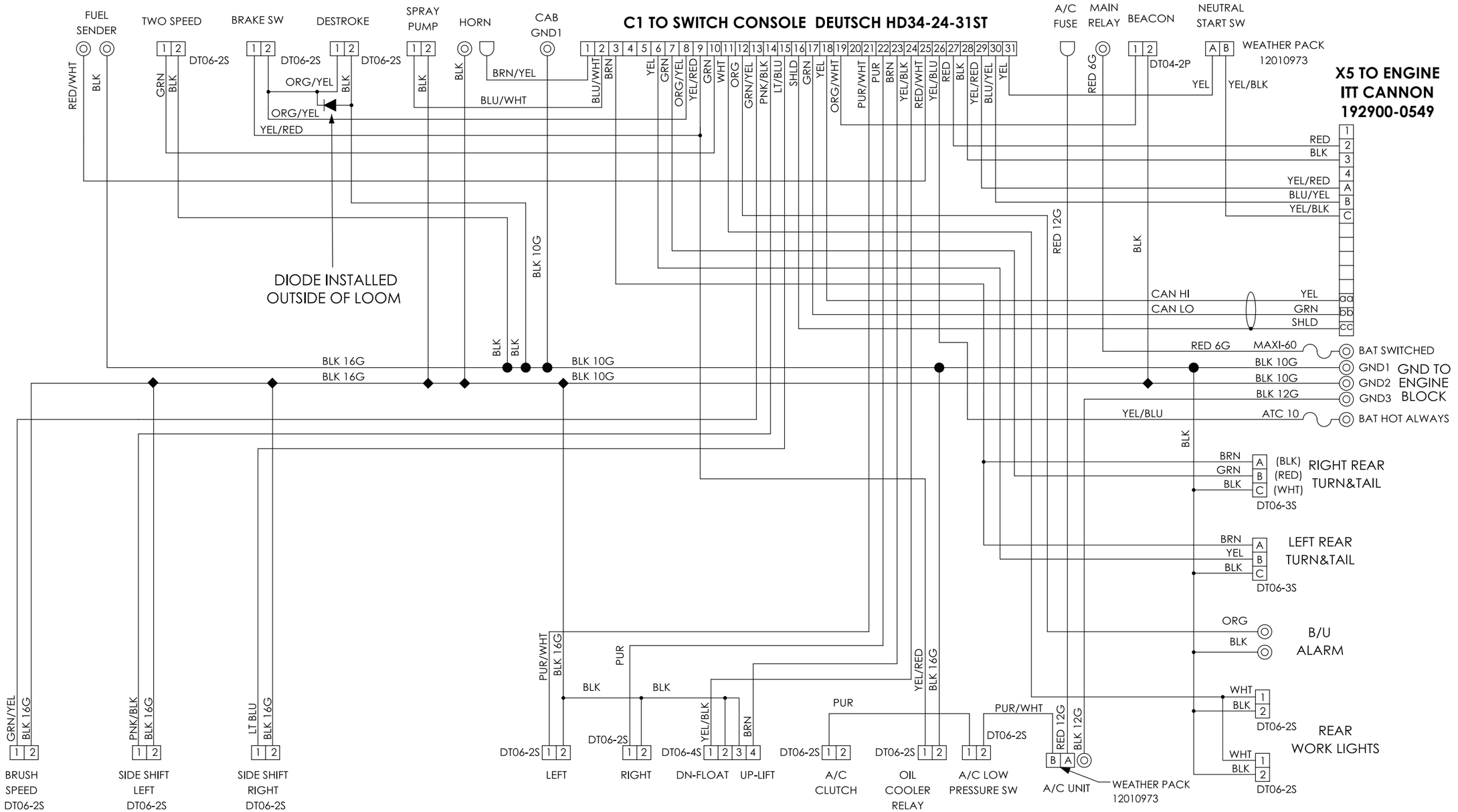


Figure 6-7. Main Wiring Harness (2 of 2)

NOTES

CAB HARNESS (1 OF 2)

Schematic #1016077

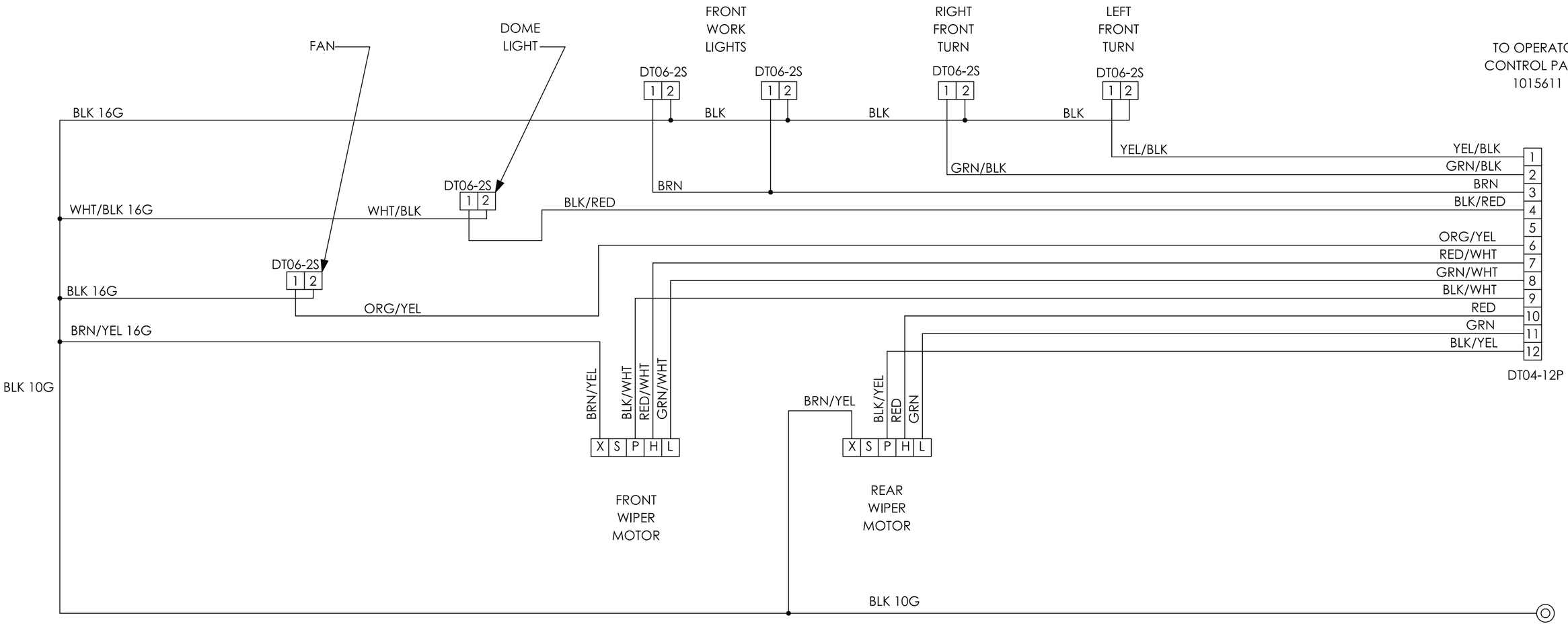


Figure 6-8. Cab Harness (1 of 2)

NOTES

CAB HARNESS (2 OF 2)

Schematic #1016077

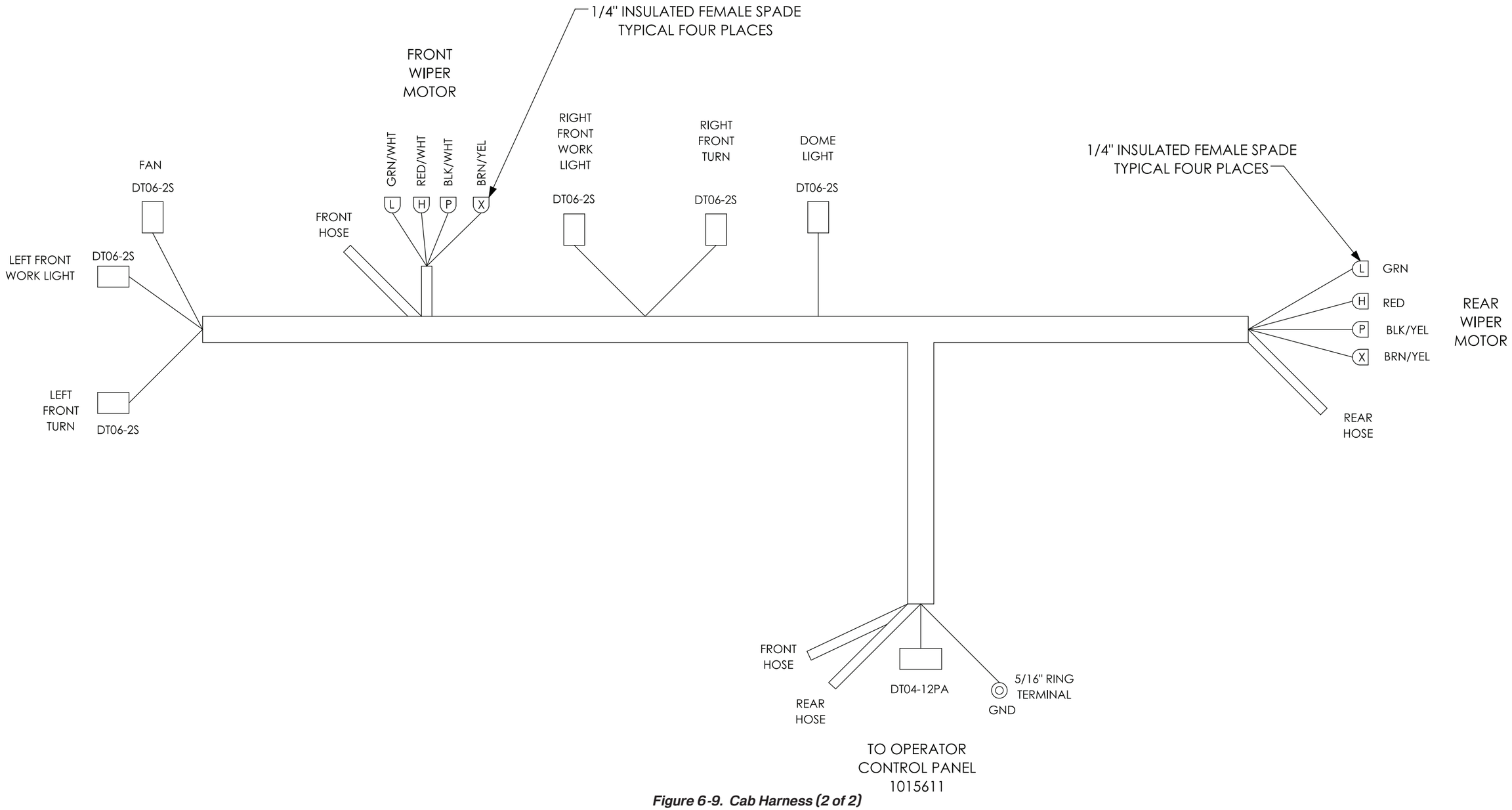


Figure 6-9. Cab Harness (2 of 2)

NOTES

CONTROLS CONSOLE (1 OF 2)

Schematic #1021630

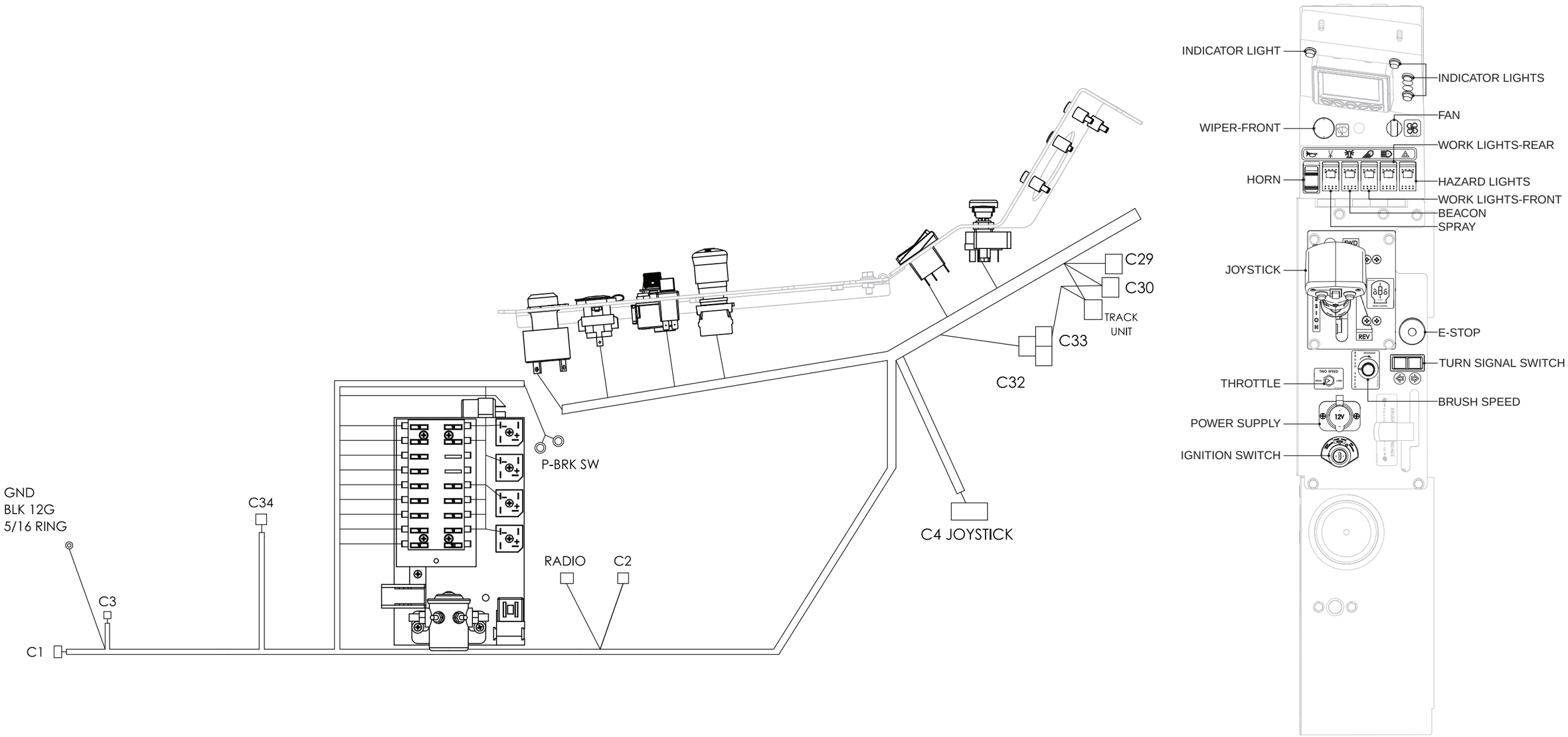


Figure 6-10. Controls Console (1 of 2)

NOTES

CONTROLS CONSOLE (2 OF 2)

Schematic #1021630

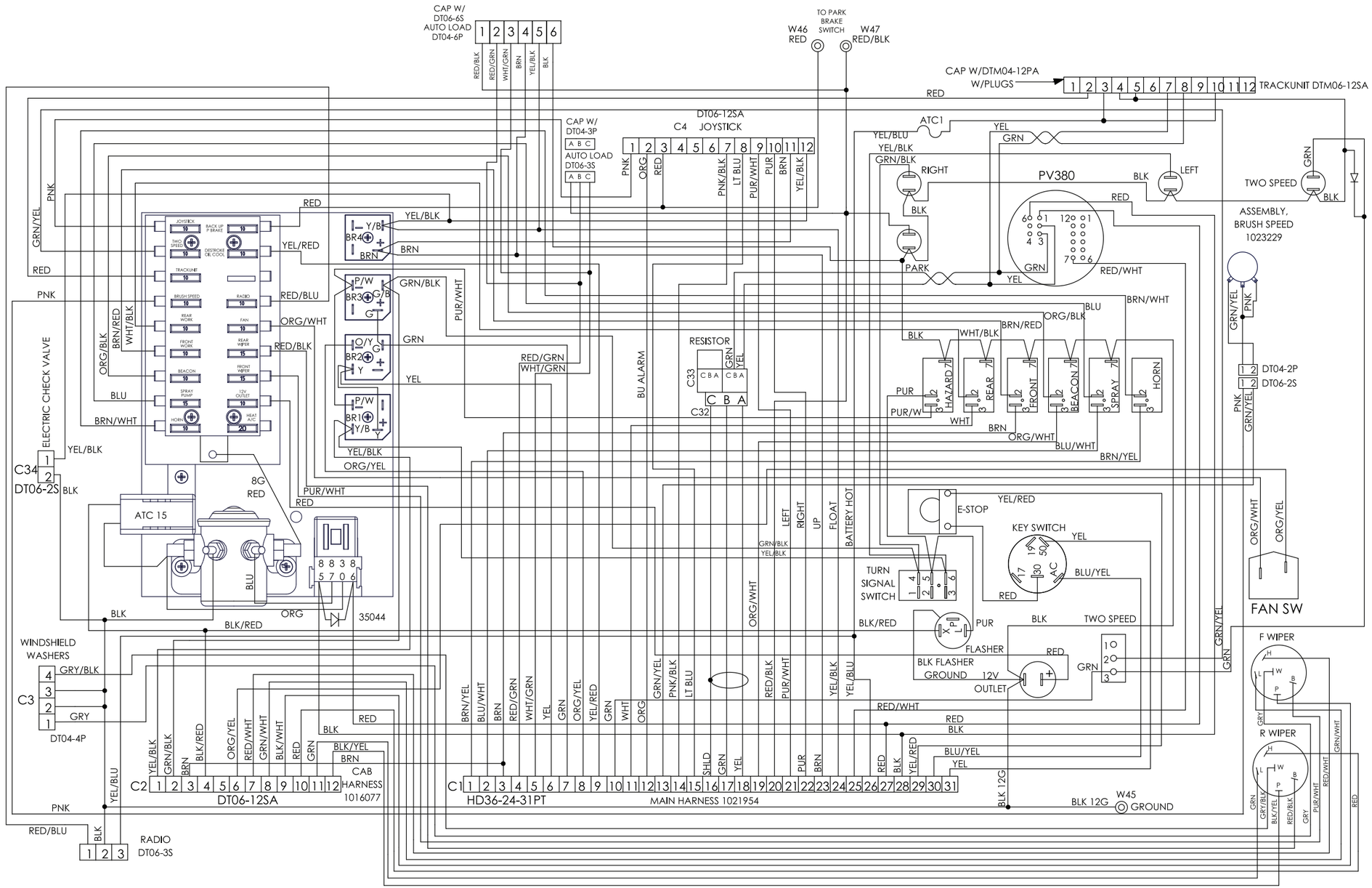


Figure 6-11. Controls Console (2 of 2)

NOTES

BEACON HARNESS

Schematic #1016701

BEACON HARNESS		
2S SOCKETS	COLOR	3S SOCKETS
1	ORG/WHT	A
2	BLK	B

Figure 6-12. Beacon Harness

NOTES

OIL COOLER SCHEMATIC/HARNESS

Schematic #1016141

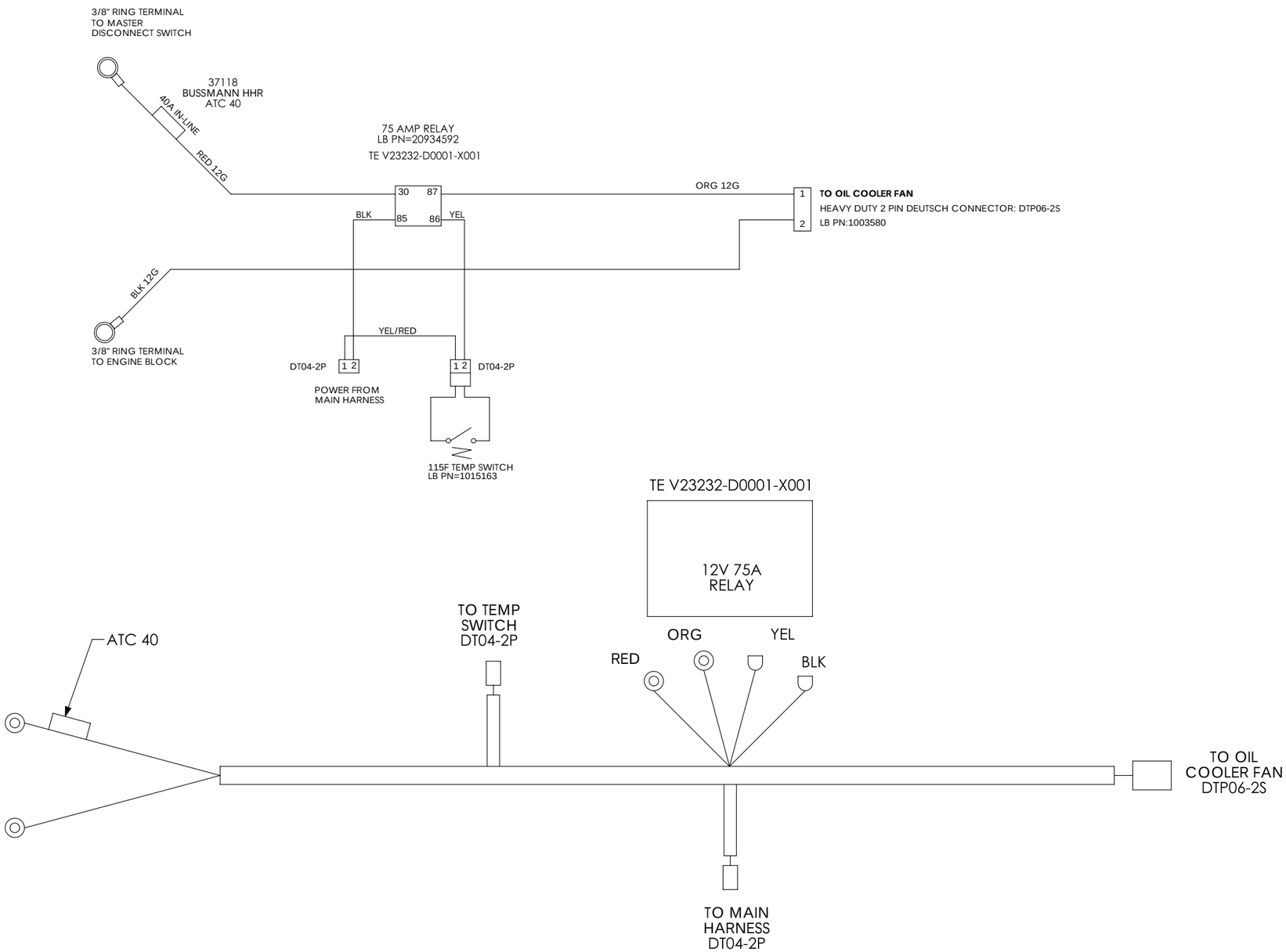
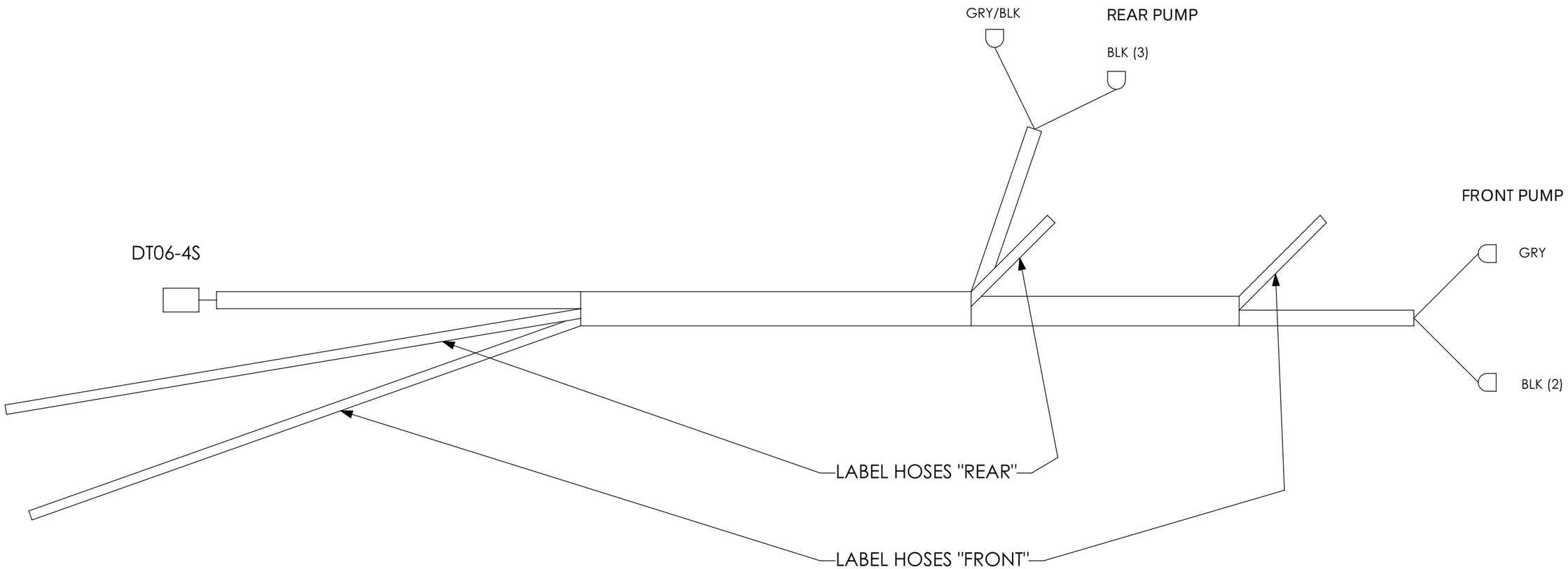


Figure 6-13. Oil Cooler Harness/Schematic

NOTES

OPTION - WINDSHIELD WASHER PUMP HARNESS

Schematic #1016079



DT06-4S TO CONTROL CONSOLE		
PIN	COLOR	FUNCTION
1	GRA	FRONT PUMP
2	BLK (2)	GND
3	BLK (3)	GND
4	GRA/BLK	REAR PUMP

1/4" INSULATED FEMALE SPADE TERMINAL

Figure 6-14. Option - Windshield Washer Pump Harness

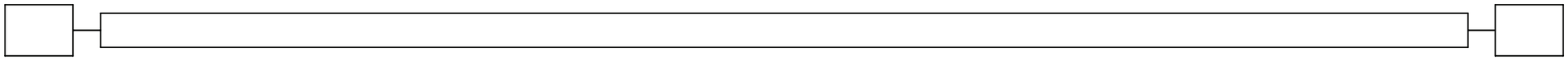
NOTES

OPTION - SPRAY PUMP HARNESS

Schematic #1016078

TO MAIN HARNESS
1016076
IN STEERING PEDESTAL
DT04-2P

TO SPRAY PUMP
DT06-2S



HARNESS, SPRAY PUMP EXTENSION		
2P PIN	COLOR	2S SOCKET
1	BLU/WHT	1
2	BLK	2

Figure 6-15. Option - Spray Pump Harness

NOTES

OPTION - CURB BRUSH HARNESS

Schematic #1022418

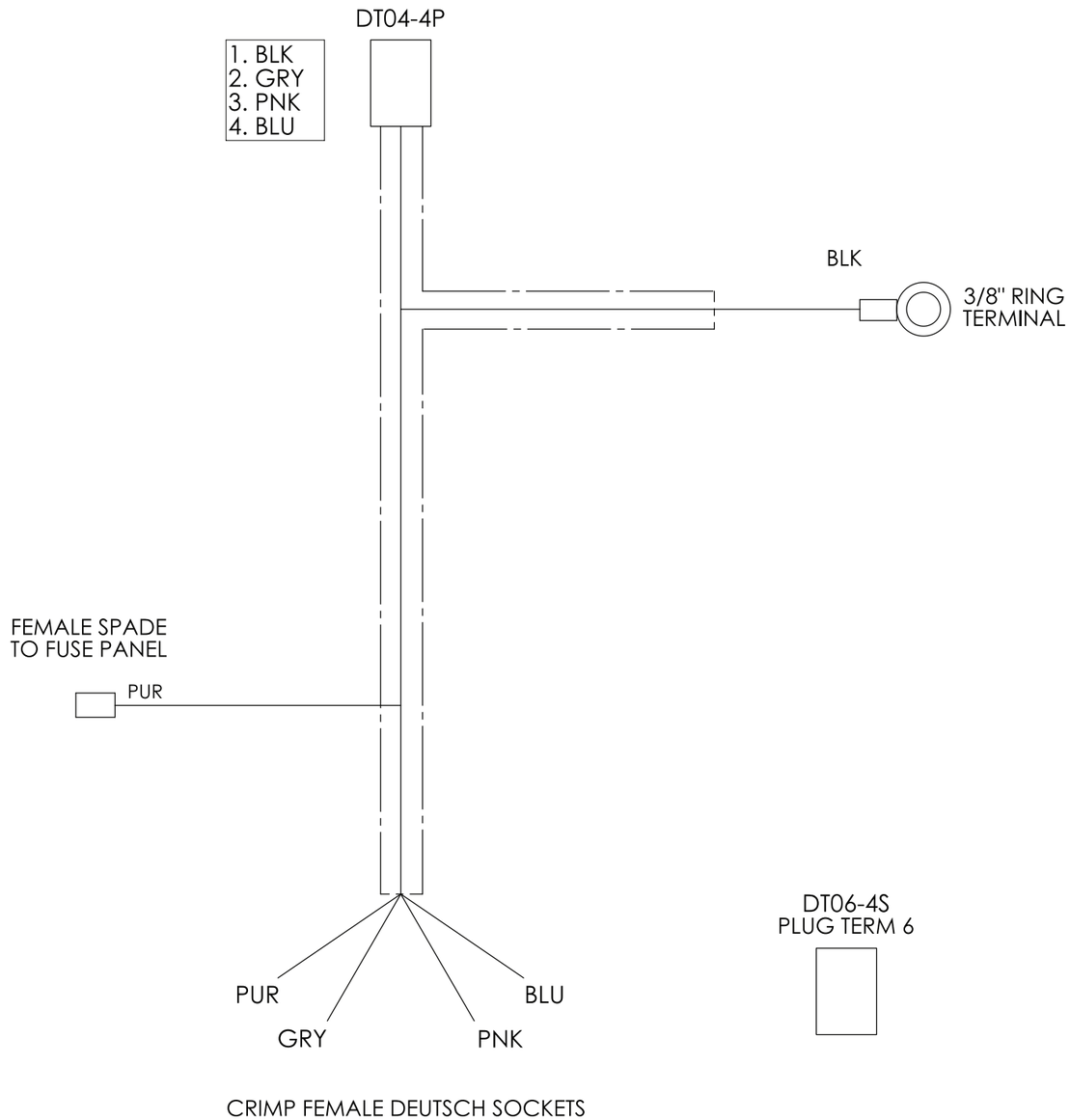


Figure 6-16. Option - Curb Brush Harness

NOTES

OPTION - SNOW PLOW HARNESS

Schematic #1022419

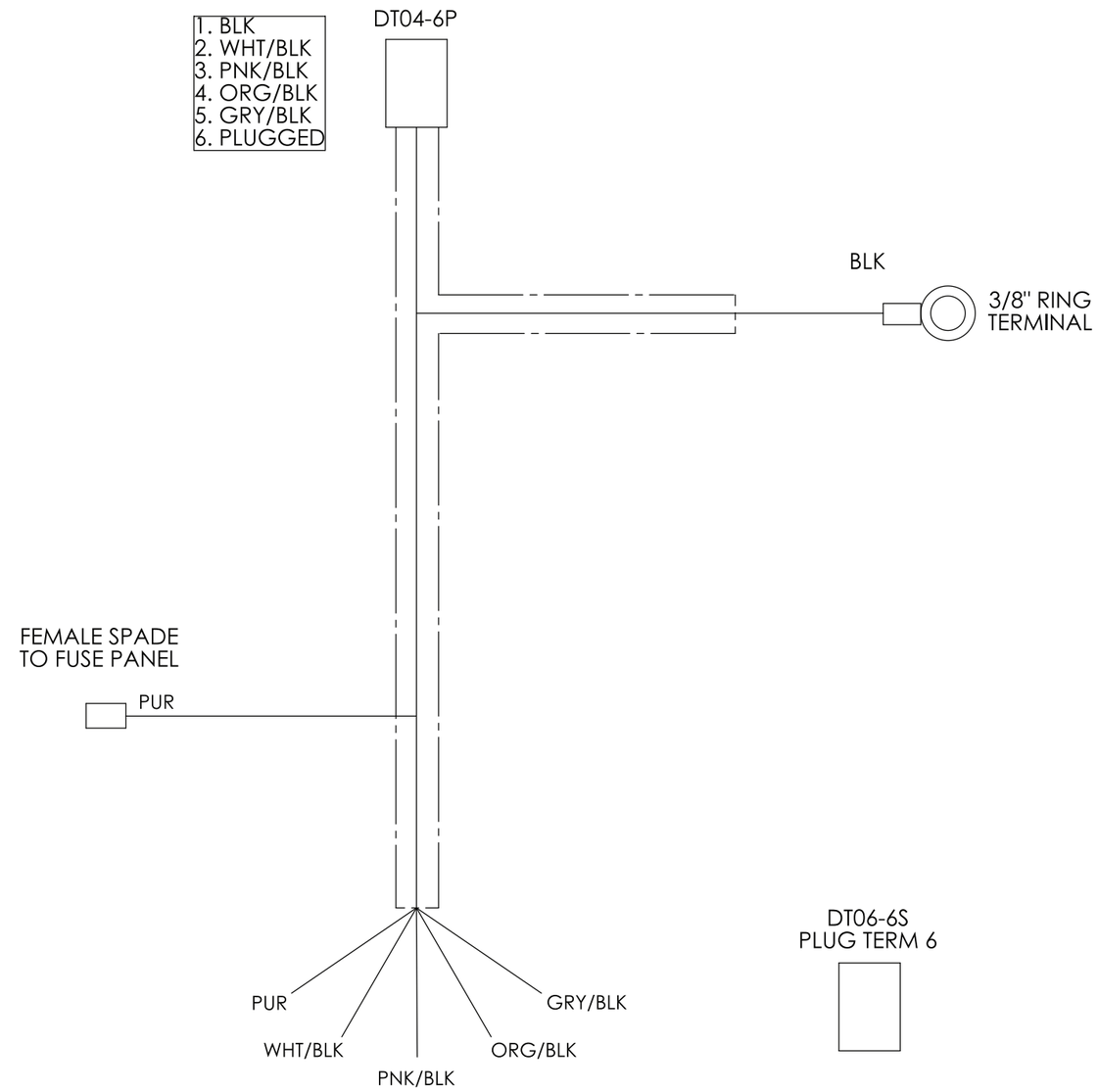


Figure 6-17. Option - Snow Plow Harness

NOTES

OPTION - FNR PEDAL MAIN HARNESS (1 OF 2)

Schematic #1023520

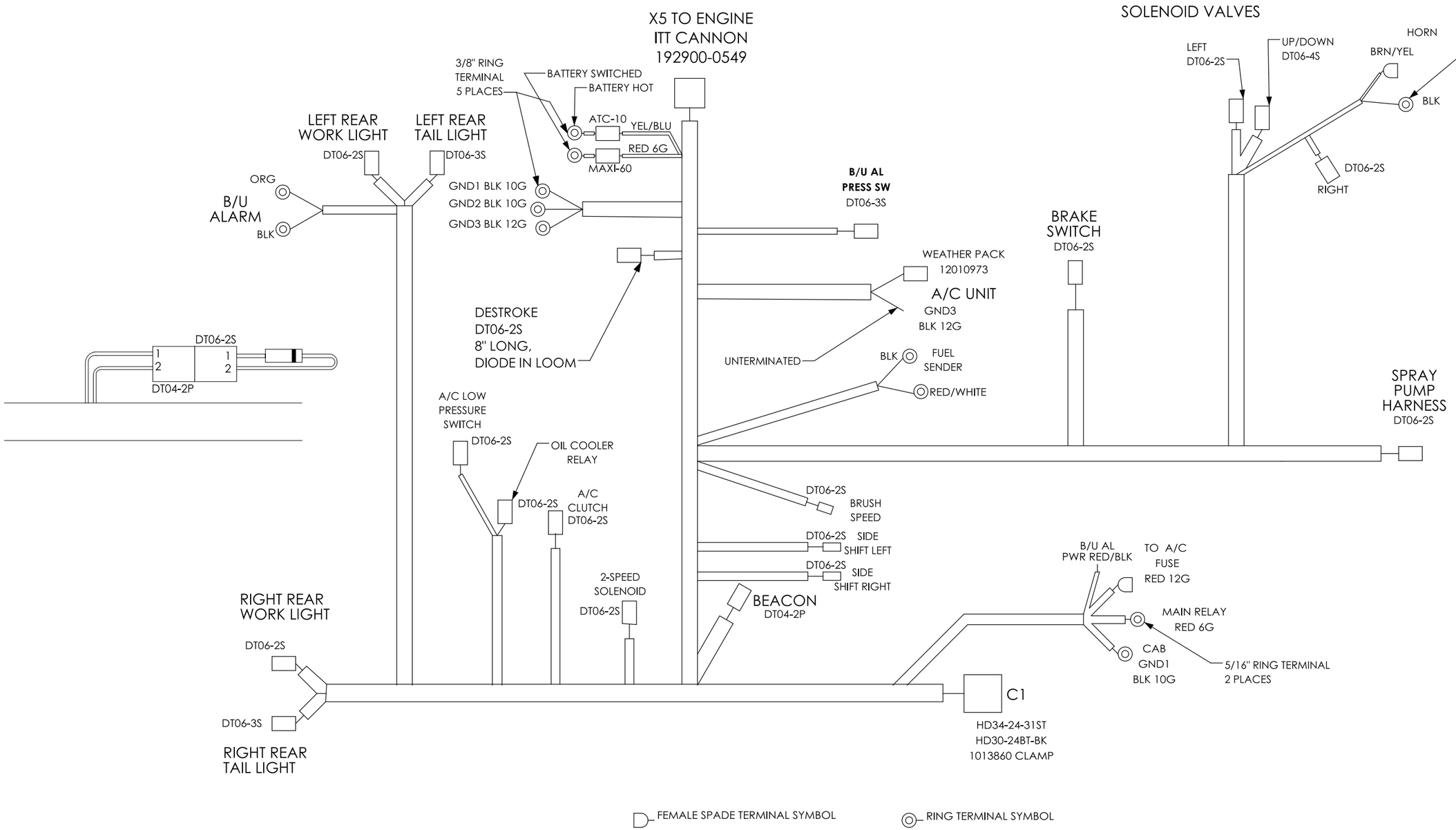


Figure 6-18. Option - FNR Pedal Main Harness (1 of 2)

NOTES

OPTION - FNR PEDAL MAIN HARNESS (2 OF 2)

Schematic #1023520

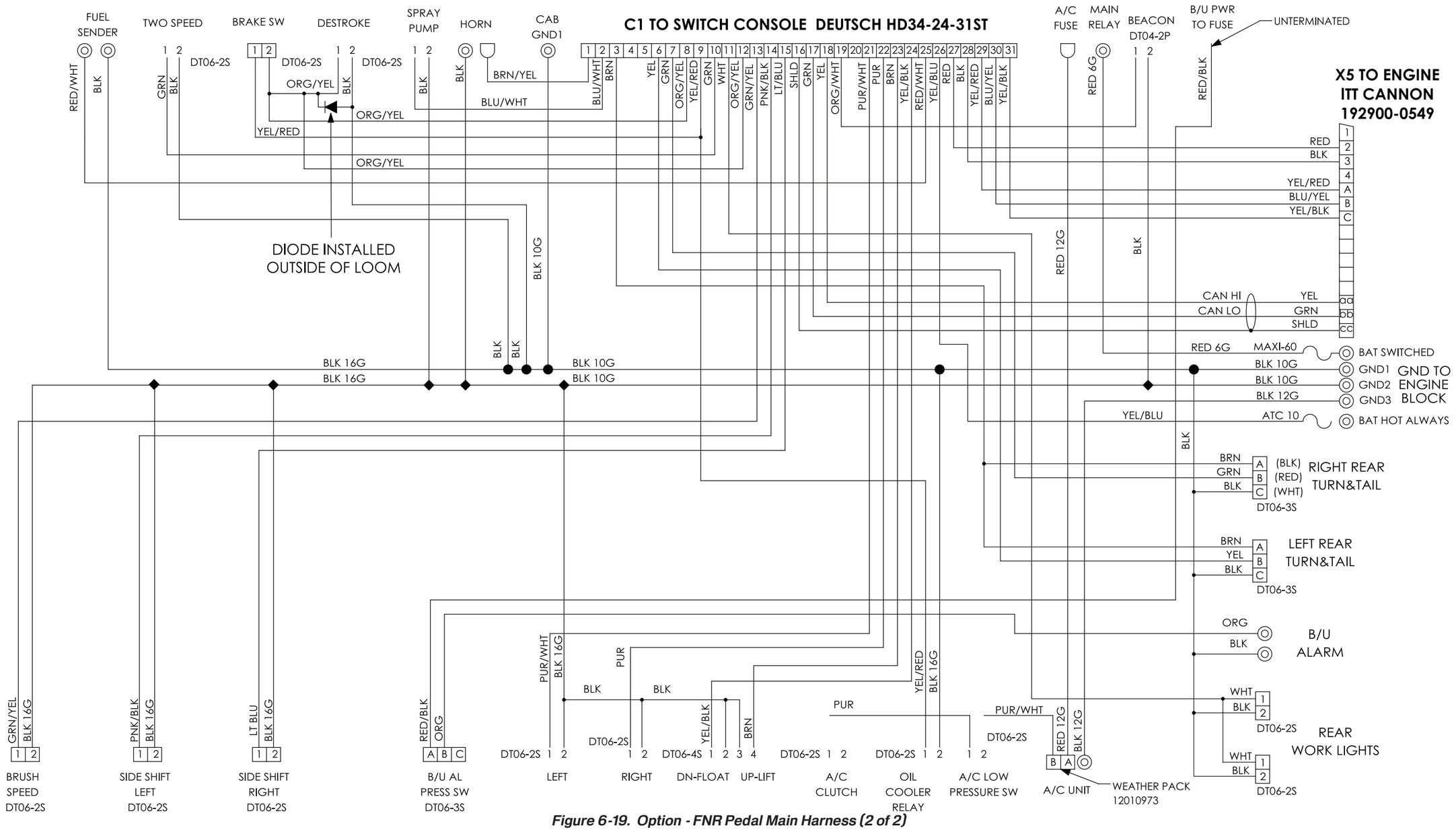


Figure 6-19. Option - FNR Pedal Main Harness (2 of 2)

NOTES

OPTION - FNR PEDAL BROOM CONTROLS

Schematic #1023493

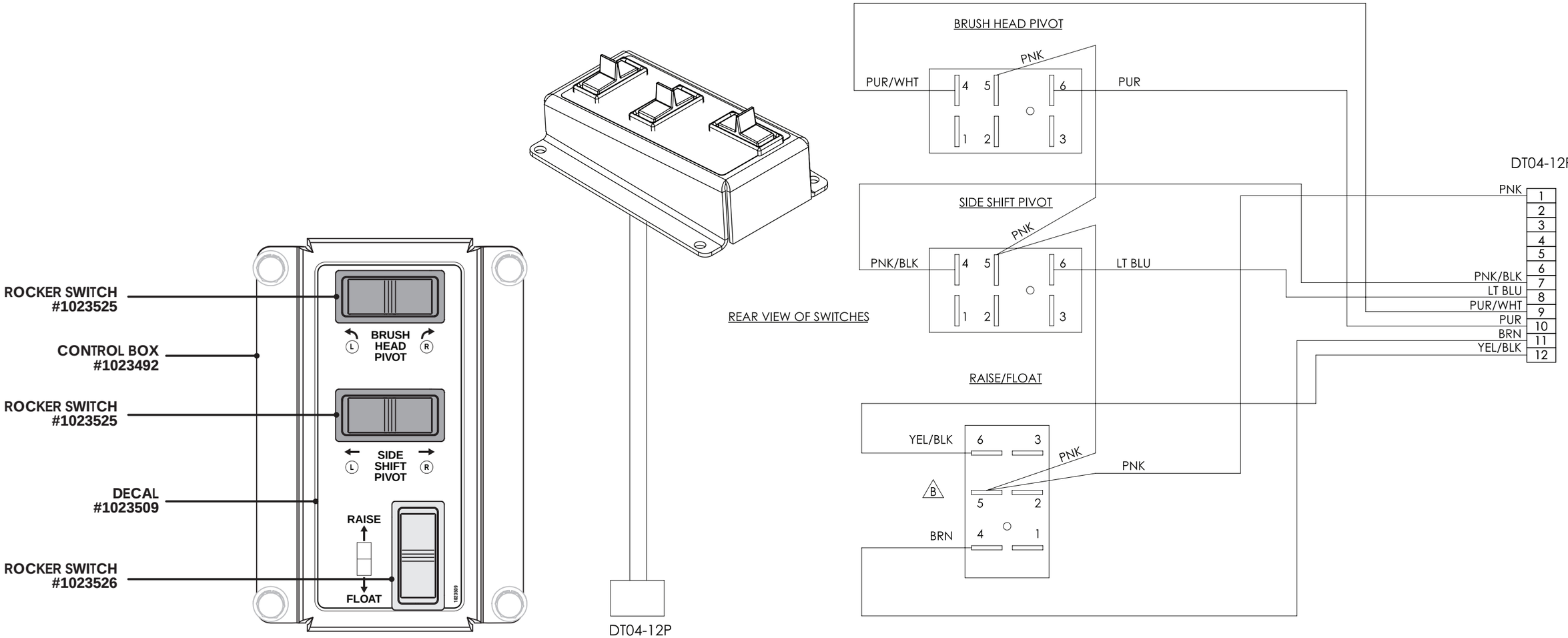
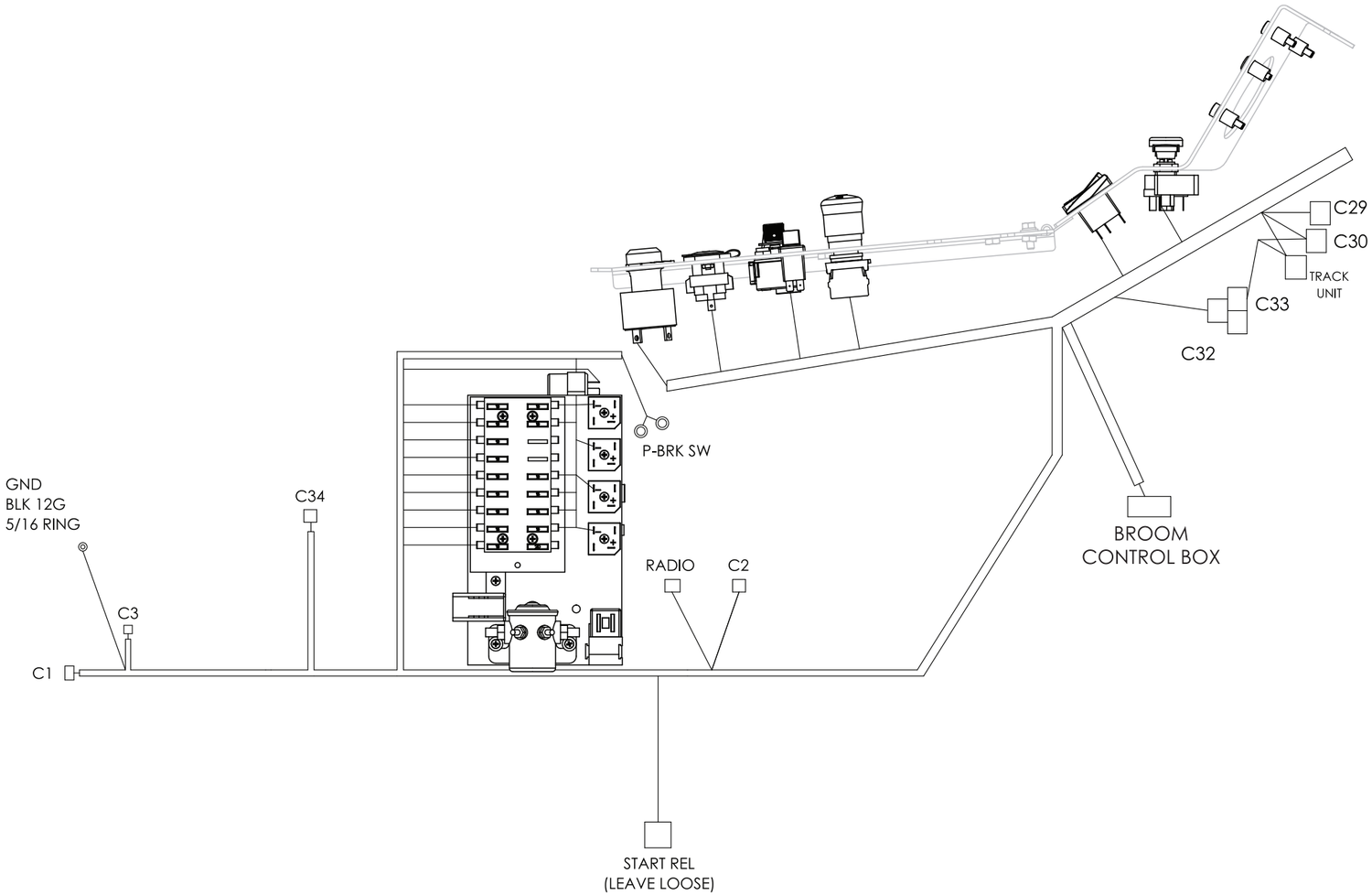


Figure 6-20. Option - FNR Pedal Broom Controls

NOTES

OPTION - FNR PEDAL CONTROL CONSOLE (1 OF 2)

Schematic #1023489



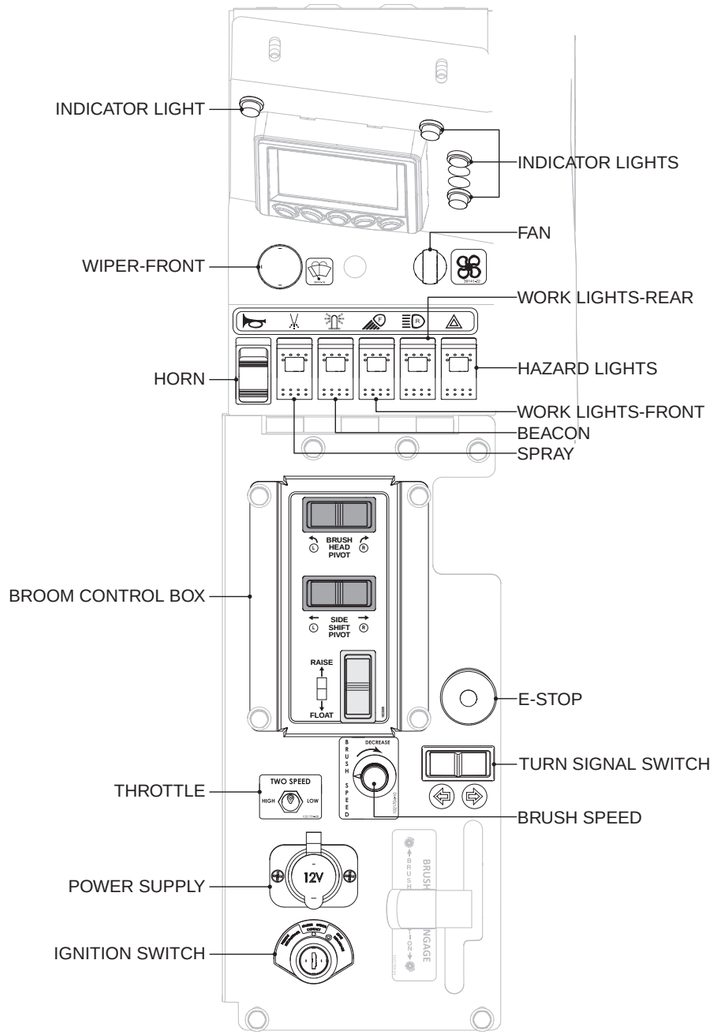


Figure 6-21. Option - FNR Pedal Control Console (1 of 2)

NOTES

OPTION - FNR PEDAL CONTROL CONSOLE (2 OF 2)

Schematic #1023489

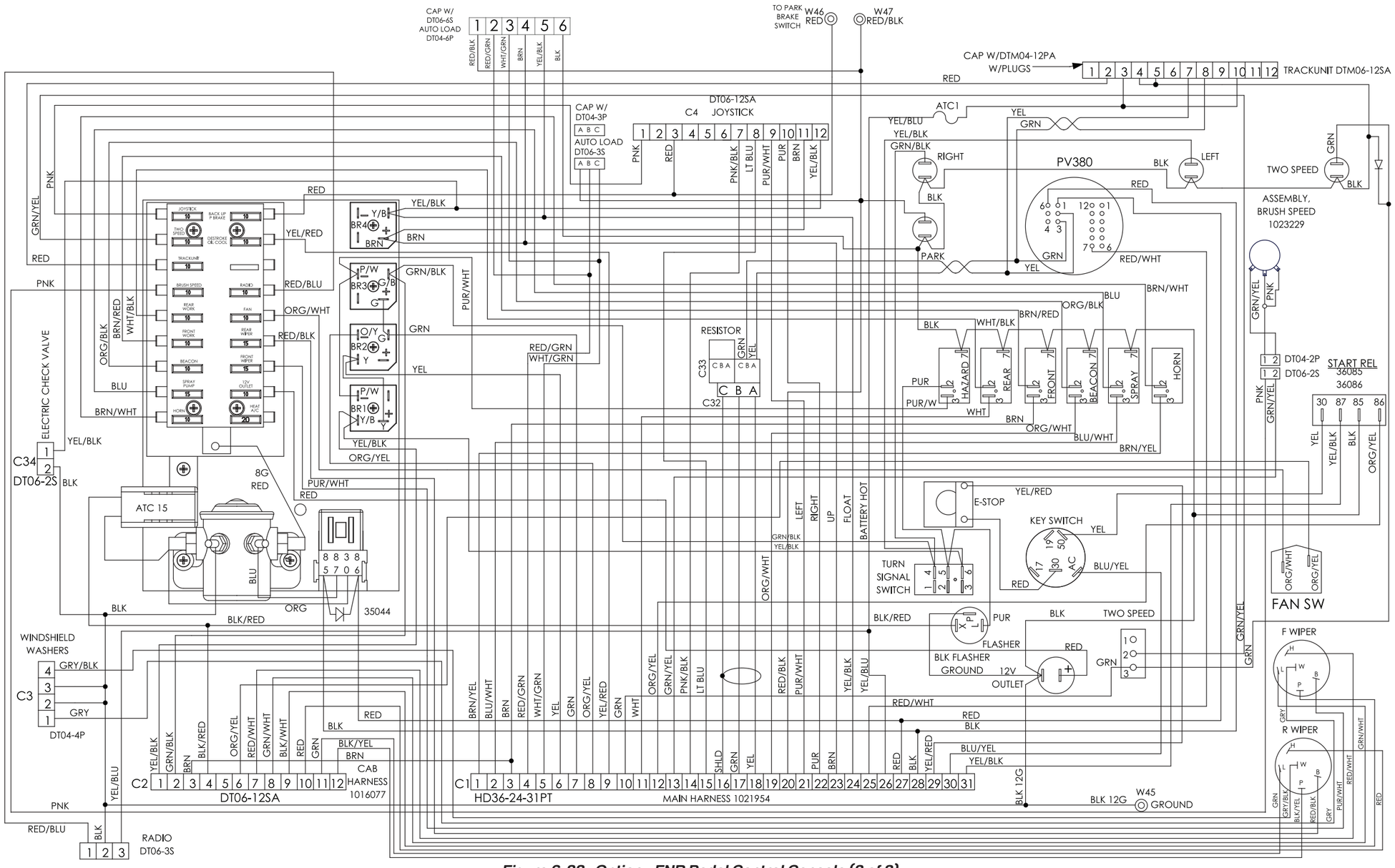
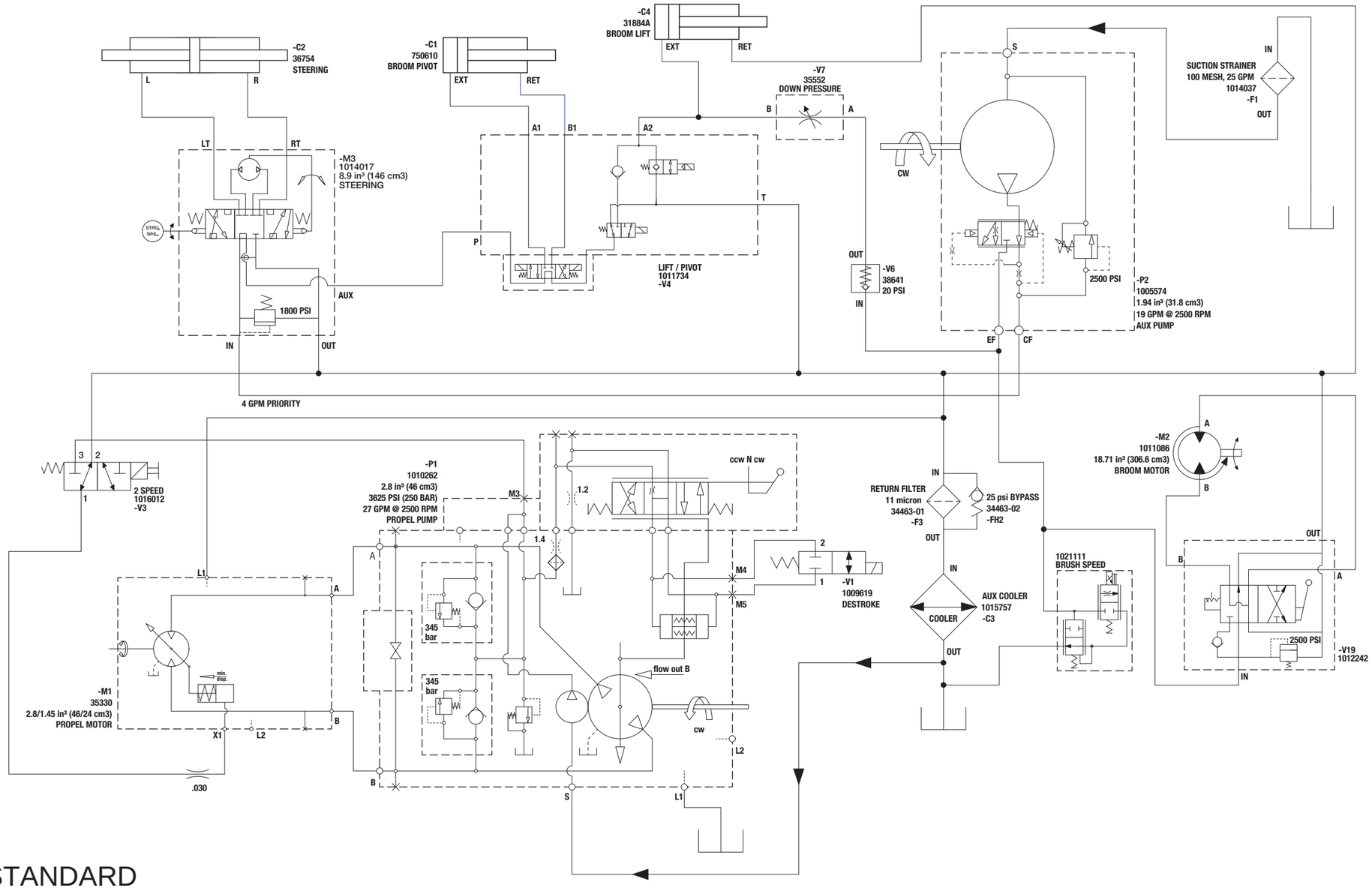


Figure 6-22. Option - FNR Pedal Control Console (2 of 2)

NOTES

HYDRAULIC SYSTEM - STANDARD (1 OF 5)

Schematic #1016638



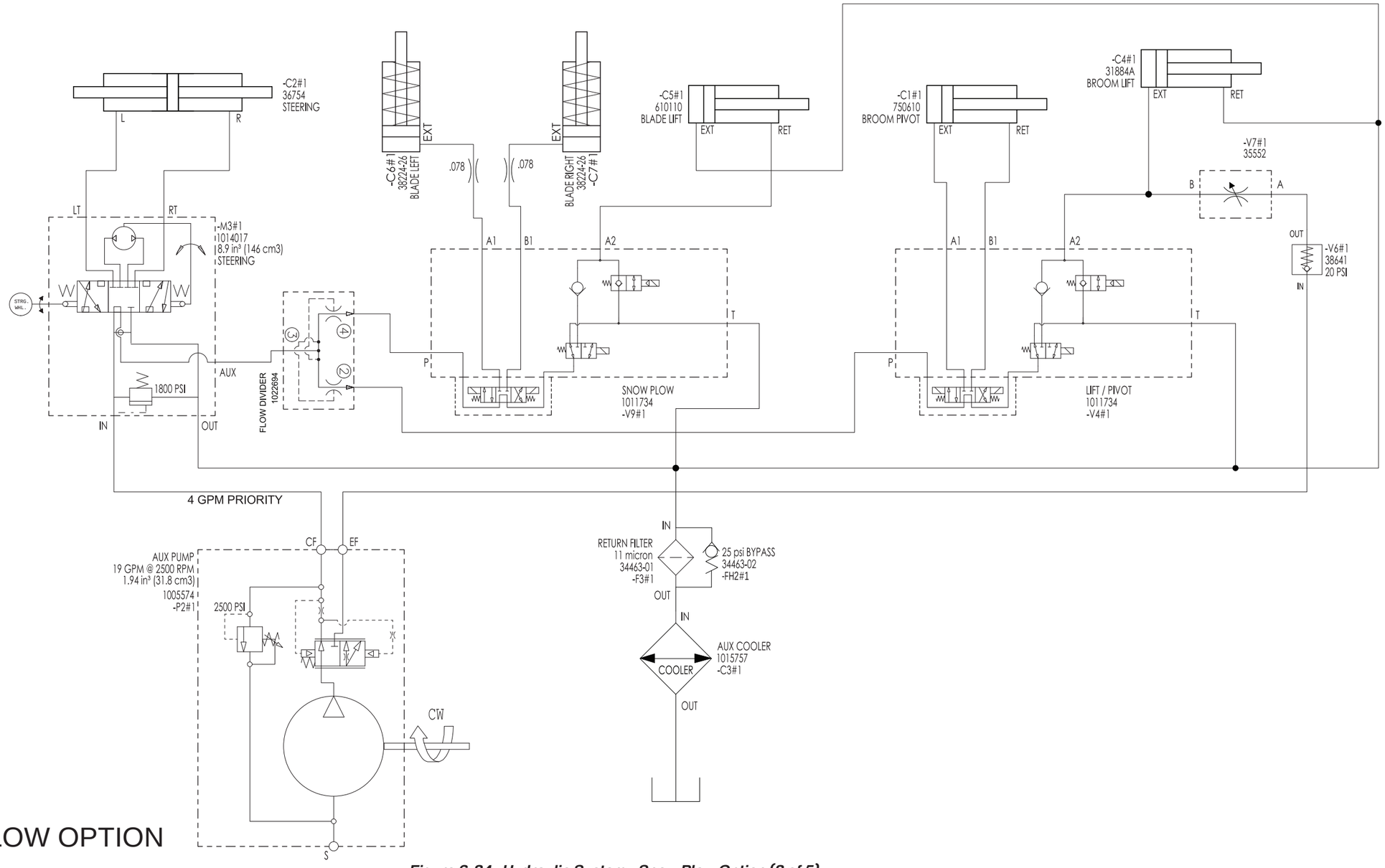
STANDARD

Figure 6-23. Hydraulic System - Standard (1 of 5)

NOTES

HYDRAULIC SYSTEM - SNOW PLOW OPTION (2 OF 5)

Schematic #1016638



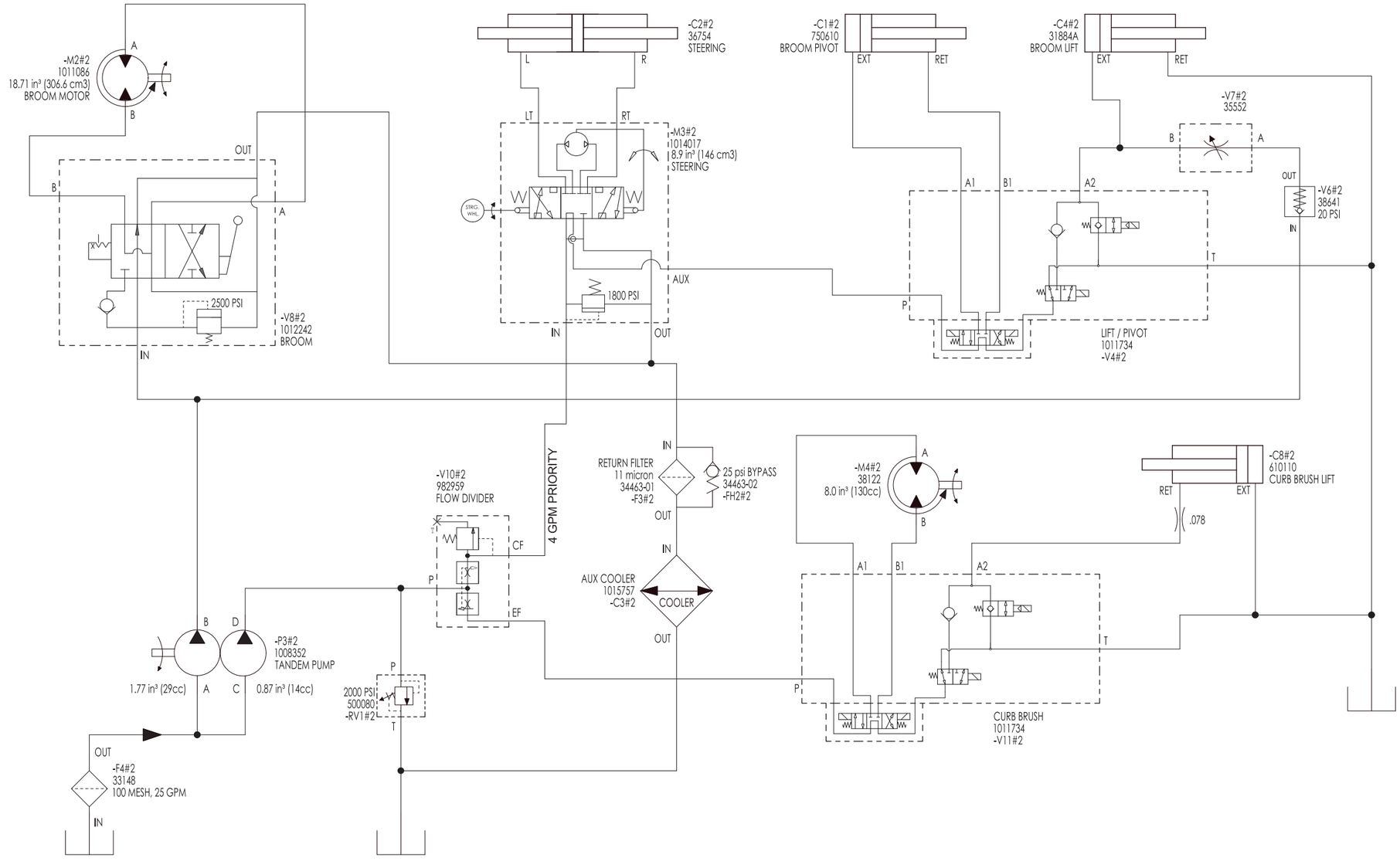
SNOW PLOW OPTION

Figure 6-24. Hydraulic System - Snow Plow Option (2 of 5)

NOTES

HYDRAULIC SYSTEM - CURB BRUSH OPTION (3 OF 5)

Schematic #1016638



CURB BRUSH OPTION

Figure 6-25. Hydraulic System - Curb Brush Option (3 of 5)

NOTES

HYDRAULIC SYSTEM - CURB BRUSH & SNOW PLOW OPTION (4 OF 5)

Schematic #1016638

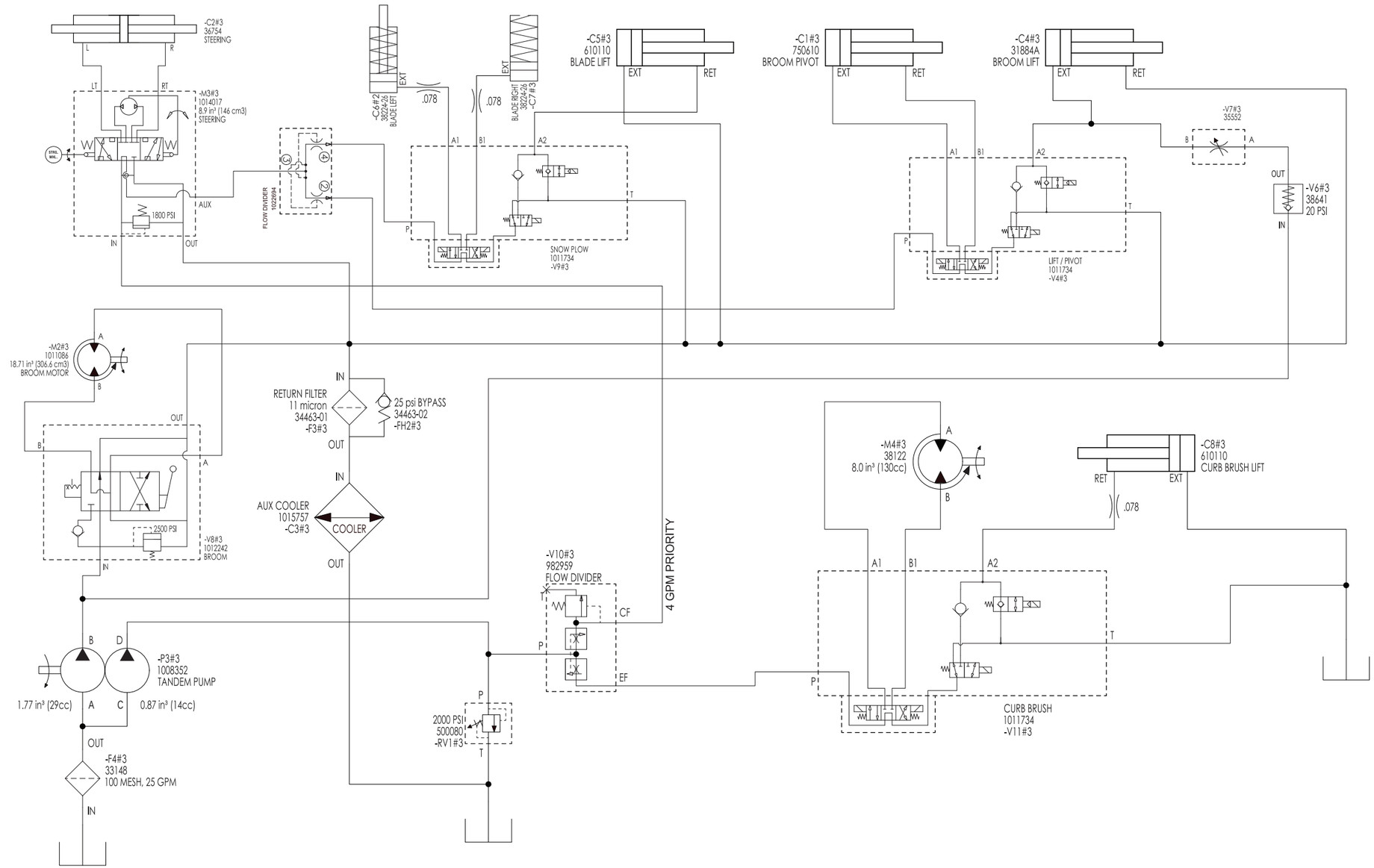
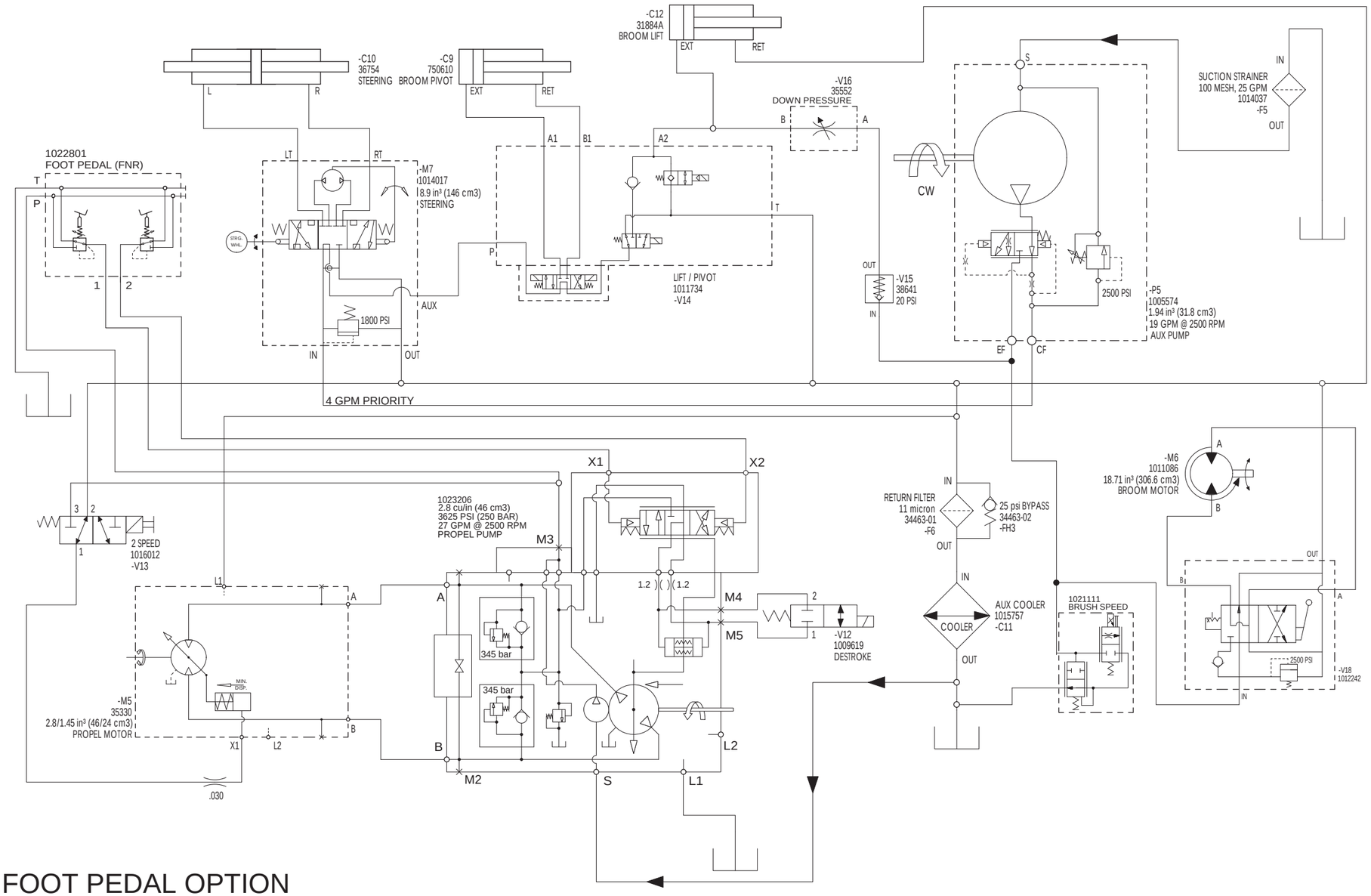


Figure 6-26. Hydraulic System - Curb Brush & Snow Plow Option (4 of 5)

NOTES

HYDRAULIC SYSTEM - FNR PEDAL OPTION (5 OF 5)

Schematic #1016638



FOOT PEDAL OPTION

Figure 6-27. Hydraulic System - FNR Pedal Option (5 of 5)

NOTES

Section 7

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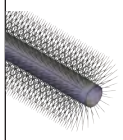
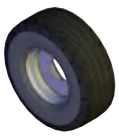
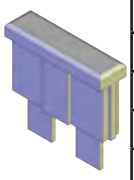
QUICK REFERENCE GUIDE

FILTERS/STRAINERS		
Engine	**Description**	**Part #**
	Oil Filter	1015794-01
	Pre-Fuel Filter	1015794-05
	Fuel Filter	1015794-06
	Air Cleaner, Primary	1015794-03
	Air Cleaner, Secondary	1015794-04
Hydraulic		
	Suction Strainer, 2-1/2-12 SAE	1014037
	Charge Filter	290030
	Charge/Return Filter Head	1017402

ENGINE COMPONENTS		
	Description	**Part #**
	Engine Belt - No A/C	1015794-07
	Engine Belt - A/C	1015794-08
	Alternator	1015794-16
	Starter	1015794-15
	A/C Compressor (Option)	1015794-30

BRUSH ASSEMBLIES		
	Description	**Part #**
	Brush -Poly Wafers	1005054
	Brush - Poly-Steel Wafers	1005055
	Brush, Tube, 32x90, Poly	38806
	Brush, Tube, 32x90, Poly/Steel	1007309

TIRE/WHEEL ASSEMBLY		
	Description	**Part #**
	Assembly, Tire & WHeel, 15x6	1009432

ELECTRICAL COMPONENTS			
Controls	**Description**	**Qty**	**Part #**
	1A Fuse, ATC	1	1015794-55
	5A Fuse, ATC	2	36746
	10A Fuse, ATC	16	36340
	15A Fuse, ATC	5	36341
	20A Fuse, ATC	1	36342
	30A Fuse, ATC	1	36218
	40A Fuse, ATC	1	115911
	Fuse Block, ATC	1	685060
	Relay, 12VDC, SPDT, 40A	1	36085
	Flasher, LED	1	1010202
	Bridge Rectifier	4	1016125
	Fuse Holder, Inline	1	37118
	Solenoid, 12V, CD	1	720290

7

AXLE - ASSEMBLIES

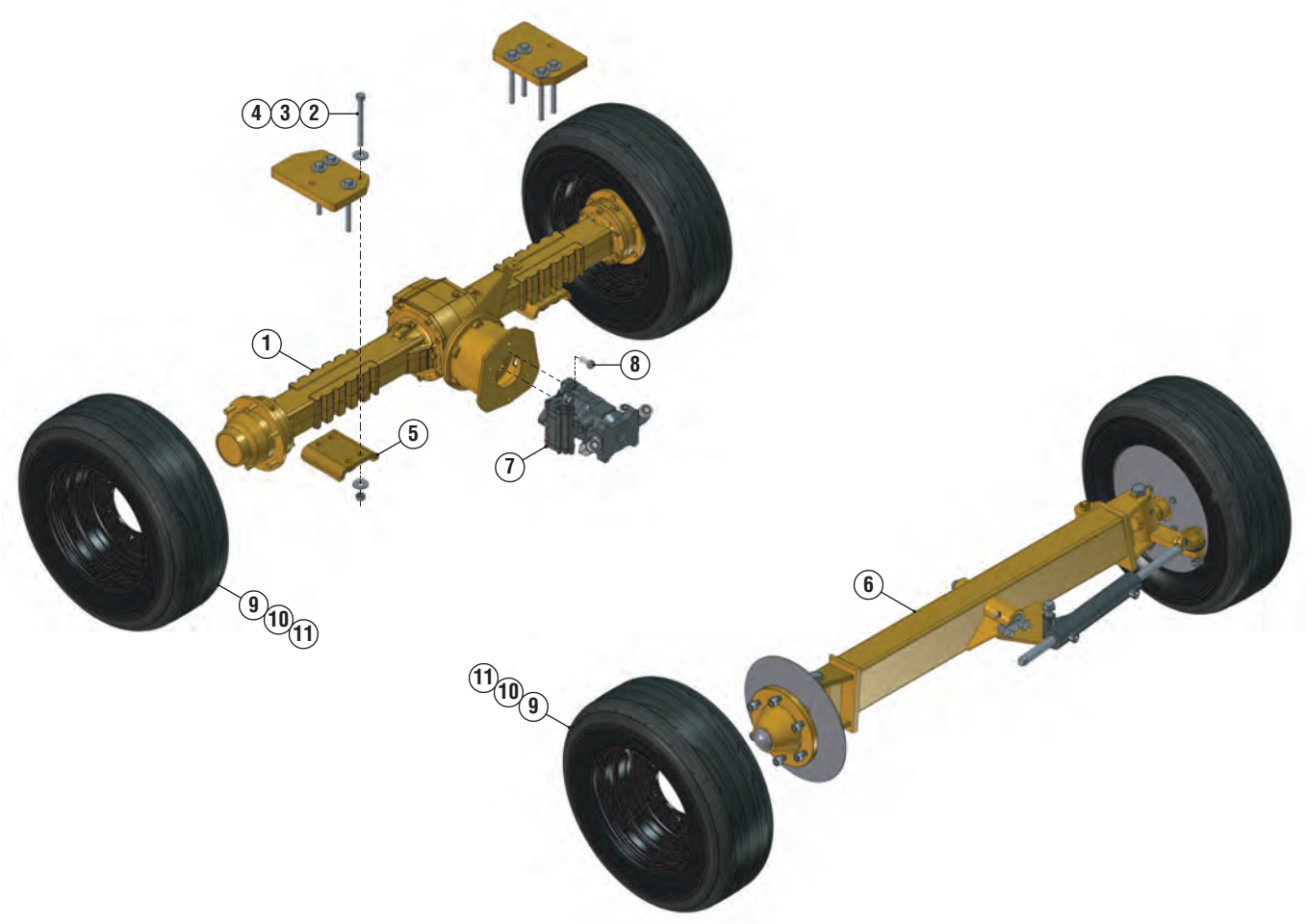


Figure 7-1. Axle - Assemblies

Axle - Assemblies

Item No.	Part Number	Qty.	Description	Remarks
GRP	1009422		Group - Axles	Includes 1-8
1	1016955	1	Drive Axle, Comer 068	See Fig. 7-2 & 7-3
2	100-10-11-104-8	8	CSHH, 5/8-11 x 6.50, GR5	
3	311-10	16	Washer, Hardened, USS, 5/8	
4	204-10-11-5	8	Nut, Lock, Stover, 5/8-11, GR5	
5	1016659	2	Axle Mounting Plate	
6	1009423	1	Assembly, Steering Axle	
REF	1009431	1	Drive Group, Comer	Includes 7-8
7	1020372	1	Assembly, Motor w/ORFS Adapters	See Fig 7-5
8	100-M14-2-35-8.8	2	CSHH, M14x2 x 35, C8.8	
ASSY	1009432	4	Assembly, Tire & Wheel, 15x6	Includes 9-11
9	35342	4	Radial Tire, ST225/75R15, E	
10	1008062	4	Wheel, 15 x 6, 6 on 205MM BC	
11	72761	4	Tire Valve, Tubeless, 2.00 LG	
12	1009294	24	Nut, Lug, M18x1.5	Not Shown

AXLE - REAR DRIVE (1 OF 3)

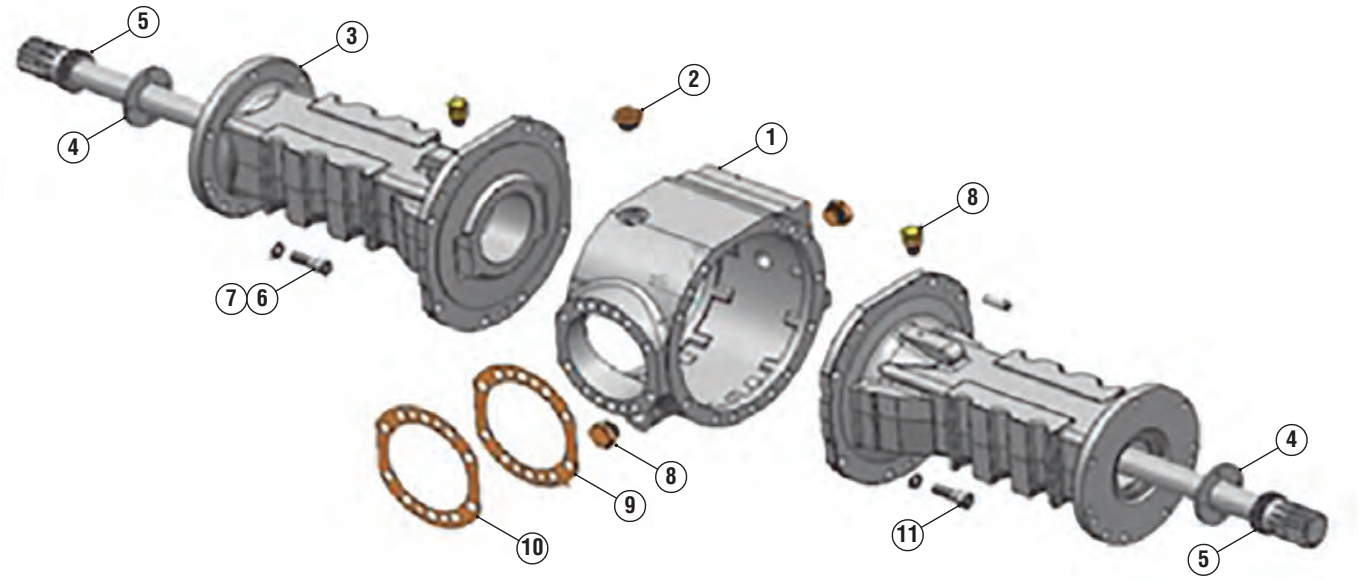


Figure 7-2. Axle - Rear Drive (1 of 3)

Axle - Rear Drive (1 of 3)

Item No.	Part Number	Qty.	Description	Remarks
REF	1016655	1	Drive Axle, Comer 068	Not All Items Shown
1	1006327-61	1	Central Housing	
2	1006327-62	1	Vent Cap M22x1.5	
3	1006327-63	2	Side Housing	
4	1006327-64	2	Shim	
5	1006327-65	2	Drive Shaft	
6	1006327-66	18	Shnorr Washer D.10	
7	1006327-67	18	Socket Head Screw M10x30	
8	1006327-68	4	Cap M22x1,5 Din 909	
9	1006327-69	2	Paper Gasket 0,2	
10	1006327-70	1	Paper Gasket 0,1	
11	1006327-71	16	Socket Head Screw 12x35	
12	1006327-72	6	Cylindrical Pin Ø10x30	

AXLE - REAR DRIVE (2 OF 3)

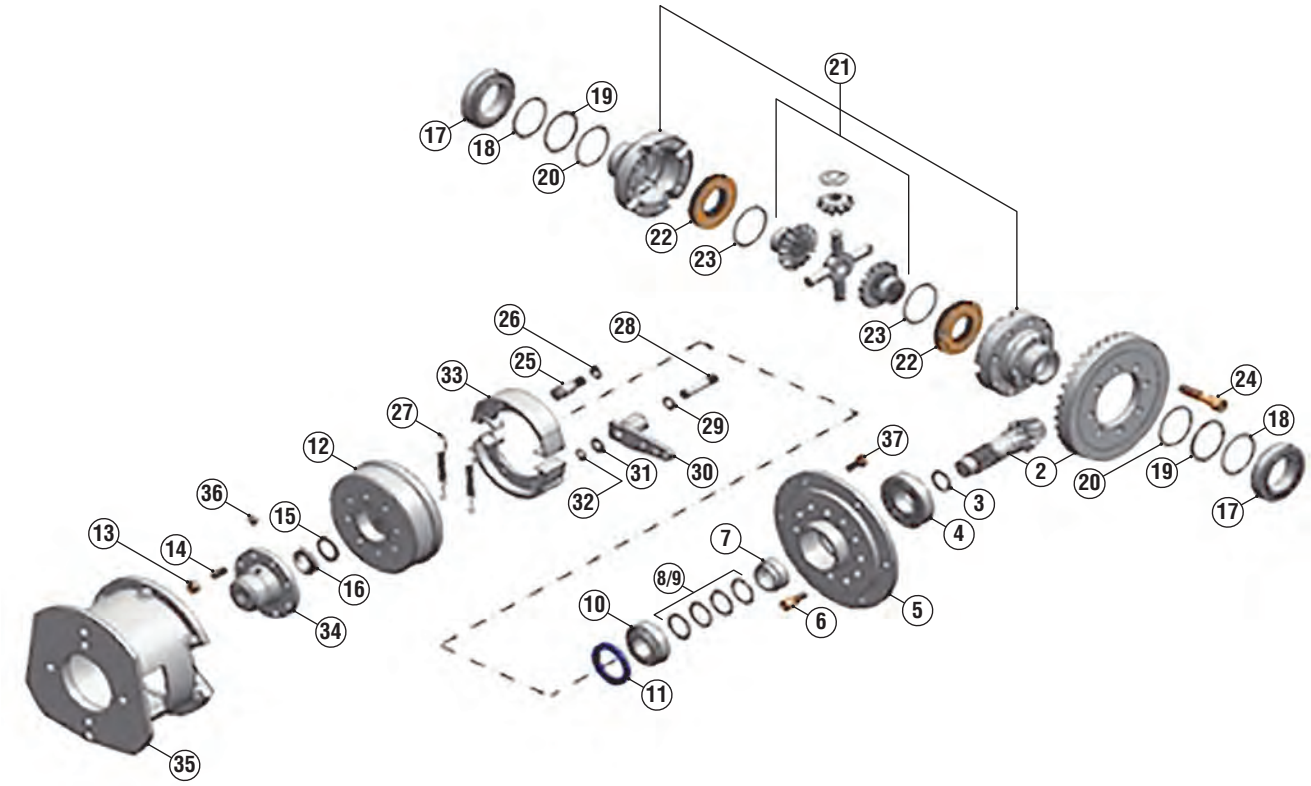


Figure 7-3. Axle - Readr Drive (2 of 3)

Axle - Rear Drive (2 of 3)

Item No.	Part Number	Qty.	Description	Remarks
REF	1016655	1	Drive Axle, Comer 068	Not All Items Shown
1	1006327-24	1	Bevel Crown Z=38	
2	1006327-25	1	Bevel Pinion Z=9	
3	1006327-26	1	O-Ring 4112	
4	1006327-27	1	Tapered Roller Bearing 30308	
5	1006327-28	1	Pinion Housing	
6	1006327-75	10	Screw T.C.E.I. M12x45 12.9	
7	1006327-29	1	Spacer	
8	1006327-30	-	Shim PS 35x45x0,1	
9	1006327-31	-	Shim PS 35x45x0,3	
10	1006327-32	1	Taper Rolling Bearing 30307	
11	1006327-33	1	Oil Seal BASL 50x72x8	
12	1006327-34	1	Flange with Brake Drum	
13	1006327-76	8	Nut AUTOBLOK M10	
14	1006327-35	8	Stud Bolt M10x25	
15	1006327-36	1	Shim AS 1528	
16	1006327-37	1	Self-Lock Nut GP M30x1,5	
17	1006327-38	2	Roller Bearing 32012	
18	1006327-39	-	Spess. ø61x70x0,1	
19	1006327-40	-	Spess. ø61x70x0,3	
20	1006327-41	-	Spess. ø61x70x0,5	
21	1006327-42	1	Differential Housing	
22	1006327-43	2	Disc Package	
23	1006327-44	4	Shim, 50x70x0.2	
24	1006327-48	8	Socket Head Screw M14x70	
25	1006327-49	1	Brake Fulcrum Pin	
26	REF	1	Washer	
27	1006327-50	2	Spring	
28	1006327-51	1	Brake Lever Pin	
29	1006327-52	2	Friction Washer	
30	1006327-53	1	Brake Lever	
31	1006327-54	1	Friction Washer	
32	1006327-55	1	Snap Ring E14 230-150H	
33	1006327-56	1	Brake Shoe Pair	
34	1006327-57	1	Hydraulic MOT Connection	
35	1006327-58	1	Hydraulic MOT Mount	
36	1006327-59	1	Greaser M10x1	
37	1006327-60	6	Screw M10x25 8.8	

AXLE - REAR DRIVE (3 OF 3)

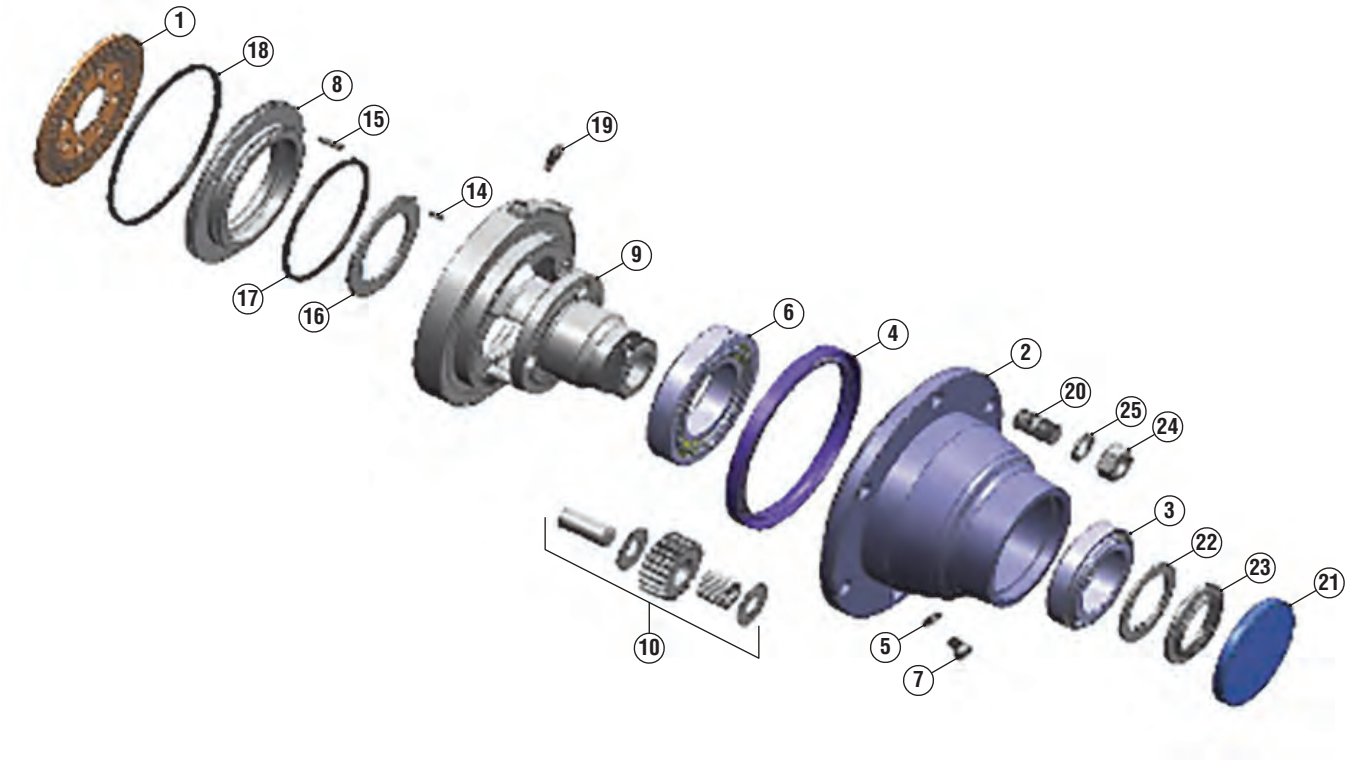


Figure 7-4. Axle - Rear Drive (3 of 3)

Axle - Rear Drive (3 of 3)

Item No.	Part Number	Qty.	Description	Remarks
REF	1016655	1	Drive Axle, Comer 068	Not All Items Shown
1	1006327-01	1	Brake Disc	
2	1006327-12	1	Wheel Hub	
3	1006327-13	1	Bearing 32012X	
4	1006327-11	1	Oil Seal BASL 145X170x12	
5	1006327-14	1	Washer 10x16x1,5	
6	1006327-10	1	Bearing 30214	
7	1006327-15	1	Cap M10x1 DIN908	
8	1006327-02	1	Brake piston	
9	1006327-09	1	Spindle	
10	1006327-16	3	Planetary Gear Pin	
11	1006327-17	6	Planetary Gear Shim	
12	1006327-18	3	Planetary Gear	
13	1006327-19	51	Roller D.4x23.8 DIN5402	
14	1006327-07	1	Parallel Pin 5X12	
15	1006327-03	3	Parallel Pin 6X20	
16	1006327-06	1	Brake Ring	
17	1006327-04	1	Square Ring 237-993-9780	
18	1006327-05	1	Square Ring 238-956-9707	
19	1006327-08	1	Brake Bleeder M10x1	
20	1006327-20	6	Stud Bolt M18x1,5 10.9	
21	1006327-21	1	Cap	
22	1006327-22	1	Lock Washer	
23	1006327-23	1	Locknut M55x2	
24	1006327-73	12	Nut M18x1.5 MF	
25	1006327-74	12	Elastic Washer D. 18	

AXLE - DRIVE MOTOR

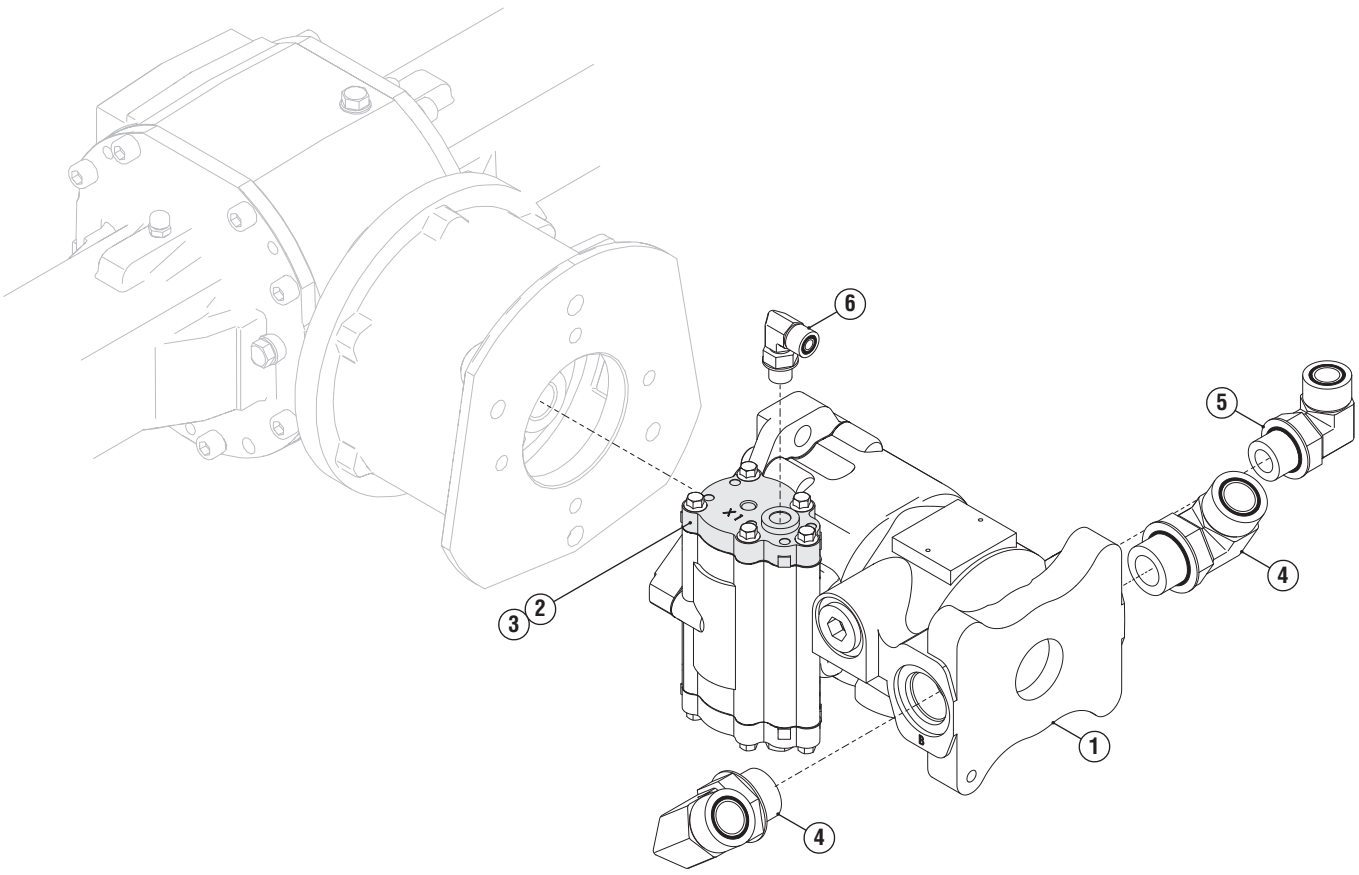


Figure 7-5. Axle - Drive Motor

Axle - Drive Motor

Item No	Part Number	Qty	Description	Remarks
GRP	1009431		Group - Motor Drive	
ASSY	1020372	1	Assembly, Motor w/ORFS Adapters	
1	35330	1	Motor, Hydraulic, M46 Variable, 2.8/1.45 CIR	
-	35330-02	A/R	Seal Kit, Hydraulic Motor, M46	Repair Part, Not Shown
2	851492	1	Cover, Servo, M46 Pump	
3	851494	1	Gasket, Servo Cover, M46 Pump	
4	FS6801-16-16-NWO-FG	2	Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	
5	FS6801-12-12-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	
6	FS6801-06-06-NWO-FG X 030	1	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring, .030 Orifice	

AXLE - FRONT STEERING

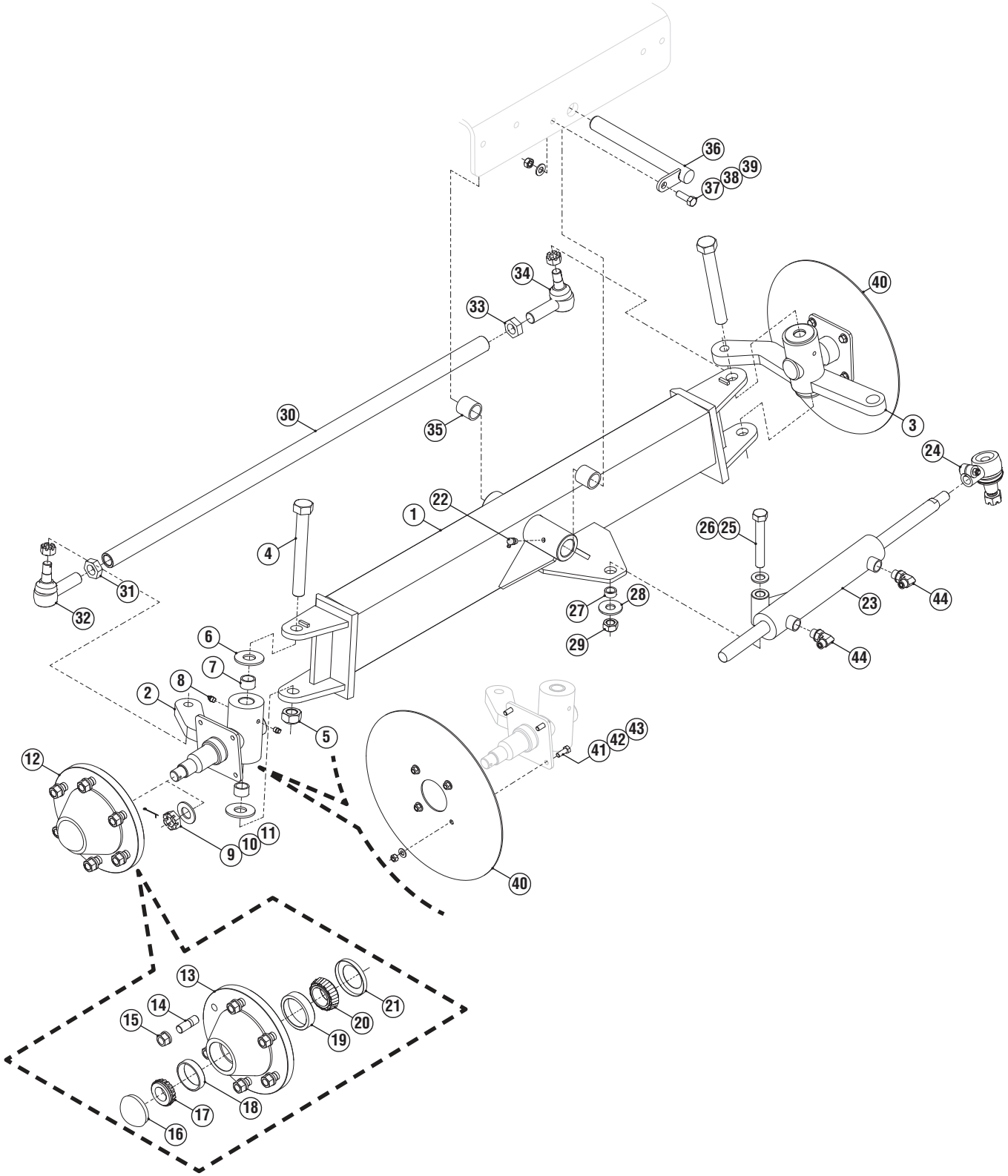


Figure 7-6. Axle - Front Steering

Axle - Front Steering

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1009423	1	Assembly, Steering Axle	Includes Items 1-36
1	28451SRV	1	Weldment, Front Axle	
2	1010689SRV	1	Weldment, Spindle - Right	
3	1010688SRV	1	Weldment, Spindle - Left	
4	100-16-8-128-5	2	CSHH, 1-8 x 8.00, GR5	
5	204-16-8-5	2	Nut, Lock, Stover, 1-8, GR5	
6	301-16	4	Washer, Flat, USS, 1	
7	35406	4	Bushing, Oilite, 1.003 ID x 1.253 OD x .75 LG	
8	33684	4	Grease Fitting, 1/8 NPT	
9	31713	2	Nut, Spindle, 6BF	
10	300-16	2	Washer, Flat, SAE, 1	
11	80332	2	Cotter Pin, .125 x 1.50	
12	1009281SRV	2	Hub Assembly, 6 Bolt, 205MM BC	Includes Items 13-24
ASSY	1011707	2	Assembly, Casted Hub, Idler Comer	Includes 13-15
13	1011706	2	Hub, Idler Casting, Comer Axle	
14	1009293	12	Stud, Wheel, M18x1.5 x 2.1" LG	
15	1009294	12	Nut, Lug, M18x1.5	
16	620200	2	Dust Cap	
17	340180	2	Bearing Cone, Wheel Hub, Outer	
18	340170	2	Bearing Cup, Wheel Hub, Outer	
19	610200	2	Bearing Cup	
20	610210	2	Bearing Cone	
21	33187	2	Seal	
22	29B-02	1	Grease Fitting, 1/8 NPT, 90°	
23	36754	1	Cylinder, Hydraulic, 2.00 x 10.75 x 1.00 Rod	
-	36754-03	A/R	Kit, Cylinder Seal	
24	36755	1	Ball Joint, Steering Cylinder End	
25	100-12-10-80-5	1	CSHH, 3/4-10 x 5.00, GR5	
26	300-12	1	Washer, Flat, SAE, 3/4	
27	16935	1	Sleeve, Steering Cylinder Mount	
28	301-12	1	Washer, Flat USS, 3/4	
29	204-12-10-5	1	Nut, Lock, Stover, 3/4-10, GR5	
30	20727	1	Tube, Tie Rod	
31	21113	1	Nut, Jam, 1-16, RH	
32	36756	1	Ball Joint - Right	
33	21114	1	Nut, Jam, 1-16, LH	
34	36757	1	Ball Joint - Left	
35	71054	2	Bushing, Oilite, 1.252 ID x 1.565 ID x 1.50 LG	
36	1016667	1	Shaft, Axle Pivot	
37	100-8-13-24-5	1	CSHH, 1/2-13 x 1.50, GR5	
38	300-8	1	Washer, Flat, SAE, 1/2	
39	204-8-13-5	1	Nut, Lock, Stover, 1/2, GR5	

AXLE - FRONT STEERING (CONT.)

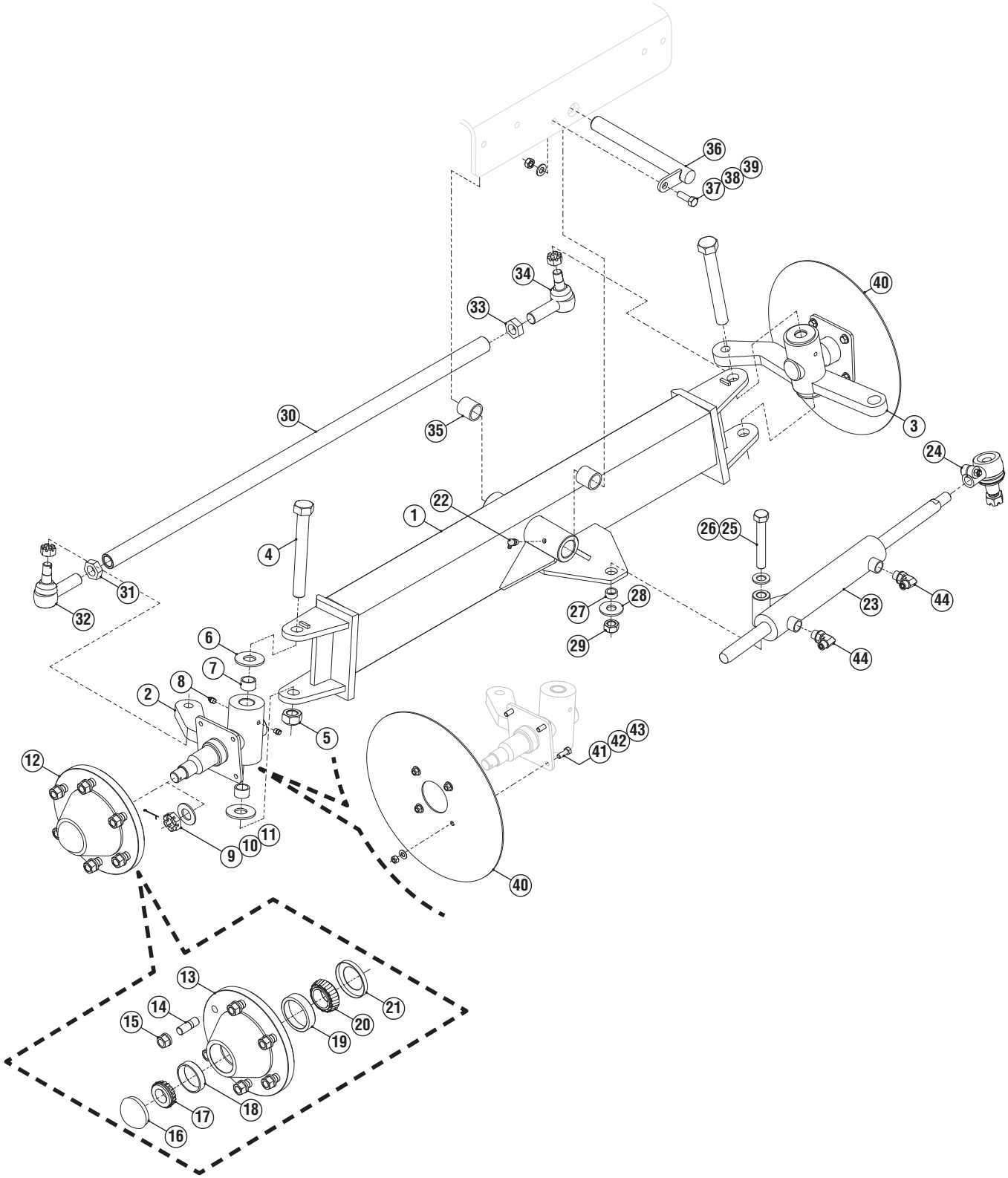


Figure 7-6. Axle - Front Steering (Cont.)

Axle - Front Steering (Cont.)

Item No.	Part Number	Qty.	Description	Remarks
40	1010979	2	Guard, Dust	
41	100-6-16-16-5F	8	CSHH, 3/8-16 x 1.00, GR5, FT	
42	300-6	8	Washer, Flat, SAE, 3/8	
43	204-6-16-5	8	Nut, Lock, Stover, 3/8-16, GR5	
44	FS6801-06-08-NWO-FG	2	Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	

BRUSH ASSEMBLY, PIVOT

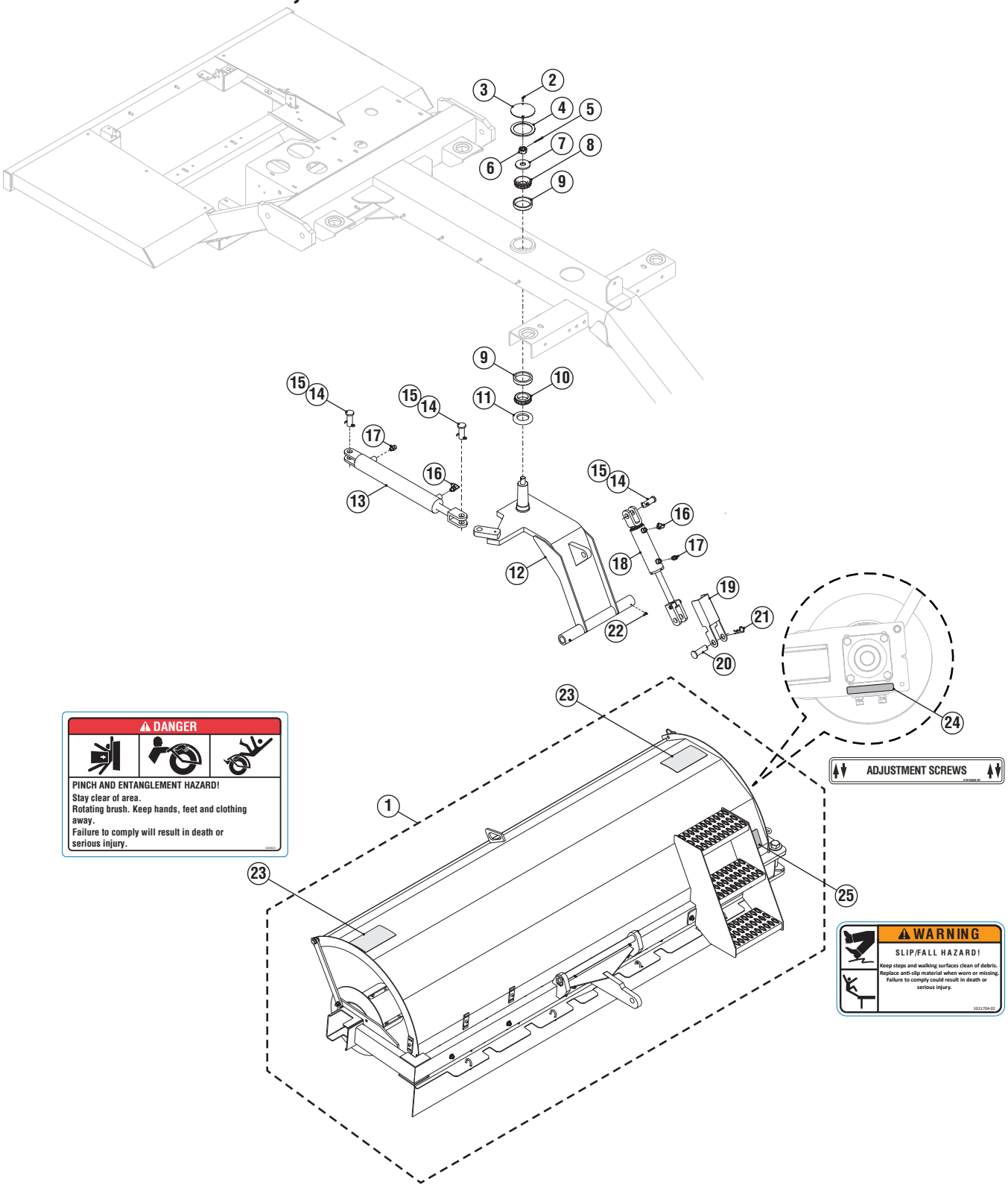


Figure 7-7. Brush Assembly, Pivot

Brush Assembly, Pivot

Item No.	Part Number	Qty.	Description	Remarks
1	REF		Brush Frame Assembly and Drive Components	See Figure 7-8
2	111-#10-24-12F	2	PHMS, Cross, #10-24 x .75, FT	
3	20891	1	Spindle Cap	
4	21342	1	Spindle Cap Spacer	
5	80331	1	Cotter Pin, 1/85" x 1-1/4"	
6	206-14-14-5	1	Slotted Hex Nut, 7/8-14, GR5	
7	20663	1	Spindle Washer	
8	36827	1	Tapered Roller Bearing, 1.875 Bore	
9	36829	2	Tapered Roller Bearing Cup, 3.50 OD x .650	
10	36828	1	Bearing Cone	
11	36830	1	Seal, 3.628 OD x 2.50 ID x .468	
12	1015652SRV	1	Brush Pivot Weldment	
ASSY	1020378	1	Assembly, Pivot Cylinder w/ORFS Adapters	Includes 13-17
13	1020385	1	Hydraulic Cylinder, 2.50 x 16.00 x 1.25	
-	981860-01	A/R	Seal Kit for 1020385	Repair Part
14	1011476	2	Clevis Pin, 1.00 x 2.50	Included with Item 13 & 18
15	80338	2	Cotter Pin, .188 x 2.00	Included with Item 13 & 18
16	FS6801-06-06-NWO-FG	2	Adapter, Elbow, 90°, -6 ORFS--6 O-Ring	
17	FS6400-06-06-O X 078	2	Adapter, Straight, -6 ORFS/-6 O-Ring, .078 Orifice	
16	31884A	1	Hydraulic Cylinder, 2.50 x 6.00 x 1.125	
17	1015845	1	Brush Pivot Pin Weldment	
ASSY	1020377	1	Assembly, Lift Cylinder w/ORFS Adapters	Includes 14-21
18	1020384	1	Hydraulic Cylinder, 2.50 x 16.00 x 1.125	
-	811374-01	A/R	Seal Kit for 1020384	Repair Part
19	1022642	1	Lift Cylinder Stop	
20	240030	1	Clevis Pin, 1.00 x 3.25	
21	95458329	1	Cotter Pin	
22	140610	1	Straight Lube Fitting, 1/4-18	
23	1008622	2	Decal, Pinch/Entanglement	
24	1015036-07	1	Decal, Adjustment Screws	
25	1021704-02	1	Decal, Warning, Slip/Fall	
REF	38806	1	Poly Tube Brush, 32 x 90	Standard - Not Shown
REF	1005054	1	Wafer Poly Brush Group	Optional - Not Shown
-	28766	1	Brush Core Assembly	Optional - Not Shown
-	6375P	54	Poly Wafer w/Spacer, 10 x 32	Optional - Not Shown
REF	1005055	1	Wafer Poly-Steel Brush Group	Optional - Not Shown
-	28766	1	Brush Core Assembly	Optional - Not Shown
-	6375P	27	Poly Wafer w/Spacer, 10 x 32	Optional - Not Shown
-	6375S	27	Steel Wafer w/Spacer, 10 x 32	Optional - Not Shown

BRUSH ASSEMBLY, DRIVE & IDLER

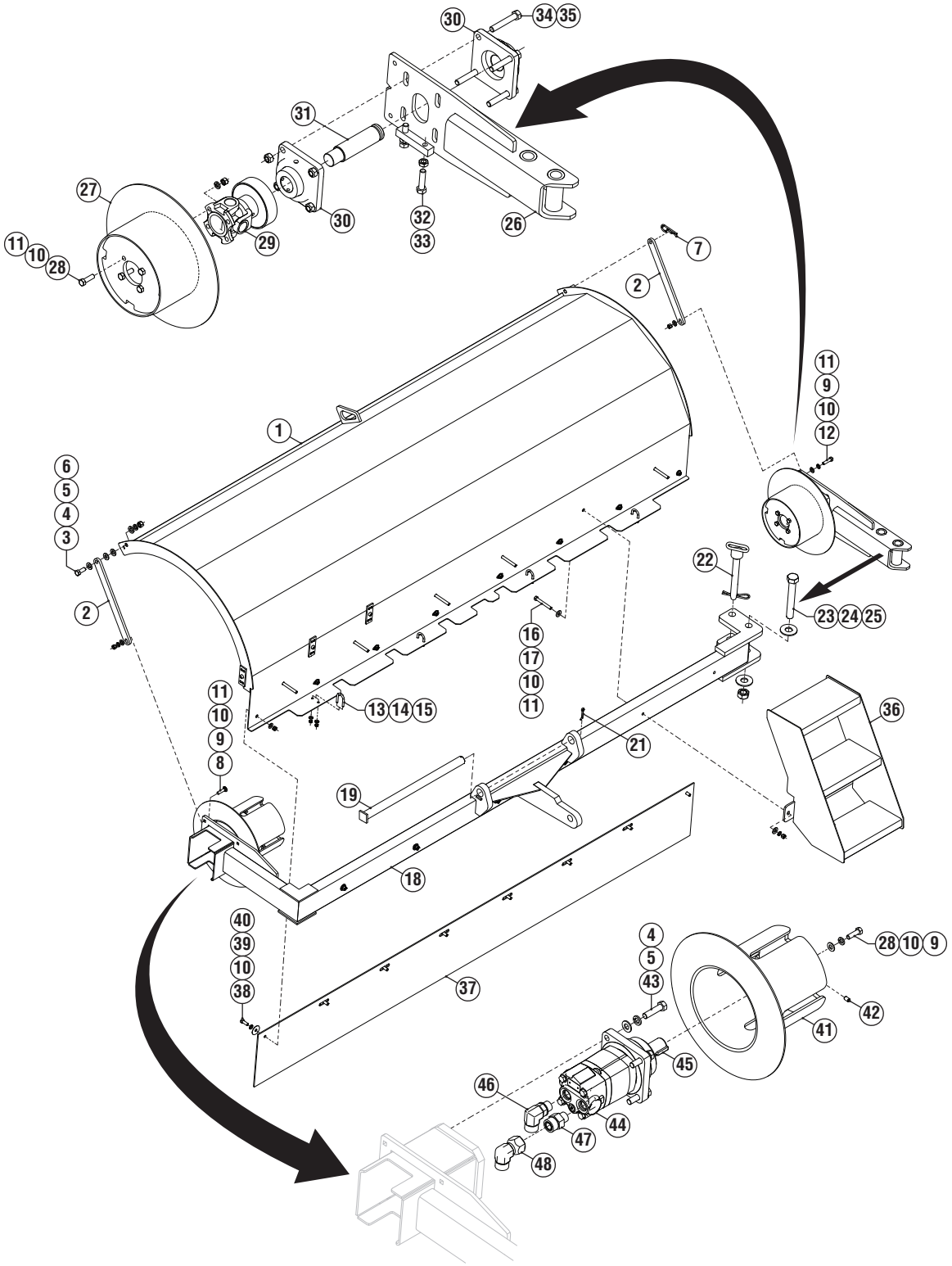


Figure 7-8. Brush Assembly, Drive & Idler

Brush Assembly, Drive & Idler

Item No.	Part Number	Qty.	Description	Remarks
GRP	1015623		Group - Brush Assembly	Includes 1-43
1	1021079	1	Brush Cover	
2	1021082	2	Strap, Brush Cover Support	
3	100-8-13-20-5F	1	CSHH, 1/2-13 x 1.25, GR5, FT	
4	300-8	8	Washer, Flat, SAE, 1/2	
5	302-8	5	Washer, Lock, 1/2	
6	200-8-13-5	1	Nut, Hex, 1/2-13, GR5	
7	36544	1	Cotter Pin, .177	
8	106-6-16-24-5F	1	CRG Bolt, 3/8-16 x 1.50, GR5	
9	300-6	11	Washer, Flat, SAE, 3/8	
10	302-6	21	Washer, Lock, 3/8	
11	200-6-16-5	16	Nut, Hex, 3/8-16, GR5	
12	100-6-16-24-5	1	CSHH, 3/8-16 x 1.50, GR5	
13	35077	4	U-Bolt, 1/4-20, 1.00 IW x 1.75 IL	
14	300-4	8	Washer, Flat, SAE, 1/4	
15	204-4-20-5	8	Nut, Lock, Stover, 1/4-20, GR5	
16	100-6-16-48-5	6	CSHH, 3/8-16 x 3.00, GR5	
17	301-6	12	Washer, Flat, USS, 3/8	
18	1017474	1	Brush Frame	
19	1015845	1	Brush Pivot Pin	
20			**NOT USED**	
21	80336	1	Cotter Pin, .188 x 2.00	
22	72840	1	Hitch Pin, ø.944 x 7.50 w/Clip	
23	100-16-8-112-5	1	CSHH, 1-8 x 7.00, GR5	
24	301-16	2	Washer, Flat, USS, 1	
25	204-1-8-5	1	Nut, Lock, Stover, 1-8, GR5	
26	1017477	1	Brush Swing Arm	
27	28828	1	Idler Hub	
28	100-6-16-20-5F	5	CSHH, 3/8-16 x 1.25, GR5, FT	
29	39138	1	Brush Drive Hub Assembly	
30	312030	2	Bearing Flange, 4 Bolt, 1.50 w/Set Screw	
31	1005200-115	1	Idle Shaft Pin	
32	100-8-13-32-5	2	CSHH, 1/2-13 x 2.00, GR5	
33	202-8-13-5	2	Nut, Hex, Jam, 1/2-13, GR5	
34	100-8-13-48-5	4	CSHH, 1/2-13 x 3.00, GR5	
35	204-8-13-5	4	Nut, Lock, Stover, 1/2-13, GR5	
36	1021084	1	Staircase	
37	38989	1	Rubber Apron	
38	100-6-16-16-5F	8	CSHH, 3/8-16 x 1.00, GR5, FT	

BRUSH ASSEMBLY, DRIVE AND IDLER (CONT.)

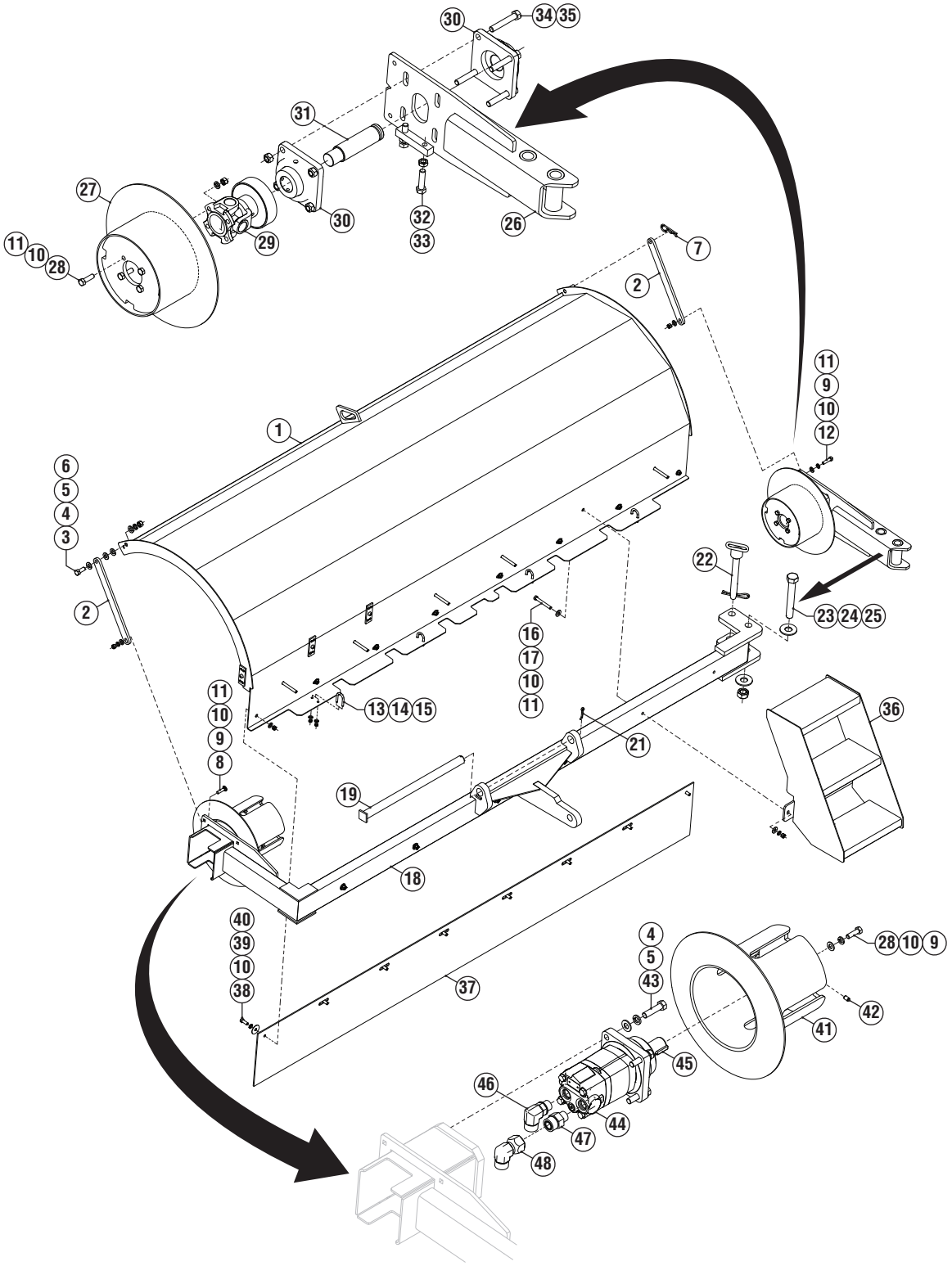


Figure 7-8 Brush Assembly, Drive and Idler (Cont.)

Brush Assembly, Drive and Idler (Cont.)

Item No.	Part Number	Qty.	Description	Remarks
39	308-6-24	8	Washer, Flat, Fender, 3/8 x 1.50	
40	204-6-16-5	8	Nut, Lock, Stover, 3/8-16, GR5	
41	28570	1	Drive Hub	
42	113-5-18-8	1	Set Screw, HKST, Cup, 5/16-1/ x .50	
43	100-8-20-32-5	4	CSHH, 1/2-20 x 2.00, GR5	
ASSY	1020376	1	Assembly, Brush Motor w/ORFS Adapters	Includes 44-48
44	1011086	1	Motor, Hydraulic Brush Drive	
45	REF	1	Square Key, 5/16 x 1.25	Included with 44
46	FS6801-12-10-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-10 O-Ring	
47	FS6400-12-10-O	1	Adapter, Straight, -12 ORFS/-10 O-Ring	
48	FS6500-12-12-FG	1	Adapter, Elbow, 90°, Swivel, -12 ORFS/-12 ORFS	

TANKS - FUEL, 30 GALLON

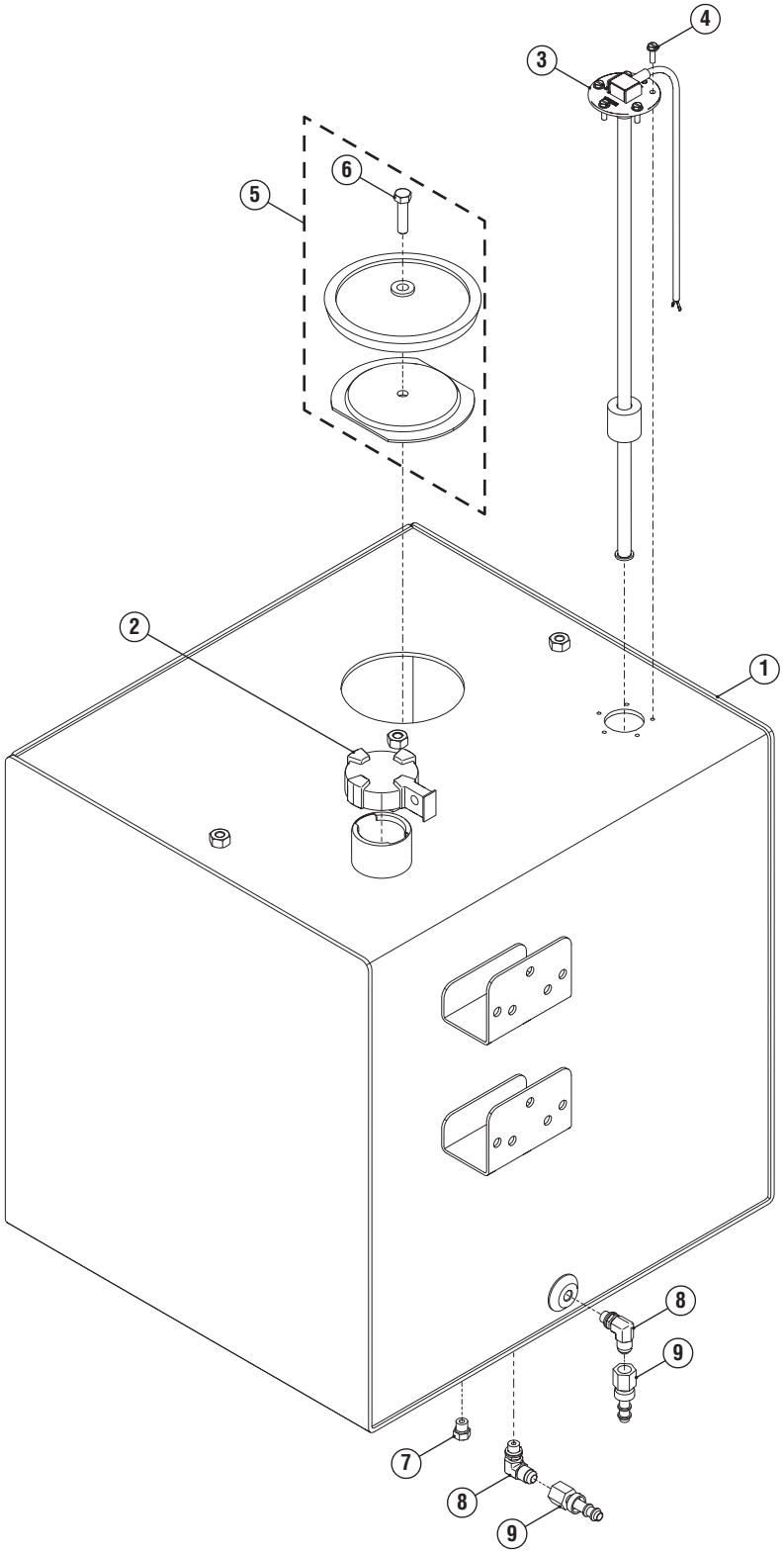


Figure 7-9. Tanks - Fuel, 30 Gallon

Tanks - Fuel, 30 Gallon

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1021896	1	Assembly, Fuel Tank, 30 GAL	
1	1021897	1	Fuel Tank, 30 GAL	
2	140030FL	1	Cap, Fuel, Lockable	
3	1021910	1	Sending Unit, Fuel, 21"	
4	122-#10-32-10F	5	PHMS, Cross, #10-32 x .625, FT	
5	1006804	1	Access Port	
6	100-6-16-24-5	1	CSHH, 3/8-16 x 1.50, GR5	
7	6408-4-O	1	Adapter, Plug, O-Ring, -4	
8	6804-6-4 NWO	2	Adapter, Elbow, 90°, -6 JIC/-4 O-Ring	
9	6LOC-6RFJX	2	Fitting, Push Lok, -6 LOC/-6 RFJX	

TANKS - HYDRAULIC, 25 GALLON

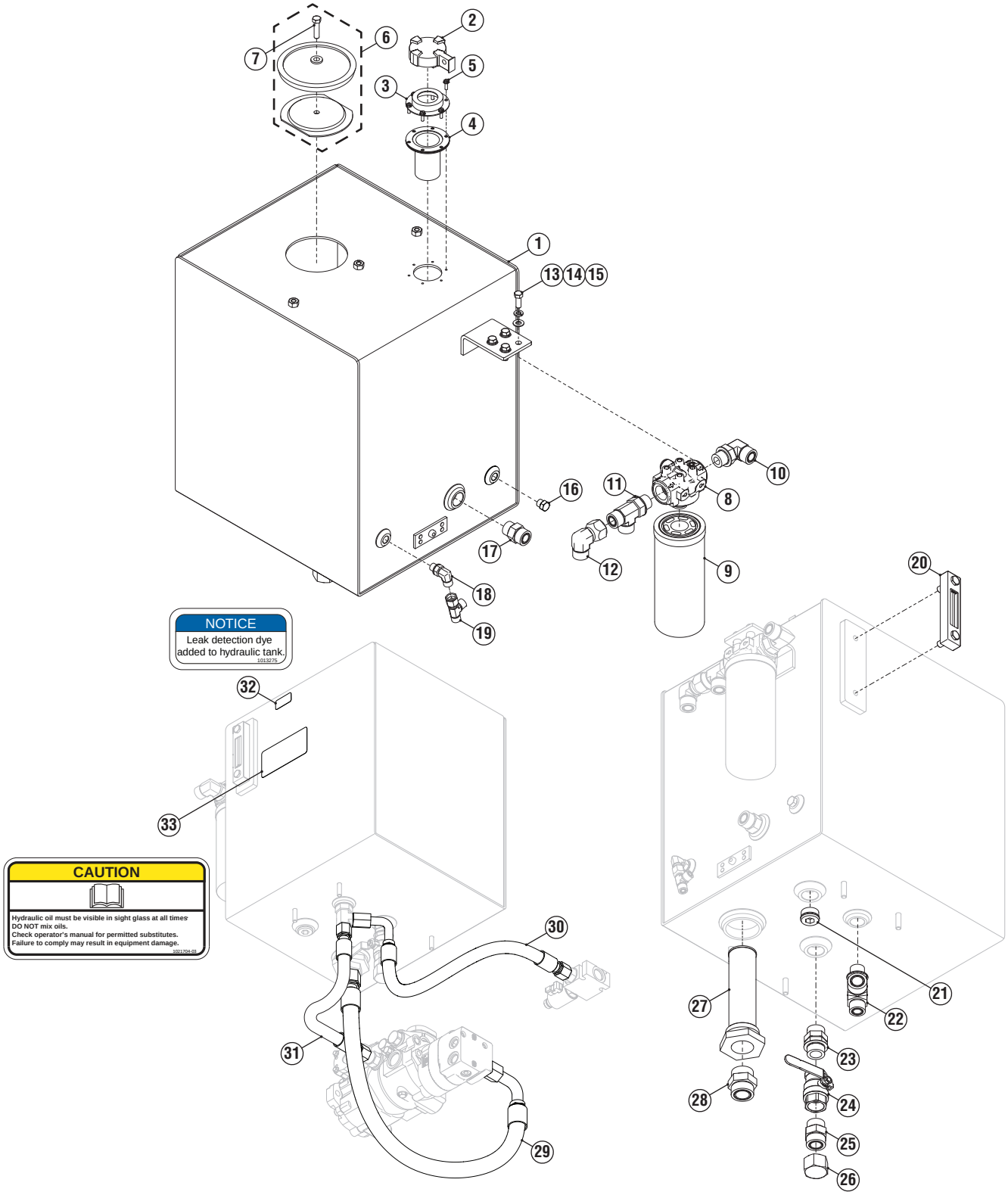


Figure 7-10. Tanks - Hydraulic, 25 Gallon

Tanks - Hydraulic, 25 Gallon

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1021145	1	Assembly, Hydraulic Tank w/ORFS Adapters	Includes Items 1-27
1	1023238	1	Hydraulic Tank, 25 GAL	
2	140030HL	1	Cap, Hydraulic, Lockable	
3	140030FN	1	Filler Neck	
4	140030GK	1	Kit, Strainer & Gasket	
5	122-#10-32-10F	6	PHMS, Cross, #10-32 x .625, FT	
6	1006804	1	Access Port	
7	100-6-16-24-5	1	CSHH, 3/8-16 x 1.50, GR5	Included with Item 6
ASSY	1020370	1	Assembly, Return Filter w/Adapters	Includes items 8-12
8	1017402	1	Filter Head, Hydraulic HMK04	
9	290030	1	Filter Element, Hydraulic, 9 Micron	
10	FS6801-12-12-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	
11	FS6804-12-12-12-NWO-FG	1	Adapter, Run Tee, -12 ORFS/-12 O-Ring/-12 ORFS	
12	FS6500-12-12-FG	1	Adapter, Elbow, 90°, Swivel, -12 ORFS/-12 ORFS	
13	100-6-16-16-5F	4	CSHH, 3/8-16 x 1.00, GR5, FT	
14	302-6	4	Washer, Lock, 3/8	
15	300-6	4	Washer, Flat, SAE, 3/8	
16	6408-6-O	1	Adapter, Plug, -6 O-Ring	
17	FS6400-12-12-O	1	Adapter, Straight, -12 ORFS/-12 O-Ring	
18	FS6801-06-06-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
19	FS6602-06-06-06-FG	1	Adapter, Run Tee, Swivel, -6 ORFS	
20	500070	1	Gauge, Sight Lever/Temp, Hydraulic Oil	
21	1011971	1	Plug, Magnetic Drain, -16 SAE ORB	
22	FS6804-12-12-16-NWO-FG	1	Adapter, Run Tee, -12 ORFs/-12 O-Ring/-16 ORFS	
23	6407-16-16 NWO	1	Adapter, Union, -16 O-Ring/-16 O-Ring	
24	1020587	1	Valve, Ball, -16 ORB/-16 ORB	
25	FS6400-16-16-O	1	Adapter, Straight, -16 ORFS/-16 O-Ring	
26	FS0304-C-16	1	Adapter, Cap, ORFS, -16	
27	1014037	1	Strainer, Suction, 100 Mesh, O-Ring Seal	
28	FS6400-20-20-O	1	Adapter, Straight, -20 ORFS/-20 O-Ring	
29	F387TCJCJ9121212-29	1	Hose Assembly, Brush Speed Valve 2 to Tank	Tag# 141
30	K811HTJCJ9161616-38	1	Hose Assembly, Tank to Pump	Tag# 119
31	F387TCJCJC121212-19	1	Hose Assembly, Pump Case Outlet to Tank	Tag# 6
32	1013275	1	Decal, Notice, Hydraulic Dye	
33	1021704-03	1	Decal, Caution, Hydraulic Oil	

ENGINE COVERS

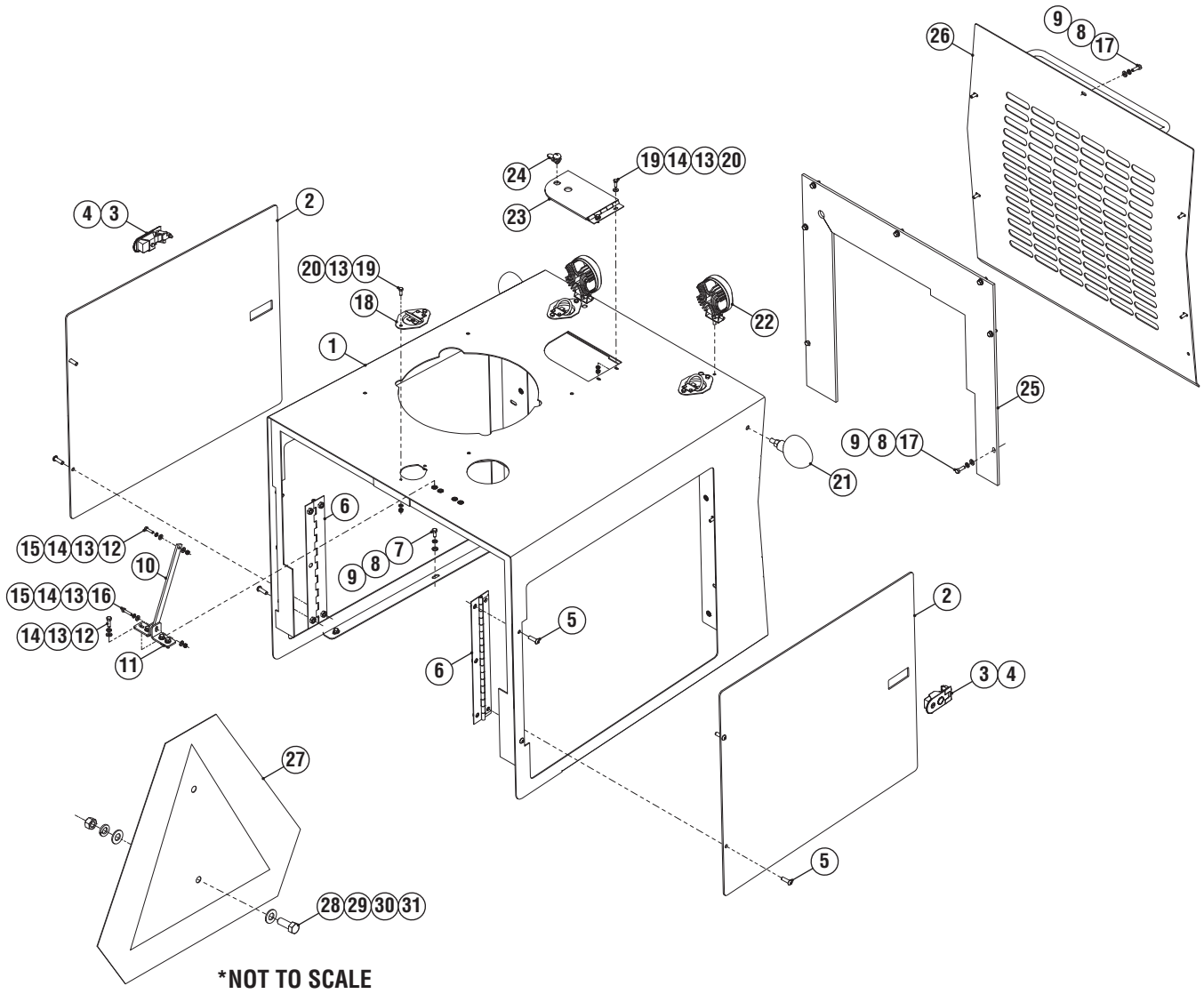


Figure 7-11. Engine Covers

Engine Covers

Item No.	Part Number	Qty.	Description	Remarks
GRP	1017187		Group - Engine Cover	
1	1021577	1	Engine Cover	
2	1019872	2	Door, Engine Compartment	
3	980460	2	Lever Latch	
4	35560	2	Key	Not Shown, Included with Item 3
5	117-6-16-20	4	CSSBH, 3/8-16 x 1.25	
6	1020511	2	Hinge, 18"	
7	100-5-18-12-5F	6	CSHH, 5/16-18 x .75, GR5, FT	
8	302-5	A/R	Washer, Lock, 5/16	
9	300-5	A/R	Washer, Flat, SAE, 5/16	
10	28818	1	Support Bar	
11	13857	2	Intake Support Bracket	
12	100-4-20-16-5	1	CSHH, 1/4-20 x 16, GR5	
13	302-4	10	Washer, Lock, 1/4	
14	300-4	6	Washer, Flat, SAE, 1/4	
15	217-4-20	2	Nut, Lock, Flexloc, 1/4-20, Full, LT	
16	100-4-20-20-5F	1	CSHH, 1/4-20 x 1.25, GR5, FT	
17	100-5-18-16-5	15	CSHH, 5/16-18 x 1.00, GR5	
18	1001057	3	Recessed Lifting Ring	
19	100-4-20-12-5F	8	CSHH, 1/4-20 x .75, GR5, FT	
20	200-4-20-5	6	Nut, Hex, 1/4-20, GR5	
21	1005940	2	Red Tail Light/Signal	
22	1012563	2	Circular LED Light	
23	1010961	1	Radiator Access Door w/Hinge	
24	25812	1	Lock w/Offset Cam	
25	1015819	1	Baffle, Radiator, Rubber	
26	1015817	1	Radiator Grill	
GRP	17898		Group - SMV Sign	Includes Items 27-31
27	35952	1	SMV Sign	
28	100-6-16-16-5F	1	CSHH, 3/8-16 x 1.00, GR5, FT	
29	300-6	2	Washer, Flat, SAE, 3/8	
30	302-6	1	Washer, Lock, 3/8	
31	200-6-16-5	1	Nut, Hex, 3/8-16, GR5	

ENGINE - RADIATOR

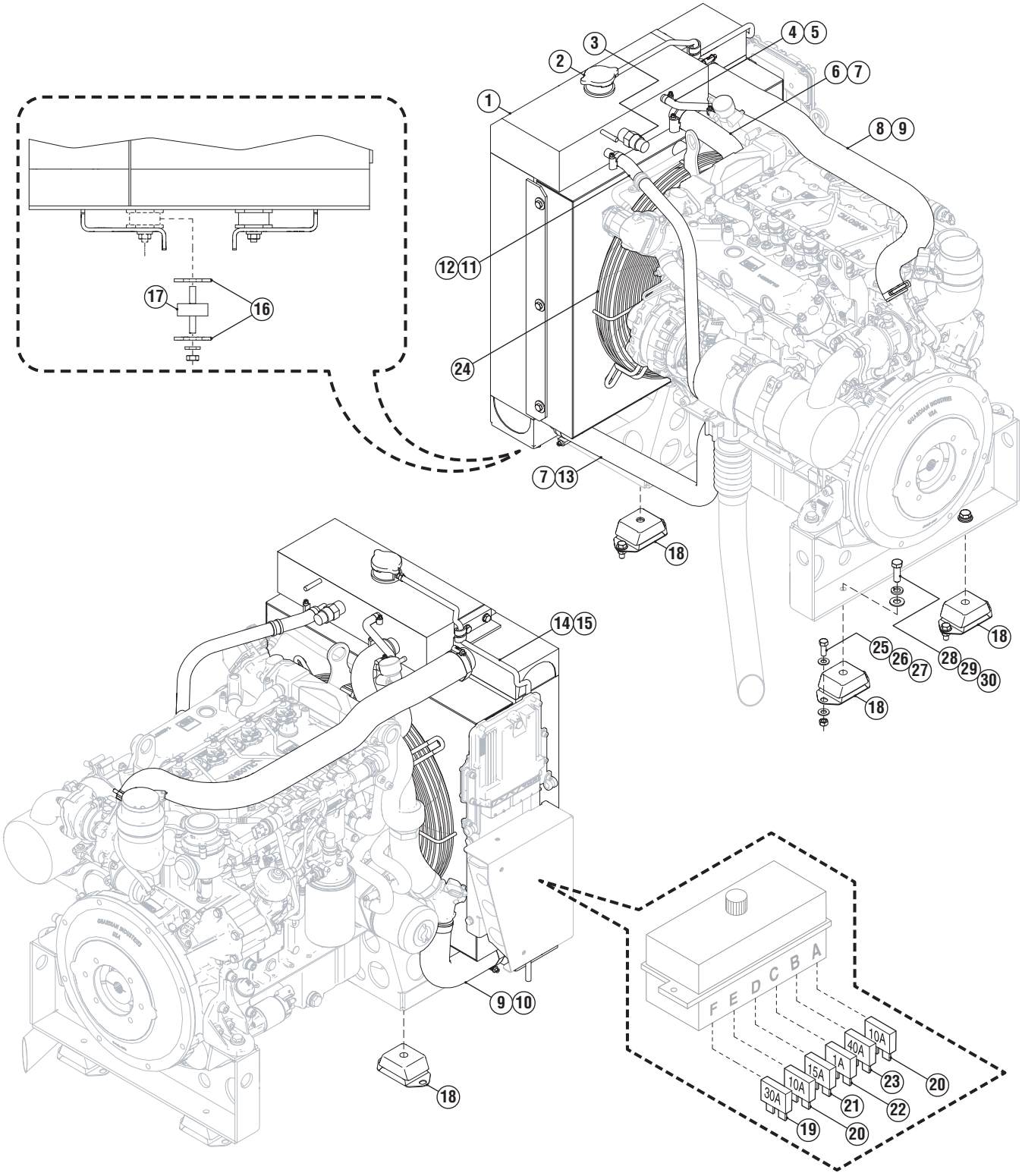


Figure 7-12. Engine - Radiator

Engine - Radiator

Item No.	Part Number	Qty.	Description	Remarks
1	1015794-20	1	Air & Water Radiator	
2	1015794-54	1	Radiator Cap	
3	1015794-53	1	Coolant Sight Glass	
4	1015794-13	1	Coolant Hose	
5	1015794-16	2	Hose Clamp, ø10-16mm	
6	1015794-10	1	Preformed Hose	
7	1015794-42	4	Hose Clamp, ø25-40mm	
8	1015794-44	1	Preformed Hose	
9	1015794-47	4	Hose Clamp, ø40-60mm	
10	1015794-45	1	Preformed Hose	
11	1015794-09	1	Coolant Hose	
12	1015794-41	2	Hose Clamp, ø12-22mm	
13	1015794-11	1	Preformed Hose	
14	1015794-12	1	Overflow Hose	
15	1015794-39	1	Hose Clamp, ø12.8	
16	1015794-38	4	Radiator Isolator Washer	
17	1015794-37	2	Radiator Mount Isolator	
18	1015794-36	4	Engine Mount Isolator	
19	36218	1	Fuse, 30A, ATC	
20	36340	2	Fuse, 10A, ATC	
21	36341	1	Fuse, 15A, ATC	
22	1015794-55	1	Fuse, 1A, ATC	
23	115911	1	Fuse, 40A, ATC	
24	TBD	1	Shroud, Fan	
25	100-6-16-16-5F	8	CSHH, 3/8-16 x 1.00. GR5, FT	
26	300-6	16	Washer, Flat, SAE, 3/8	
27	204-6-16-5	8	Nut, Lock, Stover, 3/8-16, GR5	
28	100-M12-1.75-35-8.8F	4	CSHH, M12x1.75 x 35, C8.8, FT	
29	302-8	4	Washer, Lock, 1/2	
30	300-8	4	Washer, Flat, SAE, 1/2	

ENGINE - INTAKE AND EXHAUST

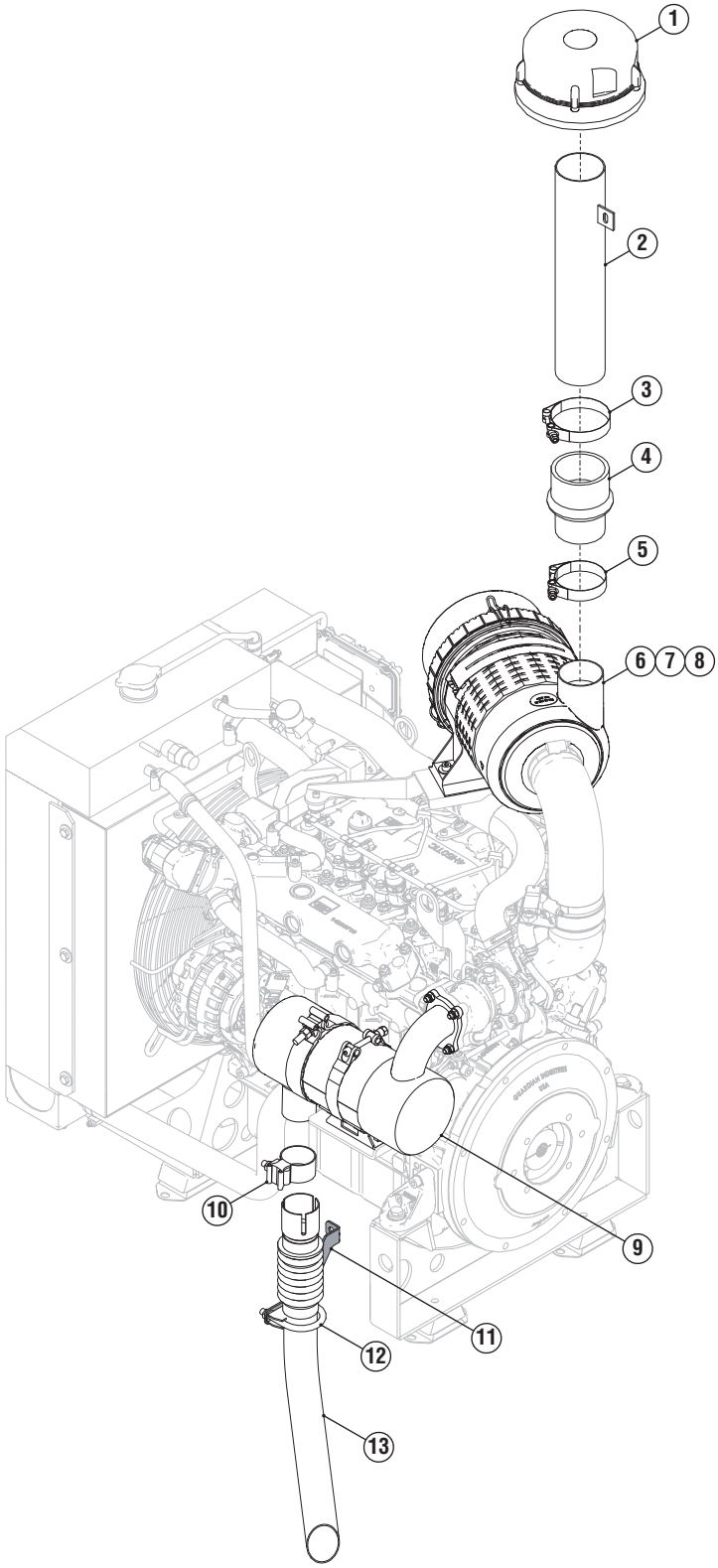


Figure 7-13. Engine - Intake and Exhaust

Engine - Intake and Exhaust

Item No.	Part Number	Qty.	Description	Remarks
1	1015872	1	Pre-Cleaner, 3.00	
2	1015788	1	Air Intake Pipe Weldment	
3	171090	1	T-Bolt Clamp, 3.31-3.59 Range	
4	34030	1	Reducer Hose, 3.00 ID to 2.50 ID	
5	36045	1	T-Bolt Clamp, 2.81-3.09 Range	
6	1015794-58	1	Air Filter Assembly	Includes Items 7 - 8
7	1015794-03	1	Main Air Filter Element	
8	1015794-04	1	Secondary Air Filter Element	
9	1015794-14	1	Catalyzer Assembly	
10	1015876	1	Accuseal Clamp, 2.00”	
11	1015790	1	Exhaust Pipe Support	
12	71172	1	Muffler Clamp, 2.00”	
13	1015867	1	Exhaust Pipe	

ENGINE - COMPONENTS

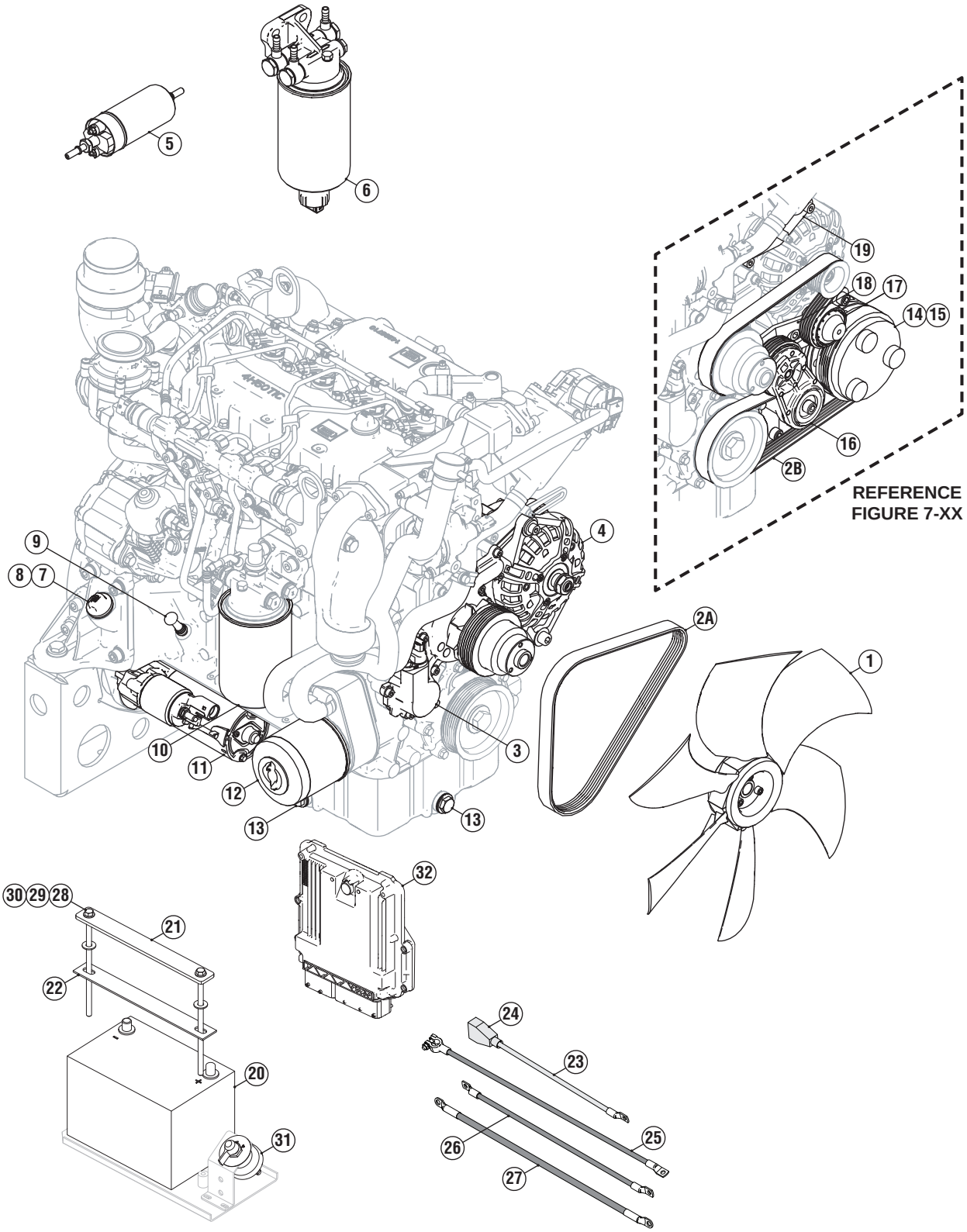


Figure 7-14. Engine - Components

Engine - Components

Item No.	Part Number	Qty.	Description	Remarks
GRP	1021576	1	Hatz Engine Assembly, T4F, No AC/Heat	
GRP	1021575	1	Hatz Engine Assembly, T4F, with AC/Heat	Option, Includes Items 2b, 14-19
1	1015794-22	1	Fan	
2a	1015794-07	1	Poly-V Belt - No AC	
2b	1015794-08	1	Poly-V Belt - With AC	
3	1015794-18	1	Water Pump w/Belt Pulley	
REF	1015794-19	1	Water Pump Gasket	Not Shown
REF	1015794-23	1	Phase Sensor	Not Shown
4	1015794-16	1	14V/110A Alternator	
5	1015794-17	1	Electric Fuel Pump	
6	1015794-05	1	Pre-Fuel Filter	
7	1015794-24	1	Oil Filler Cap	
8	1015794-25	1	Oil Filler Cap Seal	
9	1015794-27	1	Dipstick	
10	1015794-06	1	Fuel Filter	
11	1015794-15	1	12V Starter, 2.2KW	
12	1015794-01	1	Oil Filter	
13	1015794-26	2	Magnetic Drain Plug	
14	1015794-30	1	AC Compressor	Option, See Fig. 7-30
15	1015794-33	1	AC Main Mounting Bracket	Option, See Fig. 7-30, Not Shown
16	1015794-31	1	AC Belt Tensioner	Option, See Fig. 7-30
17	1015794-32	1	AC Belt Idler	Option, See Fig. 7-30
18	1015794-34	1	AC Support Bracket	Option, See Fig. 7-30
19	1015794-35	1	AC Support Strap	Option, See Fig. 7-30
20	PURCHASE LOCALLY	1	12V Battery, 770 CCA, 950 CA, Group 34	
21	150042SRV	1	Battery Hold Down Strap	
22	410050	1	Battery Hold Down Pad	
23	800072	1	Cable, Battery (+), 16", Eye/Post	
24	70437	1	Red Battery Boot	
25	800076	1	Cable, Battery (-), 26", Eye/Post	
26	1012037	1	Cable, Battery (+), 21", Eye/Eye	
27	1016367	1	Cable, Battery (+), 24", Eye/Eye	
28	100-6-16-144-5	2	CSHH, 3/8-16 x 9.00, GR5	
29	310-6	2	Washer, Hardened, SAE, 3/8	
30	301-6	2	Washer, Flat, USS, 3/8	
31	109740	1	Switch, Master Disconnect, Battery	
32	1015794-56	1	ECU	
REF	1015794-28	1	Fuse Holder	Not Shown
REF	1015794-29	1	Oil Seal - Intermediate Housing	Not Shown
REF	1015794-57	1	Hatz 4H50TIC Engine Manual	Not Shown

ENGINE - HYDRAULIC PUMPS

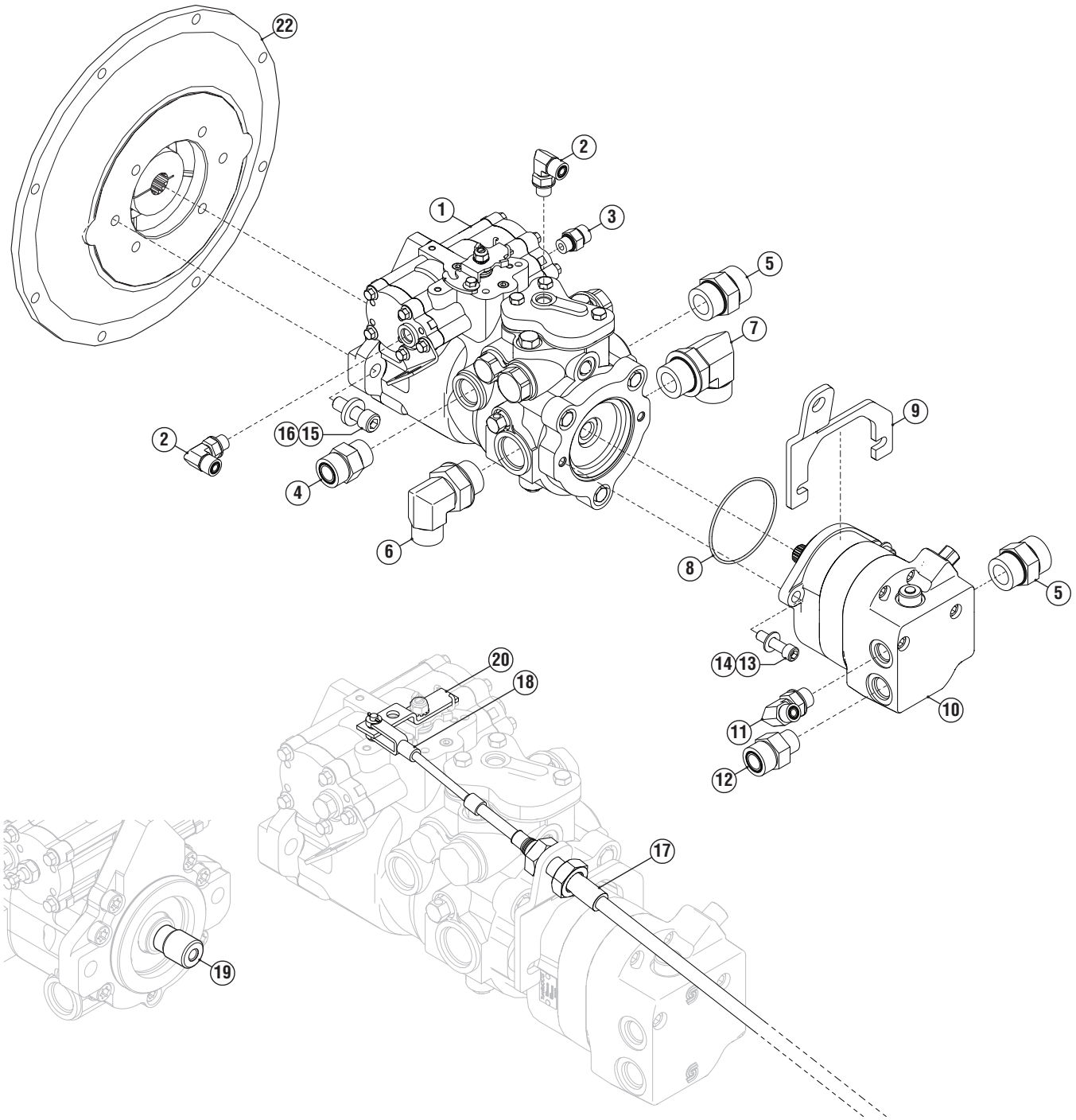


Figure 7-15. Engine - Hydraulic Pumps

Engine - Hydraulic Pumps

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1020374	1	Assembly, Pumps w/Adapters	Includes Items 1-14, 19-21
1	1010262	1	Pump, Hydraulic, Piston, 2.8 CIR	
2	FS6801-06-06-NWO-FG	2	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
3	FS6400-06-06-O	1	Adapter, Straight, -6 ORFS/-6 O-Ring	
4	FS6400-12-12-O	1	Adapter, Straight, -12 ORFS/-12 O-Ring	
5	FS6400-16-16-O	2	Adapter, Straight, -16 ORFS/-16 O-Ring	
6	FS6801-12-16-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-16 O-Ring	
7	FS6801-16-16-NWO-FG	1	Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	
8	36808	1	O-Ring, 3.237 ID x .103, SAE, 152	
9	1009333	1	Bracket, Control Cable	
10	1005574SRV	1	Pump, Hydraulic, Gear, 1.94 CIR	
11	FS6801-06-08-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	
12	FS6400-12-10-O	1	Adapter, Straight, -12 ORFS/-10 O-Ring	
13	102-6-16-24-F	2	CSSH, 3/8-16 x 1.50, FT	
14	300-6	2	Washer, Flat, SAE, 3/8	
15	102-8-13-24-F	2	CSSH, 1/2-13 x 1.50, FT	
16	312-8	2	Washer, Hardened, SAE, Thick, 1/2	
17	1016290	1	Cable, Push, Pull, 1/4-28 w/Threaded Caps	
18	989817	1	Clevis, 1/4-28 UNF x .375 ID	
19	1010262-01	1	Shaft, Drive	
20	1010262-02	1	Lever, Throttle Control	
21	1010262-03	1	Switch, Neutral Start	Not Shown
22	1015794-52	1	Assembly, Pump Drive Plate	

HYDRAULIC SYSTEM (1 OF 5)

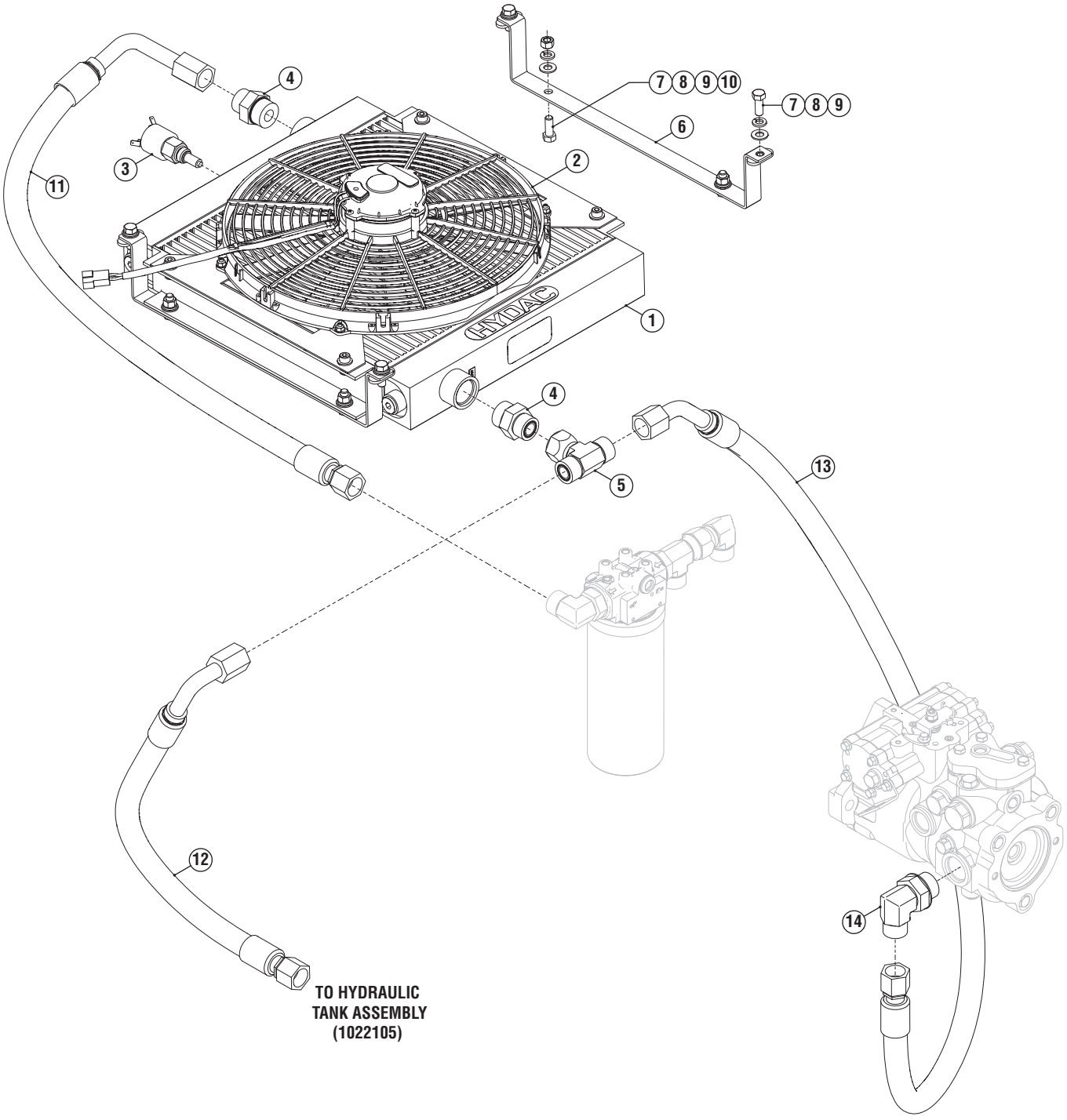


Figure 7-16. Hydraulic System (1 of 5)

Hydraulic System (1 of 5)

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1020497	1	Assembly, Oil Cooler w/Adapters	Includes 1-5
1	1015757	1	Cooler, Hydraulic Oil, 45,000 BTU	Includes 2-3
2	1015757-02	1	Fan (Pusher)	
3	1015163	1	Switch, Temperature, -8 MORB, 115°F	
4	FS6400-12-16-O	2	Adapter, Straight, -12 ORFS/-16 O-Ring	
5	FS6600-12-12-12-FG	1	Adapter, Branch Tee, Swivel, -12 ORFS	
6	1015822	2	Oil Cooler Mount	
7	100-6-16-16-5F	8	CSHH, 3/8-16 x 1.00, GR5, FT	
8	302-6	8	Washer, Lock, 3/8	
9	300-6	8	Washer, Flat, SAE,3/8	
10	200-6-16-5	4	Nut, Hex, 3/8-16, GR5	
11	F387TC06L9121212-44-AS-B-19-40.75	1	Hose Assembly - Filter To Cooler	Tag# 9
12	F387TCJCJ9121212-27	1	Hose Assembly - Cooler To Tank	Tag# 10
13	F387TCJCJ9121212-71-AS-B-19-67	1	Hose Assembly - Cooler To Charge Pump Inlet	Tag# 7
14	FS6801-12-16-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-16 O-Ring	

HYDRAULIC SYSTEM (2 OF 5)

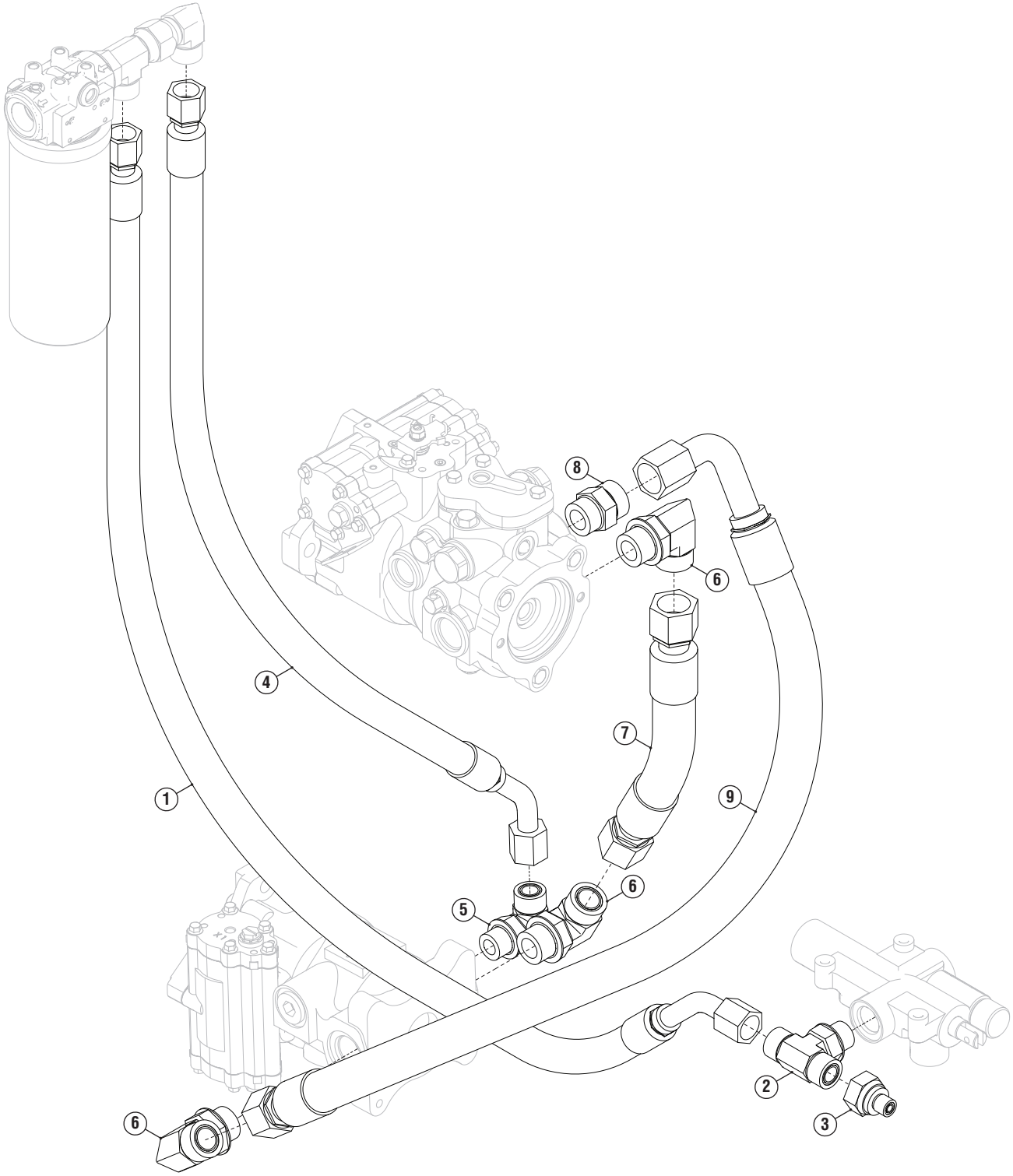


Figure 7-17. Hydrailic System (2 of 5)

Hydraulic System (2 of 5)

Item No.	Part Number	Qty.	Description	Remarks
1	F387TC0637121212-60-AS-B-19-25	1	Hose Assembly, Motor Valve to Filter	Tag# 134
2	FS6803-12-12-12-NWO-FG	1	Adapter, Branch Tee, -12 ORFS/-12 ORFS/-12 O-Ring	
3	FS2406-12-06	1	Adapter, Reducer, Straight, -12 FFORX/-6 ORFS	
4	F387TC0637121212-52	1	Hose Assembly, Motor Case Drain to Filter	Tag# 8
5	FS6801-12-12-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	
6	FS6801-16-16-NWO-FG	3	Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	
7	F797TC0606161616-19	1	Hose Assembly, Pump A to Motor !	Tag# 1
8	FS6400-16-16-O	1	Adapter, Straight, -16 ORFS/-16 O-Ring	
9	F797TC0639161616-30	1	Hose Assembly, Pump B to Motor B	Tag# 2

HYDRAULIC SYSTEM (3 OF 5)

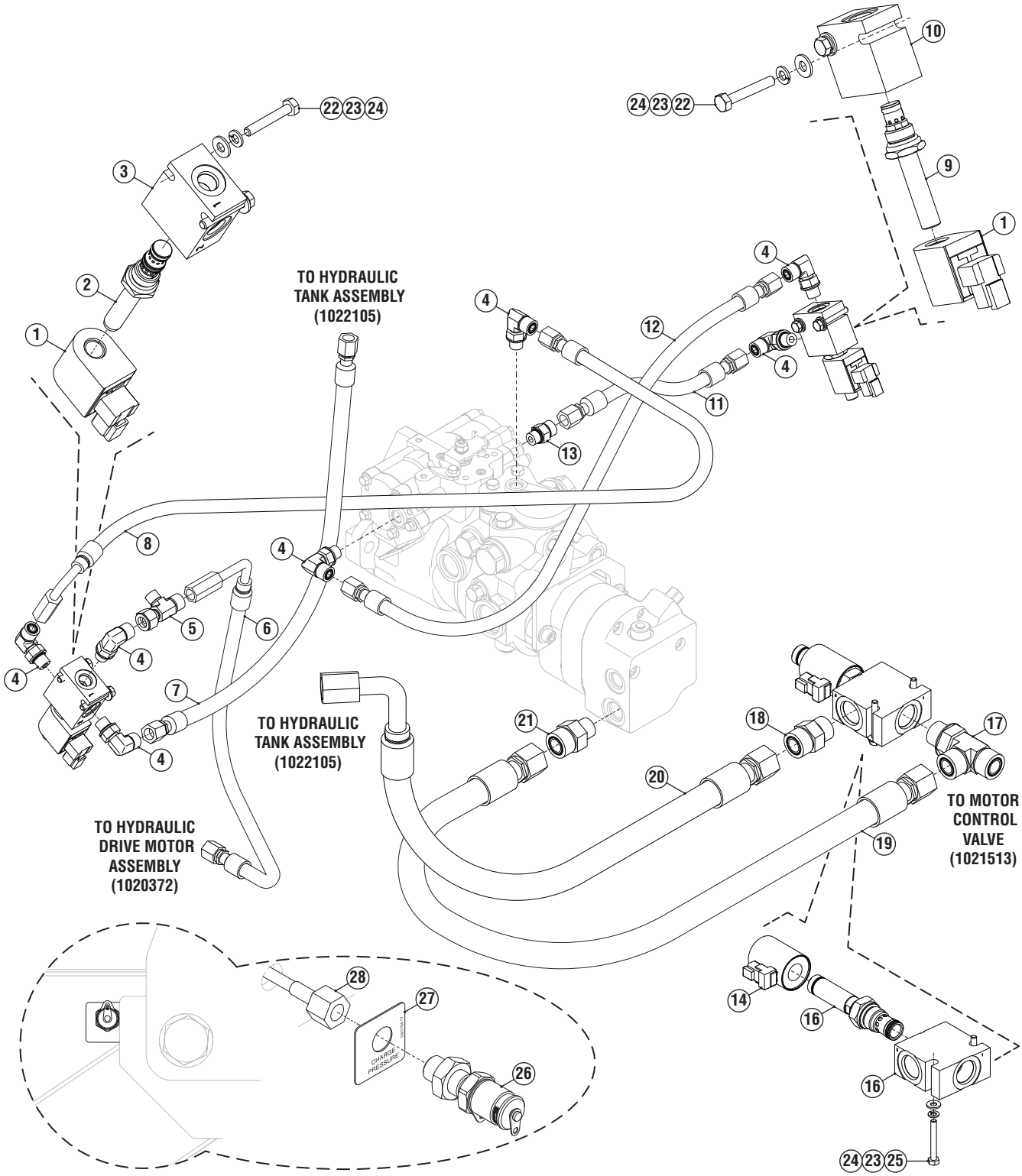


Figure 7-18. Hydraulic System (3 of 5)

Hydraulic System (3 of 5)

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1020495	1	Assembly, 2 Speed Solenoid w/ORFS Adapters	Includes 1-5
REF	1016012	1	Valve, Hydraulic Solenoid, 3-Way	Includes 1-3
1	1009619-03	2	Coil	
2	1016012-02	1	Spool, 3-Way	
3	1016012-01	1	Valve Body, 3-Way	
4	FS6801-06-06-NWO-FG	7	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
5	FS-6602-06-06-04-FG	1	Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-4 ORFS	
6	F387TCJCJ1060606-34	1	Hose Assembly, Solenoid Valve 1 to Motor X1	Tag# 106
7	F387TCJCJC060606-20	1	Hose Assembly, Solenoid Valve to Tank	Tag# 137
8	F387TCJCJ9060606-26	1	Hose Assembly, Pump Gage Port to Solenoid Valve 3	Tag# 106
ASSY	1020496	1	Assembly, Destroke Solenoid w/ORFS Adapters	Includes 1,4,9-10
REF	1009619	1	Valve, Hydraulic Solenoid, 2-Way, 2-POS	Includes 1,9-10
9	1019619-01	1	Spool, 2-Way	
10	1019619-02	1	Valve Body	
11	F387TCJCJC060606-14	1	Hose Assembly, Pump Servo Gage M4 to Destroke Valve	Tag# 4
12	F387TCJCJC060606-26	1	Hose Assembly, Pump Servo Gage M5 to Destroke Valve	Tag# 5
13	FS6400-06-06-O	1	Adapter, Straight, -6 ORFS/-6 O-Rings	
ASSY	1022095	1	Assembly, Brush Speed Solenoid w/ORFS Adapters	Includes 14-19
REF	1021111	1	Valve, Hydraulic Solenoid	Includes 14-16
14	1021111-02	1	Coil	
15	1021111-03	1	Spool	
16	1021111	1	Valve Body	
17	FS6804-12-12-12-NWO-FG	1	Adapter, Run Tee, -12 ORFS/-12 O-Ring/-12 ORFS	
18	FS6400-12-12-O	1	Adapter, Straight, -12 ORFS/-12 O-Rings	
19	F387TCJCJC121212-36	1	Hose Assembly, Aux Pump to Brush Speed Valve 1	Tag# 140
20	F387TCJCJ9121212-29	1	Hose Assembly, Brush Speed Valve 2 to Tank	Tag# 141
21	FS6400-12-10-O	1	Adapter, Straight, -12 ORFS/-10 O-Rings	
22	100-4-20-26-5	4	CSHH, 1/4-20 x 1.625, GR5	
23	302-4	6	Washer, Lock, 1/4	
24	300-4	6	Washer, Flat, SAE, 1/4	
25	100-4-20-32-5	2	CSHH, 1/4-20 x 2.00, GR5	
26	1019707	1	Adapter, Bulkhead, Test Port to -4 IRFS	Located RH Side of Vehicle
27	1021704-17	1	Decal, Test Port, Charge Pressure	
28	1019709	1	Hose Assembly, SMS Test w/-4 Female ORFS (x2)	

HYDRAULIC SYSTEM (4 OF 5)

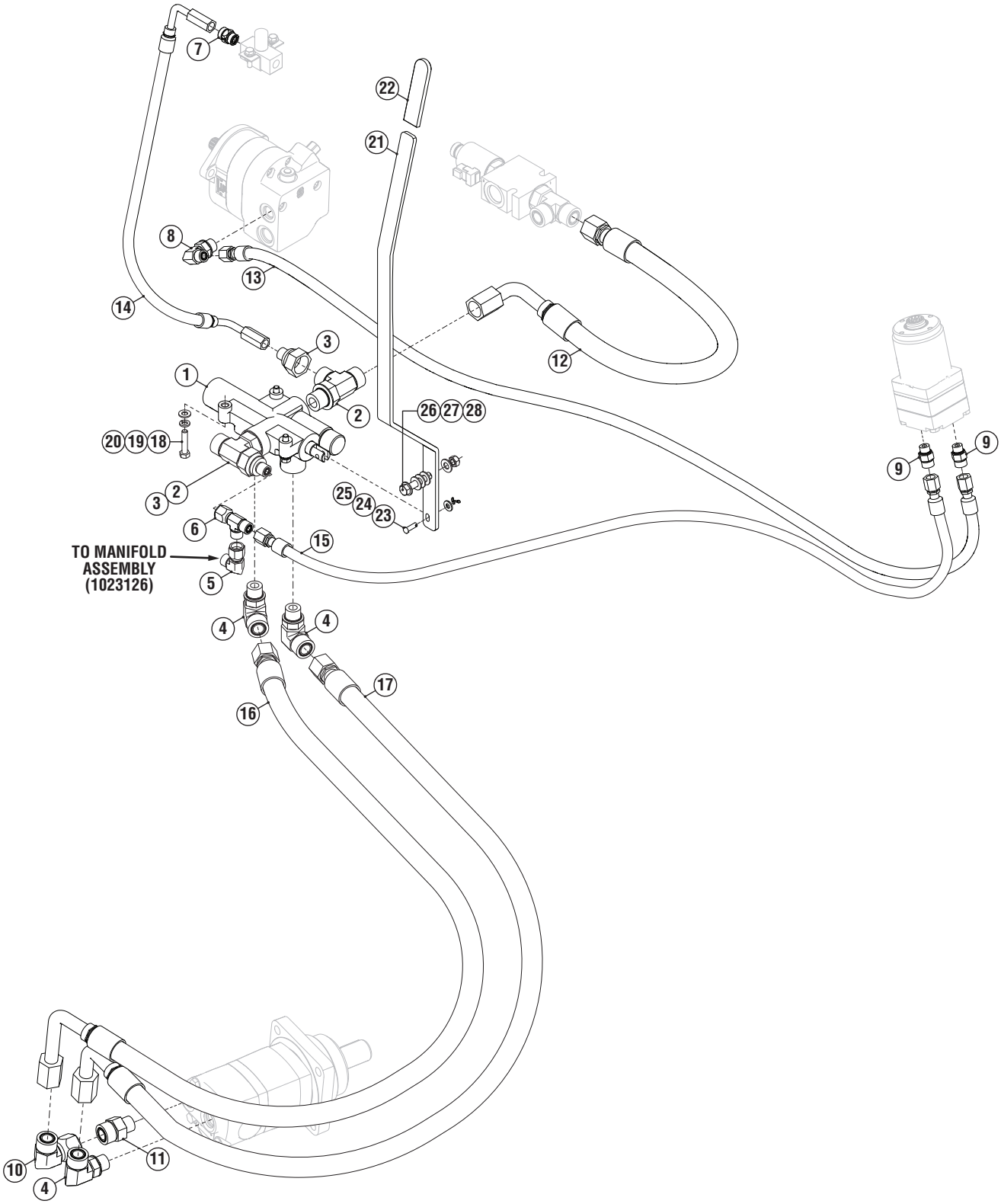


Figure 7-19. Hydraulic System (4 of 5)

Hydraulic System (4 of 5)

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1021513	1	Assembly, Brush Drive Valve w/ORFS Adapters	Includes 1-7
1	1012242	1	Valve, Motor Control	
2	FS6804-12-12-12-NWO-FG	2	Adapter, Run Tee, -12 ORFS/-12 O-Ring/-12 ORFS	
3	FS2406-12-06	1	Adapter, Reducer, Straight, -12 FFORX/-6 ORFS	
4	FS6801-12-10-NWO-FG	3	Adapter, Elbow, 90°, -12 ORFS/-10 O-Ring	
5	FS6500-06-06-FG	1	Adapter, Elbow, 90°, Swivel, -6 ORFS/-6 O-Ring	
6	FS6602-06-06-06-FG	1	Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-6 ORFS	
7	1020387	1	Valve, Check, 20 PSI, 4 GPM, -6 ORFS/-6 ORB	
8	FS6801-06-08-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	
9	FS6400-06-06-O	2	Adapter, Straight, -6 ORFS/-6 O-Ring	
10	FS6500-12-12-FG	1	Adapter, Elbow, 90°, Swivel, -12 ORFS/-12 O-Ring	
11	FS6400-12-12-O	1	Adapter, Straight, -12 ORFS/-12 O-Ring	
12	F387TCJCJ9121212-32	1	Brush Speed Valve 1 to Motor Valve	Tag# 102
13	F387TCJCJC060606-66-AS-B-13-25	1	Aux Pump to Steering Motor In	Tag# 112
14	F387TCJ5J5060606-24-0	1	Motor Valve to Needle Valve	Tag# 122
15	F387TCJCJC060606-50	1	Steering Motor Out to Motor Valve	Tag# 130
16	F387TCJCJ1121212-113-AS-B-19-35	1	Motor Valve A to Brush Motor B	Tag# 101
17	F387TCJCJ5121212-110-AS-B-19-39	1	Motor Valve B to Brush Motor A	Tag# 121
18	100-5-18-24-5	3	CSHH, 5/16-18 x 1.50, GR5	
19	302-5	3	Washer, Lock, 5/16	
20	300-2	3	Washer, Flat, SAE, 5/16	
21	1010266	1	Brush Valve, Lever	Located In Cab
22	36735	1	Grip	Located In Cab
23	404-4-12	1	Pin, Clevis, .25 x .75	
24	300-4	1	Washer, Flat, SAE, 1/4	
25	71714	1	Pin, Cotter, .094 x .75	
26	100-6-16-24-5	1	CSHH, 3/8-16 x 1.50, GR5	
27	300-6	5	Washer, Flat, SAE, 3/8	
28	204-6-16-5	1	Nut, Lock, Stover, 3/8-16, GR5	

HYDRAULIC SYSTEM (5 OF 5)

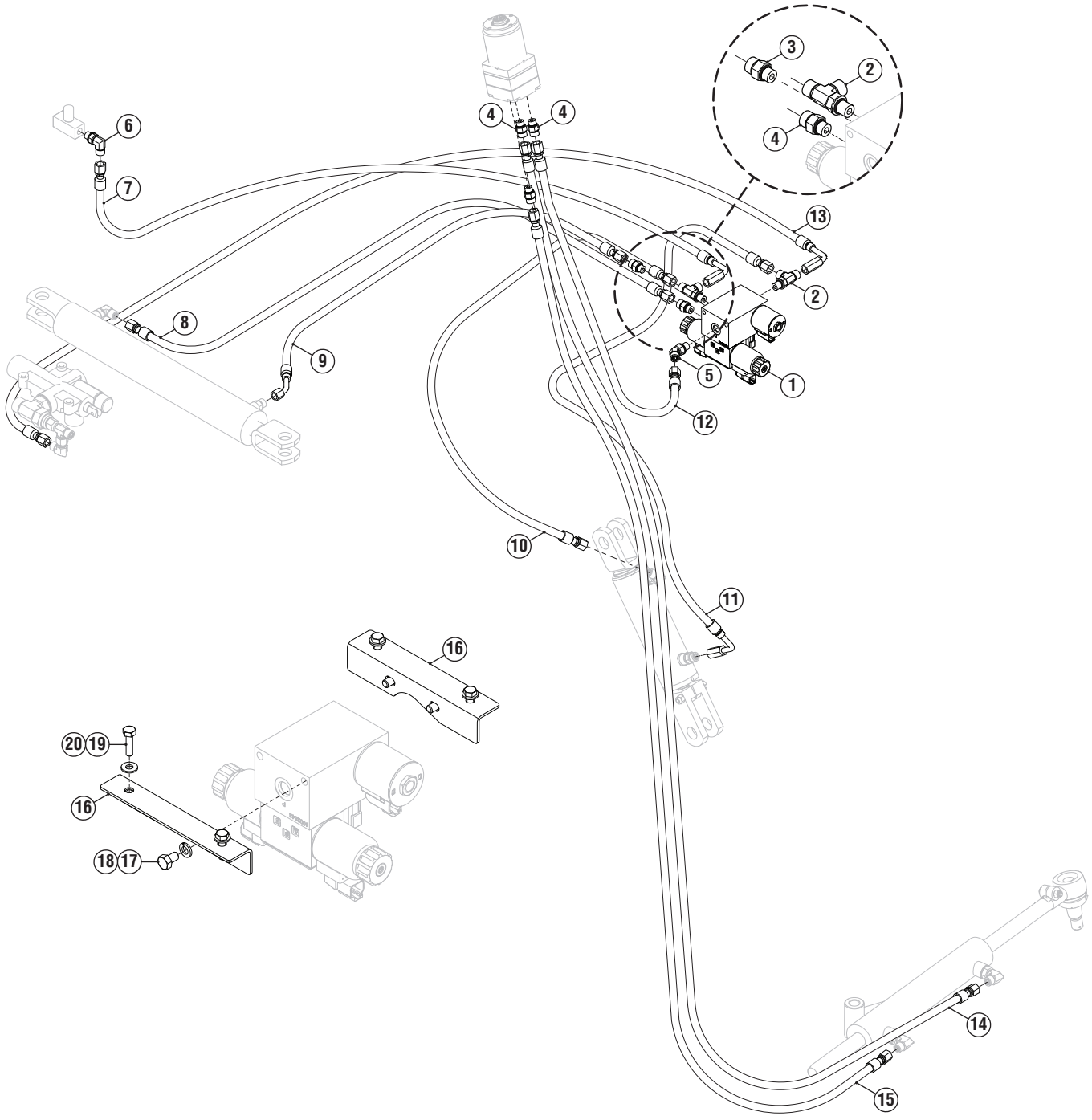


Figure 7-20. Hydraulic System (5 of 5)

Hydraulic System (5 of 5)

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1023126	1	Assembly, Manifold w/ORFS Adapters	Includes 1-5
1	1011734	1	Manifold, Hydraulic, 3 Valve w/Internal Check Valves	
-	1011734-01	A/R	Valve, Directional Solenoid	Repair Part
-	1011734-02	A/R	Coil, 12VDC	Repair Part
2	FS6804-06-06-06-NWO-FG	2	Adapter, Run Tee, -6 ORFS/-6 O-Ring/-6 ORFS	
3	FS6400-06-06-O X 078	1	Adapter, Straight, -6 ORFS/-6 O-Ring, .078 Orifice	
4	FS6400-06-06-O	3	Adapter, Straight, -6 ORFS/-6 O-Ring	
5	FS6802-06-06-NWO-FG	1	Adapter, Elbow, 45°, -6 ORFS/-6 O-Ring	
6	FS6801-06-06-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
7	F387TCJCJ9060606-67	1	Hose Assembly, Manifold A2 to Needle Valve	Tag# 123
8	F387TCJCJC060606-46	1	Hose Assembly, Manifold B1 to Pivot Cylinder	Tag# 109
9	F387TCJCJ7060606-35	1	Hose Assembly, Manifold A1 to Pivot Cylinder Rod End	Tag# 107
10	F387TCJCJC060606-68-AS-B-13-66.25	1	Hose Assembly, Manifold A2 to Lift Cylinder	Tag# 111
11	F387TCJCJ5060606-77-AS-B-13-74.75	1	Hose Assembly, Manifold T to Lift Cylinder	Tag# 108
12	F387TCJCJC060606-17-AS-B-13-15.25	1	Hose Assembly, Manifold P to Steering Motor Aux	Tag# 113
13	F387TCJCJ9060606-63	1	Hose Assembly, Manifold T to Motor Valve	Tag# 110
14	F387TCJCJC060606-117-AS-B-13-115.25	1	Hose Assembly, Steering Motor RT to Steer Cylinder	Tag# 124
15	F387TCJCJC060606-106-AS-B-13-130.25	1	Hose Assembly, Steering Motor LT to Steer Cylinder	Tag# 125
16	1008089SRV	2	Mount, Valve Block Assembly	
17	100-5-18-8-5F	4	CSHH, 5/16-18 x .50, GR5, FT	
18	302-5	4	Washer, Lock, 5/16	
19	100-4-20-16-5	4	CSHH, 1/4-20 x 1.00, GR5	
20	300-4	4	Washer, Flat, SAE, 1/4	

CAB ASSEMBLY , DETAILS (1 OF 2)

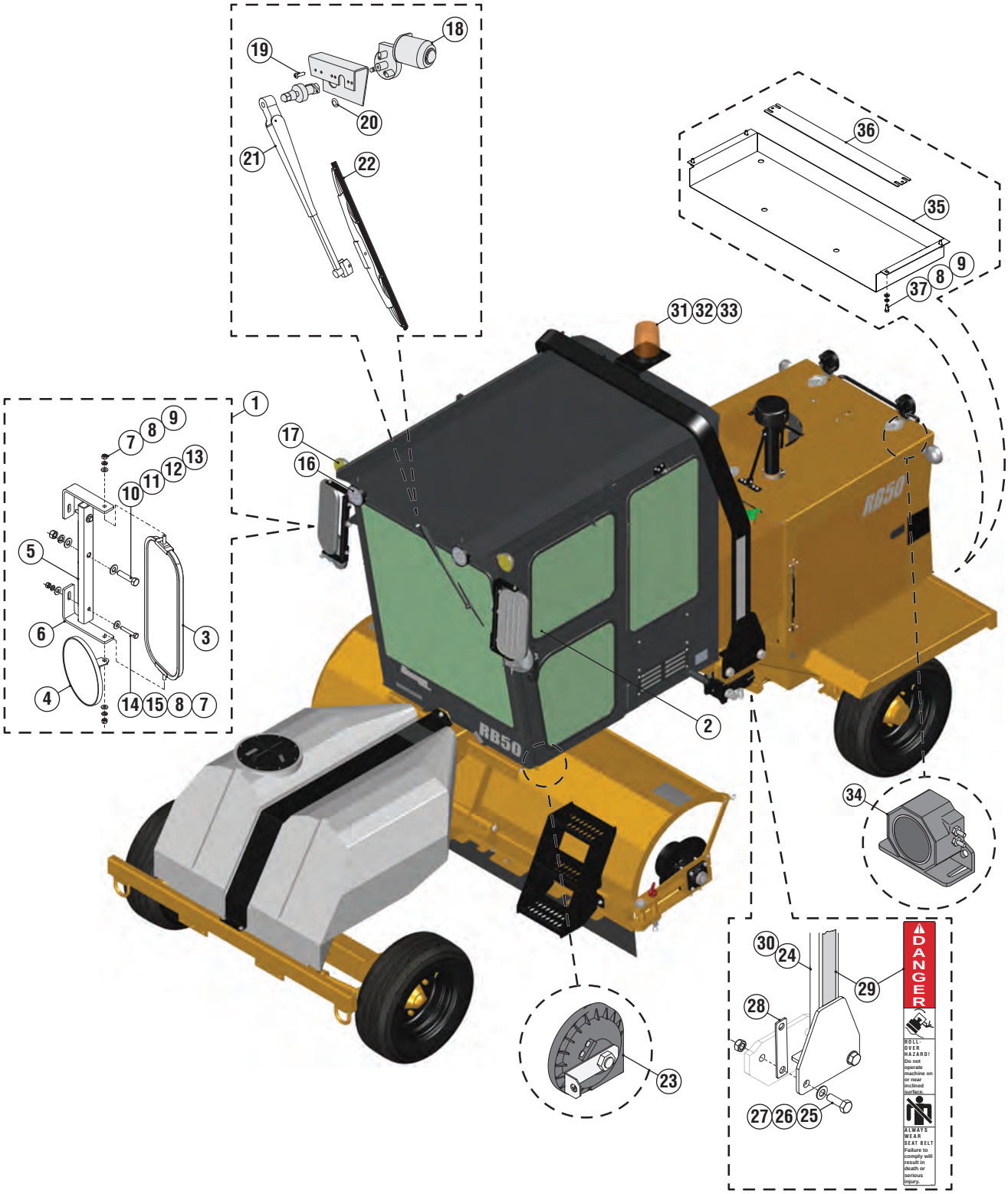


Figure 7-21. Cab Assembly, Details (1 of 2)

Cab Assembly, Details (1 of 2)

Item No.	Part Number	Qty.	Description	Remarks
1	1021679	1	Assembly, Mirror Cluster, RH	Includes Items 3-15
2	1021678	1	Assembly, Mirror Cluster, LH	Includes Items 3-15
3	852570	2	Mirror Head	(1) Per Side
4	100857	2	Mirror - ø7.50, Convex	(1) Per Side
5	856986	2	Brace, Mirror	(1) Per Side
6	856987	4	Mount, Light, Mirror	(2) Per Side
7	200-5-18-5	4	Nut, Hex, 5/16-18, GR5	(2) Per Side
8	302-5	12	Washer, Lock, 5/16	(4) Per Side
9	300-5	8	Washer, Flat, SAE, 5/16	(2) Per Side
10	100-8-13-36-5	2	CSHH, 1/2-13 x 2.25, GR5	(1) Per Side
11	300-8	4	Washer, Flat, SAE, 1/2	(2) Per Side
12	302-8	2	Washer, Lock, 1/2	(1) Per Side
13	200-8-13-5	2	Nut, Hex, 1/2-13, GR5	(1) Per Side
14	100-5-18-32-5	4	CSHH, 5/16-18 x 2.00, GR5	(2) Per Side
15	301-5	8	Washer, Flat, USS, 5/16	(4) Per Side
16	1012563	2	Light, LED Circular	(1) Per Side
17	1005941	2	LED Single Face Pedestal Lamp, Yellow	(1) Per Side
18	151170	1	Motor, Windshield Wiper	
19	81277	1	CSBHS, 1/4-20 x .875, SS	
20	35136-3	1	Plug, Hole, 3/8, Flush Mount, Plastic	
21	1021648	1	Arm, Wiper, 14"	
22	12755028	1	Blade, Wiper, 16"	
23	20190773	1	Horn, Low Pitch	
24	1015543	1	Rollbar, Complete	
25	101-16-8-48-8	4	CSHH, Heavy, 1-8 x 3.00, GR8	(2) Per Side
26	310-16	4	Washer, Hardened, SAE, 1	(2) Per Side
27	204-16-8-8	4	Nut, Lock, Stover, 1-8, GR8	(2) Per Side
28	1015784	2	Strap, Roll Bar Shipping	(1) Per Side
29	1015036-02	2	Decal - Danger, Roll-Over Hazard	Not Shown, (1) Per Side
30	1015546	6	Loom Clip, ø.50	Not Shown
31	1013231	1	Beacon Light, Amber, Strobe, LED	
32	1015786	1	Bracket, Beacon Light	
33	1016701	1	Harness, Beacon	Not Shown
34	33963	1	Alarm, Back Up	
35	1015915	1	Baffle, Lower Radiator, Formed	
36	1016327	1	Baffle, Lower Radiator	
37	100-5-18-12-5F	4	CSHH, 5/16-18 x .75, GR5, FT	

CAB ASSEMBLY, DETAILS (2 OF 2)

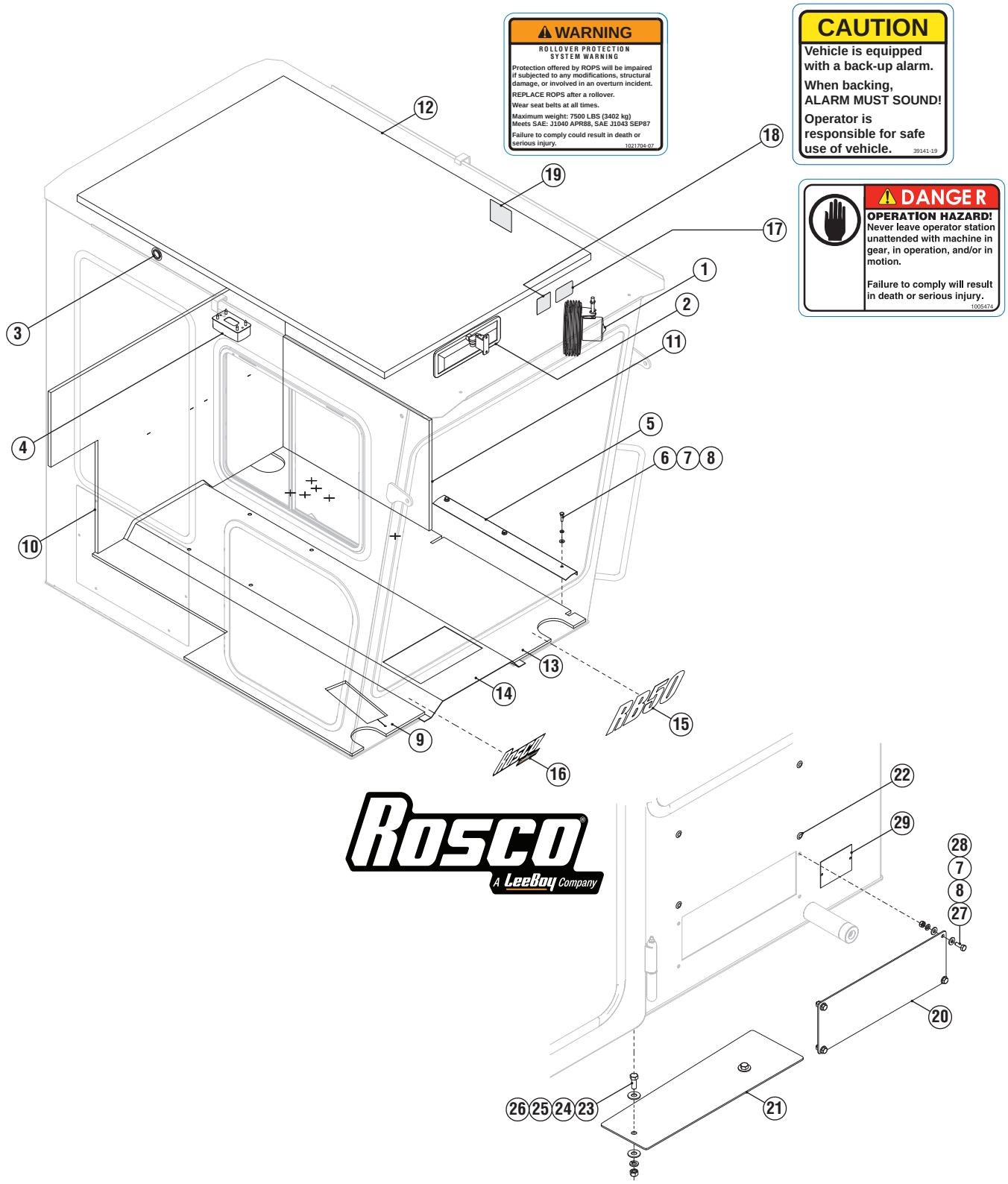


Figure 7-22. Cab Assembly, Details (2 of 2)

Cab Assembly, Details (2 of 2)

Item No.	Part Number	Qty.	Description	Remarks
1	35425	1	Defrost Fan Assembly, 12V	
2	720260	1	Mirror, Inside Rearview	
3	35136	1	Plug, Hole, 1.25, Flush Mount	
4	38840	1	Light, Dome	
5	1015622	1	Door Sill, Cab	
6	100-4-20-16-5F	3	CSHH, 1/4-20 x 1.00, GR5, FT	
7	302-4	7	Washer, Lock, 1/4	
8	300-4	11	Washer, Flat, SAE, 1/4	
KIT	1015083	1	Kit, Cab Insulation, No AC/Heat	Includes Items 9-13
9	1015083-01	1	Cab Insulation, Floor Mat - RH	
10	1015083-02	1	Cab Insulation, Rear	
11	1015083-03	1	Cab Insulation, No AC - LH	
12	1015083-04	1	Cab Insulation, Headliner	
13	1015083-05	1	Cab Insulation, Floor Mat - LH	
14	1015035-05	1	Anti-Skid, Cab, Center	
15	1015035-04	1	Decal, Deco, RB50	
16	1013790-08	1	Decal, Deco, Rosco, 8.50"	
17	1005474	1	Decal, No Unattended	Located in Cab Interior
18	39141-19	1	Decal, Caution, Alarm	Located in Cab Interior
19	1021704-04	1	Decal, Warning, ROPS	Located in Cab Interior
20	1008042	1	Plate, Access Cover	No AC/Heat Only
21	1011711	1	Cover, Plate, Cab Floor	No AC/Heat Only
22	35136-21	4	Plug, Hole, .313, Flush Mount	No AC/Heat Only
23	100-6-16-16-5F	2	CSHH, 3/8-16 x 1.00, GR5, FT	
24	301-6	4	Washer, Flat, USS, 3/8	
25	302-6	2	Washer, Lock, 3/8	
26	200-6-16-5	2	Nut, Hex, 3/8-16, GR5	
27	100-4-20-12-5F	4	CSHH, 1/4-20 x .75, GR5, FT	
28	200-4-20-5	4	Nut, Hex, 1/4-20, GR5	
29	94004-21	1	Plate, Info, Serial Number	

CAB ASSEMBLY, WINDOWS

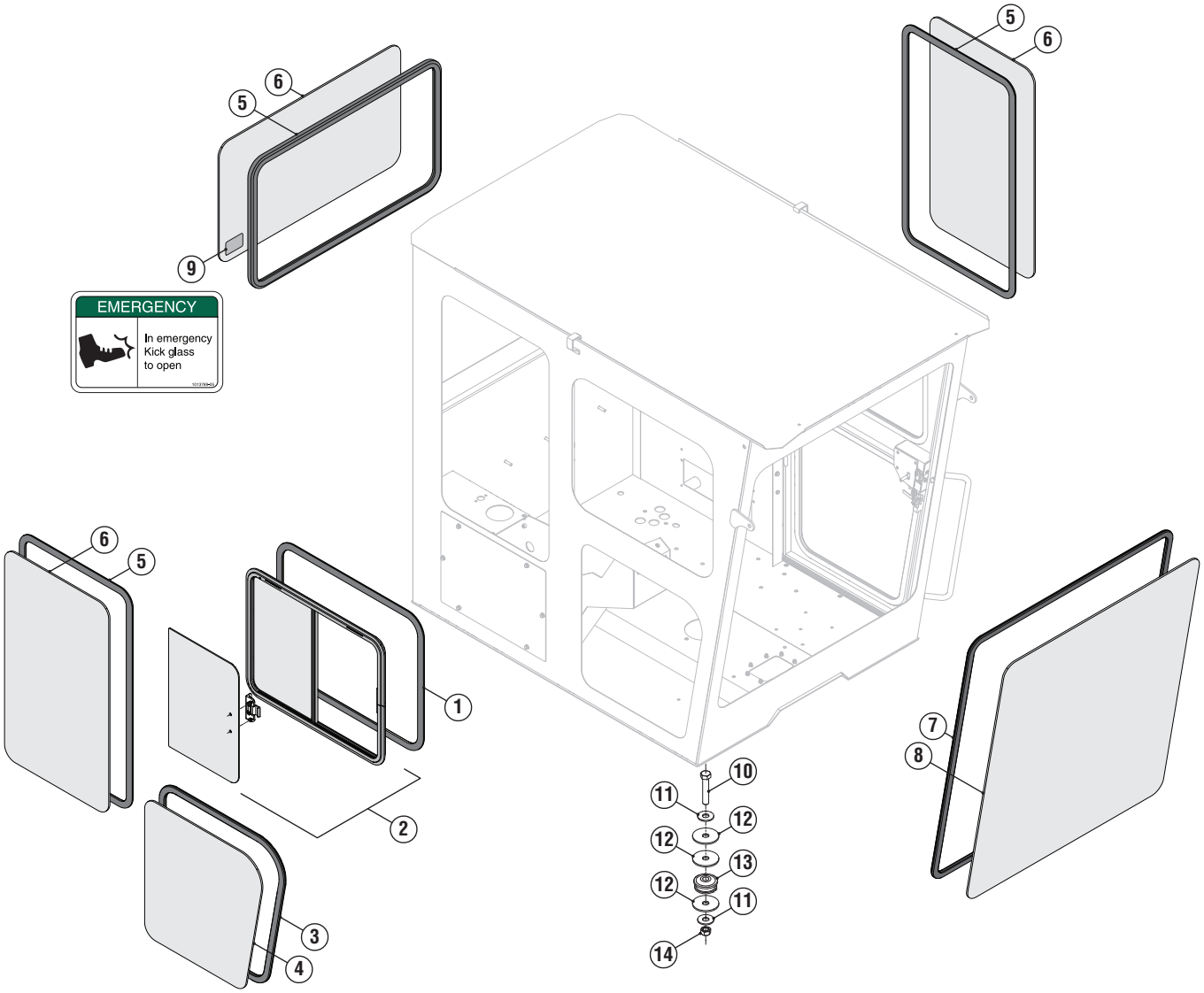


Figure 7-23. Cab Assembly, Windows

Cab Assembly, Windows

Item No.	Part Number	Qty.	Description	Remarks
1	1021669	1	Seal, Window, Upper	
2	1021714	1	Window, Side, Sliding	
3	1021668	1	Seal, Window, Lower	
4	1021666	1	Glass, Lower	
5	1015595	3	Seal, Window, Common	
6	1015593	3	Glass, Common	
7	1023442	1	Seal, Window, Front	
8	1023436	1	Glass, Front	
9	1013789-05	1	Decal, Kick Glass	
10	100-12-10-64-5	4	CSHH, 3/4-10 x 4.00, GR5	
11	301-12	8	Washer, Flat, USS, 3/4	
12	36073	12	Washer, Flat, .80 x 3.25 x .188	
13	36072	4	Mount, Rubber	
14	204-12-10-5	4	Nut, Lock, Stover, 3/4-10, GR5	

CAB ASSEMBLY, DOOR

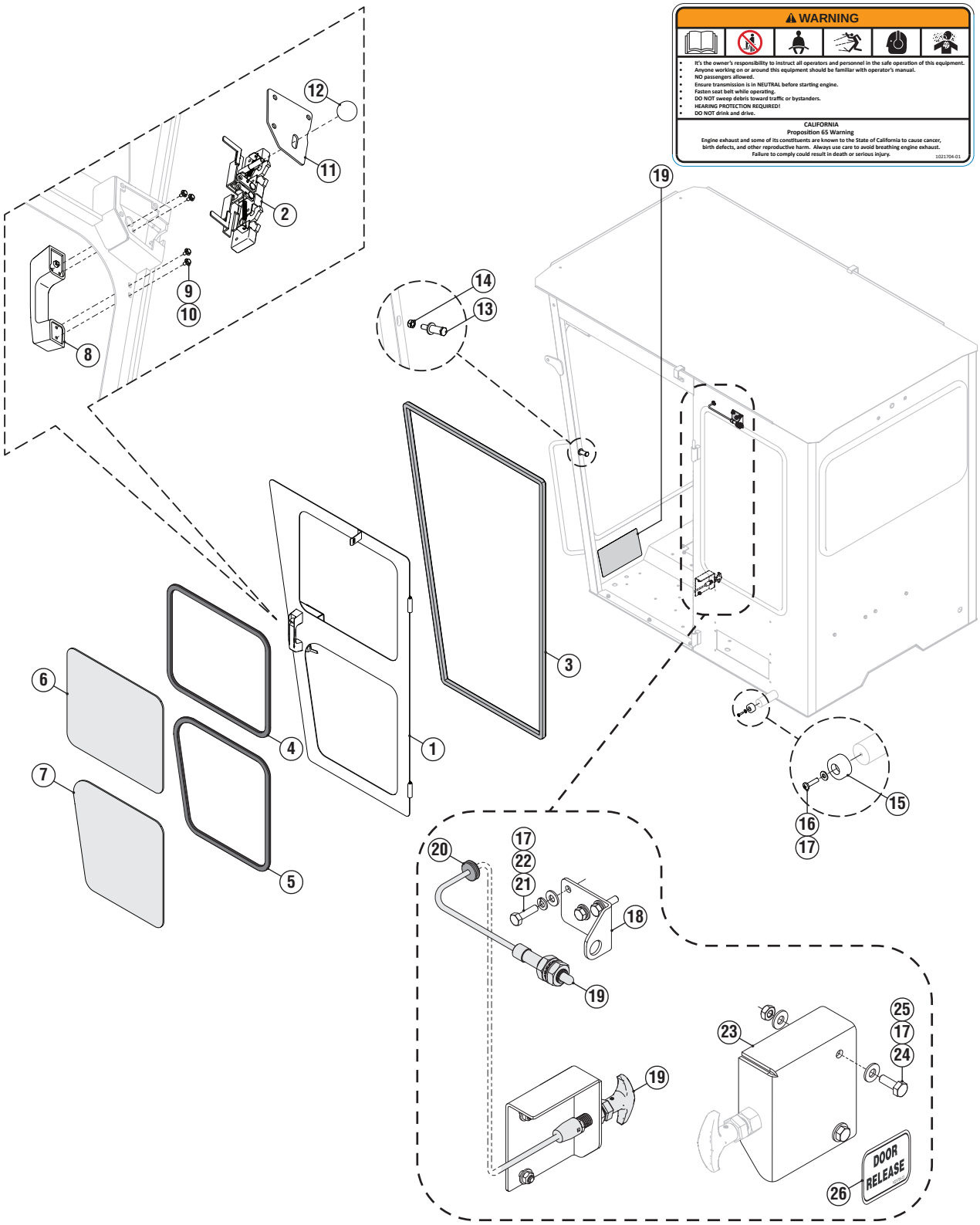


Figure 7-24. Cab Assembly, Door

Cab Assembly, Door

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1023440	1	Assembly, Cab Door	Includes 1-11
1	1023428	1	Cab Door	Includes 2
2	983423-01SRV	1	Rotary Latch	
3	1023445	1	Seal, Door Trim	
4	1023444	1	Seal, Window, Door, Upper	
5	1023443	1	Seal, Window, Door, Lower	
6	1023438	1	Glass, Door, Upper	
7	1023437	1	Glass, Door, Lower	
8	982415	1	Door Handle - Push Button	
-	982415-01	1	Key, Door Latch	Not Shown
9	100-M5-0.8-10-8.8-F	4	CSHH, M5x0.8 x 10mm, C8.8, FT	
10	300-#10	4	Washer, Flat, SAE, #10	
11	980678	1	Cover, Door Latch	
12	851156	1	Knob, Black	
13	1020242	1	Bolt, Striker, 2.75" Long	
14	205-7-14-5	1	Nut, Lock, Nylon, 7/16-14, GR5	
15	38392	1	Bumper, 1.50 OD x .75	
16	122-4-20-16F	1	PHMS, Cross, 1/4-20 x 1, FT	
17	300-4	8	Washer, Flat, SAE, 1/4	
18	1021687	1	Bracket, Door Latch	
19	1021659	1	Latch, Door, T-Handle	
20	1021720	1	Grommet, .19 ID x .63 OD x .125	
21	100-4-20-16-5F	3	CSHH, 1/4-20 x 1.00, GR5, FT	
22	302-4	3	Washer, Lock, 1/4	
23	1021721	1	Mount, Release, Cab Door	
24	100-4-20-12-5F	2	CSHH, 1/4-20 x .75, GR5, FT	
25	204-4-20-5	2	Nut, Lock, Stover, 1/4-20, GR5	
26	1021704-13	1	Decal, Door Release	
27	1021704-01	1	Decal, Warning, Equipment	

CAB ASSEMBLY, BRAKES

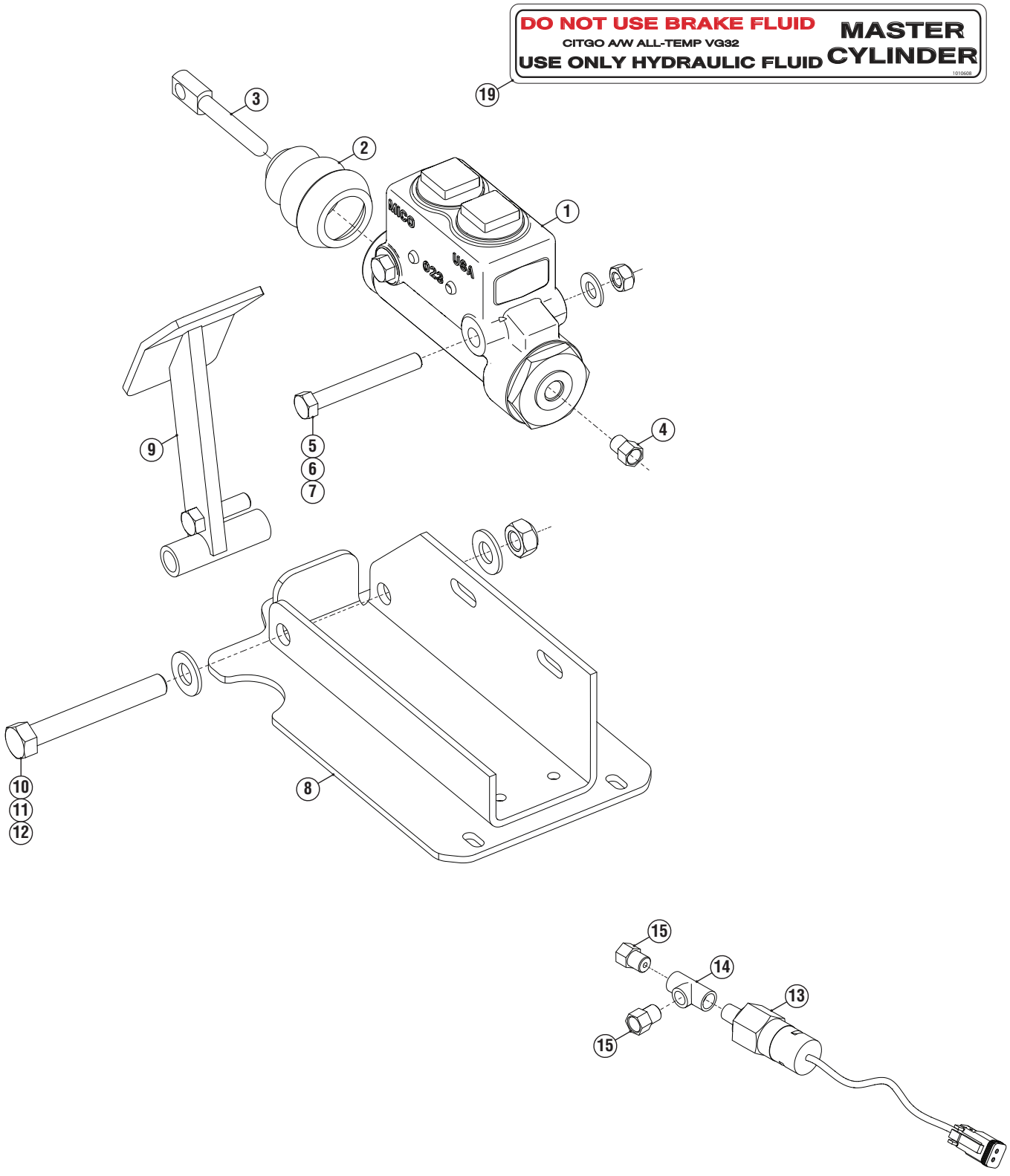


Figure 7-25. Cab Assembly, Brakes

Cab Assembly, Brakes

Item No.	Part Number	Qty.	Description	Remarks
ASSY	1023497	1	Assembly, Brake Pedal	Includes 1-12
1	1009383	1	Master Cylinder	
2	1010648	1	Boot, Master Cylinder	
3	1009786	1	Pushrod, Brake	
4	38277	1	Fitting, Master Cylinder Adapter	
5	100-6-16-48-5	2	CSHH, 3/8-16 x 3.00, GR5	
6	302-6	2	Washer, Lock, 3/8	
7	204-6-16-5	2	Nut, Lock, Stover, 3/8-16, GR5	
8	1023181	1	Mount, Brake Pedal	
9	28650SRV	1	Brake Pedal	
10	100-8-13-60-5	1	CSHH, 1/2-13 x 3.75, GR5	
11	300-8	2	Washer, Flat, SAE, 1/2	
12	204-8-13-5	1	Nut, Lock, Stover, 1/2-13, GR5	
13	1020243	1	Pressure Switch, NO, 30-120PSI, 1/8" Male NPT	
14	33557	1	Fitting, Tee 02FM, Brass	
15	39059	2	Fitting, Straight, 02MP-03IFF, Brass	
16	33953-15	3	Brake Line, .188 x 15	Not Shown
17	33953-50	1	Brake Line, .188 x 50	Not Shown
18	33953-107	2	Brake Line, .188 x 107	Not Shown
19	1010608	1	Decal, Info, Master Cylinder	Located Inside Cab

CAB ASSEMBLY, STEERING

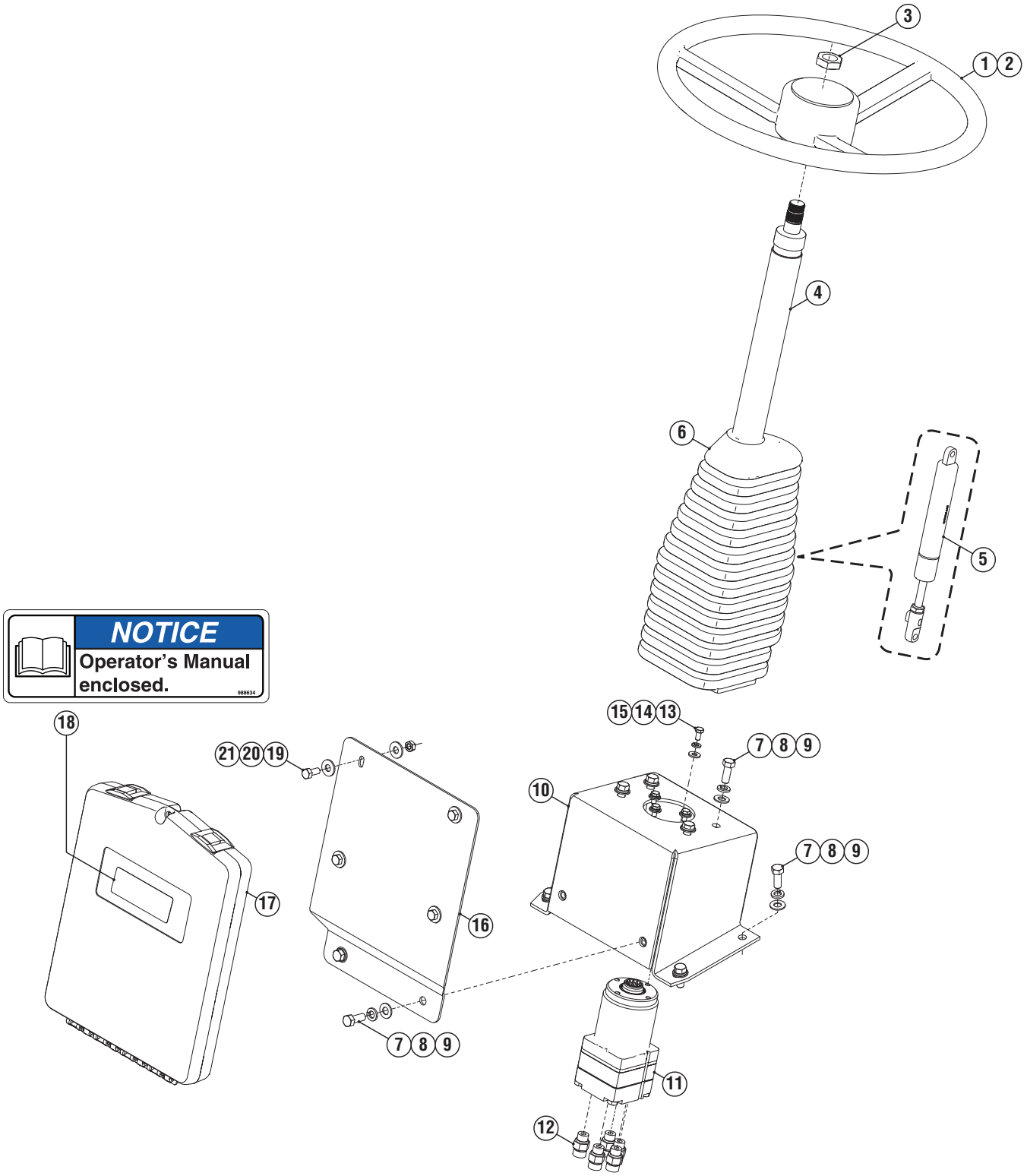


Figure 7-26. Cab Assembly, Steering

Cab Assembly, Steering

Item No.	Part Number	Qty.	Description	Remarks
1	300030	1	Steering Wheel, 17", 36 Spline	
2	300010	1	Cap, Steering Wheel	
3	300020	1	Nut, Hex, .813-20	
ASSY	1021625	1	Assembly, Tilt Steering	Includes Items 4-9
4	1021697	1	Column, Tilt Steering, Male Spline w/Boot	Includes Item 5
5	1009492-01	1	Gas Spring, Steering Column	
6	1021697-04	1	Boot	
7	100-6-16-16-5F	10	CSHH, 3/8-16 x 1.00, GR5, FT	
8	302-6	10	Washer, Lock, 3/8	
9	300-6	10	Washer, Flat, SAE, 3/8	
ASSY	1021624	1	Adapter, Tilt Steering	
10	1021699	1	Assembly, Steering Motor w/Adapters	Includes Items 10-11
11	1021698	1	Motor, Hydraulic, Power Steering, 8.9 CIR	
12	FS6400-06-06-O	5	Adapter, Straight, -6 ORFS/-6 O-Ring	
13	100-M06-1.00-12-8.5	4	CSHH, M6x1.0 x 12mm, C8.8	
14	302-4	4	Washer, Lock, 1/4	
15	300-4	4	Washer, Flat, SAE, 1/4	
16	1021788	1	Bracket, Mount, Manual Box	
17	985234	1	Manual-PAK Case	
18	988634	1	Decal - Notice, Operator's Manual	
19	100-5-18-12-5F	4	CSHH, 5/16-18 x .75, GR5, FT	
20	301-5	8	Washer, Flat, USS, 5/16	
21	204-5-18-5	4	Nut, Lock, Stover, 5/16-18, GR5	

CAB ASSEMBLY, SPRING SEAT

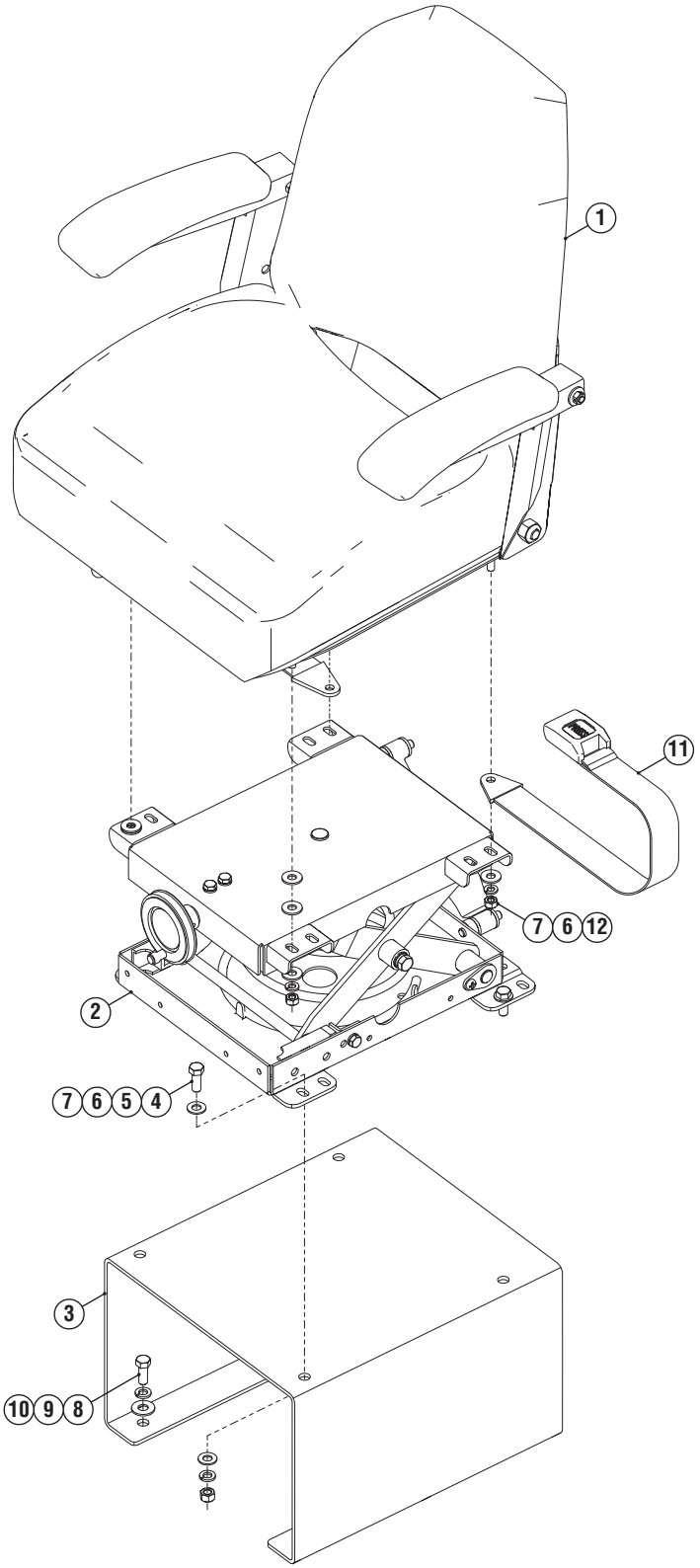


Figure 7-27. Cab Assembly, Spring Seat

Cab Assembly, Spring Seat

Item No.	Part Number	Qty.	Description	Remarks
GRP	1015620		Group, Spring Seat	
1	360010B	1	Assembly, Seat, Black w/Armrest	
2	140600	1	Suspension Bracketry, Seat, Low	
3	1015621	1	Pedestal, Spring Seat	
4	100-5-18-16-5	4	CSHH, 5/16-18 x 1.00, GR5	
5	300-5	4	Washer, Flat, SAE, 5/16	
6	302-5	8	Washer, Lock, 5/16	
7	200-5-18-5	8	Nut, Hex, 5/16-18, GR5	
8	100-6-16-16-5F	8	CSHH, 3/8-16 x 1.00, GR5, FT	
9	302-6	4	Washer, Lock, 3/8	
10	300-6	4	Washer, Flat, SAE. 3/8	
11	730-3050	1	Seat Belt, 2.00 w/Hardware	
12	301-5	8	Washer, Flat, USS, 5/16	

CAB ASSEMBLY, CONTROLS

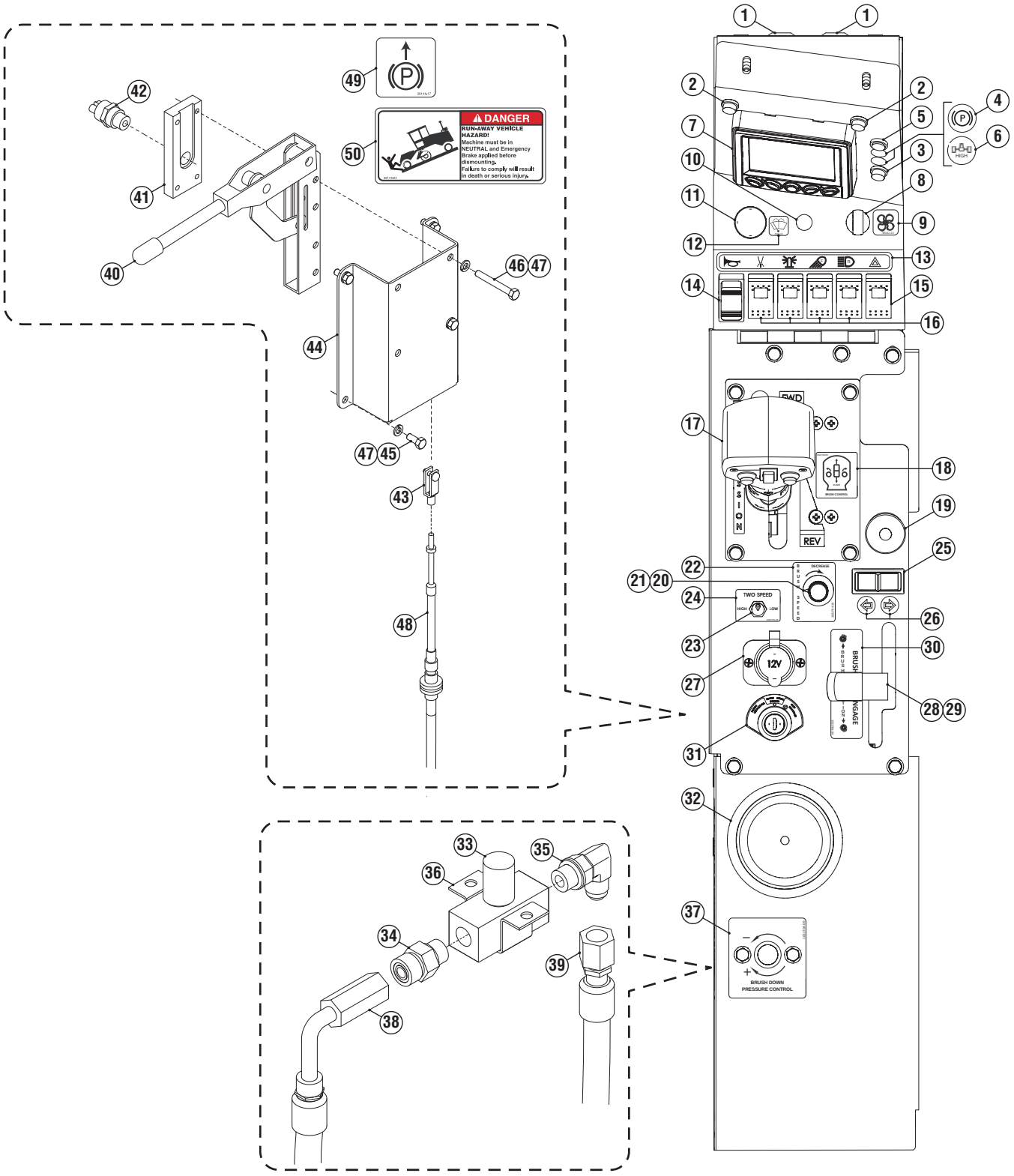


Figure 7-28. Cab Assembly, Controls

Cab Assembly, Controls

Item No.	Part Number	Qty.	Description	Remarks
1	35136-6	2	Plug, Hole, .75 Flush Mount, Plastic	
2	1016102	2	Light, Indicator, Green	
3	1009253-07	1	Light, Indicator, Red	
4	1013789-07	1	Decal - Oper, Park Brake	
5	31986	1	Light, Blue, Dash, .50 Hole	
6	1021704-16	1	Decal - Oper, High Gear	
7	1015929	1	PV380 Display	
8	35426	1	Switch, Fan, 3-SPD w/Knob	
9	39141-22	1	Decal - Oper, Fan	
10	35136-4	1	Plug, Hole, .50, Locking, Plastic	
11	853090	1	Switch, Wiper/Washer	
12	39141-16	1	Decal - Oper, Wiper/Washer	
13	1013789-01	1	Decal - Oper, Lights/Horn	
14	39140-13	1	Switch, Rocker, SPST	
15	39140-12	1	Switch, Rocker, SPST, Red	
16	39140-14	4	Switch, Rocker, SPST, Amber	
17	1019658	1	Joystick, FNR w/Brush Control	
18	1021704-06	1	Decal - Oper, Brush Control	
19	1010672	1	Switch, Emergency Stop, 1-NC	
20	35049	1	Knob, .25 Shaft	
21	1023229SRV	1	Assembly, Brush Speed Potentiometer	
22	1021704-10	1	Decal - Oper, Brush Speed	
23	851391	1	Switch, Toggle, SPST, 2 -POS	
24	1021704-08	1	Decal - Oper, 2 -Speed	
25	39140-11	1	Switch, Rocker, SPDT, Paddle	
26	1013789-06	2	Decal - Oper, Turn Signal	
27	1011147	1	Outlet, 12V	
28	1010266	1	Lever, Brush Valve	
29	36735	1	Grip, .25 x 1.00 x 4.00	
30	1021704-04	1	Decal - Oper, Brush Drive	
31	39146-14	1	Switch, Ignition w/Heat St	
-	982008-04	1	Key, Ignition	
32	982689	1	Cup Holder	
ASSY	1020371	1	Assembly, Needle Valve w/Adapters	Includes Items 31-34
33	35552	1	Valve, Needle, Hydraulic	
34	FS6400-06-06-O	1	Adapter, Straight, -6 ORFS/-6 O-Ring	
35	FS6801-06-06-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
36	17670	1	Valve Clamp	
37	1021704-09	1	Decal - Oper, Brush Crown Pressure	
38	F387TCJ5J5060606-24-0	1	Hose Assembly, Motor Valve to Needle Valve	Tag# 122

CAB ASSEMBLY, CONTROLS (CONT.)

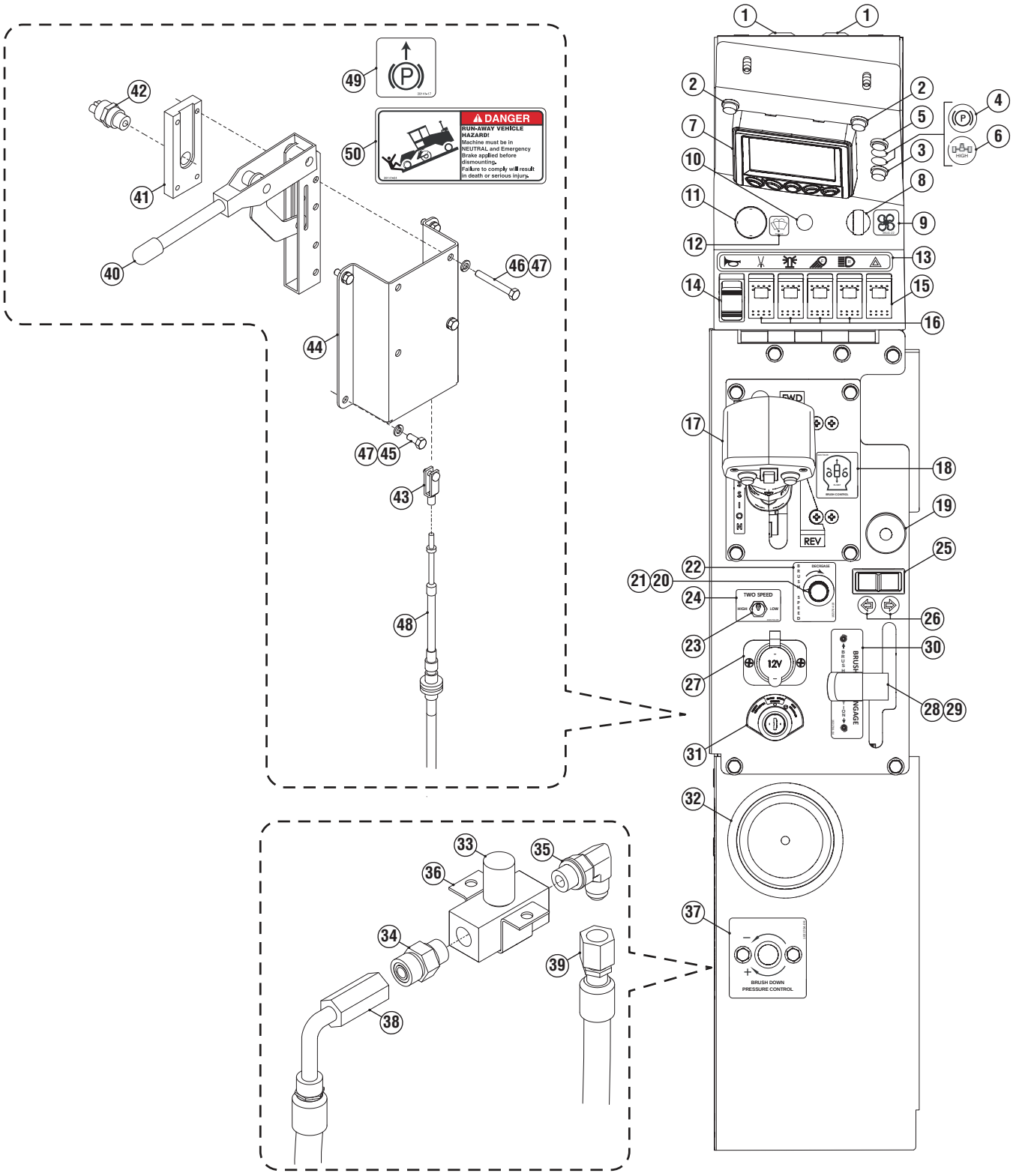


Figure 7-28. Cab Assembly, Controls (Cont.)

Cab Assembly, Controls (Cont.)

Item No.	Part Number	Qty.	Description	Remarks
39	F387TCJCJ9060606-67	1	Hose Assembly, Manifold A2 to Needle Valve	Tag# 123
GRP	1015130		Group - Park Brake	Includes Items 40-47
40	34299	1	Lever, Park Brake, 12" Handle	
41	21218	1	Mount, Brake Switch	
42	951091224	1	Switch, Safety Start	
43	6427	1	Clevis, 5/16-24 UNF w/Pin	
44	21217	1	Bracket, Brake Lever	
45	100-5-18-12-5F	4	CSHH, 5/16-18 x .75, GR5, FT	
46	100-5-18-40-5F	2	CSHH, 5/16-18 x 2.50, GR5, FT	
47	302-5	6	Washer, Lock, 5/16	
48	1010036	1	Cable, Brake	
49	39141-17	1	Decal - Park Brake	
50	39141-24	1	Decal - Danger, Run-Away Vehicle	

CAB ASSEMBLY, FUSE PANEL

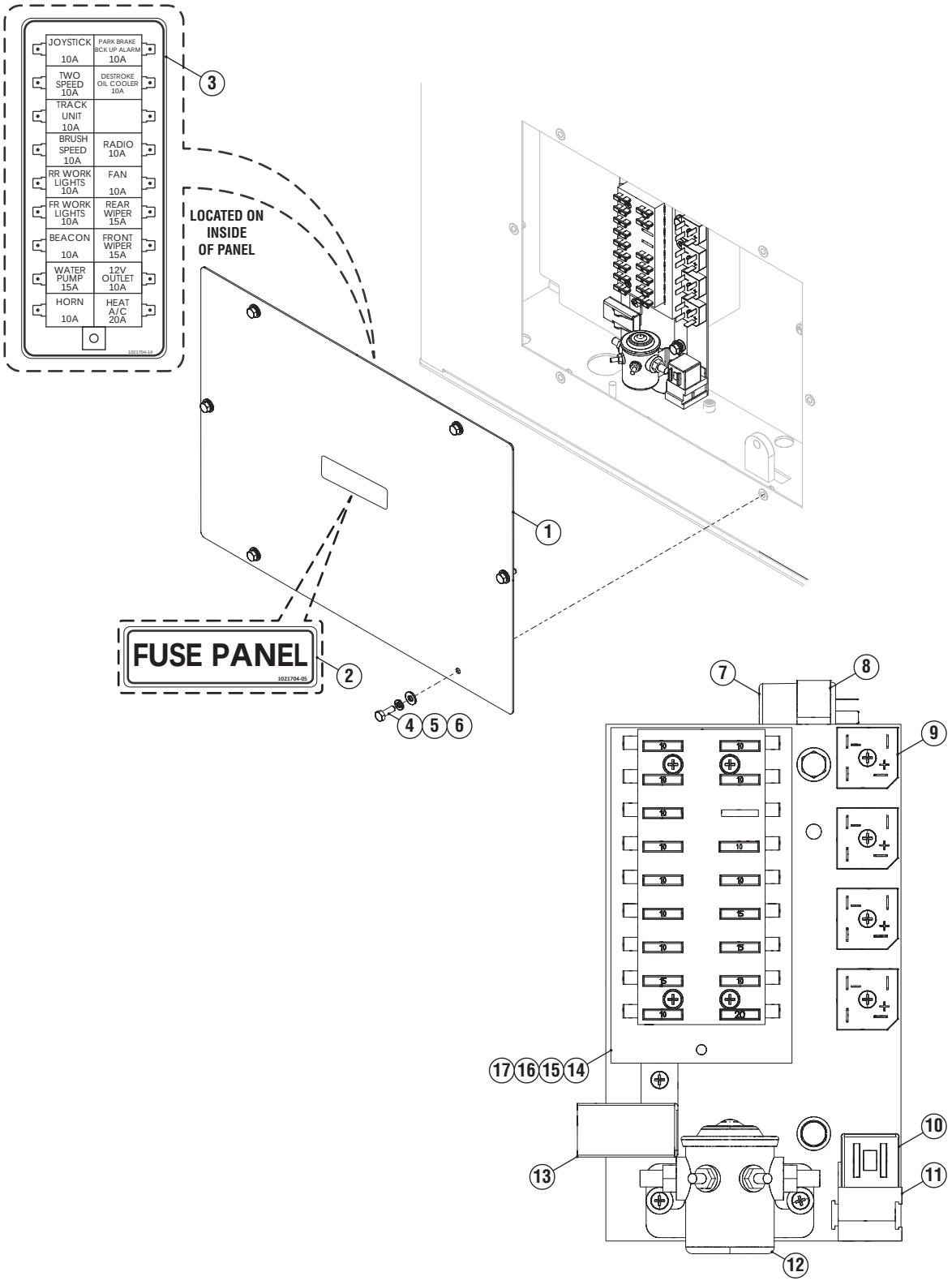


Figure 7-29. Cab Assembly, Fuse Panel

Cab Assembly, Fuse Panel

Item No.	Part Number	Qty.	Description	Remarks
1	1015587	1	Cover, Wiring Access	
2	1021704-05	1	Decal - Oper, Fuse Panel	
3	1021704-14	1	Decal - Oper, Fuse Panel	
4	100-4-20-12-5F	6	CSHH, 1/4-20 x .75, GR5, FT	
5	302-4	6	Washer, Lock, 1/4	
6	300-4	6	Washer, Flat, SAE, 1/4	
7	1010202	1	Flasher, LED	
8	34798	1	Clamp, Insulated Band, 1.25 ID	
9	1016125	4	Bridge Rectifier	
10	36085	1	Relay, 12VDC, SPDT, 40A, 5 Pin	
11	36086	1	Bracket, Relay Mounting	
12	720290	1	Solenoid, 12V, Constant Duty	
13	37118	1	Fuse Holder, Inline, ATC	
14	685060	1	Fuse Block, 18 Gang, ATC	
15	36340	13	Fuse, 10A, ATC	
16	36341	3	Fuse, 15A, ATC	
17	36342	1	Fuse, 20A, ATC	

OPTION - CAB ASSEMBLY, AC/HEATER (1 OF 2)

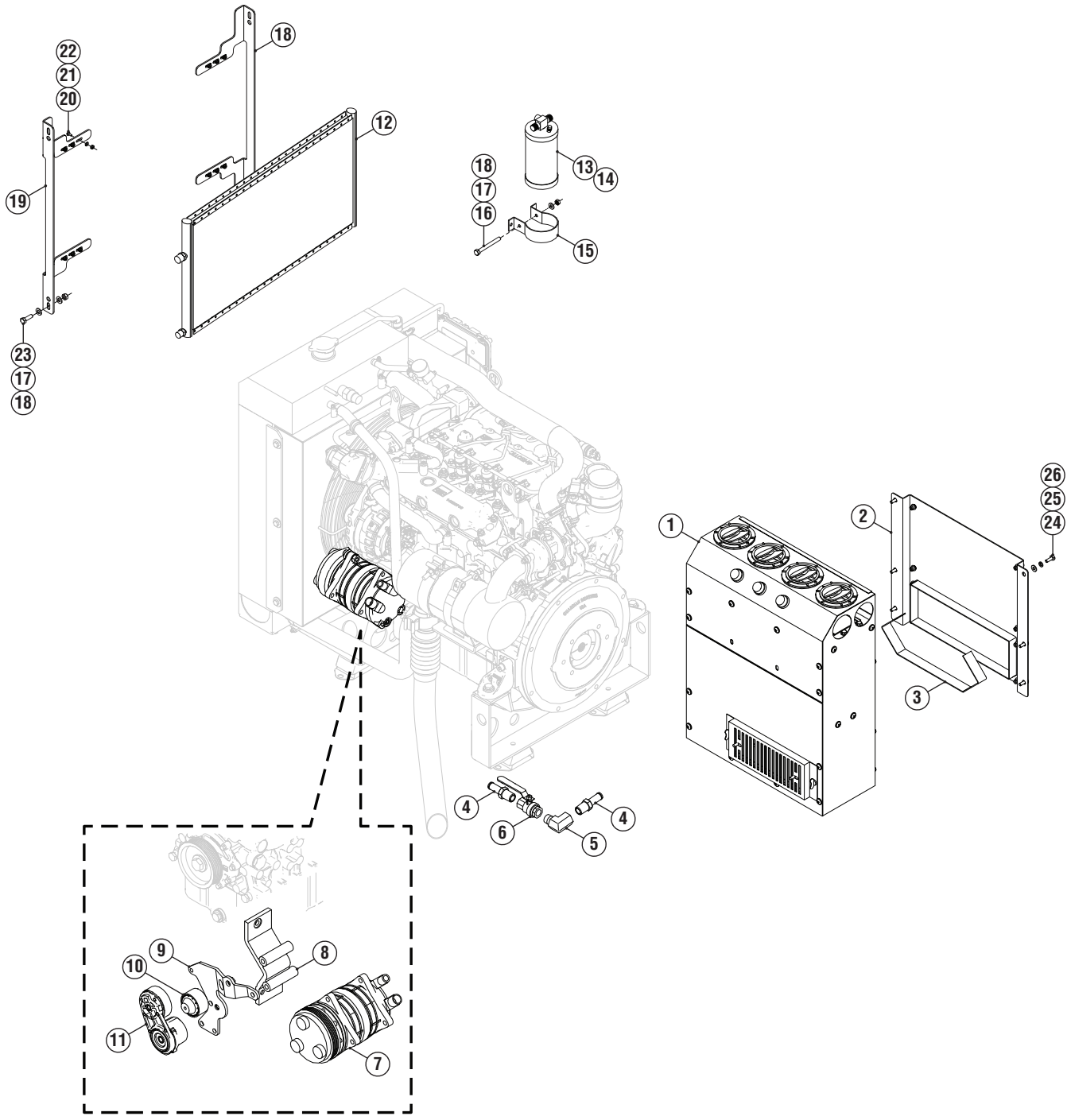


Figure 7-30. OPTION - Cab Assembly, AC/Heater (1 of 2)

OPTION - Cab Assembly, AC/Heater (1 of 2)

Item No.	Part Number	Qty.	Description	Remarks
GRP	1015031		Option - Air Conditioner/Heat	Not All Items Shown
1	1020046	1	Air Conditioner/Heater Kit	Includes 1-3,23-25
2	1015616	1	Air Conditioner/Heater Mount Weldment	
3	1016314	1	Air Conditioner/Heater Filter Spacer	
4	33492	2	Straight Fitting, 08MP-10HB, Crimped	
5	5502-8-8	1	Street Elbow Adapter, 90°, -8 NPTF / -8 NPTF	
6	32874	1	Ball Valve, 1/2 (Brass)	
REF	1015794	1	Engine Assembly, Hatz 4H50TIC w/AC	Includes 7
7	1015794-30	1	Air Conditioner Compressor	
8	1015794-33	1	AC Main Mounting Bracket	
9	1015794-34	1	AC Support Bracket	
10	1015794-32	1	AC Belt Idler	
11	1015794-31	1	AC Belt Tensioner	
12	1013712-01	1	Broom Air Conditioner Condenser	Included with 1
13	36745-07	1	Air Conditioner Dryer	Included with 1
14	36745-09	1	Switch, A/C High Pressure	Included with 13
15	36745-08	1	Receiver/Dryer Clamp	Included with 1
16	100-5-18-48-5	1	CSHH, 5/16-18 x 3.00, GR5	Included with 1
17	300-5	5	Washer, Flat, SAE, 5/16	Included with 1
18	200-5-18-5	5	Nut, Hex, 5/16-18, GR5	Included with 1
19	1013775	2	Condenser Mounting Bracket	
20	128-#6-32-6	12	Screw, HWH, Slotted, #6-32 x .375	
21	300-#6	12	Washer, Flat, SAE, #6	
22	205-#6-32-5	12	Nut, Lock, Nylon, #6-32, GR5	
23	100-5-18-16-5	2	CSHH, 5/16-18 x 1.00, GR5	
24	100-4-20-12-5F	10	CSHH, 1/4-20 x .75, GR5, FT	
25	302-4	10	Washer, Lock, 1/4	
26	300-4	10	Washer, Flat, SAE, 1/4	
REF	1016157	1	Air Conditioner/Heat Hose Kit	Not Shown

OPTION - CAB ASSEMBLY, AC/HEATER (2 OF 2)

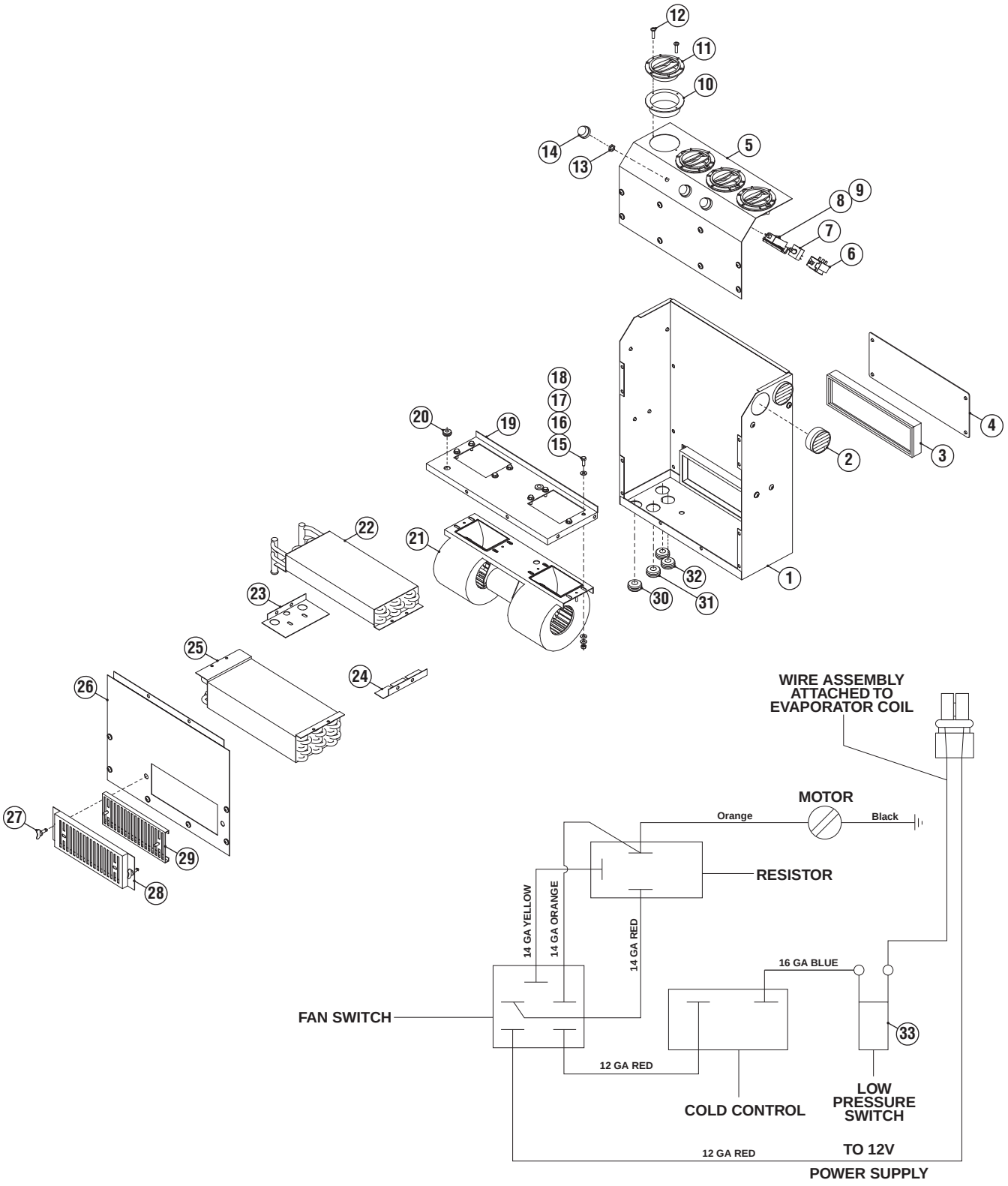


Figure 7-31. OPTION - Cab Assembly, AC/Heater (2 of 2)

OPTION - Cab Assembly,AC/Heater (2 of 2)

Item No.	Part Number	Qty.	Description	Remarks
OPT	1020046	1	Option - A/C Heat	Not All Items Shown
KIT	1023470	1	Kit, Air Conditioner/Heater, Sealed	Includes 1 - 31
1	1023470-01	1	Main Housing	
2	36745-17	2	Louver, 2.50: Round	
3	1017782	1	Filter Element, Cab Air	
4	38653-02	1	Cover, Air Filter	
5	1023470-02	1	Top Cover	
6	36745-19	1	A/C Thermostat	
7	36745-18	1	Switch, 3-Speed Fan	
8	36749-03	1	Control, Rotary Cable	
9	36749-04	1	Cable, Push/Pull	Not Shown
10	REF	4	Base, Louver	
11	36745-16	4	Louver, 3” Round	
12	122-4-20-16F	8	PHMS, Cross, 1/4-20 x 1.00, FT	
13	38525-22	3	Panel Nut	
14	37645-21	3	Knob, Fan Switch	
15	100-4-20-16-5F	8	CSHH, 1/4-20- x 1.00, GR5, FT	
16	300-4	16	Washer, Flat, SAE, 1/4	
17	302-4	8	Washer, Lock, 1/4	
18	200-4-20-5	8	Nut, Hex, 1/4-20, GR5	
19	36745-26	1	Blower Plate	
20	35465-06	2	Grommet	
21	1021528	1	Blowing Unit, Broom A/C	
22	36749-02	1	Heater Coil	
23	38653-07	1	Coil Bracket (Right)	
24	38653-06	1	Coil Bracket (Left)	
25	38653-05	1	Evaporator Coil Assembly	
26	1023470-03	1	Housing, Lower Cover	
27	36745-36	2	Fastener, Fractional Turn	
28	36745-29	1	Cover, Recirculation	
29	36745-30	1	Holder, Filter	
30	38525-34	2	Grommet	
31	38606-15	1	Grommet	
32	38606-16	1	Grommet	
33	TBD	1	Switch, Low Pressure	
34	1023211	2	Check Valve, Condensate Drain	Not Shown

OPTION - CAB ASSEMBLY, FILTRATION SYSTEM

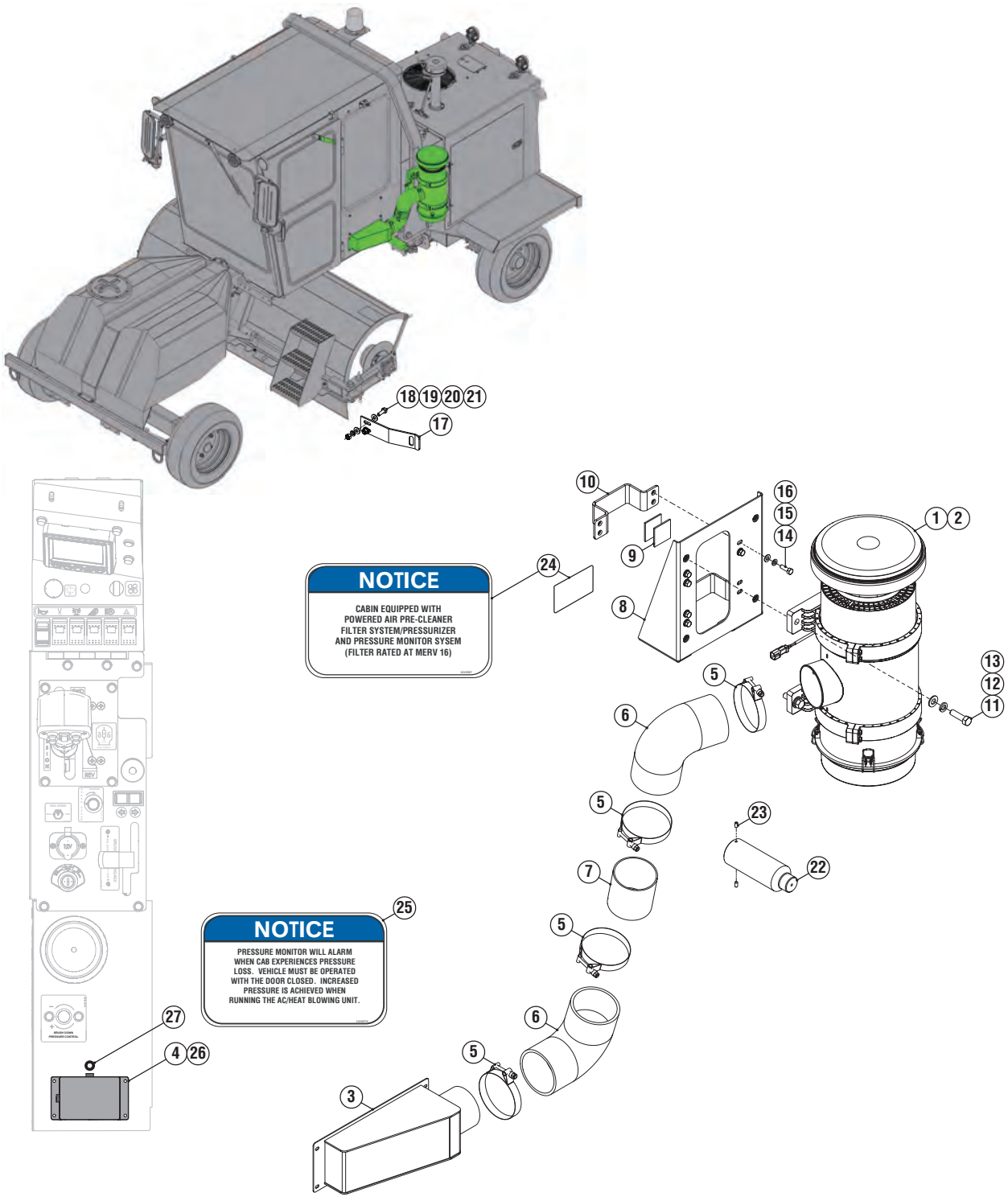


Figure 7-32. OPTION - Cab Assembly, Filtration System

OPTION - Cab Assembly, Filtration System

Item No.	Part Number	Qty.	Description	Remarks
OPT	1018731	1	Option - Cab Air Filtration	Includes Items 1-31
KIT	1018732	1	Kit, Cab Air Filtration	Includes Items 1-4
1	1018732-06	1	Filter System, Air Quality Unit	
2	1018732-07	1	Replacement Filter, Air Quality System	Not Shown
3	1018732-01	1	Plate, Cover, Air Filtration	
4	1018732-08	1	Monitor, Pressure, Air Filtration Unit	
5	953521243	4	Clamp, T-Bolt, 4.31-4.59 Range	
6	36919	2	Elbow, Rubber, 90°, 4.00 ID	
7	1018735	1	Pipe, 4.00 OD, .125 Wall x 4.00 Long	
8	1018740	1	Mounting Bracket, Air Filtration	
9	1018739	2	Bracket, Clap Mount	
10	981175	4	Rubber Pad, 2 x 2	
11	100-8-13-28-5	4	CSHH, 1/2-13 x 1.75, GR5	
12	302-8	4	Washer, Lock, 1/2	
13	300-8	4	Washer, Flat, SAE, 1/2	
14	100-5-18-16-5	8	CSHH, 5/16-18 x 1.00, GR5	
15	302-5	8	Washer, Lock, 5/16	
16	300-5	8	Washer, Flat, SAE, 5/16	
17	1023131	1	Extension, Door Latch	
18	100-4-20-12-5F	2	CSHH, 1/4-20 x .75, GR5, FT	
19	300-4	4	Washer, Flat, SAE, 1/4	
20	302-4	2	Washer, Lock, 1/4	
21	200-4-20-5	2	Nut, Hex, 1/4-20, GR5	
22	1019992	1	Extension, Door Stop	
23	113-4-20-8	2	Set Screw, HSKT, Cup, 1/4-20 x .50	
24	1019367	1	Decal - Oper, Cab Filter System	
25	1019376	1	Decal - Oper, Pressure Monitor	
26	116-#8-12	4	Screw, Self-Drilling, HWH, #8 x .75	Not Shown
27	1019362	1	Grommet, Rubber, .25 ID, .125 Panel, .438 Hole	
28	36746	1	Fuse, 5A ATC	Not Shown
29	36341	1	Fuse, 15A ATC	Not Shown
30	1018736	1	Harness, Air Filtration System	Not Shown
31	1019328	1	Sealer, Gasket, Maker, Loctite 518, 6ML	Not Shown

OPTION - FRONT WINSHIELD SPRAY WASHER

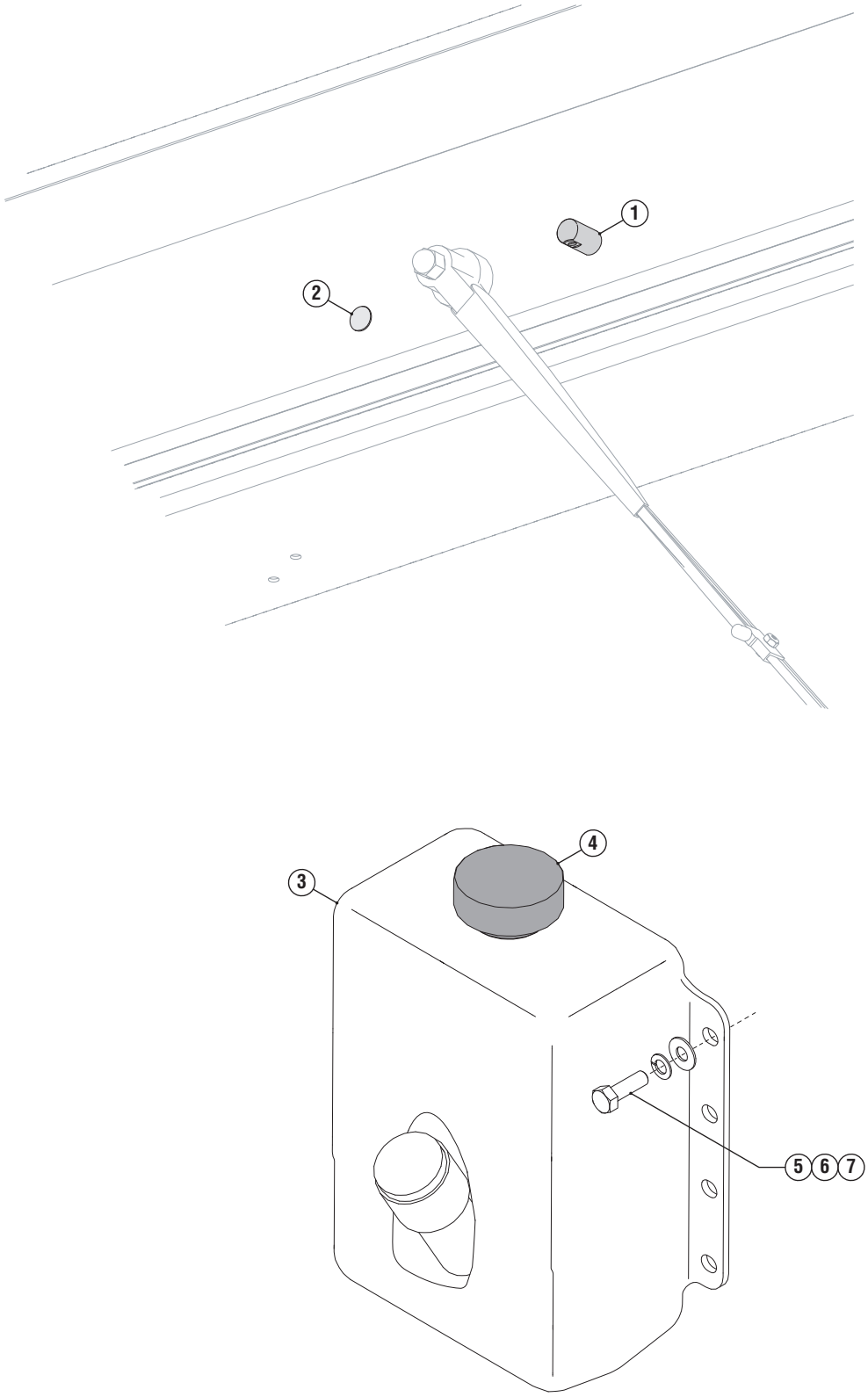


Figure 7-33. OPTION - Front Windshield Spray Washer

OPTION - Front Winshield Spray Washer

Item No.	Part Number	Qty.	Description	Remarks
OPT	1016181SRV	1	Option - Windshield Washer, Front	Includes Items 1-7
1	852810	1	Nozzle Kit, Windshield Washer	
2	35465-07	1	Grommet. Insulation, 3/4" ID	
3	33745-1	1	Washer, Tank and Pump	
4		1	Cap, Tank	
5	100-4-20-12-5F	2	CSHH, 1/4-20 x .75, GR5, FT	
6	302-4	2	Washer, Lock, 1/4	
7	301-4	2	Washer, Flat, USS, 1/4	
8	1016079	1	Harness, Windshield Washer Pumps	Not Shown
REF	35550	35	Hose, Windshield Washer, ø.188 ID (Per Foot)	Not Shown
REF	36340	1	Fuse, 10A, ATC	Not Shown

OPTION - CAB REAR WIPER/SPRAY WASHER

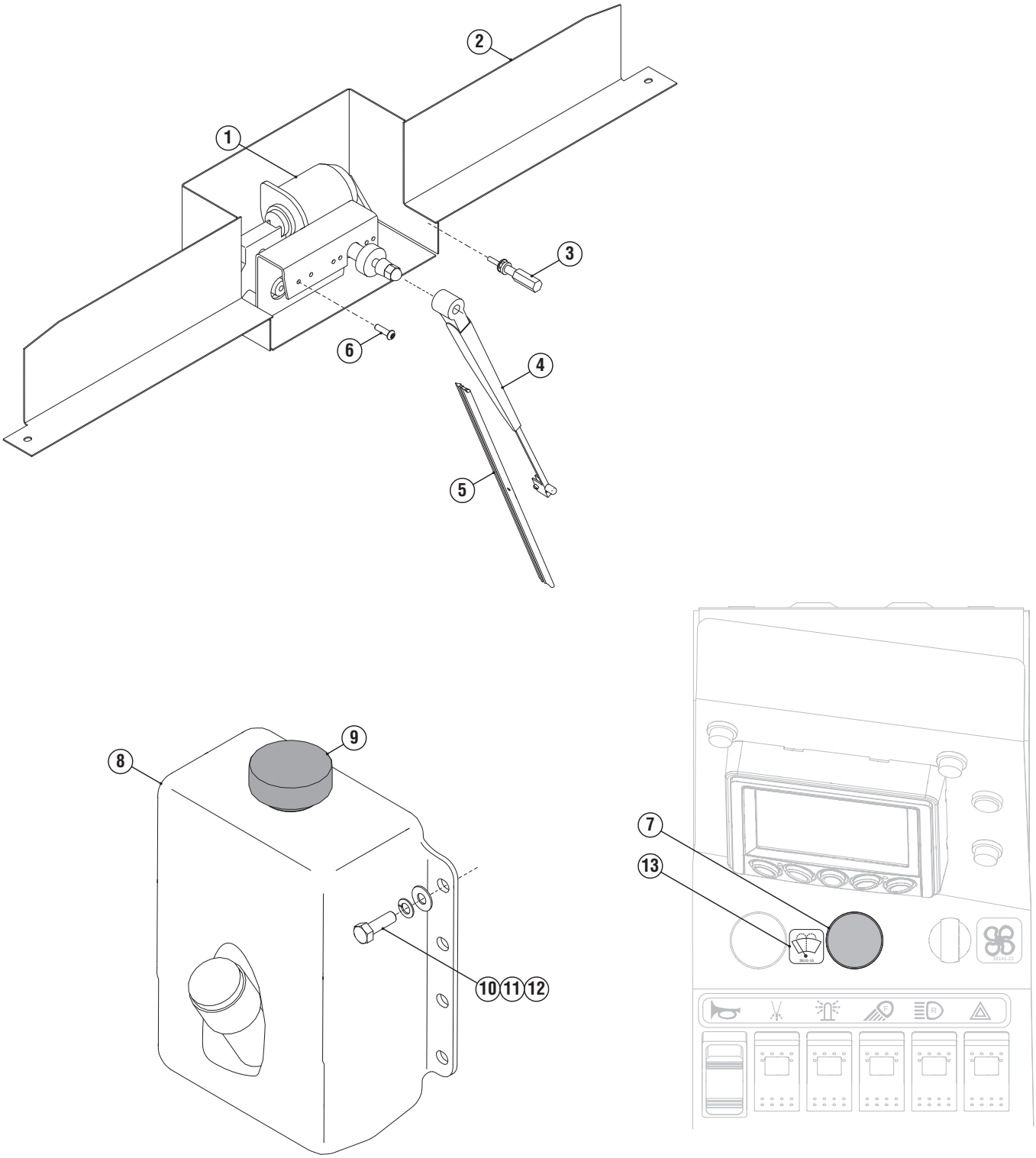


Figure 7-34. OPTION - Cab Rear Wiper/Spray Washer

OPTION - Cab Rear Wiper/Spray Washer

Item No.	Part Number	Qty.	Description	Remarks
OPT	1016182	1	Option - Windshield Wiper/Washer, Rear	Includes Items 1-12
1	151170	1	Motor, Windshield Wiper	
2	1015612	1	Weldment, Cab Rear Wiper Cover	
3	852810	1	Nozzle Kit, Windshield Washer	
4	1021646	1	Arm, Wiper, 14"	
5	1021649	1	Blade, Wiper, 16"	
6	81277	1	CSBHS, 1/4-20 x .875, SS	
7	853090	1	Switch, Wiper/Washer	
8	33745-1	1	Washer, Tank and Pump	
9		1	Cap, Tank	
10	100-4-20-12-5F	2	CSHH, 1/4-20 x .75, GR5, FT	
11	302-4	2	Washer, Lock, 1/4	
12	301-4	2	Washer, Flat, USS, 1/4	
13	35465-04	1	Grommet, Insulation, .75 ID	Not Shown
14	39141-16	1	Decal, Wipers	

OPTION - FNR BRUSH ROTATION

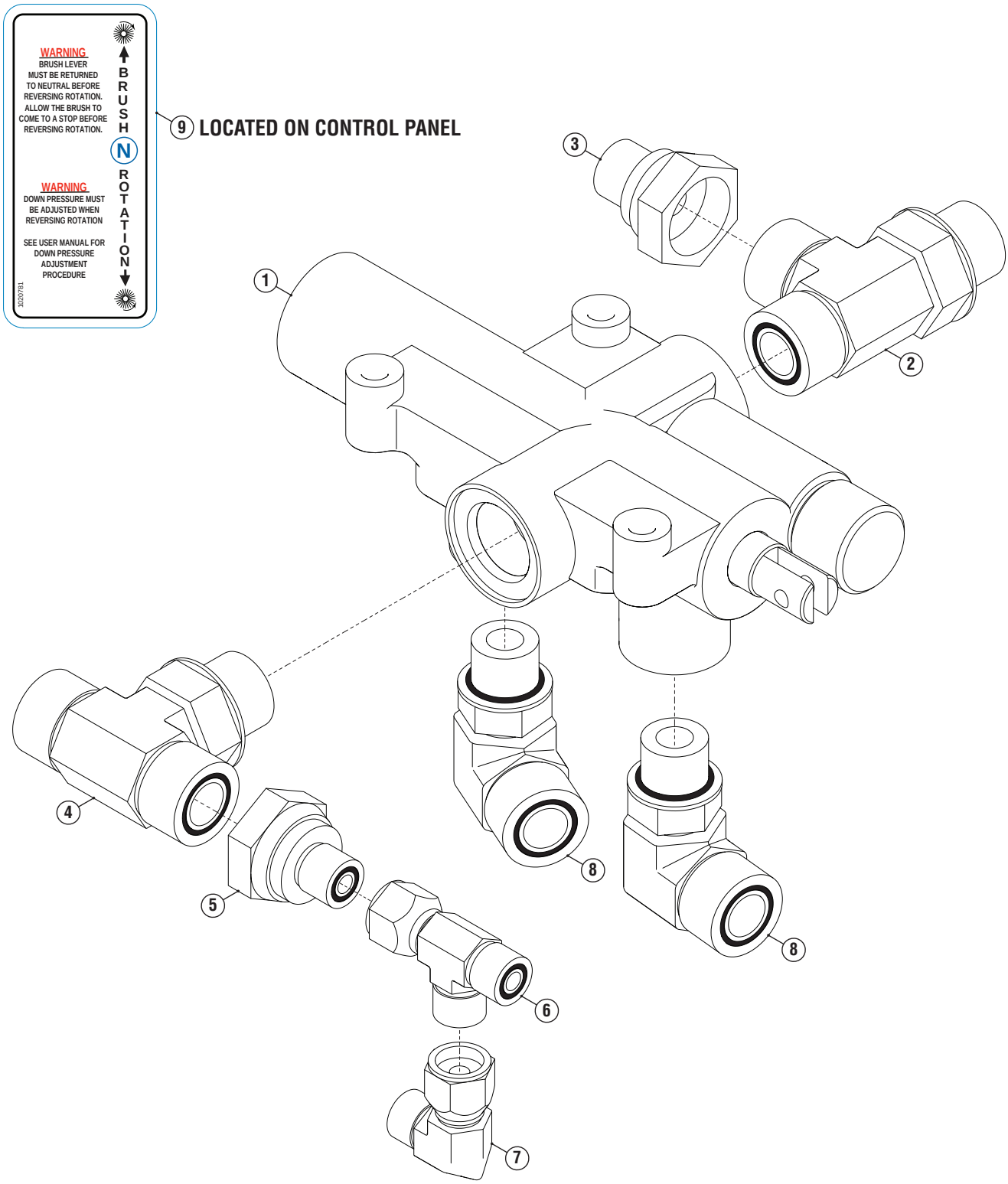


Figure 7-35. OPTION - FNR Brush Rotation

OPTION - FNR Brush Rotation

Item No.	Part Number	Qty.	Description	Remarks
OPT	1020778	1	Option - FNR Brush Rotation	
1	1022555	1	Valve, Motor Control, FNR	
2	FS6804-12-12-12-NWO-FG	1	Adapter, Run Tee, -12 ORFS/-12 O-Ring/12 ORFS	
3	FS2406-12-08	1	Adapter, Reducer, Straight, -12 FFORX/-8 ORFS	
4	FS6803-12-12-12-NWO-FG	1	Adapter, Branch Tee, -12 ORFS/-12 ORFS/-12 O-Ring	
5	FS2406-12-06	1	Adapter, Reducer, Straight, -12 FFORX/-6 ORFS	
6	FS6602-06-06-06-FG	1	Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-6 ORFS	
7	FS6500-06-06-FG	1	Adapter, Elbow, 90°, -6 ORFS/-6 ORFS	
8	FS6801-12-10-NWO-FG	2	Adapter, Elbow, 90°, -12 ORFS/-10 ORFS	
9	1020781	1	Decal, Oper, FNR, Brush Rotation	

OPTION - FOOT PEDAL CONTROLS

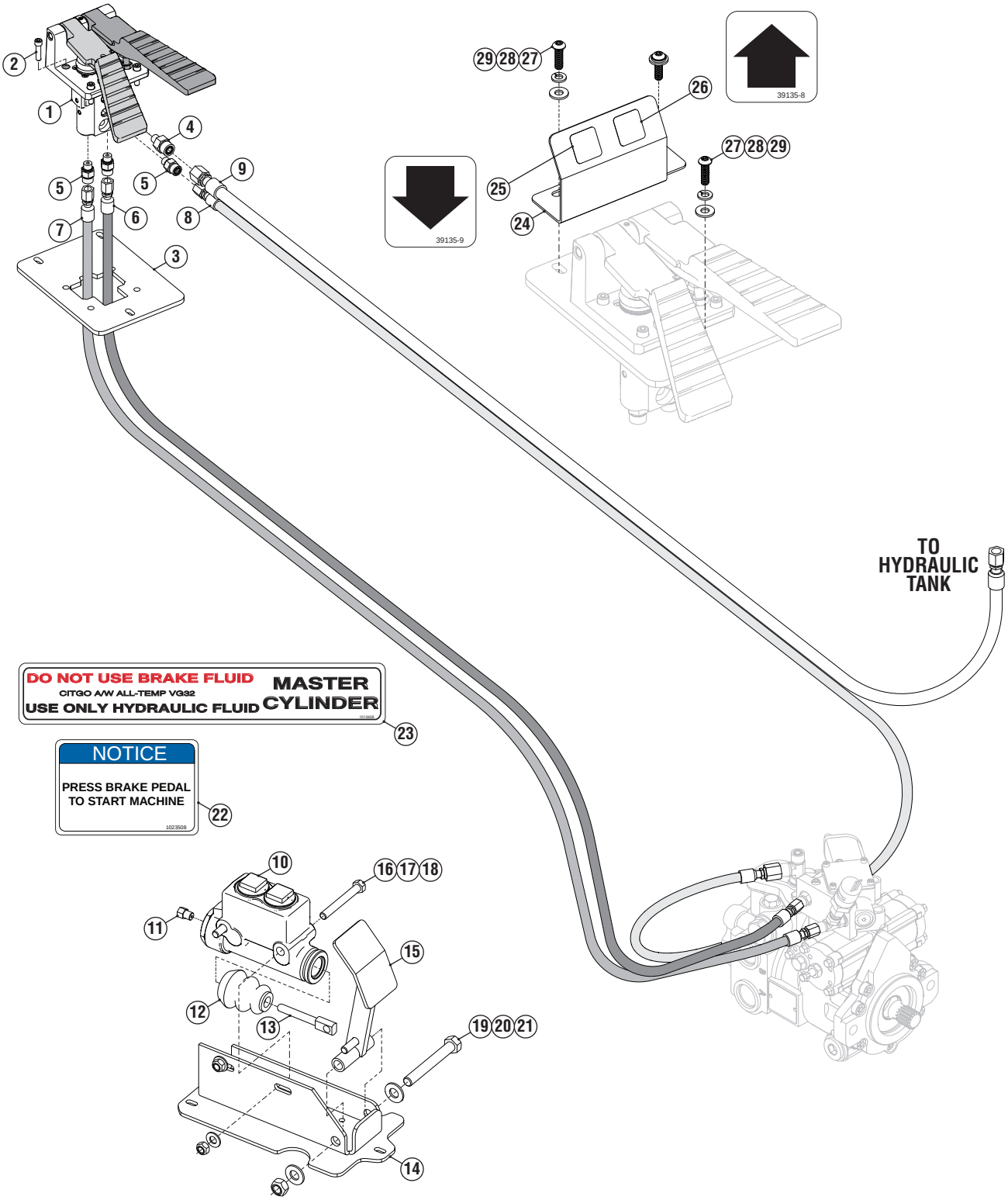


Figure 7-36. OPTION - Foot Pedal Controls

OPTION - Foot Pedal Controls

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023511	1	Option - Foot Pedal, FNR	Not All Items Shown
ASSY	1023496	1	Assembly, Foot Pedal w/ORFS Adapters	Includes 1-5
1	1022801	1	Pedal, FNR, Hydraulic	
2	102-4-20-16-F	4	CSSH, 1/4-20 x 1.00, FT	
3	1023183	1	Mount, Foot Pedal, FNR	
4	FS6400-06-04-O	1	Adapter, Straight, -6 ORFS/-4 O-Ring	
5	FS6400-04-04-O	3	Adapter, Straight, -4 ORFS/-4 O-Ring	
KIT	1023243	1	Kit, Hydraulic Hose, Foot Pedal	Includes 6-9
6	F387TCJCJC040404-94	1	Hose Assembly, Pump X1 to Pedal 1	Tag# 1
7	F387TCJCJC040404-95	1	Hose Assembly, Pump X2 to Pedal 2	Tag# 2
8	F387TCJCJC040604-85	1	Pump Gage Port to Pedal P	Tag# 3
9	F387TCJCJC060606-85	1	Pedal to Hydraulic Tank	Tag# 4
ASSY	1023497	1	Assembly, Brake Pedal	Includes 10-21
10	1009383	1	Master Cylinder	
11	38277	1	Fitting, Master Cylinder Adapter	
12	1010648	1	Boot, Master Cylinder	
13	1009786	1	Pushrod, Brake	
14	1023181	1	Mount, Brake Pedal	
15	28650SRV	1	Brake Pedal	
16	100-6-16-48-5	2	CSHH, 3/8-16 x 3.00, GR5	
17	300-6	2	Washer, Flat, SAE, 3/8	
18	204-6-16-5	2	Nut, Lock, Stover, 3/8-16, GR5	
19	100-8-13-60-5	1	CSHH, 1/2-13 x 3.75, GR5	
20	300-8	2	Washer, Flat, SAE, 1/2	
21	204-8-13-5	1	Nut, Lock, Stover, 1/2-13, GR5	
22	1023508	1	Decal, Notice, Press Brake	Located Inside Cab
23	1010608	1	Decal, Info, Master Cylinder	Located Inside Cab
24	1024402	1	Plate, Formed, Decal Mount	
25	39135-09	1	Decal, Info, Arrow, Down	
26	39135-08	1	Decal, Info, Arrow, Up	
27	81106	3	CSBHS, 1/4-20 x .875, SS	
28	302-4	3	Washer, Lock, 1/4	
29	300-4	3	Washer, Flat, SAE, 1/4	

OPTION - FOOT PEDAL CONTROLS, PUMPS

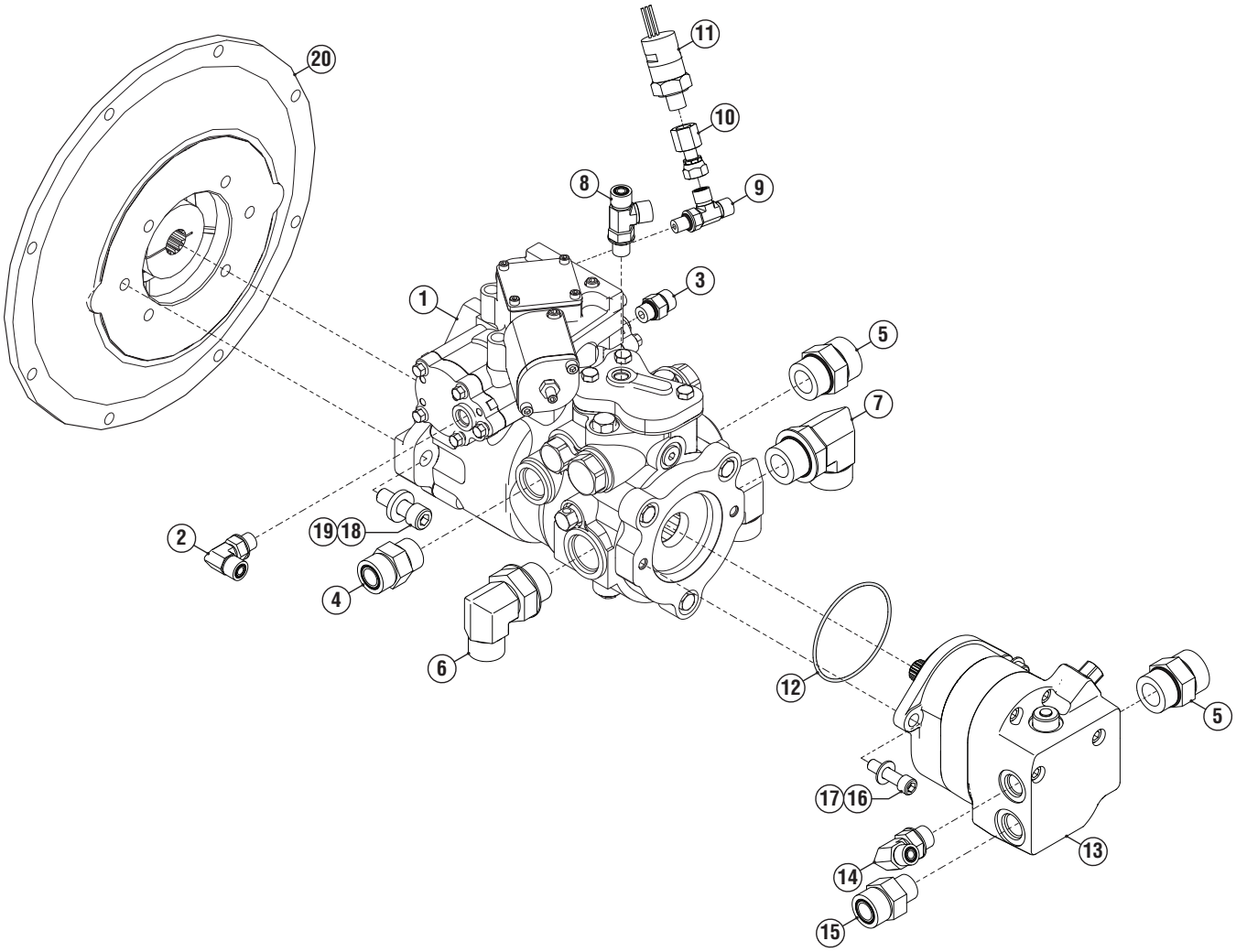


Figure 7-37. OPTION - Foot Pedal Controls, Pumps

OPTION - Foot Pedal Controls, Pumps

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023511	1	Option - Foot Pedal, FNR	Not All Items Shown
ASSY	1023528	1	Assembly, Pumps w/Adapters	Includes Items 1-14
1	1023206	1	Pump, Hydraulic, Piston, HDC, 2.8 CIR	
2	FS6801-06-06-NWO-FG	1	Adapter, Elbow, 90°, -06 ORFS/-06 O-Ring	
3	FS6400-06-06-O	1	Adapter, Straight, -6 ORFS/-6 O-Ring	
4	FS6400-12-12-O	1	Adapter, Straight, -12 ORFS/-12 O-Ring	
5	FS6400-16-16-O	2	Adapter, Straight, -16 ORFS/-16 O-Ring	
6	FS6801-12-16-NWO-FG	1	Adapter, Elbow, 90°, -12 ORFS/-16 O-Ring	
7	FS6801-16-16-NWO-FG	1	Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	
8	FS6804-06-06-06-NWO-FG	1	Adapter, Run Tee, -6 ORFS/-6 O-Ring/-6 ORFS	
9	FS6804-04-04-04-NWO-FG	1	Adapter, Run Tee, -4 ORFS/-4 O-Ring/-4 ORFS	
10	4_G65L-S	1	Adapter, Straight, -4/Swivel ORFS/-4 O-Ring	
11	1023495	1	Switch, Pressure, 10-35 PSI	
12	36808	1	O-Ring, 3.237 ID x .103, SAE, 152	
13	1005574SRV	1	Pump, Hydraulic, Gear, 1.94 CIR	
14	FS6801-06-08-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	
15	FS6400-12-10-O	1	Adapter, Straight, -12 ORFS/-10 O-Ring	
16	102-6-16-24-F	2	CSSH, 3/8-16 x 1.50, FT	
17	300-6	2	Washer, Flat, SAE, 3/8	
18	102-8-13-24-F	2	CSSH, 1/2-13 x 1.50, FT	
19	312-8	2	Washer, Hardened, SAE, Thick, 1/2	
20	1015794-52	1	Assembly, Pump Drive Plate	
21	1016290	1	Cable, Push, Pull, 1/4-28 w/Threaded Caps	Not Shown
22	989817	1	Clevis, 1/4-28 UNF x .375 ID	Not Shown

OPTION - FOOT PEDAL CONTROLS, DETAILS

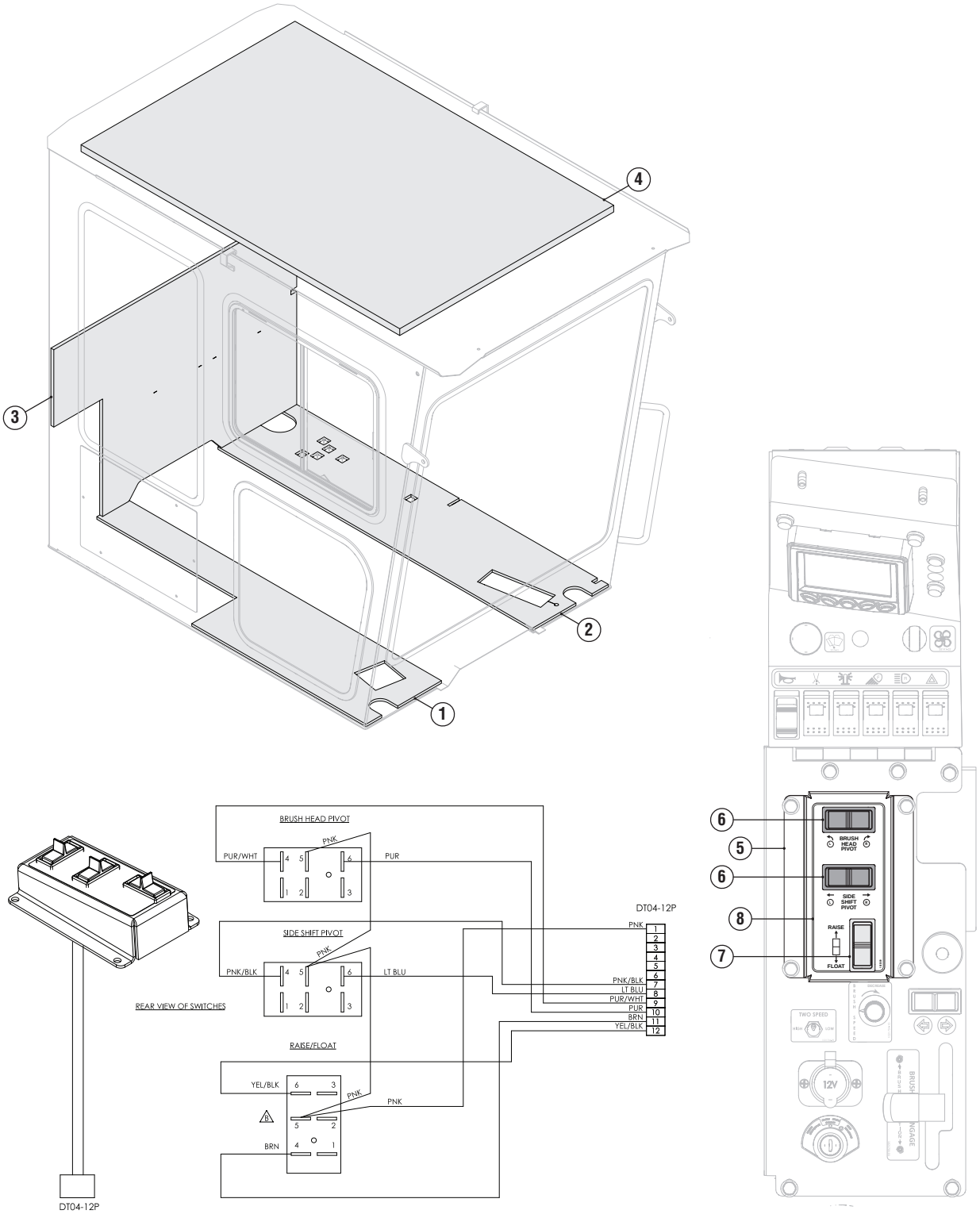


Figure 7-38. OPTION - Foot Pedal Controls, Details

OPTION - Foot Pedal Controls, Details

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023511	1	Option - Foot Pedal, FNR	Not All Items Shown
KIT	1023191	1	Kit, Insulation, Foot Pedal	
1	1023191-01	1	Cab, Insulation, Floor Mat - RH	
2	1023191-02	1	Cab, Insulation, Floor Mat - LH	
3	1015083-02	1	Cab, Insulation, Rear	
4	1015083-04	1	Cab, Insulation, Headliner	
ASSY	1023493	1	Harness, Broom Controls	Not All Items Shown
5	1023492	1	Control Box	
6	1023525	2	Switch, Rocker, MOM/OFF/MOM, Paddle	
7	1023526	1	Switch, Rocker, ON/OFF/MOM, Paddle	
8	1023509	1	Decal, Oper, Brush Controls	

OPTION - REAR CAMERA SYSTEM

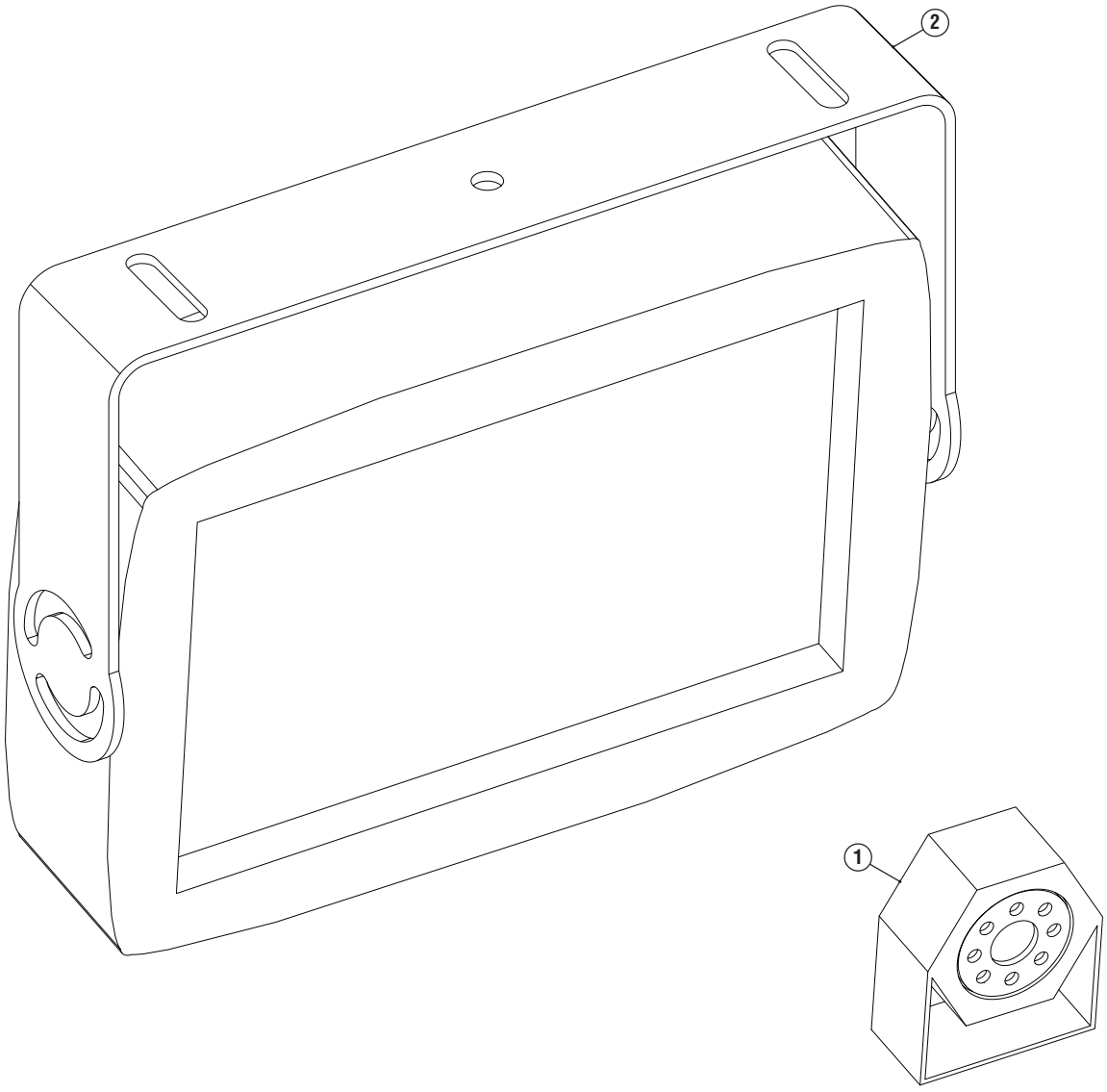


Figure 7-39. OPTION - Rear Camera System

OPTION - Rear Camera System

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023052		Option - Back-Up Camera System	
1	1003907	1	Outdoor Color Camera, 12VDC	
2	1004617	1	Camera Display, Color	
3	1022961	1	Camera System Cable, 4-Pin, 16'	Not Shown

OPTION - STEREO SYSTEM

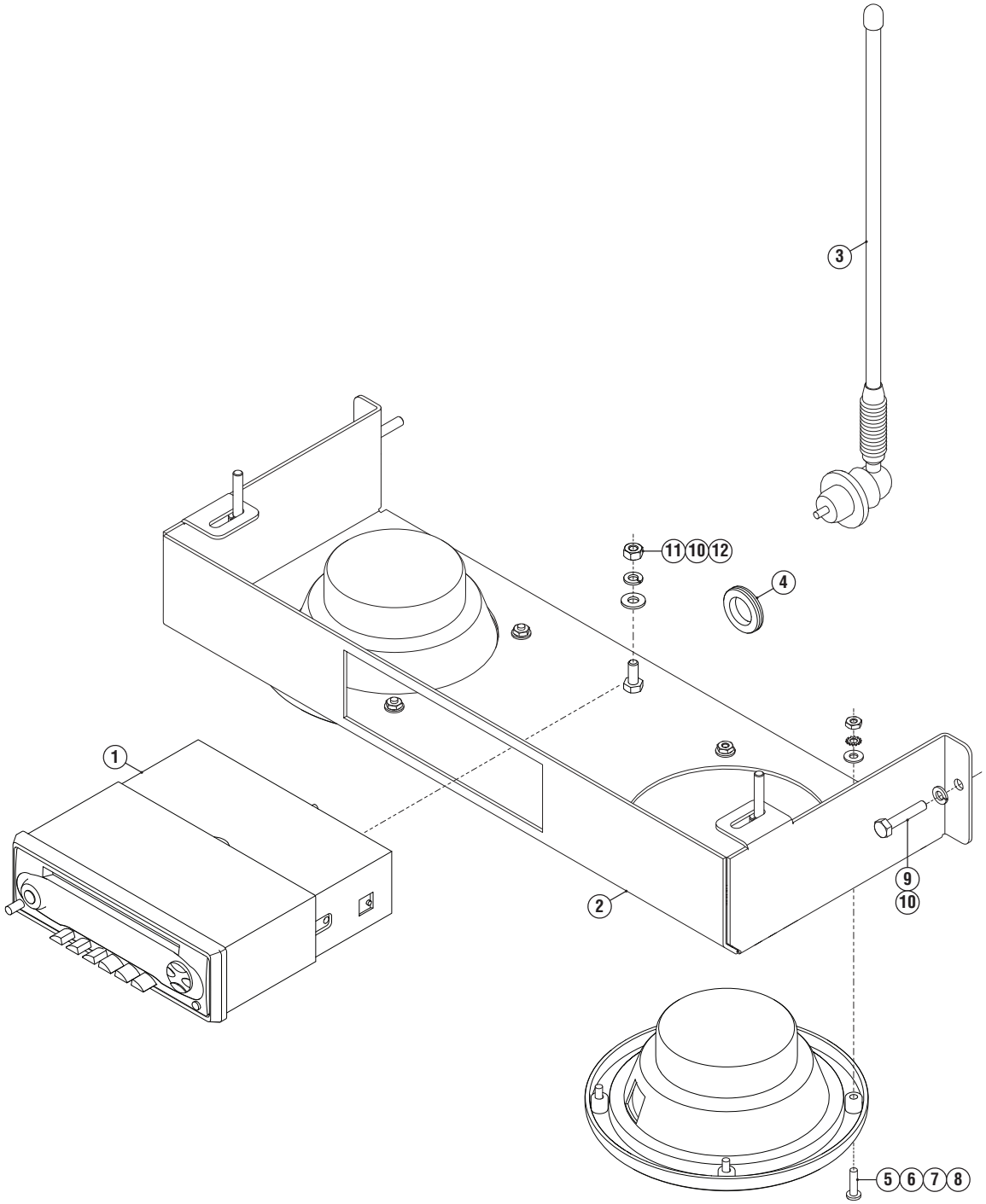


Figure 7-40. OPTION - Stereo System

OPTION - Stereo System

Item No.	Part Number	Qty.	Description	Remarks
OPT	1016867		Option - Radio/CD w/Speakers & Antenna	
1	1011314	1	Radio, AM/FM/CD w/Auxiliary Input & Speakers	
2	1008092	1	Mount, Weldment, Radio/Speaker	
3	1002021	1	Antenna, AM/FM, Flexible	
4	35465-07	1	Grommet, Insulation, .75 ID x 1.00 OD x .063	
5	122-#10-24-12F	8	PHMS, Cross, #10-24 x .75, FT	
6	300-#10	8	Washer, Flat, SAE, #10	
7	303-#10	8	Washer, Lock, External Tooth, #10	
8	209-#10-24	8	Nut, Hex, Machine, #10-24	
9	100-4-28-12-5F	4	CSHH, 1/4-28 x .75, GR5, FT	
10	302-4	5	Washer, Lock, 1/4	
11	200-4-20-5	1	Nut, Hex, 1/4-20, GR5	
12	300-4	4	Washer, Flat, SAE, 1/4	
REF	35136-6	2	Plug, Hole, ø.75, Flush Mount, Plastic	Not Shown

OPTION - BRUSH WATER SPRAY DOWN

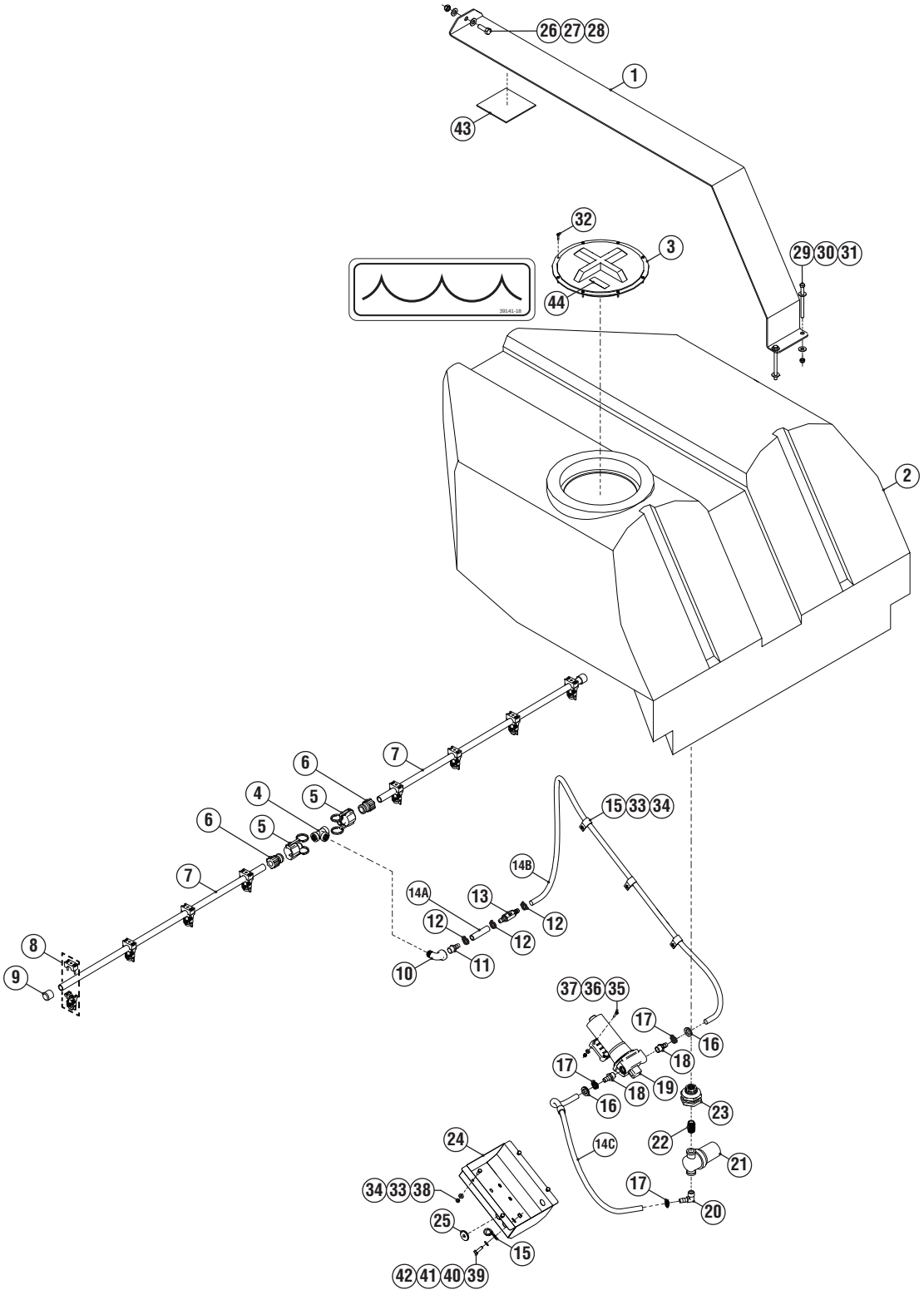


Figure 7-41. OPTION - Brush Water Spray Down

OPTION - Brush Water Spray Down

Item No.	Part Number	Qty.	Description	Remarks
OPT	1015862		Option - Water Spray Down	
1	1015544	1	Tank Strap	
ASSY	1012308SRV	1	Assembly, Water Tank, 150 GAL	Includes 2-3, 23,32
2	36727	1	Tank, Water, 150 GAL	
3	1012310	1	Water Tank Cap w/Ring, 1/8 Turn	
4	99845	1	Galvanized Pipe Tee, 08FP	
5	1007326	2	Cam Lock Fitting, 3/4" F to 1/2" M Pipe	
6	1007325	2	Cam Lock Fitting, 3/4" M to 1/2" F Pipe	
7	1007328	2	HD Brush Spray Pipe	
8	1008851	8	Nozzle Assembly, 08 Pipe, Straight	
9	91152	2	Galvanized Pipe Cap, .500	
10	38843	1	Galvanized Pipe Street Elbow, 90°, .500	
11	X427	1	Straight Fitting, 08MP-08HB, Crimped	
12	33163	3	Hose Clamp, #08 (.50-.91)	
13	36883	1	Check Valve, 08HB, 5 PSI, Poly	
14	6352	-	1/2 Low Pressure Hose - Black (Per Foot)	
14A	REF	0.25	1/2 Low Pressure Hose - Black x 3.00"	
14B	REF	6	1/2 Low Pressure Hose - Black x 72.00"	
14C	REF	3	1/2 Low Pressure Hose - Black x 36.00"	
15	871111602	5	Insulated Band Clamp, 3/4"	
16	35465-07	2	Insulation Grommet, .75 ID	
17	33164	3	Hose Clamp, #10 (.56-1.06)	
18	38093	1	Straight Port Kit, 08 - 2 pcs.	
19	1004598SRV	1	Diaphragm Water Pump w/Pressure Switch	
20	70319	1	90° Fitting, 08MP-08HB, Poly	
21	36926	1	Nylon Strainer Assembly, 1/2PT, 100 Mesh	
22	36810	1	Close Pipe Nipple, .500, PVC	
23	36731	1	Bulkhead Pipe w/Nut & Gasket	
24	1015545	1	Water Pump Cover	
25	35465-09	1	Insulation Grommet, .375 ID x 1.00 Groove	
26	100-8-13-24-5	1	CSHH, 1/2-13 x 1.50, GR5	
27	300-8	2	Washer, Flat, SAE, 1/2	
28	217-8-13	1	Nut, Lock, Flexloc, 1/2-13	
29	100-6-16-80-5	2	CSHH, 3/8-16 x 5.00, GR5	
30	301-6	4	Washer, Flat, USS, 3/8	
31	217-6-16	2	Nut, Lock, Flexloc, 3/8-16	
32	116-#8-12	8	Screw, Self-Drilling, HWH, #8 x .75	
33	300-4	8	Washer, Flat, SAE, 1/4	
34	217-4-20	8	Nut, Lock, Flexloc, 1/4-20	
35	122-#10-32-16F	4	PHMS, Cross, #10-32, GR5	

OPTION - BRUSH WATER SPRAY DOWN ASSEMBLY (CONT.)

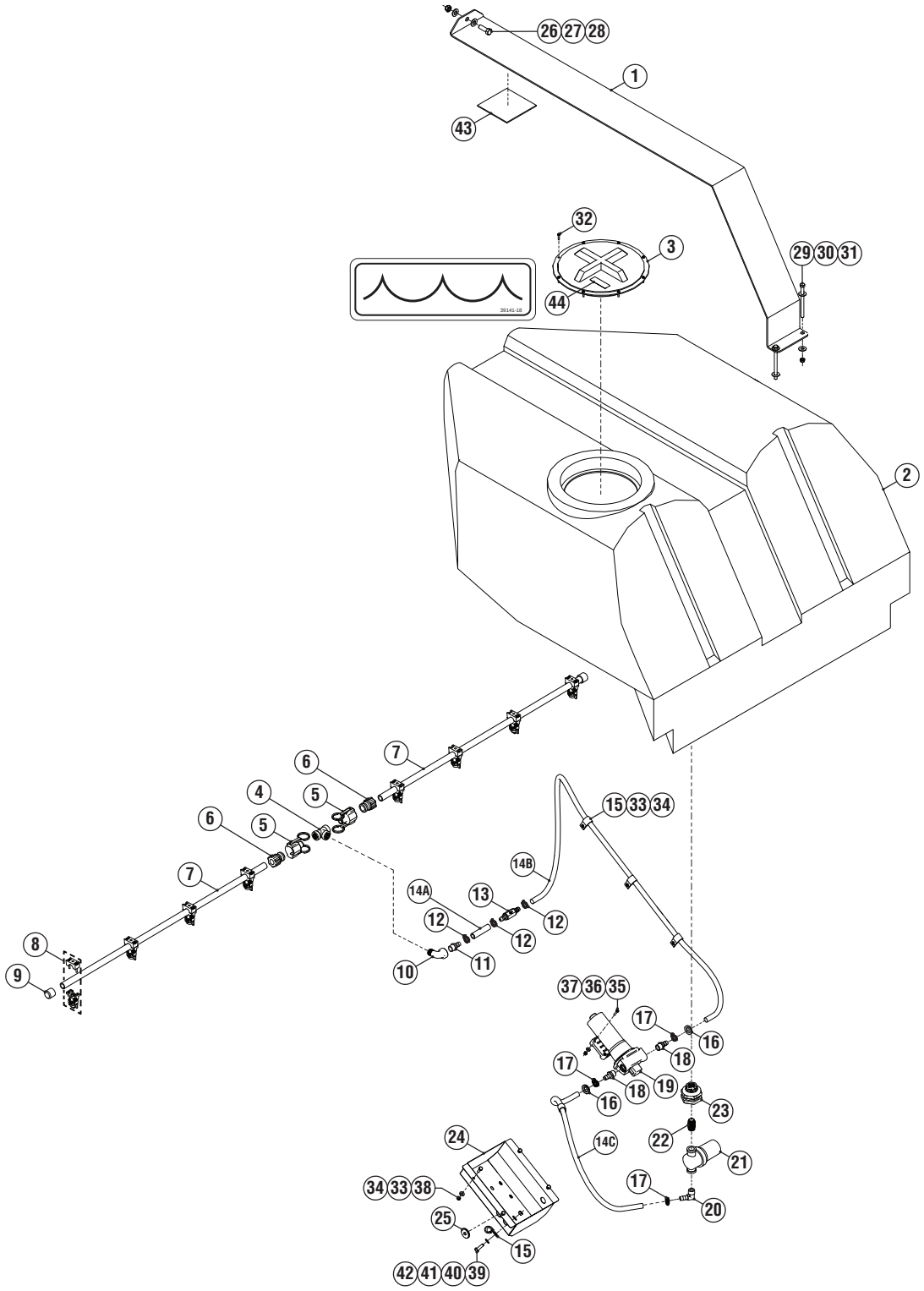


Figure 7-41. OPTION - Brush Water Spray Down Assembly (Cont.)

OPTION - Brush Water Spray Down Assembly (Cont.)

Item No.	Part Number	Qty.	Description	Remarks
36	300-#10	4	Washer, Flat, SAE, #10	
37	205-#10-32-5	4	Nut, Lock, Nylon, #10-32, GR5	
38	100-4-20-12-5F	4	CSHH, 1/4-20 x .75, GR5, FT	
39	100-5-18-16-5	1	CSHH, 5/16-18 x 1.00, GR5	
40	300-5	2	Washer, Flat, SAE, 5/16	
41	302-5	1	Washer, Lock, 5/16	
42	200-5-16-5	1	Nut, Hex, 5/16-18, GR5	
43	1021245	1	Rubber Pad	
44	39141-18	1	Decal, Oper, Water Tank	

OPTION - CURB BRUSH

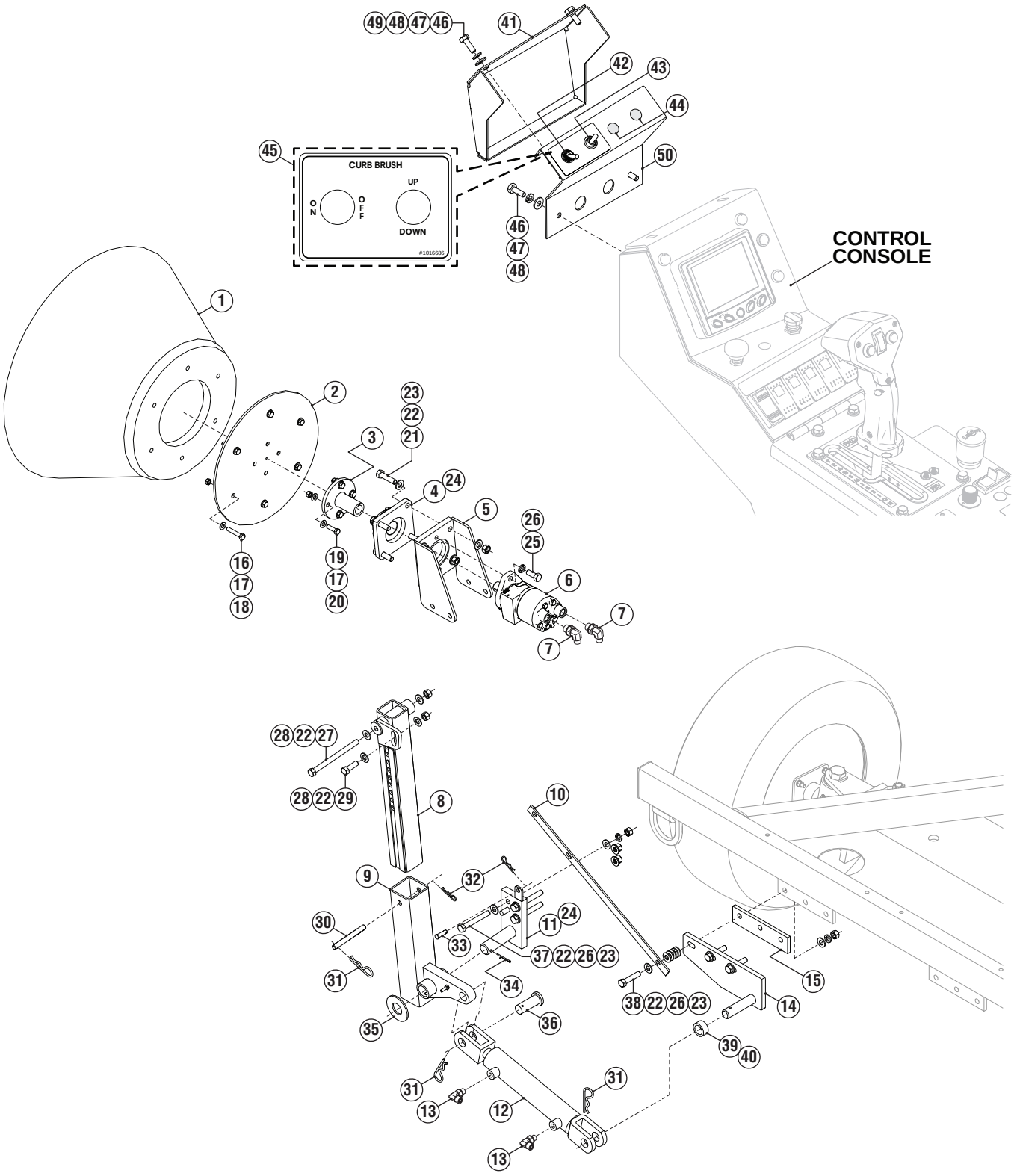


Figure 7-42. OPTION - Curb Brush

OPTION - Curb Brush

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023225	1	Option - Curb Brush Assembly	
1	39000	1	Gutter Brush	
2	28925	1	Brush Motor Plate	
3	28928	1	Brush Motor Mount Weldment	
4	312030	1	Bearing Flange, 4 Bolt, 1.50	
5	28931	1	Motor Mount Weldment	
6	38122	1	Hydraulic Motor, LSHT, 8.0 DISP	
7	FS6801-06-08-NWO-FG	2	Elbow Adapter, 90°, -6 ORFS / -8 O-Ring	
8	28943	1	Adjustment Plate Weldment	
9	28937	1	Length Adjustment Tube Weldment	
10	29121	1	Raised Bar Lock	
11	28941SRV	1	Bumper Mount Weldment	
12	610110	1	Hydraulic Cylinder, 2.00 x 8.00 x 1.00 Rod	
-	910145-01	1	Seal Kit	Repair Part
13	FS6801-06-06-NWO-FG	2	Elbow Adapter, 90°, -6 ORFS / -6 O-Ring	
14	28946	1	Center Bumper Mount Weldment	
15	1016712	1	Curb Brush Mount Spacer	
16	100-6-16-32-5	6	CSHH, 3/8-16 x 2.00, GR5	
17	300-6	10	Washer, Flat, SAE, 3/8	
18	204-6-16-5	6	Nut, Lock, Stover, 3/8-16, GR5	
19	100-6-16-20-5	4	CSHH, 3/8-16 x 1.25, GR5	
20	200-6-16-5	4	Nut, Hex, 3/8-16, GR5	
21	100-8-13-32-5	4	CSHH, 1/2-13 x 2.00, GR5	
22	300-8	20	Washer, Flat, SAE, 1/2	
23	200-8-13-5	10	Nut, Hex, 1/2-13, GR5	
24	33684	2	Short Straight Lube Fitting, 02MP	
25	100-8-13-20-5F	2	CSHH, 1/2-13 x 1.25, GR5, FT	
26	302-8	8	Washer, Lock, 1/2	
27	100-8-13-104-5	1	CSHH, 1/2-13 x 6.50, GR5	
28	204-8-13-5	2	Nut, Lock, Stover, 1/2-13, GR5	
29	100-8-13-24-5	1	CSHH, 1/2-13 x 1.50, GR5	
30	29130	1	Adjustment Pin	
31	36544	3	Cotter Pin, ø.177, 7 GA	
32	37108	2	Cotter Pin, ø.094 x .250 x .500	
33	81214	1	Clevis Pin, .375 x 1.25, 1.015 Grip	
34	405-3-24	1	Cotter Pin, ø.188 x 1.50	
35	301-20	1	Washer, Flat, USS, 1-1/4	
36	1011476	1	Clevis Pin, 1.00 x 2.50 HDM	
37	100-8-13-60-5	3	CSHH, 1/2-13 x 3.75, GR5	
38	100-8-13-36-5	3	CSHH, 1/2-13 x 2.25, GR5	

OPTION - CURB BRUSH (CONT.)

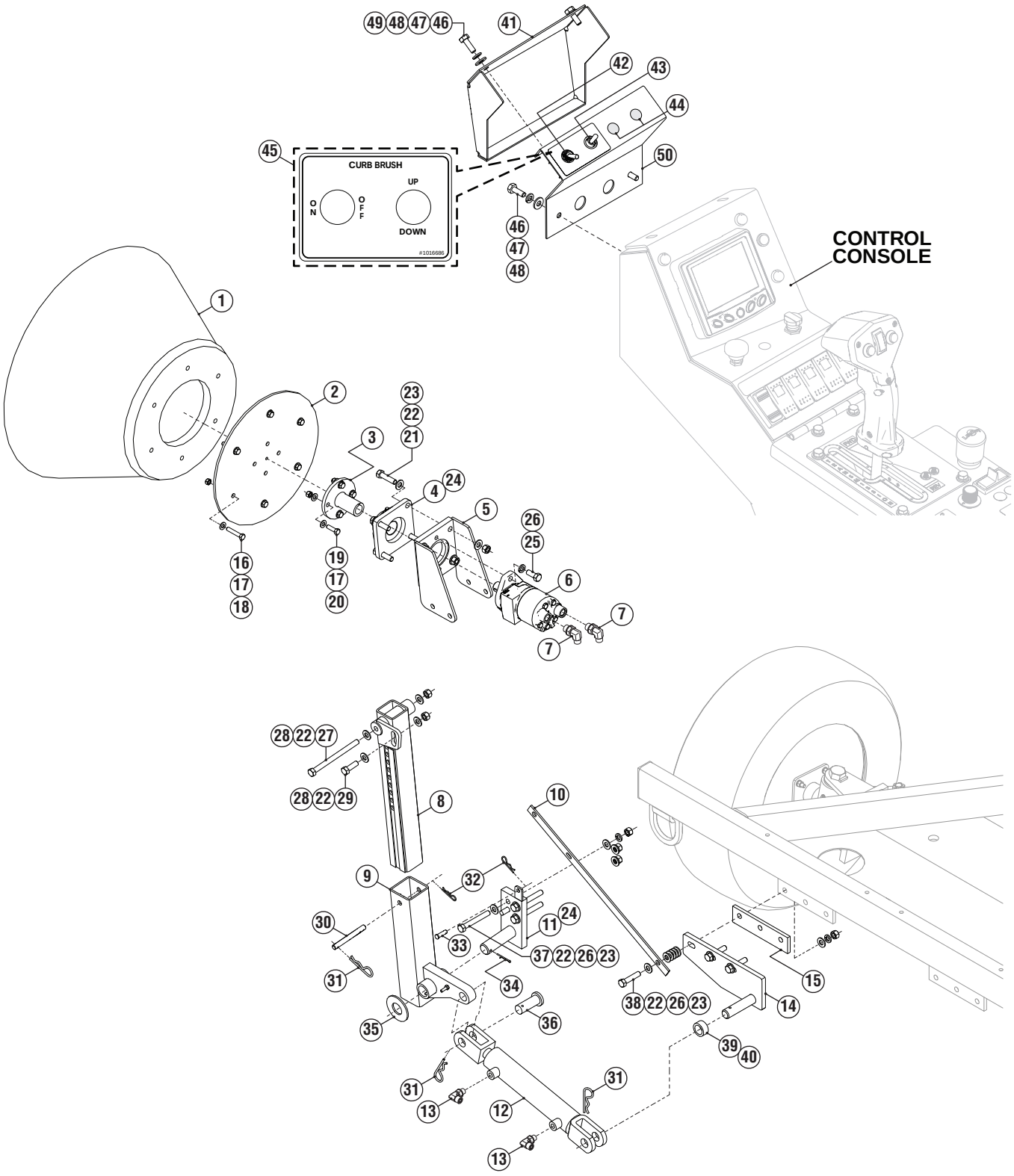


Figure 7-42. OPTION - Curb Brush (Cont.)

OPTION - Curb Brush (Cont.)

Item No.	Part Number	Qty.	Description	Remarks
39	28932	1	Spacer Tube	
40	300-16	1	Washer, Flat, SAE, 1.00	Not Shown
41	1016637	1	Control Box	
42	38157	1	Toggle Switch, DPST, 2 Position	
43	38961	1	Toggle Switch, DPDT, 3 Position	
44	35136-4	2	Flush Mount Hole Plug, .500	
45	1016686	1	Decal, Curb Brush	
46	100-4-20-12-5	4	CSHH, 1/4-20 x .75, GR5	
47	302-4	4	Washer, Lock, 1/4	
48	300-4	8	Washer, Flat, SAE, 1/4	
49	200-4-20-5	4	Nut, Hex, 1/4-20, GR5	Not Shown
50	1021739	1	Control Panel, Curb Brush/Snow Plow	
51	1016123	1	Harness, Valves, Curb Brush	Not Shown
52	1018050	1	Harness, Controls, Curb Brush	Not Shown

OPTION - CURB BRUSH, HYDRAULICS (1 OF 2)

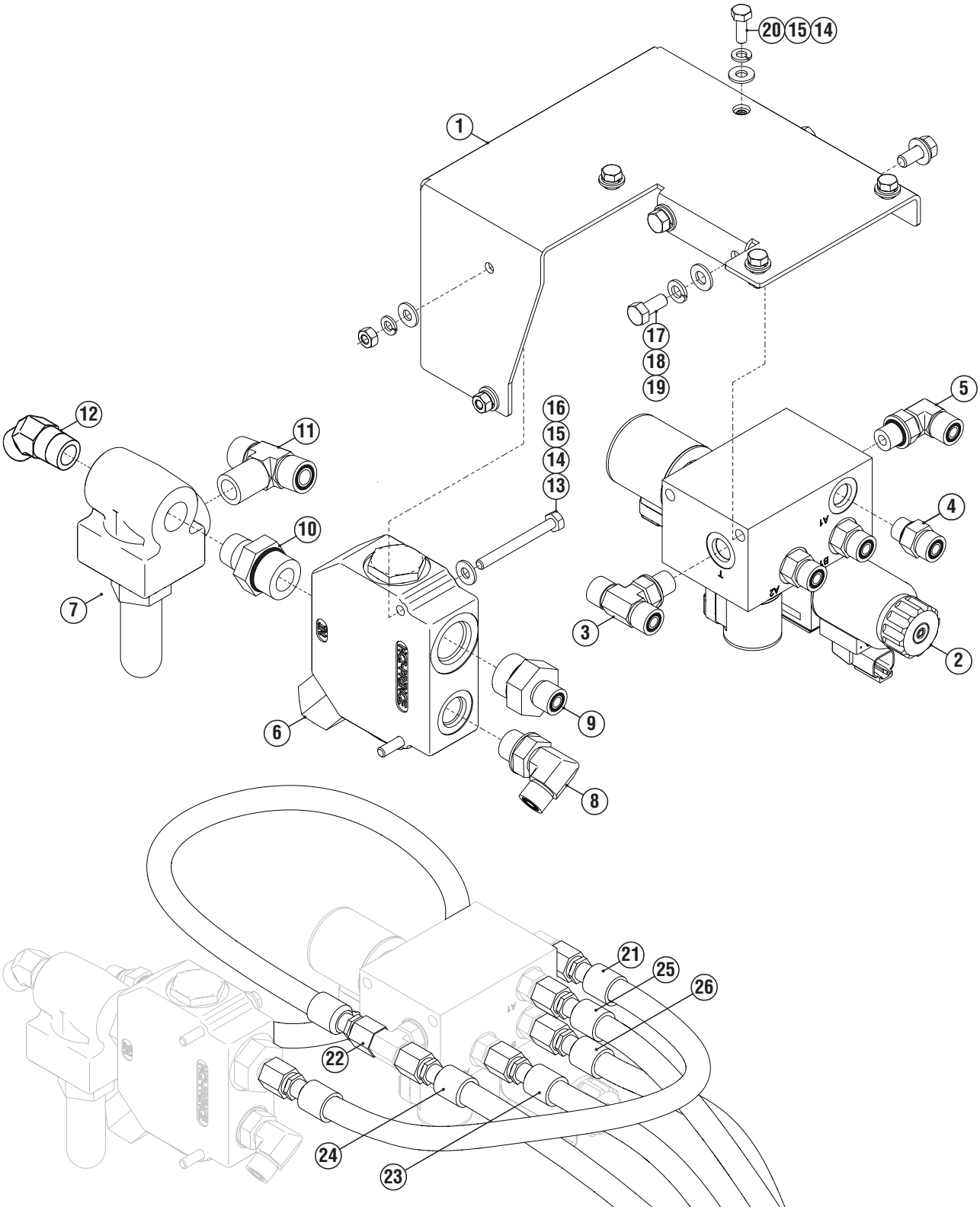


Figure 7-43. OPTION - Curb Brush, Hydraulics (1 of 2)

OPTION - Curb Brush, Hydraulics (1 of 2)

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023225	1	Option - Curb Brush Assembly	Not All Items Shown
ASSY	1020531	1	Assembly, Curb Brush Controls	Includes 1-13
1	1016688	1	Mounting Bracket, Curb Brush Valve	
ASSY	1020532	1	Assembly, Curb Brush Valve w/ORFS Adapters	Includes 2-5
2	1011734	1	Manifold, Hydraulic, 3-Valve w/Internal Checks	
3	FS6803-06-06-06-NWO-FG	1	Adapter, Branch Tee, -6 ORFS/-6 ORFS/-6 O-Ring	
4	FS6400-06-06-O	3	Adapter, Straight, -6 ORFS/-6 O-Ring	
5	FS6801-06-06-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
ASSY	1020533	1	Assembly, Flow Divider & Relief Valve w/ORFS Adapters	Includes 6-12
6	982959	1	Flow Divider, Fixed, w/Pressure Relief	
7	500080	1	Valve, Relief, 2000 PSI	
8	FS6801-06-08-NWO-FG	1	Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	
9	FS6400-06-12-O	1	Adapter, Straight, -6 ORFS/-12 O-Ring	
10	6401-12-8-O	1	Adapter, Straight, -12 O-Ring/-8 NPTF	
11	FS2601-08-08-08-FG	1	Adapter, Branch Tee, -8 ORFS/-8 ORFS/-8 NPT	
12	FS2503-08-08-FG	1	Adapter, Elbow, 45°, -8 ORFS/-8 NPTF	
13	100-4-20-40-5	2	CSHH, 1/4-20 x 2.50, GR5	
14	300-4	8	Washer, Flat, SAE, 1/4	
15	302-4	6	Washer, Lock, 1/4	
16	200-4-20-5	2	Nut, Hex, 1/4-20, GR5	
17	100-5-18-12-5F	4	CSHH, 5/16-18 x .75, GR5, FT	
18	302-5	4	Washer, Lock, 5/16	
19	300-5	4	Washer, Flat, SAE, 5/16	
20	100-4-20-12-5F	4	CSHH, 1/4-20 x .75, GR5, FT	
21	F387TCJCJC060606-22	1	Hose Assembly, Flow Divider Excess Flow to Curb Brush Valve P	Tag# 1
22	F387TCJCJS060806-32	1	Hose Assembly, Curb Brush Valve T to Relief Valve Return	Tag# 2
23	F387TCJCJC060606-146	1	Hose Assembly, Curb Brush Valve, A2 to Lift Cylinder Rod End	Tag# 6
24	F387TCJCJC060606-136	1	Hose Assembly, Curb Brush Valve T to Lift Cylinder Back End	Tag# 7
25	F387TCJCJC060606-184	1	Hose Assembly, Curb Brush Valve A1 to Brush Motor A	Tag# 8
26	F387TCJCJC060606-184	1	Hose Assembly, Curb Brush Valve B1 to Brush Motor B	Tag# 9

OPTION - CURB BRUSH, HYDRAULICS (2 OF 2)

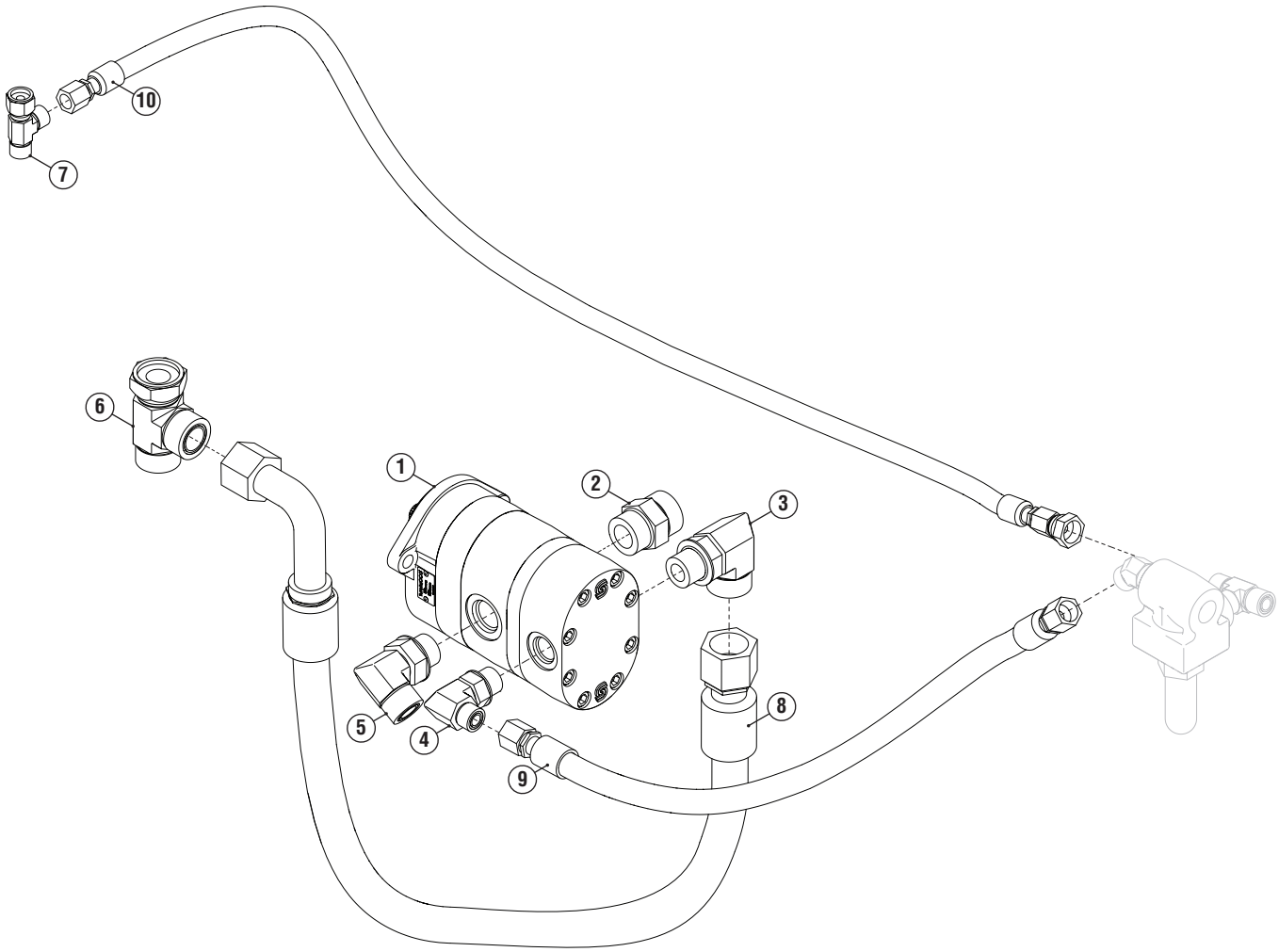


Figure 7-44. OPTION - Curb Brush, Hydraulics (2 of 2)

OPTION - Curb Brush, Hydraulics (2 of 2)

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023225	1	Option - Curb Brush Assembly	Not All Items Shown
ASSY	1020530	1	Assembly, Pump w/ORFS Adapters	Includes 1-5
1	1008352	1	Pump, Hydraulic, Tandem	
2	FS6400-16-16-O	1	Adapter, Straight, -16 ORFS/-16 O-Ring	
3	FS6801--16-12-NWO-FS	1	Adapter, Elbow, 90°, -16 ORFS/-12 O-Ring	
4	FS6801--08-10-NWO-FS	1	Adapter, Elbow, 90°, -8 ORFS/-10 O-Ring	
5	FS6801--12-12-NWO-FS	1	Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	
6	FS6602-16-16-16-FG	1	Adapter, Run Tee, Swivel, -16 ORFS/-16 ORFS/-16 ORFS	
7	FS6602-06-06-06-FG	1	Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-6 ORFS	
8	F387TCJCJ9161616-35	1	Hose Assembly, Hydraulic Tank to Aux Pump Inlet	Tag# 3
9	F387TCJCJC080808-21	1	Hose Assembly, Aux Pump Rear Outlet to Relief Valve Inlet	Tag# 4
10	F387TCJCJC060606-45	1	Relief Valve Out to Hydraulic Tank	Tag# 5

OPTION - SNOW PLOW

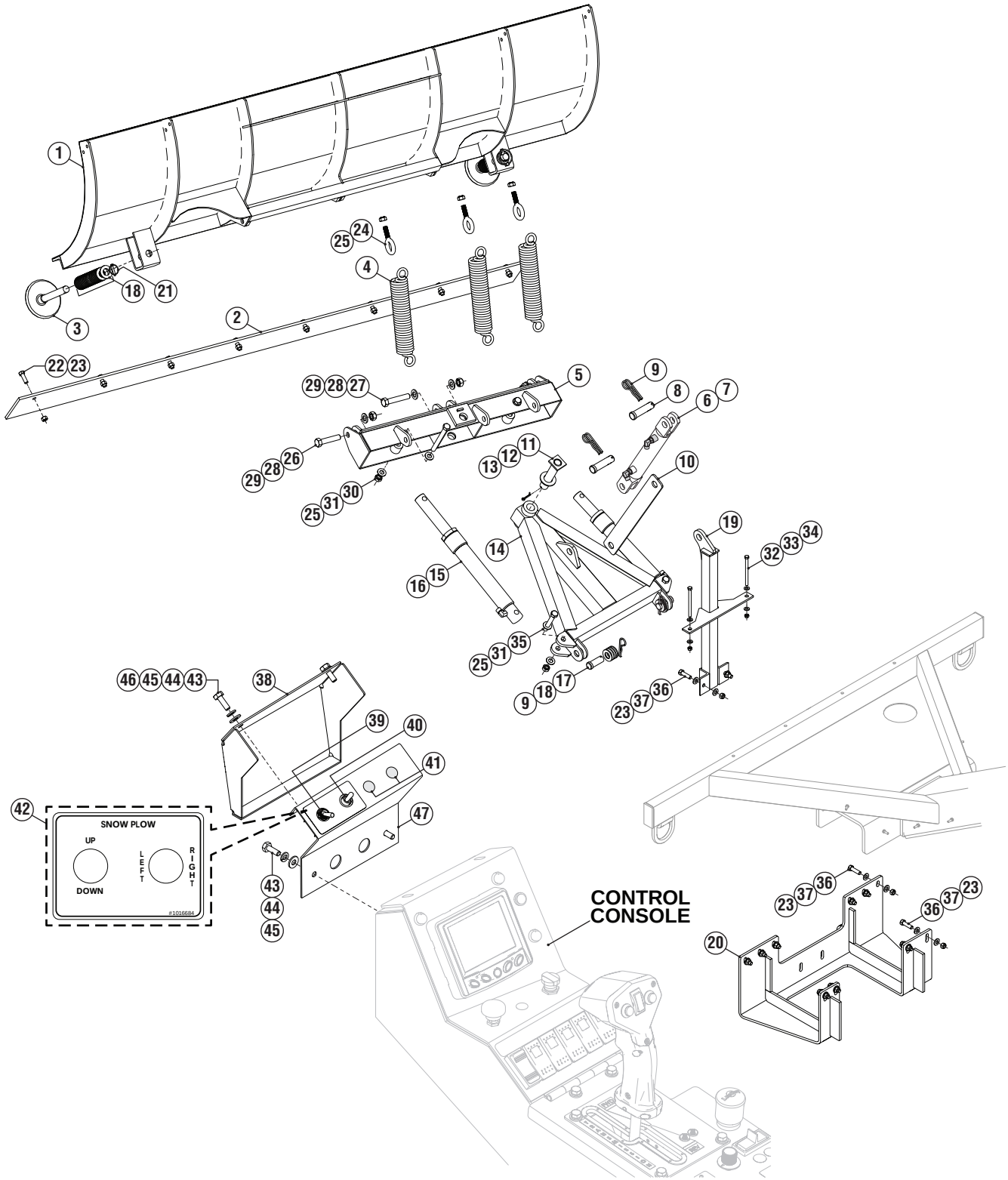


Figure 7-45. OPTION - Snow Plow

OPTION - Snow Plow

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023226		Option - Snow Plow, ORFS	Not All Items Shown
1	1019745	1	Snow Plow Blade - Standard Duty, 7'-6"	
2	38224-03	1	Snow Plow Cutting Edge, 7'-6"	
3	38224-05	2	Assembly, Plow Shoe	Includes 18,21
4	38224-32	3	Trip Spring	
5	28718SRV	1	Front Bracket	
ASSY	1020527	1	Assembly, Lift Cylinder w/ORFS Adapters	Includes 6-9
6	610110	1	Hydraulic Cylinder, 2.00 x 8.00 x 1.00 Rod	
-	610110-01	A/R	Seal Kit for 610110 Hydraulic Cylinder	Repair Part
7	FS6801-06-06-NOW-FG	2	Adapter Elbow, 90°, -6 ORFS/-6 O-Ring	Not Shown
8	1016641	2	Clevis Pin, 1.00 x 3.25 w/ Second Hole	
9	870307	6	Cotter Pin, .177, 7 GA	
10	28948	1	Hydraulic Cylinder Lock	
11	28758SRV	1	Pivot Pin	
12	300-20	1	Washer, Flat, SAE, 1-1/4	
13	80336	1	Cotter Pin, .188 x 1.50	
14	1016646	1	Snow Plow Lift Frame	
ASSY	1020528	2	Assembly, Swing Cylinder w/ORFS Adapters	Includes 15-16
15	38224-26	2	Cylinder, Hydraulic, Snow Plow	
-	TBD	A/R	Seal Kit for 38224-26	Repair Part
16	FS2501-06-06-FG x 078	2	Adapter Elbow, 90°, -6 ORFS/-6 NPTF, .078 Orifice	
17	1011476	2	Clevis Pin, 1.00 x 2.50 HDM	
18	300-16	A/R	Washer, Flat, SAE, 1	
19	1016647	1	Snow Plow Lift Mount	
20	1016645	1	Snow Plow Mount	
21	989272-21	2	Lynch Pin, 1/4	
22	106-8-13-24-5	8	Carriage Bolt, 1/2-13 x 1.50, GR5	
23	204-8-13-5	20	Nut, Lock, Stover, 1/2-13, GR5	
24	38967	3	Eye bot, 5/8-11 x 1.75 x 1.38 ID	
25	204-10-11-5	7	Nut, Lock, Stover, 5/8-11, GR5	
26	100-12-10-56-5	2	CSHH, 3/4-10 x 3.50, GR5	
27	100-12-10-64-5	1	CSHH, 3/4-10 x 4.00, GR5	
28	300-12	4	Washer, Flat, SAE, 1	
29	204-12-10-5	3	Nut, Lock, Stover, 3/4-10, GR5	
30	100-10-11-96-5	2	CSHH, 5/8-11 x 6.00, GR5	
31	300-10	12	Washer, Flat, SAE, 5/8	
32	100-6-16-80-5	2	CSHH, 3/8-16 x 5.00, GR5	
33	300-6	4	Washer, Flat, SAE, 3/8	
34	217-6-16	2	Nut, Lock, Flexloc, 3/8-16	
35	100-10-11-64-5	2	CSHH, 5/8-11 x 4.00, GR5	

OPTION - SNOW PLOW (CONT.)

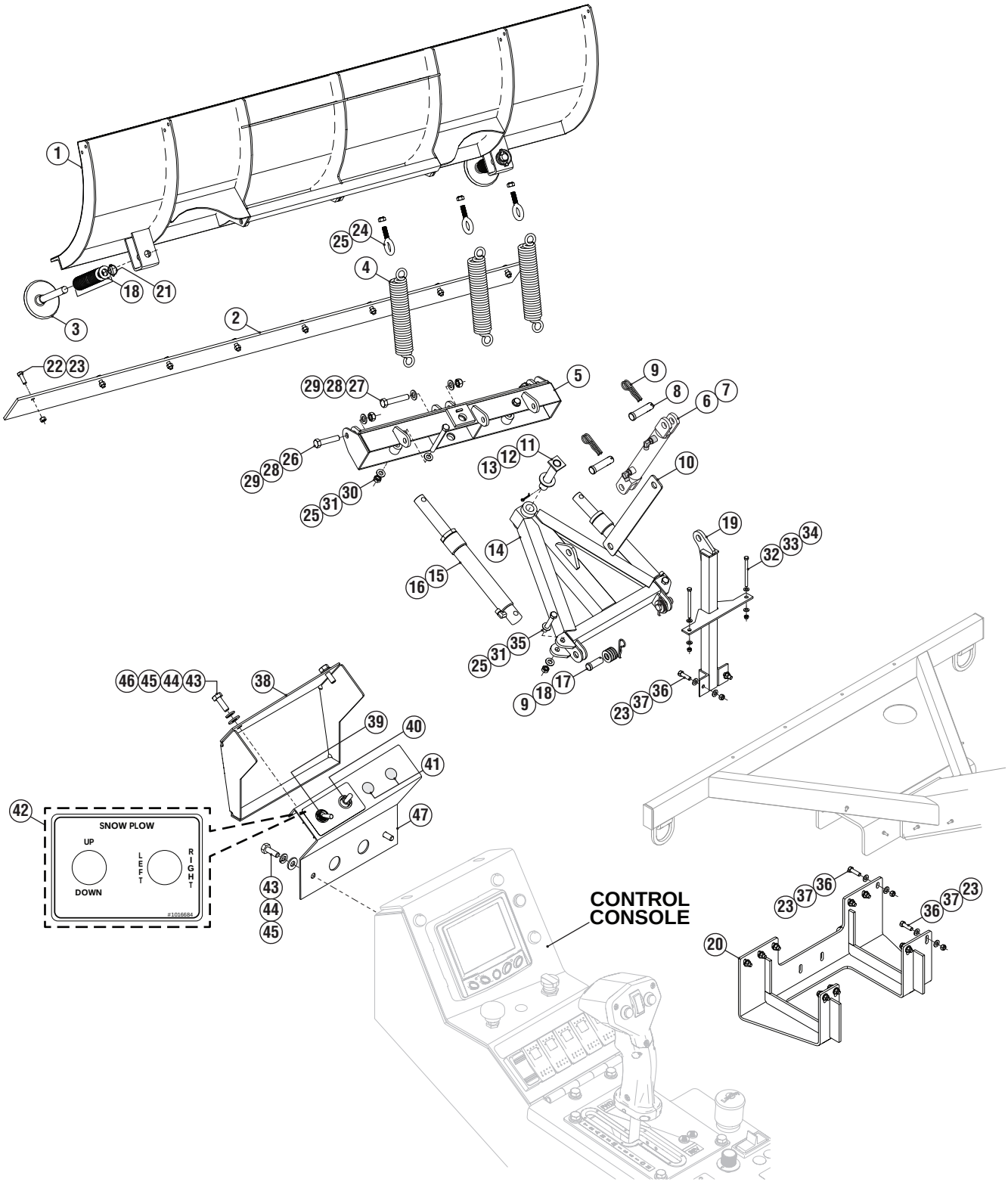


Figure 7-45. OPTION - Snow Plow (Cont.)

OPTION - Snow Plow (Cont.)

Item No.	Part Number	Qty.	Description	Remarks
36	100-8-13-28-5F	2	CSHH, 1/2-13 x 1.75, GR5, FT	
37	300-8	20	Washer, Flat, SAE, 1/2	
38	1016637	1	Control Box	
39	37516	1	Toggle Switch, SPDT, 3 Position, MOM, CO	
40	38961	1	Toggle Switch, DPDT, 3 Position	
41	35136-4	2	Flush Mount Hole Plug, .500	
42	1016684	1	Decal, Control Box, Snow Plow	
43	100-4-20-12-5	4	CSHH, 1/4-20 x .75, GR5	
44	302-4	4	Washer, Lock, 1/4	
45	300-4	8	Washer, Flat, SAE, 1/4	
46	200-4-20-5	4	Nut, Hex, 1/4-20, GR5	Not Shown
47	1021739	1	Control Panel, Curb Brush/Snow Plow	
48	1016080	1	Harness, Valves, Snow Plow	Not Shown
49	1017667	1	Harness, Controls, Snow Plow	Not Shown

OPTION - SNOW PLOW, HYDRAULICS

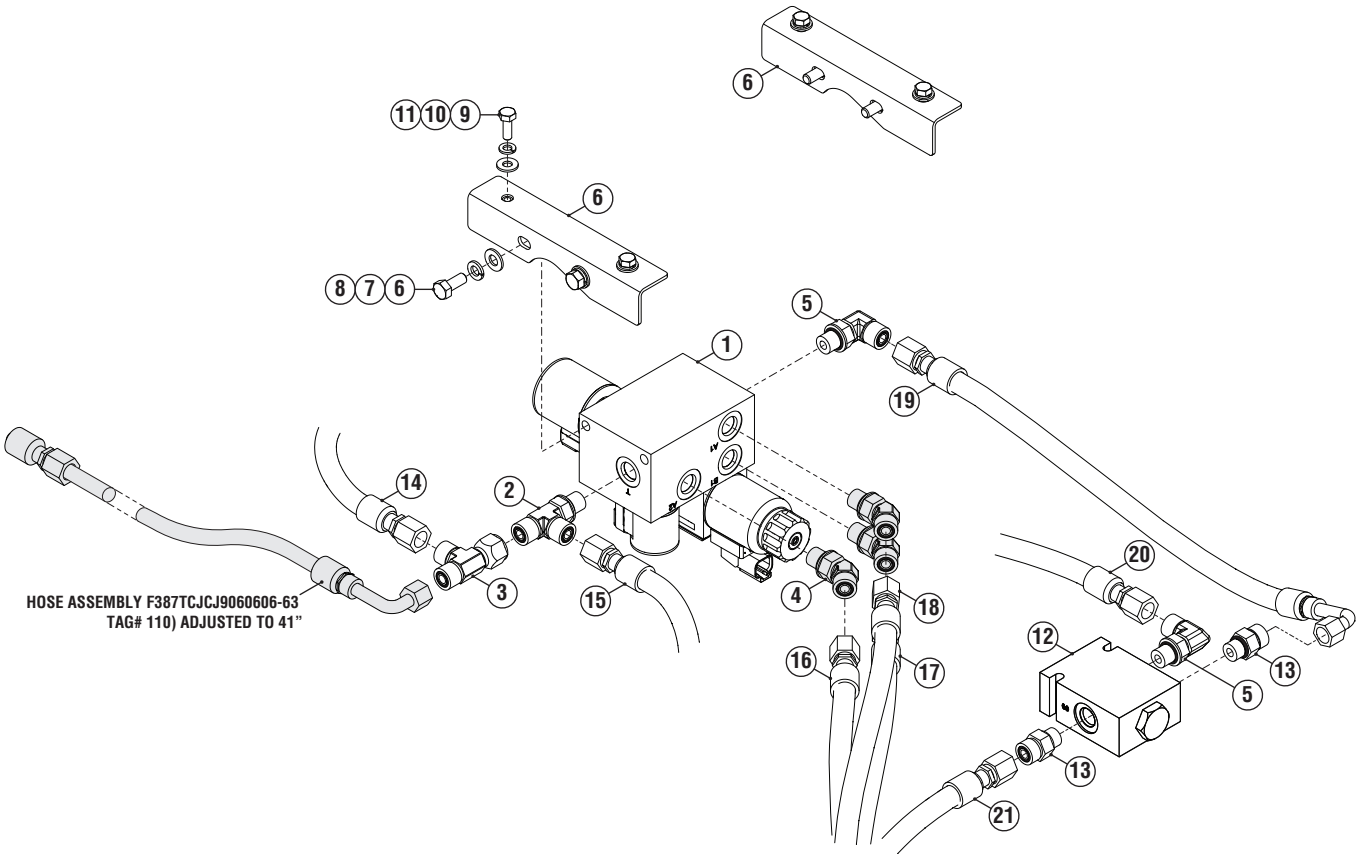


Figure 7-46. OPTION - Snow Plow, Hydraulics

OPTION - Snow Plow, Hydraulics

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023226	1	Option - Snow Plow, ORFS	Not All Items Shown
ASSY	1020526	1	Assembly, Valve w/ORFS Adapters, Snow Plow	Includes 1-11
1	1011734	1	Manifold, Hydraulic, 3-Valve w/Internal Checks	
2	FS6804-06-06-06-NWO-FG	1	Adapter, Run Tee, -6 ORFS/-6 O-Ring/-6 ORFS	
3	FS6602-06-06-06-FG	1	Adapter, Run Tee, -6 ORFS/-6 ORFS/-6 ORFS	
4	FS6802-06-06-NWO-FG	3	Adapter, Elbow, 45°, -6 ORFS/-6 O-Ring	
5	FS6801-06-06-NWO-FG	2	Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	
6	1008089SRV	2	Mount, Valve Block Assembly	
6	100-5-18-12-5F	4	CSHH, 5/16-18 x .75, GR5, FT	
7	302-5	4	Washer, Lock, 5/16	
8	300-5	4	Washer, Flat, SAE, 5/16	
9	100-4-20-12-5F	4	CSHH, 1/4-20 x .75, GR5, FT	
10	302-4	4	Washer, Lock, 1/4	
11	300-4	4	Washer, Flat, SAE, 1/4	
ASSY	1024785	1	Assembly, Flow Divider w/ORFS Adapters	Includes 5, 12-13
12	1022694	1	Flow Divider, 4 GPM, 5000 PSI	
13	FS6400-06-06-O	2	Adapter, Straight, -6 ORFS/-6 O-Ring	
KIT	1020525	1	Hose Kite, Hydraulic, Snow Plow Option	Includes 14-21
14	F387TCJCJ7060606-33	1	Hose Assembly, Main Manifold T to Snow Plow Manifold T	Tag# 8
15	F387TCJCJC060606-130	1	Hose Assembly, Lift Cylinder back End to Hydraulic Tank	Tag# 2
16	F387TCJCJC060606-128	1	Hose Assembly, Snow Plow Valve A2 to Lift Cylinder Rod End	Tag# 3
17	F387TCJCJC060606-142	1	Hose Assembly, Snow Plow Valve B1 to Right Pivot Cylinder	Tag# 5
18	F387TCJCJC060606-156	1	Hose Assembly, Snow Plow Valve A1 to Left Pivot Cylinder	Tag# 4
19	F387TCJCJ9060606-20	1	Hose Assembly, Flow Divider 4 to Snow Plow Manifold P	Tag# 7
20	F387TCJCJ9060606-26	1	Hose Assembly, Flow Divider 2 to Main Manifold P	Tag# 6
21	F387TCJCJC060606-17	1	Hose Assembly, Steering Motor Aux to Flow Divider 3	Tag# 113

OPTION - FRONT FENDERS

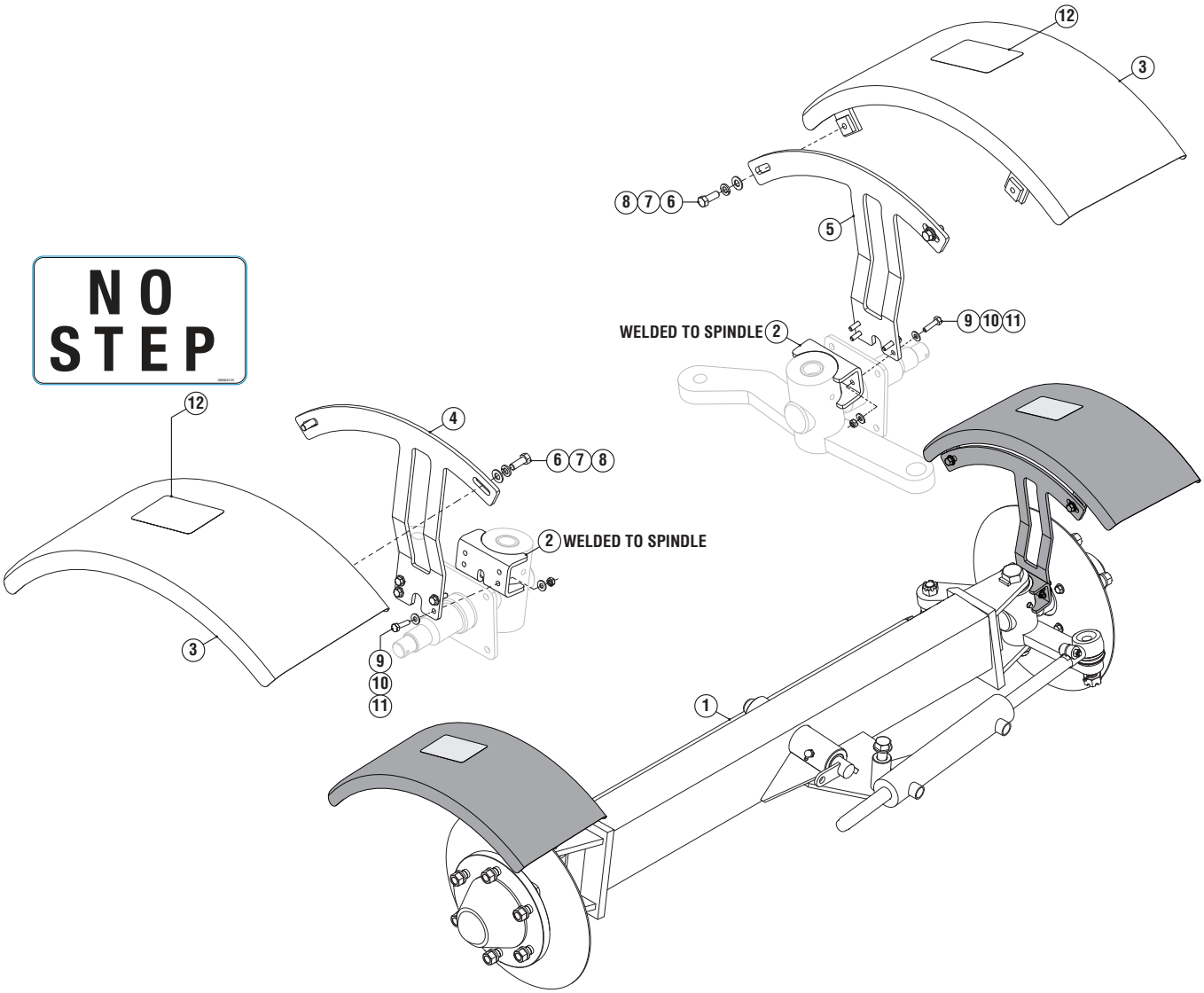


Figure 7-47. OPTION - Front Fenders

OPTION - Front Fender

Item No.	Part Number	Qty.	Description	Remarks
OPT	1008024	1	Option - Front Fender	Includes 1-12
ASSY	1024307	1	Assembly, Axle, Front Fenders	Includes 1-2
1	1009423	1	Assembly, Steering Axle	See Fig. 7-6 for Parts Breakdown
2	1008020	2	Plate, Spindle Mount, Fender Option	
3	1008025	2	Fender Weldment	
4	1008023	1	Plate, RH Fender Bracket, Fender Option	
5	1008021	1	Plate, LH Fender Bracket, Fender Option	
6	100-6-24-16-5F	4	CSHH, 3/8-24 x 1.00, GR5, FT	
7	302-6	4	Washer, Lock, 3/8	
8	300-6	4	Washer, Flat, SAE, 3/8	
9	100-4-20-16-5F	8	CSHH, 1/4-20 x 1.00, GR5, FT	
10	300-4	16	Washer, Flat, SAE, 1/4	
11	204-4-20-5	8	Nut, Lock, Stover, 1/4-20, GR5	
12	1000623-01	2	Decal, Info, No Step	

OPTION - LIGHTBAR

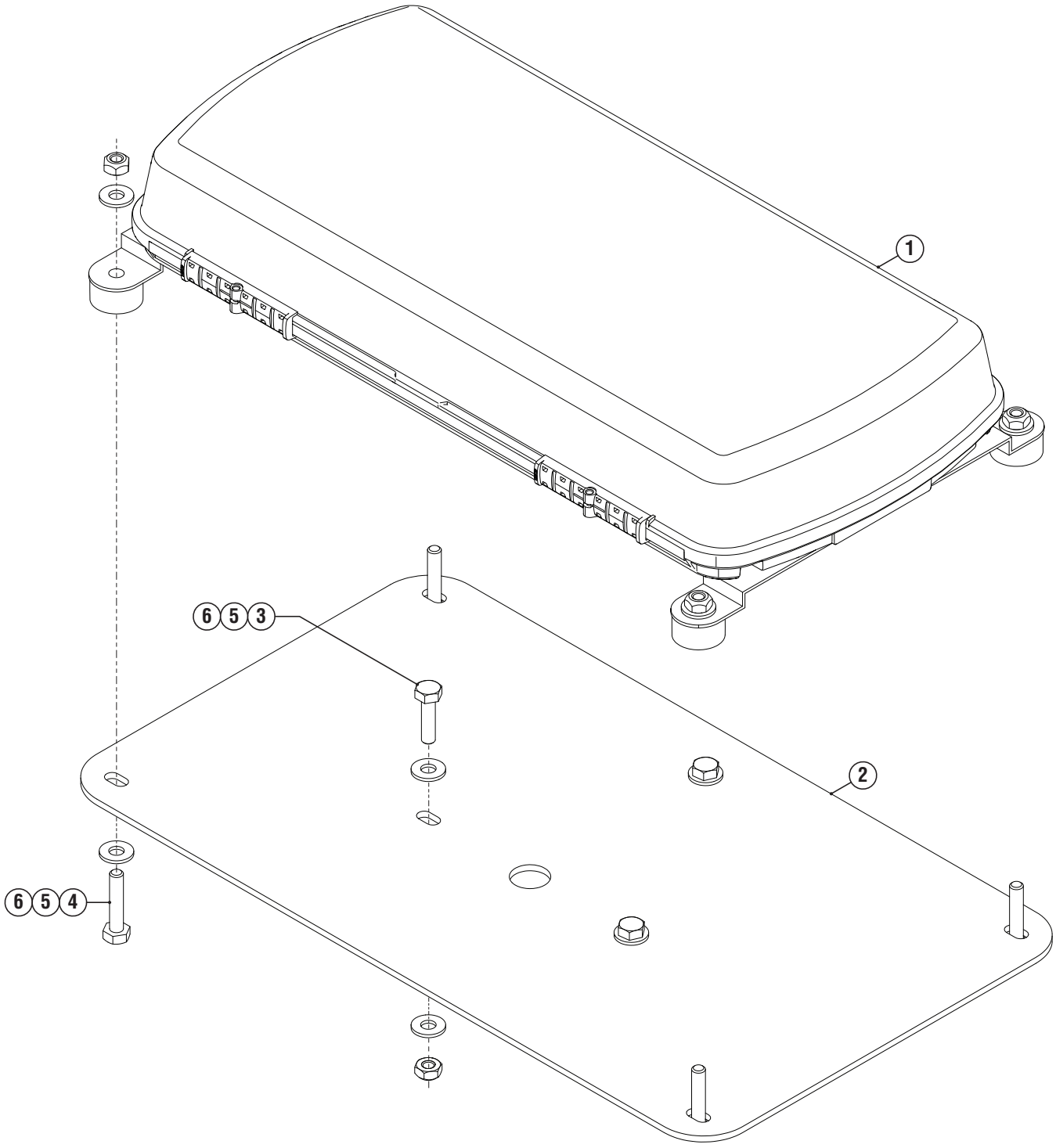


Figure 7-48. OPTION - Lightbar

OPTION - Lightbar

Item No.	Part Number	Qty.	Description	Remarks
OPT	1018575	1	Option - Kit, Lightbar, LED, 16"	
1	1018573	1	Amber Lightbar, LED, 16"	
2	1018574	1	Mount, Lightbar, LED, 16"	
3	100-4-20-16-5F	3	CSHH, 1/4-20 x 1.00, GR5, FT	
4	100-4-20-20-5	4	CSHH, 1/4-20 x 1.25, GR5	
5	300-4	14	Washer, Flat, SAE, 1/4	
6	204-4-20-5	7	Nut, Lock, Stover, 1/4-20, GR5	

OPTION - EXTENDED AIR CLEANER

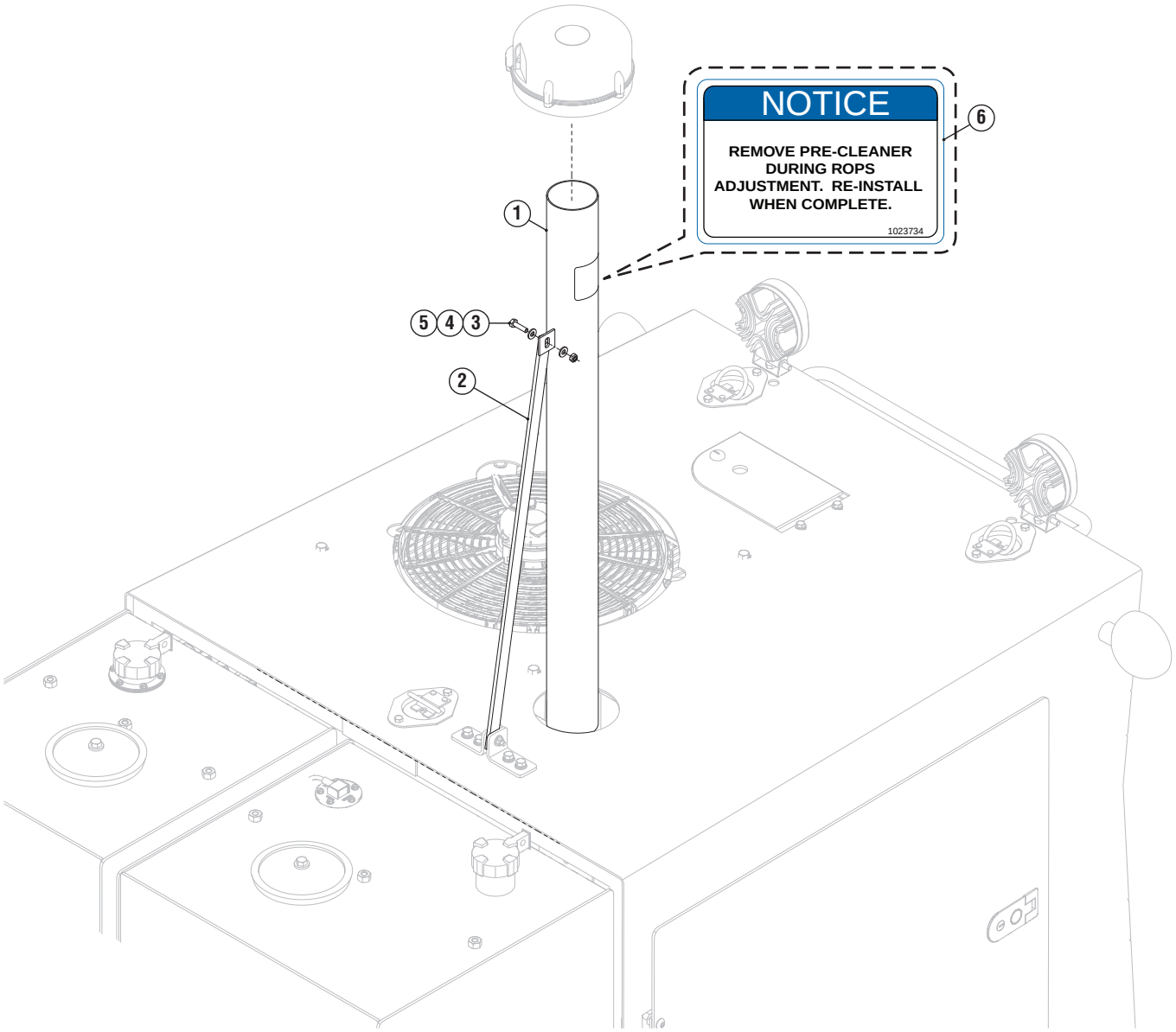


Figure 7-49. OPTION - Extended Air Cleaner

OPTION - Extended Air Cleaner

Item No.	Part Number	Qty.	Description	Remarks
OPT	1023482	1	Option - Extended Air Cleaner	
1	1023083	1	Air Intake, Extended	
2	1023082	1	Bar, Intake Support, Extended	
3	100-4-20-16-5	1	CSHH, 1/4-20 x 1.00, GR5	
4	300-4	2	Washer, Flat, SAE, 1/4	
5	204-4-20-5	1	Nut, Lock, Stover, 1/4-20, GR5	
6	1023734	1	Decal, Temporary, Remove Pre-Cleaner	

OPTION - SPARE TIRE MOUNT

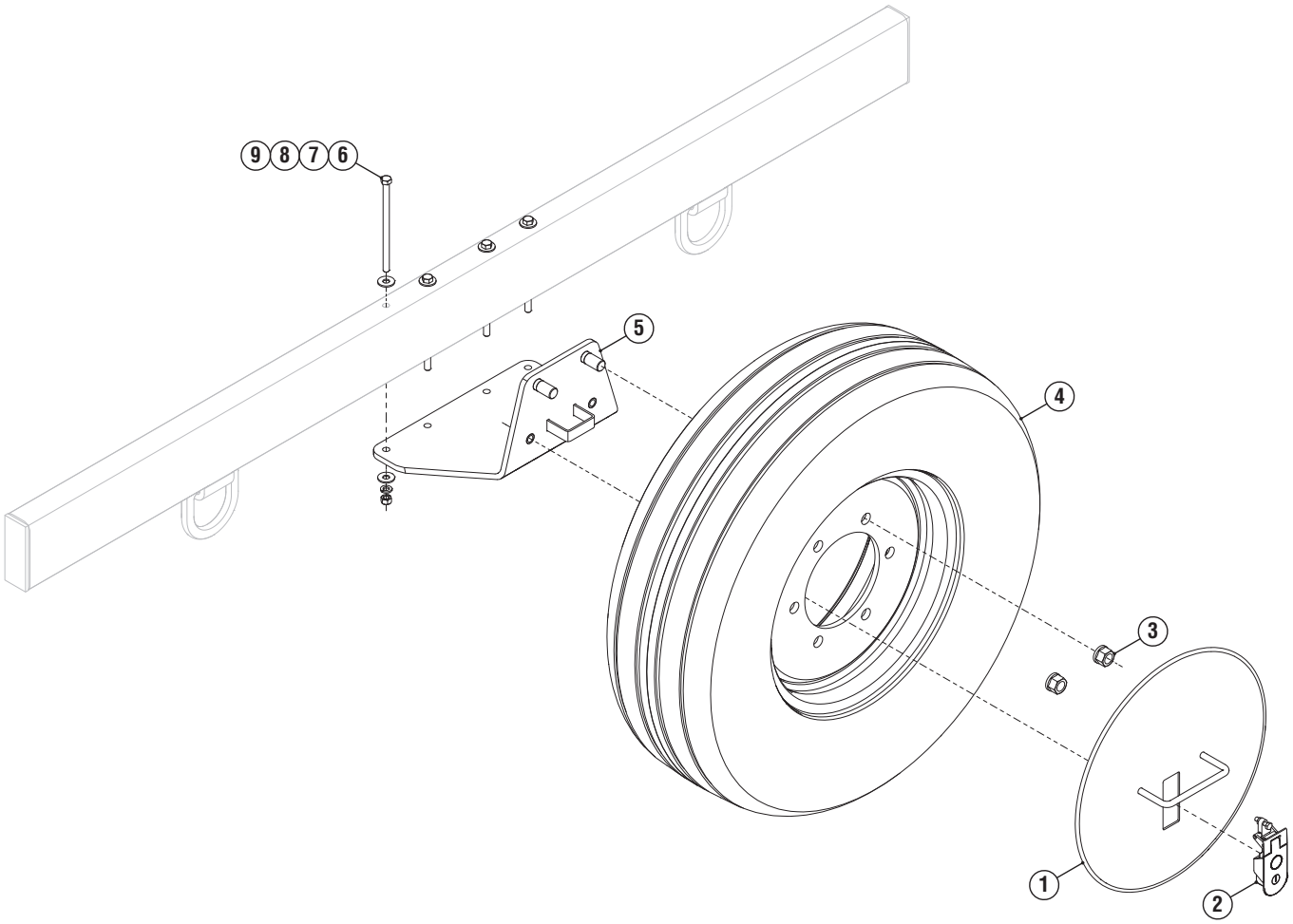


Figure 7-50. OPTION - Spare Tire Mount

OPTION - Spare Tire Mount

Item No.	Part Number	Qty.*	Description	Remarks
OPT	1011742	1	Option - Spare Tire	
1	1017980SRV	1	Assembly, Vandalism Cover, Spare Tire	
2	980460	1	Lever Latch	
-	35560	A/R	Replacement Key	Not Shown
3	1009294	2	Lug Nut, M18X1.5	
4	1009432	1	Assembly, Tire & Wheel	
5	1011743	1	Mount, Spare Tire	
6	100-6-16-104-8	4	CSHH, 3/8-16 x 6.50, GR8	
7	311-6	8	Washer, Hardened, USS, 3/8	
8	302-6	4	Washer, Lock, 3/8	
9	200-6-16-8	4	Nut, Hex, 3/8-16, GR8	

DECAL SCHEDULE

Item	Qty.	Part Number	Description
KIT	**1**	**1022713**	**DECAL KIT - SAFETY & OPERATIONS**
	2	1013789-06	DECAL, OPER, TURN SIGNAL
	1	983125-23	DECAL, OPER, BATTERY
	1	39141-21	DECAL, DIESEL FUEL
	1	1021704-03	DECAL, CAUTION, HYDRAULIC OIL
	1	1021704-14	DECAL, OPER, FUSE PANEL
	1	1010090	DECAL, ULTRA LOW SULFUR FUEL ONLY
	2	940002	DECAL, DECO, FLAG
	1	1010089	DECAL, CJ-4 OIL ONLY
	1	1010091	DECAL, CERTIFIED EMISSION ENGINE INSTALLATION
	1	1013789-05	DECAL, KICK GLASS
	2	1015036-02	DECAL, DANGER, ROLL-OVER HAZARD
	2	983125-10	DECAL, WARNING, SEVER HAZARD, FAN
	1	1021704-05	DECAL, OPER, FUSE PANEL
	1	1021704-04	DECAL, OPER, BRUSH DRIVE
	1	1010608	DECAL, INFO, MASTER CYLINDER
	1	39141-17	DECAL, PARK BRAKE
	1	39141-24	DECAL, DANGER, RUNAWAY VEHICLE
	1	39141-19	DECAL, CAUTION, ALARM
	1	1005474	DECAL, NO UNATTENDED
	1	1021704-07	DECAL, WARNING, ROPS
	1	1021704-01	DECAL, WARNING, EQUIPMENT
	1	1021704-02	DECAL, WARNING, SLIP/FALL
	2	1008622	DECAL, PINCH/ENTANGLEMENT
	1	988634	DECAL, NOTICE OPERATOR'S MANUAL
	1	1013789-10	DECAL, OPER, HEATER SHUTOFF
	1	1015036-04	DECAL, GREASE DAILY
	1	59020040	DECAL: PRESSURE HAZARD WARNING
	1	1021704-09	DECAL, OPER, BRUSH DOWN
	1	1021704-06	DECAL, BRUSH CONTROL
	1	1015036-07	DECAL, ADJUSTMENT SCREWS
	1	1013275	DECAL, NOTICE, HYDRAULIC DYE
	1	39141-18	DECAL, OPER, WATER TANK
	1	1021704-10	DECAL, OPER, BRUSH SPEED
	1	1013789-01	DECAL, OPER, LIGHT SWITCHES/HORN
	1	1013789-07	DECAL, OPER, PARK BRAKE
	1	1021704-08	DECAL, OPER, TWO SPEED
	1	1021704-16	DECAL, OPER, HIGH GEAR
	1	1021704-17	DECAL, TEST PORT, CHARGE PRESSURE
	1	1021704-13	DECAL, OPER, DOOR RELEASE
	2	1023749	DECAL, INSPECT/CLEAN RADIATOR

Item	Qty.	Part Number	Description
KIT	**1**	**1015035**	**DECAL KIT - DECO**
	1	1015035-01	DECAL, DECO, RB50 SIDE, RIGHT
	2	1013790-08	DECAL, DECO, ROSCO, LEEBOY COMPANY, 8.5"
	3	1015035-04	DECAL, DECO, RB50
	1	1015035-02	DECAL, DECO, RB50 SIDE, LEFT
	1	1015035-05	ANTI-SKID, CAB, CENTER

HYDRAULIC HOSE SCHEDULE - STANDARD UNIT

Hose No.	Qty.	Part Number	Description
GRP		1022904	MAIN HOSE KIT
1	1	F797TCJCJC161616-19	PUMP A TO MOTOR A
2	1	F797TCJCJ9161616-30	PUMP B TO MOTOR B
4	1	F387TCJCJC060606-14	PUMP SERVO GAGE M4 TO DESTROKE VALVE
5	1	F387TCJCJC060606-26	PUMP SERVO GAGE M5 TO DESTROKE VALVE
6	1	F387TCJCJC121212-19	PUMP CASE OUTLET TO TANK
7	1	F387TCJCJ7121212-71-AS-B-19-67	COOLER TO CHARGE PUMP INLET
8	1	F387TCJCJ7121212-52	MOTOR CASE DRAIN TO FILTER
9	1	F387TCJCJ5121212-44-AS-B-19-40.75	FILTER TO COOLER
10	1	F387TCJCJ9121212-27	COOLER TO TANK
101	1	F387TCJCJ1121212-113-AS-B-19-35	MOTOR VALVE A TO BRUSH MOTOR B
102	1	F387TCJCJ9121212-32	BRUSH SPEED VALVE 1 TO MOTOR VALVE
105	1	F387TCJCJ9060606-26	PUMP GAGE PORT TO SOLENOID VALVE 3
106	1	F387TCJCJ1060606-34	SOLENOID VALVE 1 TO MOTOR X1
107	1	F387TCJCJ7060606-35	MANIFOLD A1 TO PIVOT CYLINDER ROD END
108	1	F387TCJCJ5060606-77-AS-B-13-74.75	MANIFOLD T TO LIFT CYLINDER
109	1	F387TCJCJC060606-46	MANIFOLD B1 TO PIVOT CYLINDER
110	1	F387TCJCJ9060606-63	MANIFOLD T TO MOTOR VALVE
111	1	F387TCJCJC060606-68-AS-B-13-66.25	MANIFOLD A2 TO LIFT CYLINDER
112	1	F387TCJCJC060606-66-AS-B-13-25	AUX PUMP TO STEERING MOTOR IN
113	1	F387TCJCJ9060606-17-AS-B-13-15	MANIFOLD P TO STEERING MOTOR AUX
119	1	K811HTJCJ9161616-38	TANK TO PUMP
121	1	F387TCJCJ5121212-110-AS-B-19-39	MOTOR VALVE B TO BRUSH MOTOR A
122	1	F387TCJ5J5060606-24-0	MOTOR VALVE TO NEEDLE VALVE
123	1	F387TCJCJ9060606-67	MANIFOLD A2 TO NEEDLE VALVE
124	1	F387TCJCJ9060606-117-AS-B-13-113.75	STEERING MOTOR RT TO STEER CYLINDER
125	1	F387TCJCJC060606-106-AS-B-13-104.25	STEERING MOTOR LT TO STEER CYLINDER
130	1	F387TCJCJ9060606-50	STEERING MOTOR OUT TO MOTOR VALVE
134	1	F387TCJCJ7121212-60-AS-B-19-25	MOTOR VALVE TO FILTER
137	1	F387TCJCJC060606-20	SOLENOID VALVE 2 TO TANK
140	1	F387TCJCJC121212-36	AUX PUMP TO BRUSH SPEED VALVE 1
141	1	F387TCJCJ9121212-29	BRUSH SPEED VALVE 2 TO TANK

HYDRAULIC HOSE SCHEDULE - OPTIONS

Hose No.	Qty.	Part Number	Description
GRP		1023243	FNR PEDAL HOSE KIT
1	1	F387TCJCJC040404-94	PUMP X1 TO PEDAL 1
2	1	F387TCJCJC040404-95	PUMP X2 TO PEDAL 2
3	1	F387TCJCJC040604-85	PUMP GAGE PORT TO PEDAL P
4	1	F387TCJCJC060606-85	PEDAL T TO TANK

Hose No.	Qty.	Part Number	Description
GRP		1020525	SNOW PLOW HOSE KIT
2	1	F387TCJCJC060606-130	LIFT CYLINDER BACK END TO TANK
3	1	F387TCJCJC060606-128	SNOW PLOW VALVE A2 TO LIFT CYLINDER ROD END
4	1	F387TCJCJC060606-156	SNOW PLOW VALVE A1 TO LEFT PIVOT CYLINDER
5	1	F387TCJCJC060606-142	SNOW PLOW VALVE B1 TO RIGHT PIVOT CYLINDER
6	1	F387TCJCJ9060606-26	FLOW DIVIDER 2 TO MAIN MANIFOLD P
7	1	F387TCJCJ9060606-20	FLOW DIVIDER 4 TO SNOW PLOW MANIFOLD P
8	1	F387TCJ7JC060606-33	MAIN MANIFOLD T TO SNOW PLOW MANIFOLD T
113	1	F387TCJCJC060606-17	STEERING MOTOR AUX TO FLOW DIVIDER 3

Hose No.	Qty.	Part Number	Description
GRP		1020529	CURB BRUSH HOSE KIT
1	1	F387TCJCJC060606-22	HOSE, HYDRAULIC, FLOW DIVIDER EXCESS FLOW TO CURB BRUSH VALVE P
2	1	F387TCJCJS060806-32	HOSE, HYDRAULIC, CURB BRUSH VALVE T TO RELIEF VAVLE RETURN
3	1	F387TCJCJ9161616-35	HOSE, HYDRAULIC, TANK TO AUX PUMP INLET
4	1	F387TCJCJC080808-21	HOSE, HYDRAULIC, AUX PUMP REAR OUTLET TO RELIEF VALVE INLET
5	1	F387TCJCJC060606-45	HOSE, HYDRAULIC, RELIEF VALVE OUT TO TANK
6	1	F387TCJCJC060606-146	HOSE, HYDRAULIC, CURB BRUSH VALVE A2 TO LIFT CYCLINDER ROD END
7	1	F387TCJCJC060606-136	HOSE, HYDRAULIC, CURB BRUSH VALVE T TO LIFT CYLINDER BACK END
8	1	F387TCJCJC060606-184	HOSE, HYDRAULIC, CURB BRUSH VALVE A1 TO BRUSH MOTOR A
9	1	F387TCJCJC060606-184	HOSE, HYDRAULIC, CURB BRUSH VALVE B1 TO BRUSH MOTOR B

ALPHABETICAL PARTS INDEX

Description	Part No.	Figure No.	Item No.
1/2 Low Pressure Hose - Black (Per Foot)	6352	7-41	14
1/2 Low Pressure Hose - Black x 3.00"	REF	7-41	14A
1/2 Low Pressure Hose - Black x 36.00"	REF	7-41	14C
1/2 Low Pressure Hose - Black x 72.00"	REF	7-41	14B
12V Battery, 770 CCA, 950 CA, Group 34	PURCHASE LOCALLY	7-14	20
12V Starter, 2.2KW	1015794-15	7-14	11
14V/110A Alternator	1015794-16	7-14	4
90° Fitting, 08MP-08HB, Poly	70319	7-41	20
-A-			
A/C Thermostat	36745-19	7-31	6
AC Belt Idler	1015794-32	7-14	17
AC Belt Idler	1015794-32	7-30	10
AC Belt Tensioner	1015794-31	7-14	16
AC Belt Tensioner	1015794-31	7-30	11
AC Compressor	1015794-30	7-14	14
AC Main Mounting Bracket	1015794-33	7-14	15
AC Main Mounting Bracket	1015794-33	7-30	8
AC Support Bracket	1015794-34	7-14	18
AC Support Bracket	1015794-34	7-30	9
AC Support Strap	1015794-35	7-14	19
Access Port	1006804	7-9	5
Access Port	1006804	7-10	6
Accuseal Clamp, 2.00"	1015876	7-13	10
Adapter Elbow, 90°, -6 ORFS/-6 NPTF, .078 Orifice	FS2501-06-06-FG x 078	7-45	16
Adapter Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-45	7
Adapter, Branch Tee, -12 ORFS/-12 ORFS/-12 O-Ring	FS6803-12-12-12-NWO-FG	7-17	2
Adapter, Branch Tee, -12 ORFS/-12 ORFS/-12 O-Ring	FS6803-12-12-12-NWO-FG	7-35	4
Adapter, Branch Tee, -6 ORFS/-6 ORFS/-6 O-Ring	FS6803-06-06-06-NWO-FG	7-43	3
Adapter, Branch Tee, -8 ORFS/-8 ORFS/-8 NPT	FS2601-08-08-08-FG	7-43	11
Adapter, Branch Tee, Swivel, -12 ORFS	FS6600-12-12-12-FG	7-16	5
Adapter, Bulkhead, Test Port to -4 IRFS	1019707	7-18	26
Adapter, Cap, ORFS, -16	FS0304-C-16	7-10	26
Adapter, Elbow, 45°, -6 ORFS/-6 O-Ring	FS6802-06-06-NWO-FG	7-46	4
Adapter, Elbow, 45°, -6 ORFS/-6 O-Ring	FS6802-06-06-NWO-FG	7-20	5

Alphabetical Parts Index

Description	Part No.	Figure No.	Item No.
Adapter, Elbow, 45°, -8 ORFS/-8 NPTF	FS2503-08-08-FG	7-43	12
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-46	5
Adapter, Elbow, 90°, -06 ORFS/-06 O-Ring	FS6801-06-06-NWO-FG	7-37	2
Adapter, Elbow, 90°, -12 ORFS/-10 ORFS	FS6801-12-10-NWO-FG	7-35	8
Adapter, Elbow, 90°, -12 ORFS/-10 O-Ring	FS6801-12-10-NWO-FG	7-8	46
Adapter, Elbow, 90°, -12 ORFS/-10 O-Ring	FS6801-12-10-NWO-FG	7-19	4
Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	FS6801-12-12-NWO-FG	7-5	5
Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	FS6801-12-12-NWO-FG	7-10	10
Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	FS6801-12-12-NWO-FG	7-17	5
Adapter, Elbow, 90°, -12 ORFS/-12 O-Ring	FS6801--12-12-NWO-FS	7-44	5
Adapter, Elbow, 90°, -12 ORFS/-16 O-Ring	FS6801-12-16-NWO-FG	7-15	6
Adapter, Elbow, 90°, -12 ORFS/-16 O-Ring	FS6801-12-16-NWO-FG	7-16	14
Adapter, Elbow, 90°, -12 ORFS/-16 O-Ring	FS6801-12-16-NWO-FG	7-37	6
Adapter, Elbow, 90°, -16 ORFS/-12 O-Ring	FS6801--16-12-NWO-FS	7-44	3
Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	FS6801-16-16-NWO-FG	7-5	4
Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	FS6801-16-16-NWO-FG	7-15	7
Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	FS6801-16-16-NWO-FG	7-17	6
Adapter, Elbow, 90°, -16 ORFS/-16 O-Ring	FS6801-16-16-NWO-FG	7-37	7
Adapter, Elbow, 90°, -6 JIC/-4 O-Ring	6804-6-4 NWO	7-9	8
Adapter, Elbow, 90°, -6 ORFS/-6 ORFS	FS6500-06-06-FG	7-35	7
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-10	18
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-15	2
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-18	4
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-20	6
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-28	35
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring	FS6801-06-06-NWO-FG	7-43	5
Adapter, Elbow, 90°, -6 ORFS/-6 O-Ring, .030 Orifice	FS6801-06-06-NWO-FG X 030	7-5	6
Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	FS6801-06-08-NWO-FG	7-6	44
Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	FS6801-06-08-NWO-FG	7-15	11
Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	FS6801-06-08-NWO-FG	7-19	8
Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	FS6801-06-08-NWO-FG	7-37	14
Adapter, Elbow, 90°, -6 ORFS/-8 O-Ring	FS6801-06-08-NWO-FG	7-43	8
Adapter, Elbow, 90°, -6 ORFS--6 O-Ring	FS6801-06-06-NWO-FG	7-7	16
Adapter, Elbow, 90°, -8 ORFS/-10 O-Ring	FS6801--08-10-NWO-FS	7-44	4
Adapter, Elbow, 90°, Swivel, -12 ORFS/-12 ORFS	FS6500-12-12-FG	7-8	48
Adapter, Elbow, 90°, Swivel, -12 ORFS/-12 ORFS	FS6500-12-12-FG	7-10	12
Adapter, Elbow, 90°, Swivel, -12 ORFS/-12 O-Ring	FS6500-12-12-FG	7-19	10

Alphabetical Parts Index

Description	Part No.	Figure No.	Item No.
Adapter, Elbow, 90°, Swivel, -6 ORFS/-6 O-Ring	FS6500-06-06-FG	7-19	5
Adapter, Plug, -6 O-Ring	6408-6-O	7-10	16
Adapter, Plug, O-Ring, -4	6408-4-O	7-9	7
Adapter, Reducer, Straight, -12 FFORX/-6 ORFS	FS2406-12-06	7-17	3
Adapter, Reducer, Straight, -12 FFORX/-6 ORFS	FS2406-12-06	7-19	3
Adapter, Reducer, Straight, -12 FFORX/-6 ORFS	FS2406-12-06	7-35	5
Adapter, Reducer, Straight, -12 FFORX/-8 ORFS	FS2406-12-08	7-35	3
Adapter, Run Tee, -12 ORFS/-12 O-Ring/12 ORFS	FS6804-12-12-12-NWO-FG	7-35	2
Adapter, Run Tee, -12 ORFS/-12 O-Ring/-12 ORFS	FS6804-12-12-12-NWO-FG	7-10	11
Adapter, Run Tee, -12 ORFS/-12 O-Ring/-12 ORFS	FS6804-12-12-12-NWO-FG	7-18	17
Adapter, Run Tee, -12 ORFS/-12 O-Ring/-12 ORFS	FS6804-12-12-12-NWO-FG	7-19	2
Adapter, Run Tee, -12 ORFs/-12 O-Ring/-16 ORFS	FS6804-12-12-16-NWO-FG	7-10	22
Adapter, Run Tee, -4 ORFS/-4 O-Ring/-4 ORFS	FS6804-04-04-04-NWO-FG	7-37	9
Adapter, Run Tee, -6 ORFS/-6 ORFS/-6 ORFS	FS6602-06-06-06-FG	7-46	3
Adapter, Run Tee, -6 ORFS/-6 O-Ring/-6 ORFS	FS6804-06-06-06-NWO-FG	7-20	2
Adapter, Run Tee, -6 ORFS/-6 O-Ring/-6 ORFS	FS6804-06-06-06-NWO-FG	7-37	8
Adapter, Run Tee, -6 ORFS/-6 O-Ring/-6 ORFS	FS6804-06-06-06-NWO-FG	7-46	2
Adapter, Run Tee, Swivel, -16 ORFS/-16 ORFS/-16 ORFS	FS6602-16-16-16-FG	7-44	6
Adapter, Run Tee, Swivel, -6 ORFS	FS6602-06-06-06-FG	7-10	19
Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-4 ORFS	FS-6602-06-06-04-FG	7-18	5
Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-6 ORFS	FS6602-06-06-06-FG	7-19	6
Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-6 ORFS	FS6602-06-06-06-FG	7-35	6
Adapter, Run Tee, Swivel, -6 ORFS/-6 ORFS/-6 ORFS	FS6602-06-06-06-FG	7-44	7
Adapter, Straight, -12 ORFS/-10 O-Ring	FS6400-12-10-O	7-8	47
Adapter, Straight, -12 ORFS/-10 O-Ring	FS6400-12-10-O	7-15	12
Adapter, Straight, -12 ORFS/-10 O-Ring	FS6400-12-10-O	7-37	15
Adapter, Straight, -12 ORFS/-10 O-Rings	FS6400-12-10-O	7-18	21
Adapter, Straight, -12 ORFS/-12 O-Ring	FS6400-12-12-O	7-10	17
Adapter, Straight, -12 ORFS/-12 O-Ring	FS6400-12-12-O	7-15	4
Adapter, Straight, -12 ORFS/-12 O-Ring	FS6400-12-12-O	7-19	11
Adapter, Straight, -12 ORFS/-12 O-Ring	FS6400-12-12-O	7-37	4
Adapter, Straight, -12 ORFS/-12 O-Rings	FS6400-12-12-O	7-18	18
Adapter, Straight, -12 ORFS/-16 O-Ring	FS6400-12-16-O	7-16	4
Adapter, Straight, -12 O-Ring/-8 NPTF	6401-12-8-O	7-43	10
Adapter, Straight, -16 ORFS/-16 O-Ring	FS6400-16-16-O	7-10	25
Adapter, Straight, -16 ORFS/-16 O-Ring	FS6400-16-16-O	7-15	5
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CSHH, 3/8-16 x 9.00, GR5	100-6-16-144-5	7-14	28
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Description	Part No.	Figure No.	Item No.
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Water Tank Cap w/Ring, 1/8 Turn	1012310	7-41	3
Weldment, Cab Rear Wiper Cover	1015612	7-34	2
Weldment, Front Axle	28451SRV	7-6	1
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